Problem 6-1

Determine the force in each member of the truss and state if the members are in tension or compression.

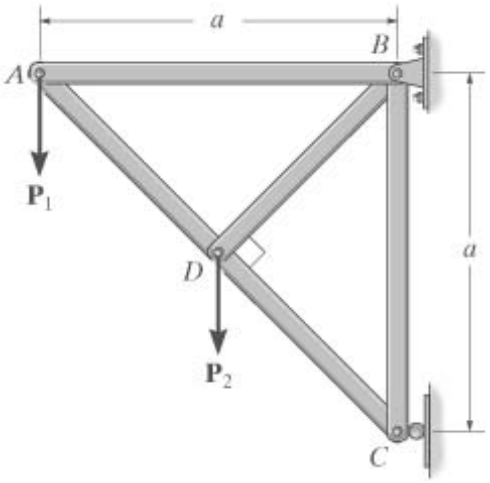
Units Used:

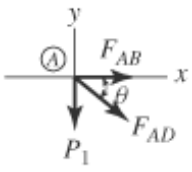
$kN = 10^3\ N$

Given:

$P_1 = 7\ kN$

$P_2 = 7\ kN$





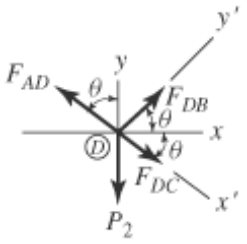
Solution:

$\theta = 45\ deg$

Initial Guesses:

$F_{AB} = 1\ kN \quad F_{AD} = 1\ kN \quad F_{DB} = 1\ kN$

$F_{DC} = 1\ kN \quad F_{CB} = 1\ kN$



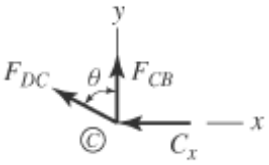
Given

Joint *A*: $F_{AB} + F_{AD}\cos(\theta) = 0$

$-P_1 - F_{AD}\sin(\theta) = 0$

Joint *D*: $F_{DB}\cos(\theta) - F_{AD}\cos(\theta) + F_{DC}\cos(\theta) = 0$

$(F_{AD} + F_{DB} - F_{DC})\sin(\theta) - P_2 = 0$



Joint *C*: $F_{CB} + F_{DC}\sin(\theta) = 0$

$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{DB} \\ F_{DC} \\ F_{CB} \end{pmatrix} = \text{Find}(F_{AB}, F_{AD}, F_{DB}, F_{DC}, F_{CB})$$

$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{DB} \\ F_{DC} \\ F_{CB} \end{pmatrix} = \begin{pmatrix} 7 \\ -9.9 \\ 4.95 \\ -14.85 \\ 10.5 \end{pmatrix} kN$$

Positive means Tension, Negative means Compression

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Problem 6-2

Determine the force in each member of the truss and state if the members are in tension or compression.

Units Used:

$kN = 10^3\ N$

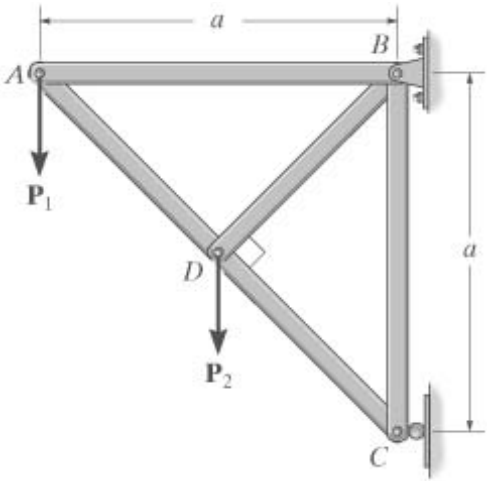
Given:

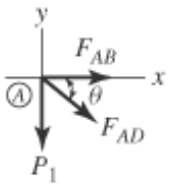
$P_1 = 8\ kN$

$P_2 = 10\ kN$

Solution:

$\theta = 45\ deg$

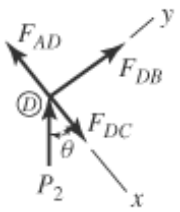




Initial Guesses:

$F_{AB} = 1\ kN \quad F_{AD} = 1\ kN \quad F_{DB} = 1\ kN$

$F_{DC} = 1\ kN \quad F_{CB} = 1\ kN$



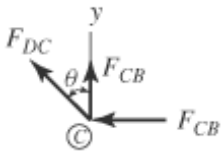
Given

Joint *A*: $F_{AB} + F_{AD}\cos(\theta) = 0$

$-P_1 - F_{AD}\sin(\theta) = 0$

Joint *D*: $F_{DB}\cos(\theta) - F_{AD}\cos(\theta) + F_{DC}\cos(\theta) = 0$

$(F_{AD} + F_{DB} - F_{DC})\sin(\theta) - P_2 = 0$



Joint *C*: $F_{CB} + F_{DC}\sin(\theta) = 0$

$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{DB} \\ F_{DC} \\ F_{CB} \end{pmatrix} = \text{Find}(F_{AB}, F_{AD}, F_{DB}, F_{DC}, F_{CB}) \qquad \begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{DB} \\ F_{DC} \\ F_{CB} \end{pmatrix} = \begin{pmatrix} 8 \\ -11.31 \\ 7.07 \\ -18.38 \\ 13 \end{pmatrix} kN$$

Positive means Tension, Negative means Compression

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Problem 6-3

The truss, used to support a balcony, is subjected to the loading shown. Approximate each joint as a pin and determine the force in each member. State whether the members are in tension or compression.

Units Used:

$\text{kip} = 10^3 \text{ lb}$

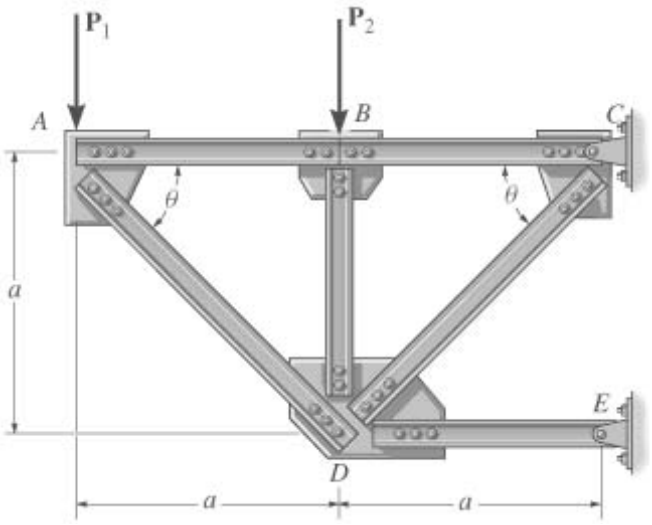
Given:

$P_1 = 600 \text{ lb}$

$P_2 = 400 \text{ lb}$

$a = 4 \text{ ft}$

$\theta = 45 \text{ deg}$



Solution:

Initial Guesses

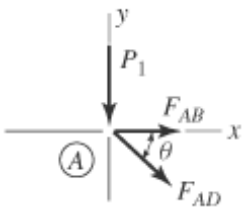
$F_{AB} = 1 \text{ lb}$ $F_{AD} = 1 \text{ lb}$ $F_{DC} = 1 \text{ lb}$

$F_{BC} = 1 \text{ lb}$ $F_{BD} = 1 \text{ lb}$ $F_{DE} = 1 \text{ lb}$

Given

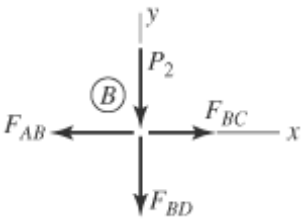
Joint *A*: $F_{AB} + F_{AD}\cos(\theta) = 0$

$-P_1 - F_{AD}\sin(\theta) = 0$



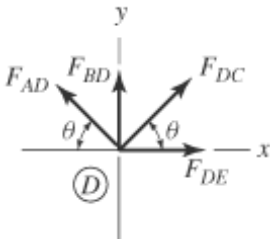
Joint *B*: $F_{BC} - F_{AB} = 0$

$-P_2 - F_{BD} = 0$



Joint *D*: $(F_{DC} - F_{AD})\cos(\theta) + F_{DE} = 0$

$(F_{DC} + F_{AD})\sin(\theta) + F_{BD} = 0$



$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{BC} \\ F_{BD} \\ F_{DC} \\ F_{DE} \end{pmatrix} = \text{Find}(F_{AB}, F_{AD}, F_{BC}, F_{BD}, F_{DC}, F_{DE})$$

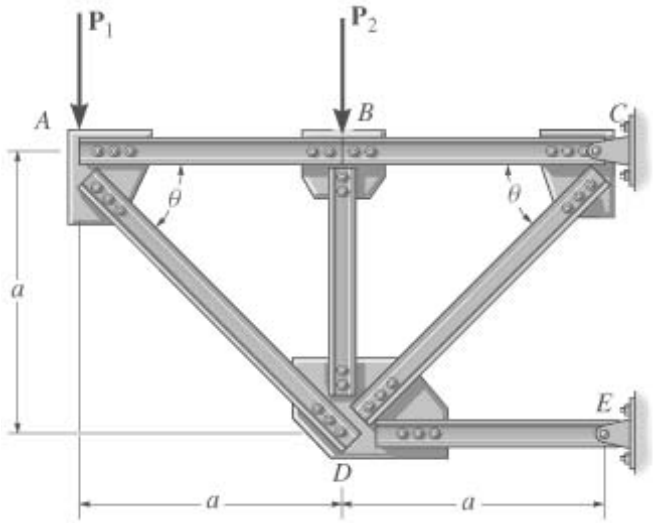
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$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{BC} \\ F_{BD} \\ F_{DC} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} 600 \\ -849 \\ 600 \\ -400 \\ 1414 \\ -1600 \end{pmatrix} \text{ lb}$$

Positive means Tension,
Negative means Compression

Problem 6-4

The truss, used to support a balcony, is subjected to the loading shown. Approximate each joint as a pin and determine the force in each member. State whether the members are in tension or compression.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$P_1 = 800 \text{ lb}$

$P_2 = 0 \text{ lb}$

$a = 4 \text{ ft}$

$\theta = 45 \text{ deg}$

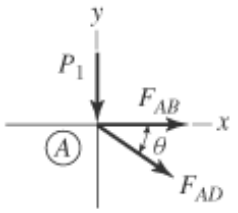
Solution:

Initial Guesses

$F_{AB} = 1 \text{ lb}$ $F_{AD} = 1 \text{ lb}$ $F_{DC} = 1 \text{ lb}$

$F_{BC} = 1 \text{ lb}$ $F_{BD} = 1 \text{ lb}$ $F_{DE} = 1 \text{ lb}$

Given



Joint *A*: $F_{AB} + F_{AD}\cos(\theta) = 0$

$-P_1 - F_{AD}\sin(\theta) = 0$

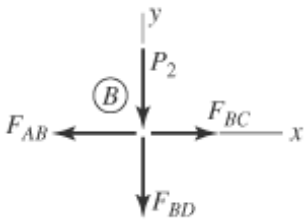
Joint *B*: $F_{BC} - F_{AB} = 0$

$-P_2 - F_{BD} = 0$

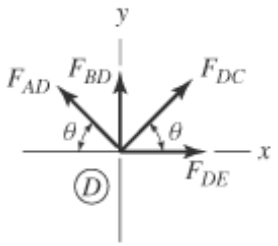
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Joint D: $\left(F_{DC} - F_{AD}\right)\cos(\theta) + F_{DE} = 0$

$\left(F_{DC} + F_{AD}\right)\sin(\theta) + F_{BD} = 0$



$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{BC} \\ F_{BD} \\ F_{DC} \\ F_{DE} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AD}, F_{BC}, F_{BD}, F_{DC}, F_{DE}\right)$$

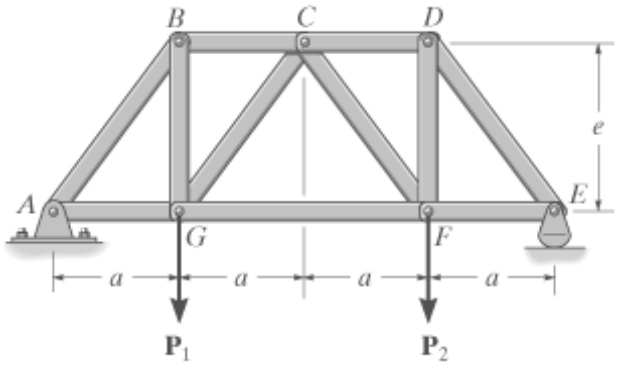


$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{BC} \\ F_{BD} \\ F_{DC} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} 800 \\ -1131 \\ 800 \\ 0 \\ 1131 \\ -1600 \end{pmatrix} \text{ lb}$$

Positive means Tension,
Negative means Compression

Problem 6-5

Determine the force in each member of the truss and state if the members are in tension or compression.



Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

$P_1 = 20 \text{ kN}$

$P_2 = 10 \text{ kN}$

$a = 1.5 \text{ m}$

$e = 2 \text{ m}$

Solution: $\theta = \text{atan}\left(\frac{e}{a}\right)$

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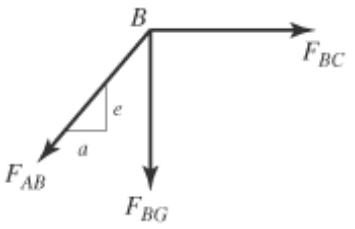
Initial Guesses:

$F_{AB} = 1\ \text{kN}\quad F_{AG} = 1\ \text{kN}\quad F_{CF} = 1\ \text{kN}$

$F_{BC} = 1\ \text{kN}\quad F_{BG} = 1\ \text{kN}\quad F_{DE} = 1\ \text{kN}$

$F_{CG} = 1\ \text{kN}\quad F_{FG} = 1\ \text{kN}\quad F_{EF} = 1\ \text{kN}$

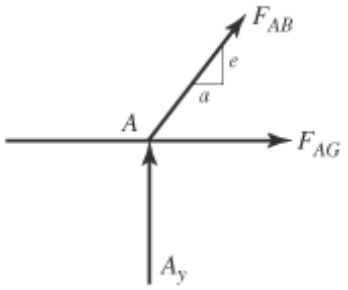
$F_{CD} = 1\ \text{kN}\quad F_{DF} = 1\ \text{kN}$



Given

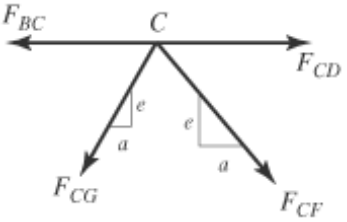
Joint B $F_{BC} - F_{AB}\cos(\theta) = 0$

$-F_{BG} - F_{AB}\sin(\theta) = 0$



Joint G $F_{FG} + F_{CG}\cos(\theta) - F_{AG} = 0$

$F_{CG}\sin(\theta) + F_{BG} - P_1 = 0$

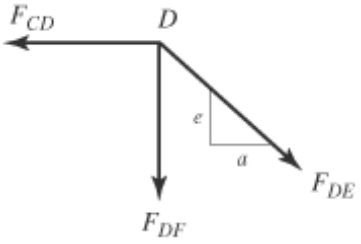


Joint C $-F_{BC} + F_{CD} + \left(F_{CF} - F_{CG}\right)\cos(\theta) = 0$

$-\left(F_{CG} + F_{CF}\right)\sin(\theta) = 0$

Joint D $-F_{CD} + F_{DE}\cos(\theta) = 0$

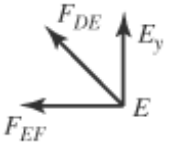
$-F_{DF} - F_{DE}\sin(\theta) = 0$

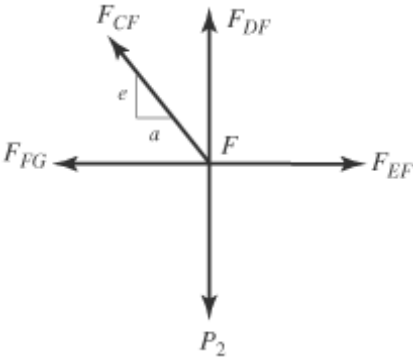


Joint F $F_{EF} - F_{FG} - F_{CF}\cos(\theta) = 0$

$F_{DF} + F_{CF}\sin(\theta) - P_2 = 0$

Joint E $-F_{DE}\cos(\theta) - F_{EF} = 0$





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$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CG} \\ F_{CD} \\ F_{AG} \\ F_{BG} \\ F_{FG} \\ F_{DF} \\ F_{CF} \\ F_{DE} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BC}, F_{CG}, F_{CD}, F_{AG}, F_{BG}, F_{FG}, F_{DF}, F_{CF}, F_{DE}, F_{EF}\right)$$

$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CG} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} -21.88 \\ -13.13 \\ 3.13 \\ -9.37 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{AG} \\ F_{BG} \\ F_{FG} \\ F_{DF} \end{pmatrix} = \begin{pmatrix} 13.13 \\ 17.5 \\ 11.25 \\ 12.5 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{CF} \\ F_{DE} \\ F_{EF} \end{pmatrix} = \begin{pmatrix} -3.13 \\ -15.62 \\ 9.37 \end{pmatrix} \text{kN}$$

Positive means Tension, Negative means Compression

Problem 6-6

Determine the force in each member of the truss and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3 \text{ N}$

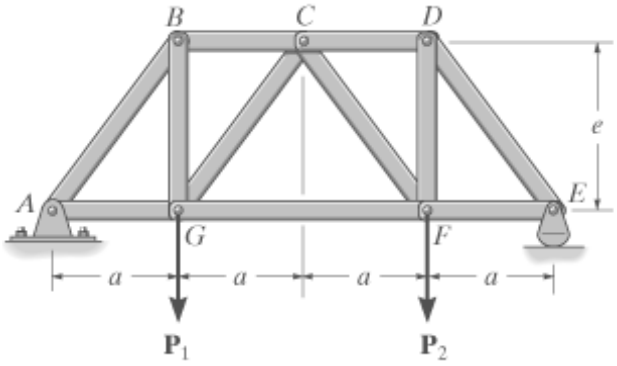
Given:

$P_1 = 40 \text{ kN}$

$P_2 = 20 \text{ kN}$

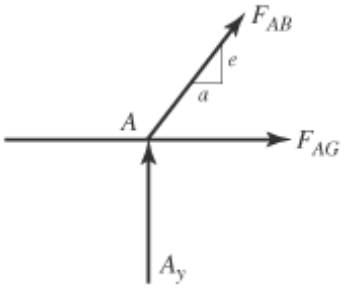
$a = 1.5 \text{ m}$

$e = 2 \text{ m}$



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Solution: $\theta = \operatorname{atan}\left(\frac{e}{a}\right)$



Initial Guesses:

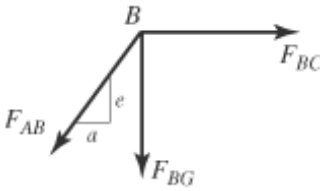
$F_{AB} = 1\ \text{kN}$ $F_{AG} = 1\ \text{kN}$ $F_{CF} = 1\ \text{kN}$

$F_{BC} = 1\ \text{kN}$ $F_{BG} = 1\ \text{kN}$ $F_{DE} = 1\ \text{kN}$

$F_{CG} = 1\ \text{kN}$ $F_{FG} = 1\ \text{kN}$ $F_{EF} = 1\ \text{kN}$

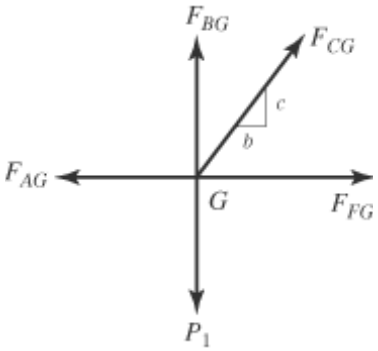
$F_{CD} = 1\ \text{kN}$ $F_{DF} = 1\ \text{kN}$

Given



Joint B $F_{BC} - F_{AB}\cos(\theta) = 0$

$-F_{BG} - F_{AB}\sin(\theta) = 0$

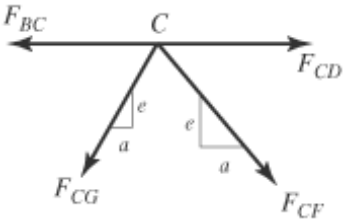


Joint G $F_{FG} + F_{CG}\cos(\theta) - F_{AG} = 0$

$F_{CG}\sin(\theta) + F_{BG} - P_1 = 0$

Joint C $-F_{BC} + F_{CD} + \left(F_{CF} - F_{CG}\right)\cos(\theta) = 0$

$-\left(F_{CG} + F_{CF}\right)\sin(\theta) = 0$



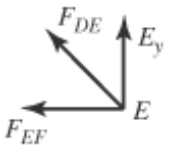
Joint D $-F_{CD} + F_{DE}\cos(\theta) = 0$

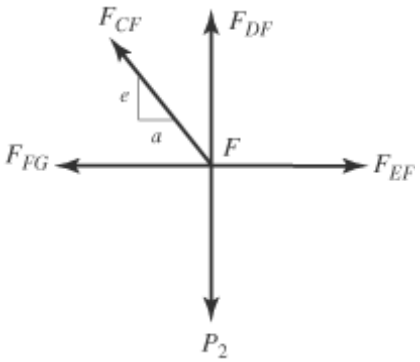
$-F_{DF} - F_{DE}\sin(\theta) = 0$

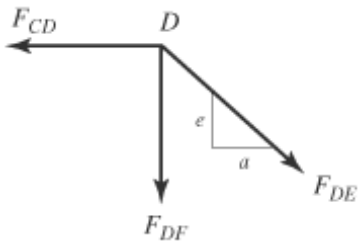
Joint F $F_{EF} - F_{FG} - F_{CF}\cos(\theta) = 0$

$F_{DF} + F_{CF}\sin(\theta) - P_2 = 0$

Joint E $-F_{DE}\cos(\theta) - F_{EF} = 0$







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$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CG} \\ F_{CD} \\ F_{AG} \\ F_{BG} \\ F_{FG} \\ F_{DF} \\ F_{CF} \\ F_{DE} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BC}, F_{CG}, F_{CD}, F_{AG}, F_{BG}, F_{FG}, F_{DF}, F_{CF}, F_{DE}, F_{EF}\right)$$

$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CG} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} -43.75 \\ -26.25 \\ 6.25 \\ -18.75 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{AG} \\ F_{BG} \\ F_{FG} \\ F_{DF} \end{pmatrix} = \begin{pmatrix} 26.25 \\ 35 \\ 22.5 \\ 25 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{CF} \\ F_{DE} \\ F_{EF} \end{pmatrix} = \begin{pmatrix} -6.25 \\ -31.25 \\ 18.75 \end{pmatrix} \text{kN}$$

Positive means Tension, Negative means Compression

Problem 6-7

Determine the force in each member of the truss and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3 \text{ N}$

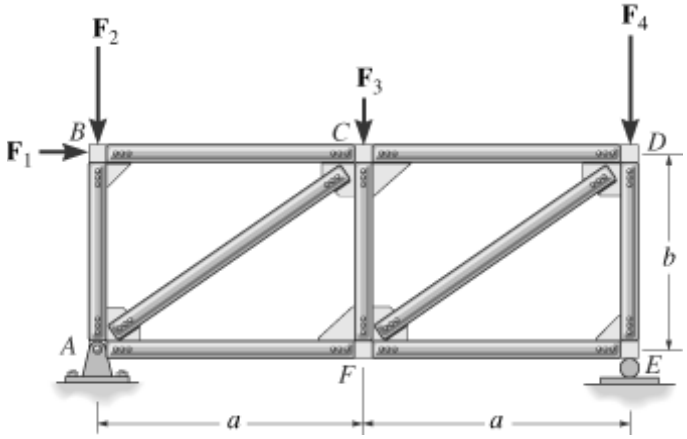
Given:

$F_1 = 3 \text{ kN}$

$F_2 = 8 \text{ kN}$

$F_3 = 4 \text{ kN}$

$F_4 = 10 \text{ kN}$

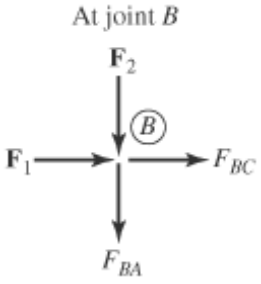


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$a = 2$ m

$b = 1.5$ m

Solution: $\theta = \operatorname{atan}\left(\frac{b}{a}\right)$

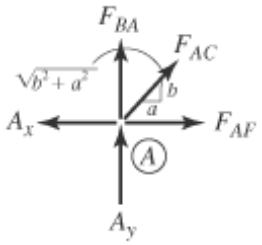


Initial Guesses

$F_{BA} = 1$ kN $F_{BC} = 1$ kN $F_{AC} = 1$ kN

$F_{AF} = 1$ kN $F_{CD} = 1$ kN $F_{CF} = 1$ kN

$F_{DF} = 1$ kN $F_{ED} = 1$ kN $F_{EF} = 1$ kN



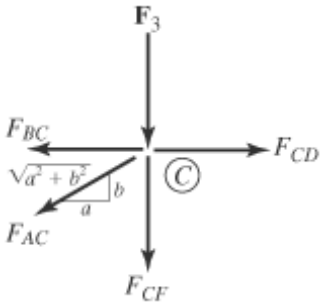
Given

Joint B $F_1 + F_{BC} = 0$

$-F_2 - F_{BA} = 0$

Joint C $F_{CD} - F_{BC} - F_{AC}\cos(\theta) = 0$

$-F_3 - F_{AC}\sin(\theta) - F_{CF} = 0$



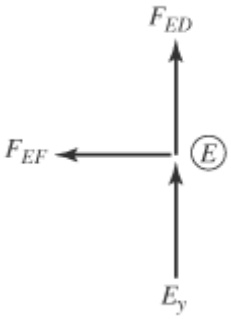
Joint E $-F_{EF} = 0$

Joint D $-F_{CD} - F_{DF}\cos(\theta) = 0$

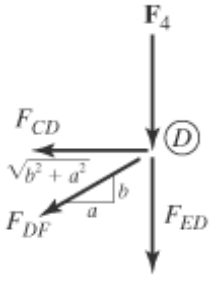
$-F_4 - F_{DF}\sin(\theta) - F_{ED} = 0$

Joint F $-F_{AF} + F_{EF} + F_{DF}\cos(\theta) = 0$

$F_{CF} + F_{DF}\sin(\theta) = 0$

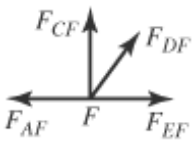


$$\begin{pmatrix} F_{BA} \\ F_{AF} \\ F_{DF} \\ F_{BC} \\ F_{CD} \\ F_{ED} \\ F_{AC} \\ F_{CF} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{BA}, F_{AF}, F_{DF}, F_{BC}, F_{CD}, F_{ED}, F_{AC}, F_{CF}, F_{EF}\right)$$



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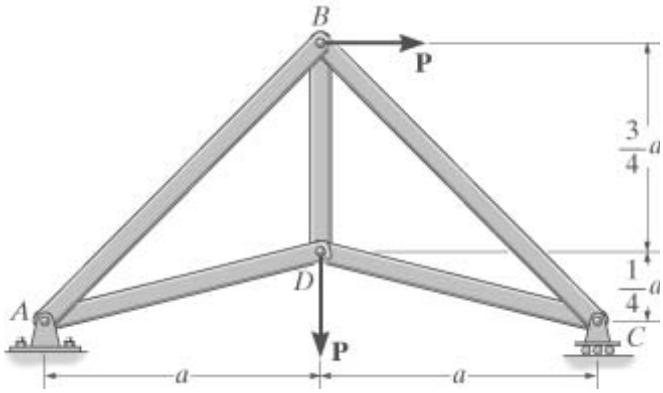
$$\begin{pmatrix} F_{BA} \\ F_{AF} \\ F_{DF} \\ F_{BC} \\ F_{CD} \\ F_{ED} \\ F_{AC} \\ F_{CF} \\ F_{EF} \end{pmatrix} = \begin{pmatrix} -8 \\ 4.167 \\ 5.208 \\ -3 \\ -4.167 \\ -13.125 \\ -1.458 \\ -3.125 \\ 0 \end{pmatrix} \text{kN}$$



Positive means tension,
Negative means compression.

Problem 6-8

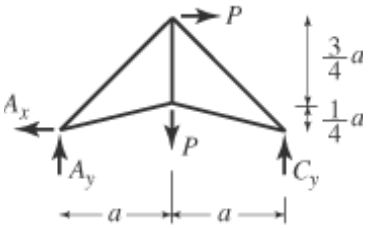
Determine the force in each member of the truss in terms of the external loading and state if the members are in tension or compression.



Solution:

$\Sigma M_A = 0;$ $\quad -P\ a + C_y 2a - P\,a = 0$

$$C_y = P$$



Joint *C*:

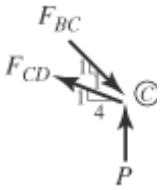
$\Sigma F_x = 0;$ $\quad \frac{1}{\sqrt{2}} F_{BC} - \frac{4}{\sqrt{17}} F_{CD} = 0$

$\Sigma F_y = 0;$ $\quad P + \frac{1}{\sqrt{17}} F_{CD} - \frac{1}{\sqrt{2}} F_{BC} = 0$

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$$F_{BC} = \frac{4\sqrt{2}P}{3} = 1.886\ P \quad \text{(C)}$$

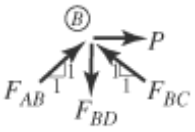
$$F_{CD} = \frac{\sqrt{17}P}{3} = 1.374\ P \quad \text{(T)}$$



Joint *B*:

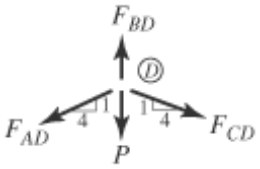
$\Sigma F_x = 0;\quad P - \frac{1}{\sqrt{2}}F_{CD} + \frac{1}{\sqrt{2}}F_{AB} = 0$

$\Sigma F_y = 0;\quad \frac{1}{\sqrt{2}}F_{CD} + \frac{1}{\sqrt{2}}F_{AB} - F_{BD} = 0$



$$F_{AB} = \frac{\sqrt{2}P}{3} = 0.471P \quad \text{(C)}$$

$$F_{BD} = \frac{5P}{3} = 1.667P \quad \text{(T)}$$

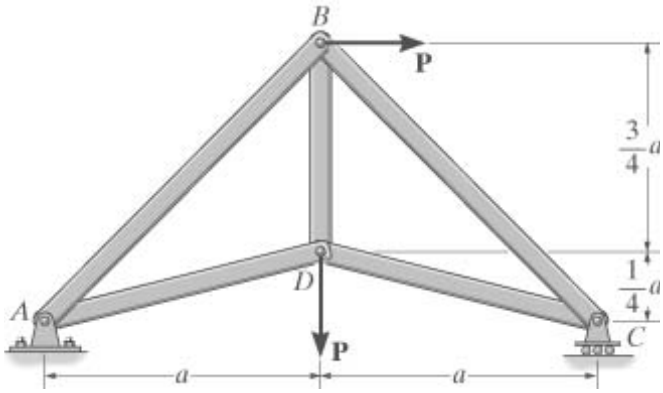


Joint *D*:

$\Sigma F_x = 0;\quad F_{DA} = F_{CD} = 1.374P \quad \text{(T)}$

Problem 6-9

The maximum allowable tensile force in the members of the truss is T_{max}, and the maximum allowable compressive force is C_{max}. Determine the maximum magnitude *P* of the two loads that can be applied to the truss.



Given:

$T_{max} = 1500$ lb

$C_{max} = 800$ lb

Solution:

Set $P = 1$ lb

Initial Guesses

$F_{AB} = 1$ lb $\quad F_{AD} = 1$ lb $\quad F_{BD} = 1$ lb

$F_{BC} = 1$ lb $\quad F_{CD} = 1$ lb

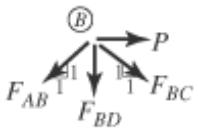


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Given

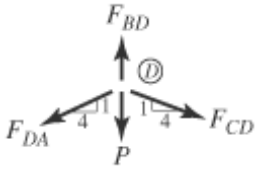
Joint B $\left(F_{BC} - F_{AB}\right)\frac{1}{\sqrt{2}} + P = 0$

$-F_{BD} - \left(F_{AB} + F_{BC}\right)\frac{1}{\sqrt{2}} = 0$



Joint D $\left(F_{CD} - F_{AD}\right)\frac{4}{\sqrt{17}} = 0$

$F_{BD} - P - \left(F_{AD} + F_{CD}\right)\frac{1}{\sqrt{17}} = 0$



Joint C $-F_{BC}\frac{1}{\sqrt{2}} - F_{CD}\frac{4}{\sqrt{17}} = 0$

$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{AD} \\ F_{CD} \\ F_{BD} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BC}, F_{AD}, F_{CD}, F_{BD}\right) \qquad \begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{AD} \\ F_{CD} \\ F_{BD} \end{pmatrix} = \begin{pmatrix} -0.471 \\ -1.886 \\ 1.374 \\ 1.374 \\ 1.667 \end{pmatrix} \text{lb}$$

Now find the critical load

$$P_1 = P\frac{T_{max}}{\max\left(F_{AB}, F_{BC}, F_{AD}, F_{CD}, F_{BD}\right)} \qquad P_1 = 900\,\text{lb}$$

$$P_2 = P\frac{C_{max}}{\left|\min\left(F_{AB}, F_{BC}, F_{AD}, F_{CD}, F_{BD}\right)\right|} \qquad P_2 = 424.264\,\text{lb}$$

$$P = \min\left(P_1, P_2\right) \qquad P = 424.3\,\text{lb}$$

Problem 6-10

Determine the force in each member of the truss and state if the members are in tension or compression.

Given:

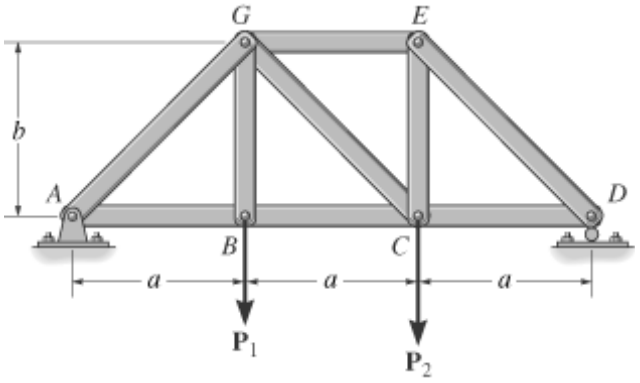
$P_1 = 0$ lb

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$P_2 = 1000$ lb

$a = 10$ ft

$b = 10$ ft



Solution:

$$\theta = \operatorname{atan}\left(\frac{b}{a}\right)$$

Initial Guesses:

$F_{AB} = 1$ lb $F_{AG} = 1$ lb $F_{BG} = 1$ lb

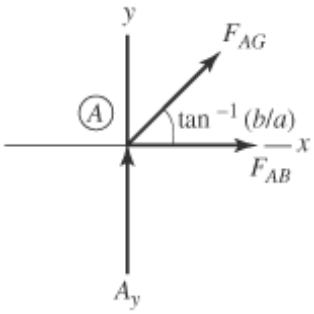
$F_{BC} = 1$ lb $F_{DC} = 1$ lb $F_{DE} = 1$ lb

$F_{EG} = 1$ lb $F_{EC} = 1$ lb $F_{CG} = 1$ lb

Given

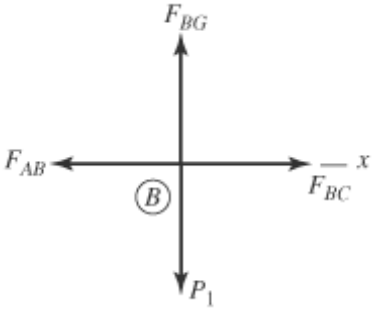
Joint B $F_{BC} - F_{AB} = 0$

$F_{BG} - P_1 = 0$



Joint G $\left(F_{CG} - F_{AG}\right)\cos(\theta) + F_{EG} = 0$

$-\left(F_{CG} + F_{AG}\right)\sin(\theta) - F_{BG} = 0$



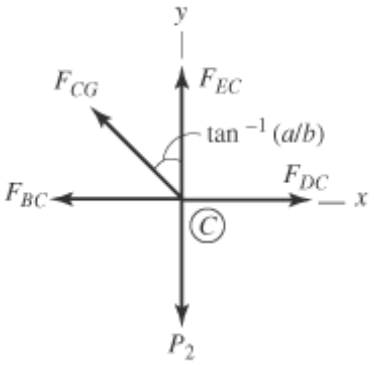
Joint C $F_{DC} - F_{BC} - F_{CG}\cos(\theta) = 0$

$F_{EC} + F_{CG}\sin(\theta) - P_2 = 0$

Joint E $F_{DE}\cos(\theta) - F_{EG} = 0$

$-F_{EC} - F_{DE}\sin(\theta) = 0$

Joint D $-F_{DE}\cos(\theta) - F_{DC} = 0$

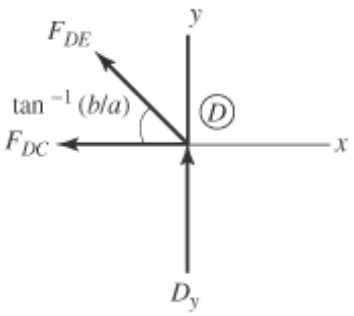


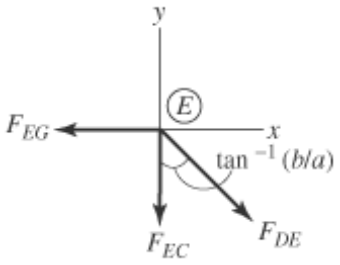
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$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{EG} \\ F_{AG} \\ F_{DC} \\ F_{EC} \\ F_{BG} \\ F_{DE} \\ F_{CG} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BC}, F_{EG}, F_{AG}, F_{DC}, F_{EC}, F_{BG}, F_{DE}, F_{CG}\right)$$

$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{EG} \\ F_{AG} \\ F_{DC} \\ F_{EC} \\ F_{BG} \\ F_{DE} \\ F_{CG} \end{pmatrix} = \begin{pmatrix} 333 \\ 333 \\ -667 \\ -471 \\ 667 \\ 667 \\ 0 \\ -943 \\ 471 \end{pmatrix} \text{lb}$$

Positive means tension,
Negative means compression.





Problem 6-11

Determine the force in each member of the truss and state if the members are in tension or compression.

Given:

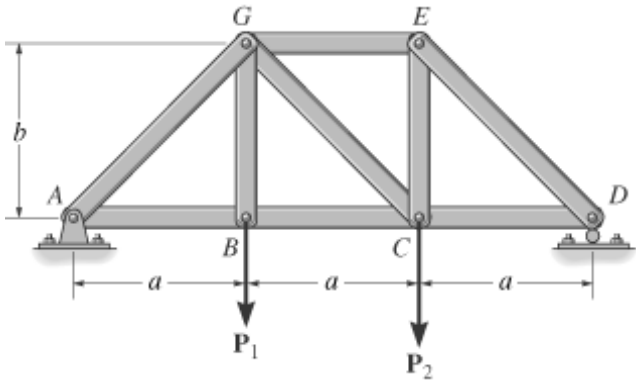
$P_1 = 500$ lb

$P_2 = 1500$ lb

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$a = 10$ ft

$b = 10$ ft



Solution:

$$\theta = \operatorname{atan}\left(\frac{b}{a}\right)$$

Initial Guesses:

$F_{AB} = 1$ lb $F_{AG} = 1$ lb $F_{BG} = 1$ lb

$F_{BC} = 1$ lb $F_{DC} = 1$ lb $F_{DE} = 1$ lb

$F_{EG} = 1$ lb $F_{EC} = 1$ lb $F_{CG} = 1$ lb

Given

Joint B $F_{BC} - F_{AB} = 0$

$F_{BG} - P_1 = 0$

Joint G $\left(F_{CG} - F_{AG}\right)\cos(\theta) + F_{EG} = 0$

$-\left(F_{CG} + F_{AG}\right)\sin(\theta) - F_{BG} = 0$

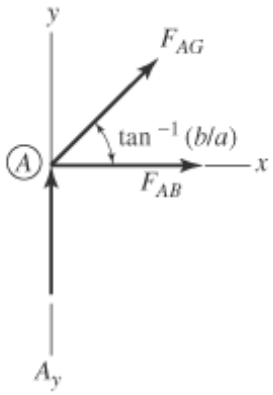
Joint C $F_{DC} - F_{BC} - F_{CG}\cos(\theta) = 0$

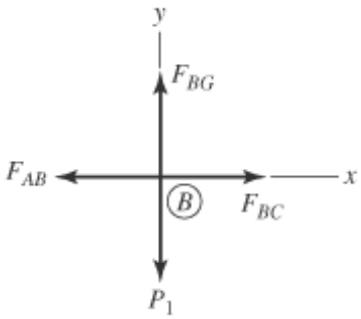
$F_{EC} + F_{CG}\sin(\theta) - P_2 = 0$

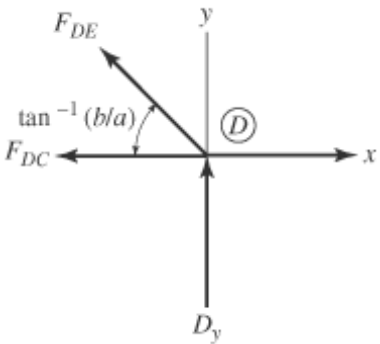
Joint E $F_{DE}\cos(\theta) - F_{EG} = 0$

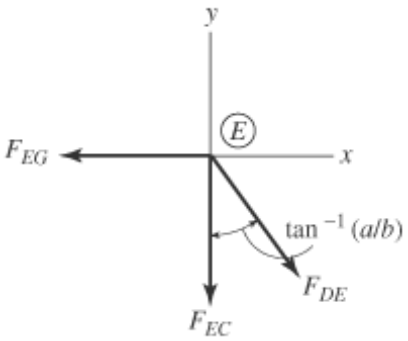
$-F_{EC} - F_{DE}\sin(\theta) = 0$

Joint D $-F_{DE}\cos(\theta) - F_{DC} = 0$







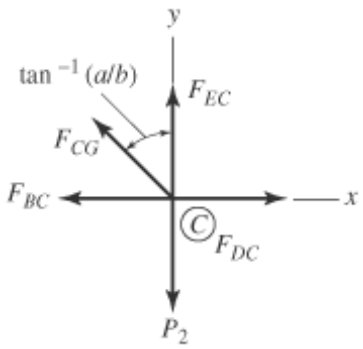


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$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{EG} \\ F_{AG} \\ F_{DC} \\ F_{EC} \\ F_{BG} \\ F_{DE} \\ F_{CG} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BC}, F_{EG}, F_{AG}, F_{DC}, F_{EC}, F_{BG}, F_{DE}, F_{CG}\right)$$

$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{EG} \\ F_{AG} \\ F_{DC} \\ F_{EC} \\ F_{BG} \\ F_{DE} \\ F_{CG} \end{pmatrix} = \begin{pmatrix} 833 \\ 833 \\ -1167 \\ -1179 \\ 1167 \\ 1167 \\ 500 \\ -1650 \\ 471 \end{pmatrix} \text{lb}$$

Positive means tension,
Negative means compression.



Problem 6-12

Determine the force in each member of the truss and state if the members are in tension or compression.

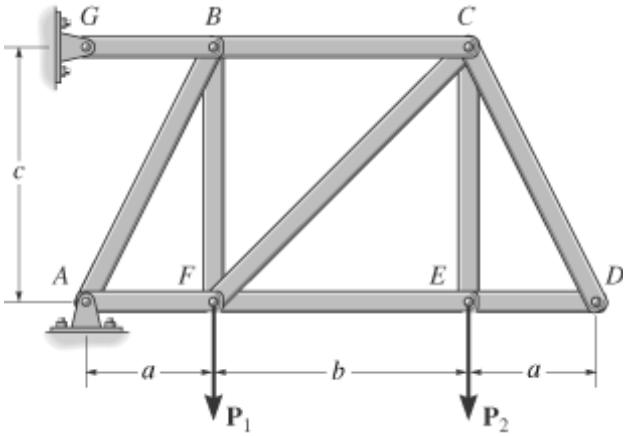
Units Used:

$$\text{kN} = 10^3 \text{ N}$$

Given:

$$P_1 = 10 \text{ kN}$$

$$P_2 = 15 \text{ kN}$$



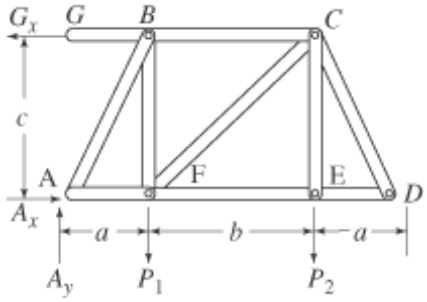
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$a = 2$ m

$b = 4$ m

$c = 4$ m

Solution: $\alpha = \operatorname{atan}\left(\frac{c}{a}\right)$ $\beta = \operatorname{atan}\left(\frac{c}{b}\right)$



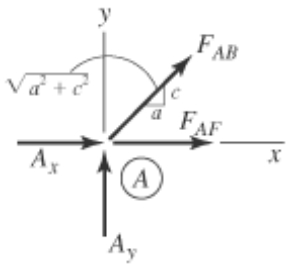
Initial Guesses:

$F_{AB} = 1$ kN $F_{AF} = 1$ kN $F_{GB} = 1$ kN

$F_{BF} = 1$ kN $F_{FC} = 1$ kN $F_{FE} = 1$ kN

$F_{BC} = 1$ kN $F_{EC} = 1$ kN $F_{CD} = 1$ kN

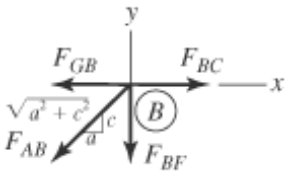
$F_{ED} = 1$ kN



Given

Joint B $-F_{GB} + F_{BC} - F_{AB}\cos(\alpha) = 0$

$-F_{AB}\sin(\alpha) - F_{BF} = 0$



Joint F $-F_{AF} + F_{FE} + F_{FC}\cos(\beta) = 0$

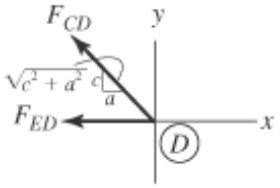
$F_{BF} + F_{FC}\sin(\beta) - P_1 = 0$

Joint C $-F_{BC} - F_{FC}\cos(\beta) + F_{CD}\cos(\alpha) = 0$

$-F_{FC}\sin(\beta) - F_{CD}\sin(\alpha) - F_{EC} = 0$

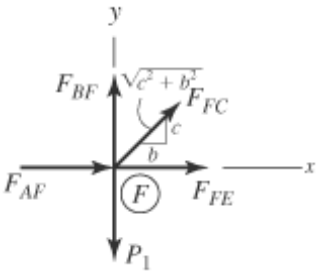
Joint E $-F_{FE} + F_{ED} = 0$

$F_{EC} - P_2 = 0$



Joint D $-F_{CD}\cos(\alpha) - F_{ED} = 0$

$F_{CD}\sin(\alpha) = 0$



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$$\begin{pmatrix} F_{AB} \\ F_{BF} \\ F_{BC} \\ F_{ED} \\ F_{AF} \\ F_{FC} \\ F_{EC} \\ F_{GB} \\ F_{FE} \\ F_{CD} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BF}, F_{BC}, F_{ED}, F_{AF}, F_{FC}, F_{EC}, F_{GB}, F_{FE}, F_{CD}\right)$$

$$\begin{pmatrix} F_{AB} \\ F_{BF} \\ F_{BC} \\ F_{ED} \\ F_{AF} \end{pmatrix} = \begin{pmatrix} -27.951 \\ 25 \\ 15 \\ 0 \\ -15 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{FC} \\ F_{EC} \\ F_{GB} \\ F_{FE} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} -21.213 \\ 15 \\ 27.5 \\ 0 \\ 0 \end{pmatrix} \text{kN}$$

Positive means Tension,
Negative means Compression

Problem 6-13

Determine the force in each member of the truss and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

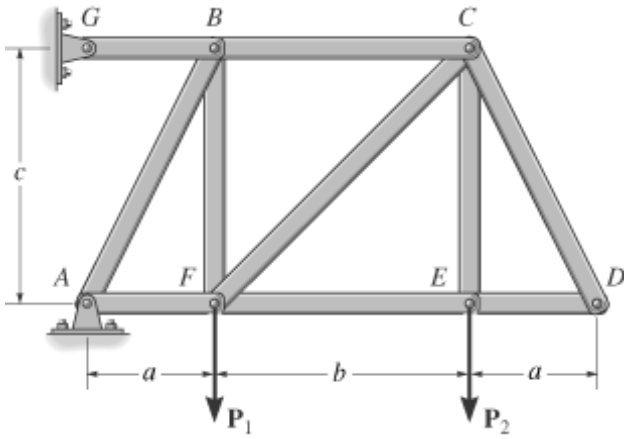
$P_1 = 0 \text{ kN}$

$P_2 = 20 \text{ kN}$

$a = 2 \text{ m}$

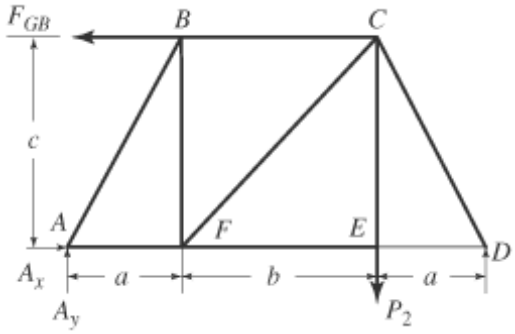
$b = 4 \text{ m}$

$c = 4 \text{ m}$



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Solution: $\alpha = \operatorname{atan}\left(\frac{c}{a}\right)$ $\beta = \operatorname{atan}\left(\frac{c}{b}\right)$



Initial Guesses:

$F_{AB} = 1$ kN $F_{AF} = 1$ kN $F_{GB} = 1$ kN

$F_{BF} = 1$ kN $F_{FC} = 1$ kN $F_{FE} = 1$ kN

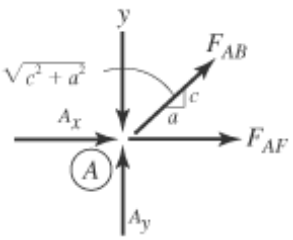
$F_{BC} = 1$ kN $F_{EC} = 1$ kN $F_{CD} = 1$ kN

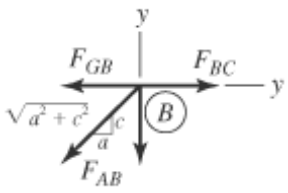
$F_{ED} = 1$ kN

Given

Joint B $-F_{GB} + F_{BC} - F_{AB}\cos(\alpha) = 0$

$-F_{AB}\sin(\alpha) - F_{BF} = 0$





Joint F $-F_{AF} + F_{FE} + F_{FC}\cos(\beta) = 0$

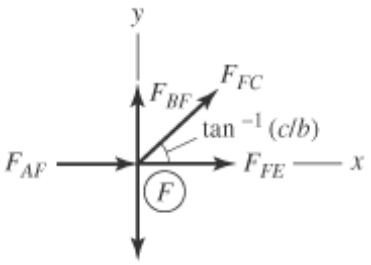
$F_{BF} + F_{FC}\sin(\beta) - P_1 = 0$

Joint C $-F_{BC} - F_{FC}\cos(\beta) + F_{CD}\cos(\alpha) = 0$

$-F_{FC}\sin(\beta) - F_{CD}\sin(\alpha) - F_{EC} = 0$

Joint E $-F_{FE} + F_{ED} = 0$

$F_{EC} - P_2 = 0$

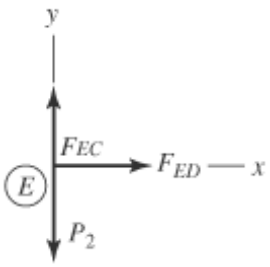


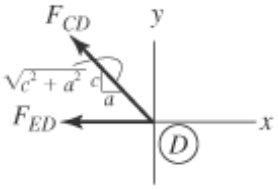
Joint D $-F_{CD}\cos(\alpha) - F_{ED} = 0$

$F_{CD}\sin(\alpha) = 0$

$$\begin{pmatrix} F_{AB} \\ F_{BF} \\ F_{BC} \\ F_{ED} \\ F_{AF} \\ F_{FC} \\ F_{EC} \\ F_{GB} \\ F_{FE} \\ F_{CD} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BF}, F_{BC}, F_{ED}, F_{AF}, F_{FC}, F_{EC}, F_{GB}, F_{FE}, F_{CD}\right)$$

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$$\begin{pmatrix} F_{AB} \\ F_{BF} \\ F_{BC} \\ F_{ED} \\ F_{AF} \end{pmatrix} = \begin{pmatrix} -22.361 \\ 20 \\ 20 \\ 0 \\ -20 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{FC} \\ F_{EC} \\ F_{GB} \\ F_{FE} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} -28.284 \\ 20 \\ 30 \\ 0 \\ 0 \end{pmatrix} \text{kN}$$

Positive means Tension,
Negative means Compression

Problem 6-14

Determine the force in each member of the truss and state if the members are in tension or compression.

Given:

$P_1 = 100$ lb

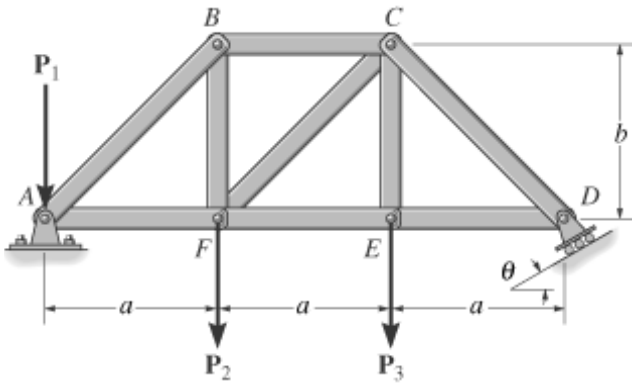
$P_2 = 200$ lb

$P_3 = 300$ lb

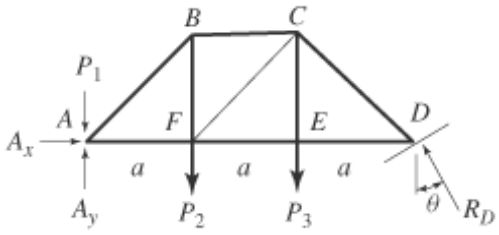
$a = 10$ ft

$b = 10$ ft

$\theta = 30$ deg



Solution: $\phi = \text{atan}\left(\frac{b}{a}\right)$



Initial Guesses:

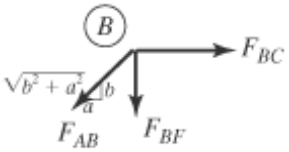
$F_{AB} = 1$ lb $\quad F_{AF} = 1$ lb $\quad F_{BC} = 1$ lb

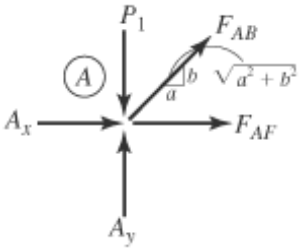
$F_{BF} = 1$ lb $\quad F_{FC} = 1$ lb $\quad F_{FE} = 1$ lb

$F_{ED} = 1$ lb $\quad F_{EC} = 1$ lb $\quad F_{CD} = 1$ lb

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Given



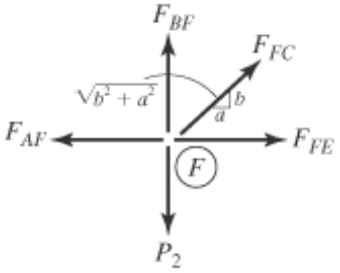


Joint B $F_{BC} - F_{AB}\cos(\phi) = 0$

$-F_{BF} - F_{AB}\sin(\phi) = 0$

Joint F $-F_{AF} + F_{FE} + F_{FC}\cos(\phi) = 0$

$-P_2 + F_{BF} + F_{FC}\sin(\phi) = 0$



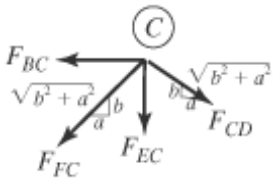
Joint C $-F_{BC} + F_{CD}\cos(\phi) - F_{FC}\cos(\phi) = 0$

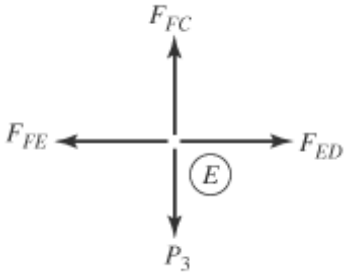
$-F_{EC} - F_{CD}\sin(\phi) - F_{FC}\sin(\phi) = 0$

Joint E $-F_{FE} + F_{ED} = 0$

$F_{EC} - P_3 = 0$

Joint D $-F_{ED}\cos(\theta) - F_{CD}\cos(\phi + \theta) = 0$





$$\begin{pmatrix} F_{AB} \\ F_{AF} \\ F_{BC} \\ F_{BF} \\ F_{FC} \\ F_{FE} \\ F_{ED} \\ F_{EC} \\ F_{CD} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AF}, F_{BC}, F_{BF}, F_{FC}, F_{FE}, F_{ED}, F_{EC}, F_{CD}\right)$$

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$$\begin{pmatrix} F_{AB} \\ F_{AF} \\ F_{BC} \\ F_{BF} \\ F_{FC} \\ F_{FE} \\ F_{ED} \\ F_{EC} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} -330.0 \\ 79.4 \\ -233.3 \\ 233.3 \\ -47.1 \\ 112.7 \\ 112.7 \\ 300.0 \\ -377.1 \end{pmatrix} \text{lb}$$

Positive means Tension,
Negative means Compression

Problem 6-15

Determine the force in each member of the truss and state if the members are in tension or compression.

Given:

$P_1 = 400$ lb

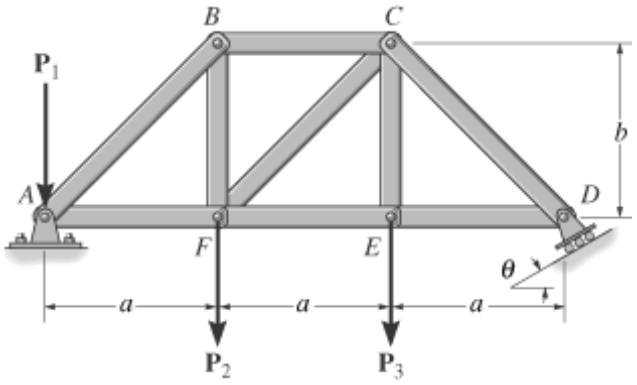
$P_2 = 400$ lb

$P_3 = 0$ lb

$a = 10$ ft

$b = 10$ ft

$\theta = 30$ deg



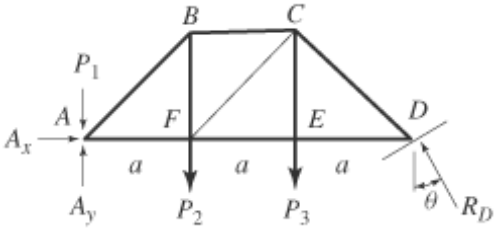
Solution: $\phi = \text{atan}\left(\frac{b}{a}\right)$

Initial Guesses:

$F_{AB} = 1$ lb $F_{AF} = 1$ lb $F_{BC} = 1$ lb

$F_{BF} = 1$ lb $F_{FC} = 1$ lb $F_{FE} = 1$lb

$F_{ED} = 1$ lb $F_{EC} = 1$ lb $F_{CD} = 1$ lb



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Given

Joint B $F_{BC} - F_{AB}\cos(\phi) = 0$

$-F_{BF} - F_{AB}\sin(\phi) = 0$

Joint F $-F_{AF} + F_{FE} + F_{FC}\cos(\phi) = 0$

$-P_2 + F_{BF} + F_{FC}\sin(\phi) = 0$

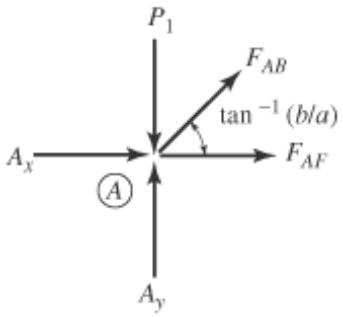
Joint C $-F_{BC} + F_{CD}\cos(\phi) - F_{FC}\cos(\phi) = 0$

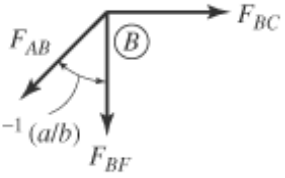
$-F_{EC} - F_{CD}\sin(\phi) - F_{FC}\sin(\phi) = 0$

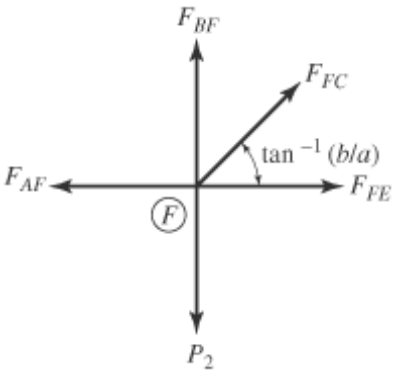
Joint E $-F_{FE} + F_{ED} = 0$

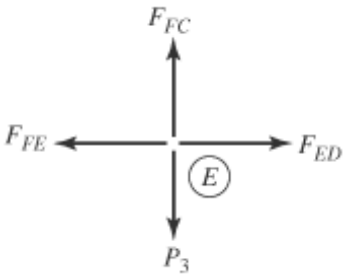
$F_{EC} - P_3 = 0$

Joint D $-F_{ED}\cos(\theta) - F_{CD}\cos(\phi + \theta) = 0$







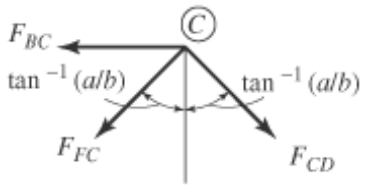


$$\begin{pmatrix} F_{AB} \\ F_{AF} \\ F_{BC} \\ F_{BF} \\ F_{FC} \\ F_{FE} \\ F_{ED} \\ F_{EC} \\ F_{CD} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AF}, F_{BC}, F_{BF}, F_{FC}, F_{FE}, F_{ED}, F_{EC}, F_{CD}\right)$$

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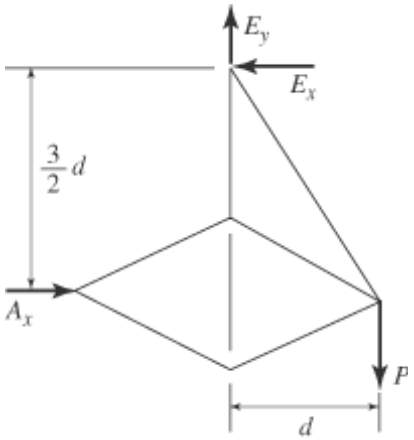
$$\begin{pmatrix} F_{AB} \\ F_{AF} \\ F_{BC} \\ F_{BF} \\ F_{FC} \\ F_{FE} \\ F_{ED} \\ F_{EC} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} -377.1 \\ 189.7 \\ -266.7 \\ 266.7 \\ 188.6 \\ 56.4 \\ 56.4 \\ 0.0 \\ -188.6 \end{pmatrix} \text{ lb}$$

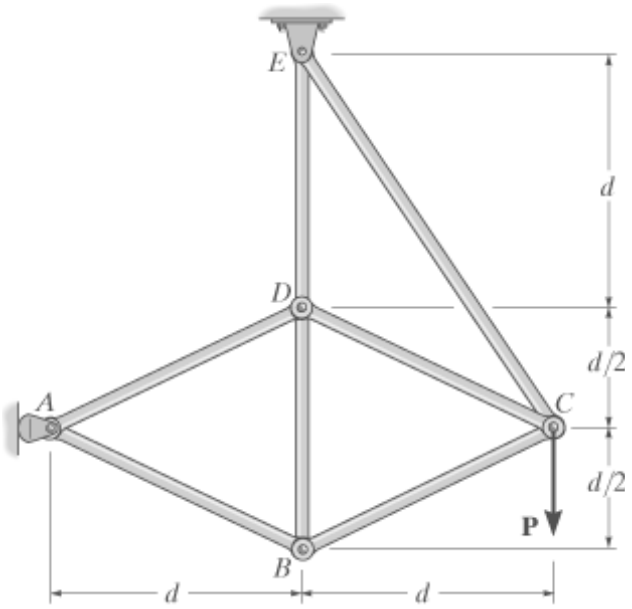
Positive means Tension,
Negative means Compression



Problem 6-16

Determine the force in each member of the truss in terms of the load *P* and state if the members are in tension or compression.





Solution:

Support reactions:

$$\Sigma M_E = 0; \qquad A_x \frac{3}{2} d - P\,d = 0 \qquad A_x = \frac{2P}{3}$$

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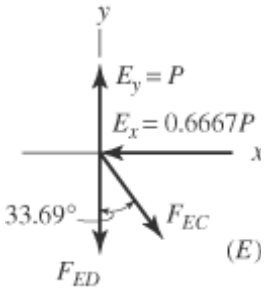
$\Sigma F_x = 0; \quad A_x - E_x = 0 \quad E_x = \frac{2P}{3}$

$\Sigma F_y = 0; \quad E_y - P = 0 \quad E_y = P$

Joint E:

$\Sigma F_x = 0; \quad F_{EC}\frac{2}{\sqrt{13}} - E_x = 0 \quad F_{EC} = \sqrt{\frac{13}{3}}P = 1.20P$ (T)

$\Sigma F_y = 0; \quad P - F_{ED} - F_{EC}\frac{3}{\sqrt{13}} = 0 \quad F_{ED} = 0$



Joint A:

$\Sigma F_y = 0; \quad F_{AB}\frac{1}{\sqrt{5}} - F_{AD}\frac{1}{\sqrt{5}} = 0 \quad F_{AB} = F_{AD}$

$\Sigma F_x = 0; \quad A_x - 2F_{AB}\frac{2}{\sqrt{5}} = 0 \quad F_{AB} = F_{AD} = \frac{\sqrt{5}}{6}P = 0.373\text{P}$ (C)

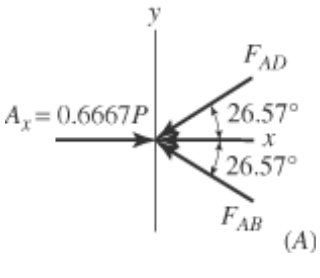
Joint D:

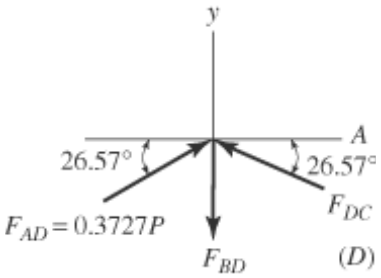
$\Sigma F_x = 0; \quad F_{AD}\frac{2}{\sqrt{5}} - F_{DC}\frac{2}{\sqrt{5}} = 0 \quad F_{DC} = \frac{\sqrt{5}}{6}P = 0.373\text{P}$ (C)

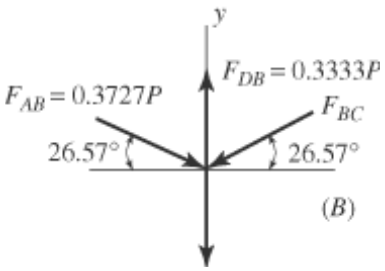
$\Sigma F_y = 0; \quad 2F_{AD}\frac{1}{\sqrt{5}} - F_{DB} = 0 \quad F_{DB} = \frac{P}{3}$ (T)

Joint B:

$\Sigma F_x = 0; \quad F_{AB}\frac{1}{\sqrt{5}} - F_{BC}\frac{1}{\sqrt{5}} = 0 \quad F_{BC} = \sqrt{\frac{5}{6}}P = 0.373P$ (C)







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Problem 6-17

The maximum allowable tensile force in the members of the truss is T_{max} and the maximum allowable compressive force is C_{max}. Determine the maximum magnitude of the load P that can be applied to the truss.

Units Used:

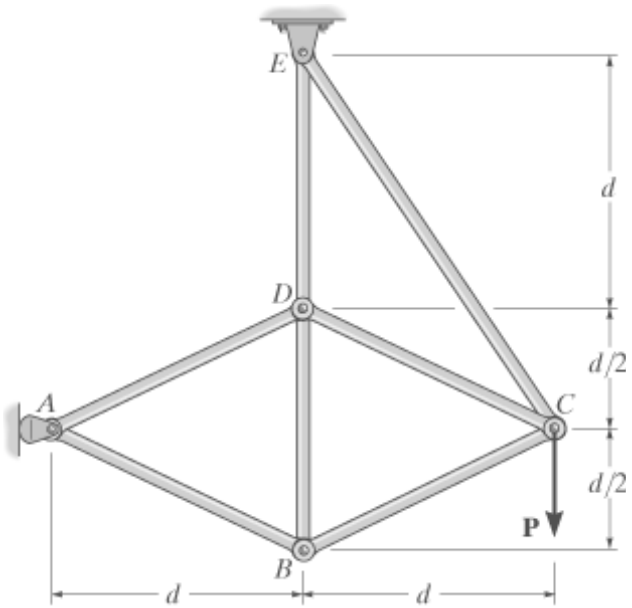
$\text{kN} = 10^3\ \text{N}$

Given:

$T_{max} = 5\ \text{kN}$

$C_{max} = 3\ \text{kN}$

$d = 2\ \text{m}$



Solution:

Set $P = 1\ \text{kN}$

Initial Guesses:

$F_{AD} = 1\ \text{kN}$ $F_{AB} = 1\ \text{kN}$

$F_{BC} = 1\ \text{kN}$ $F_{BD} = 1\ \text{kN}$

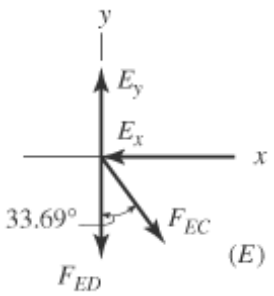
$F_{CD} = 1\ \text{kN}$ $F_{CE} = 1\ \text{kN}$ $F_{DE} = 1\ \text{kN}$

Given

Joint A $F_{AD}\frac{1}{\sqrt{5}} - F_{AB}\frac{1}{\sqrt{5}} = 0$

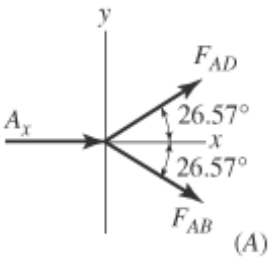
Joint B $F_{BC}\frac{2}{\sqrt{5}} - F_{AB}\frac{2}{\sqrt{5}} = 0$

$\left(F_{BC} + F_{AB}\right)\frac{1}{\sqrt{5}} + F_{BD} = 0$



Joint D $\left(F_{CD} - F_{AD}\right)\frac{2}{\sqrt{5}} = 0$

$F_{DE} - F_{BD} - \left(F_{AD} + F_{CD}\right)\frac{1}{\sqrt{5}} = 0$

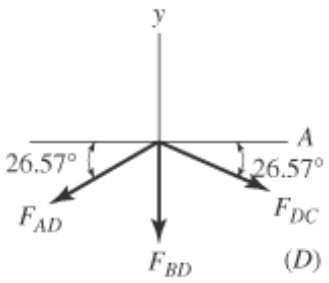


Joint C $-\left(F_{CD} + F_{BC}\right)\frac{2}{\sqrt{5}} - F_{CE}\frac{2}{\sqrt{13}} = 0$

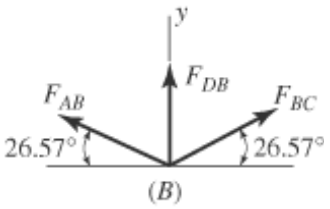
$\left(F_{CD} - F_{BC}\right)\frac{1}{\sqrt{5}} + F_{CE}\frac{3}{\sqrt{13}} - P = 0$

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$$\begin{pmatrix} F_{AD} \\ F_{AB} \\ F_{BC} \\ F_{BD} \\ F_{CD} \\ F_{CE} \\ F_{DE} \end{pmatrix} = \text{Find}\left(F_{AD}, F_{AB}, F_{BC}, F_{BD}, F_{CD}, F_{CE}, F_{DE}\right)$$



$$\begin{pmatrix} F_{AD} \\ F_{AB} \\ F_{BC} \\ F_{BD} \\ F_{CD} \\ F_{CE} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} -0.373 \\ -0.373 \\ -0.373 \\ 0.333 \\ -0.373 \\ 1.202 \\ 0 \end{pmatrix} \text{kN}$$



Now Scale the answer

$$P_1 = P\frac{T_{max}}{\max\left(F_{AD}, F_{AB}, F_{BC}, F_{BD}, F_{CD}, F_{CE}, F_{DE}\right)}$$

$$P_2 = P\frac{C_{max}}{\left|\min\left(F_{AD}, F_{AB}, F_{BC}, F_{BD}, F_{CD}, F_{CE}, F_{DE}\right)\right|}$$

$$P = \min\left(P_1, P_2\right) \qquad P = 4.16\,\text{kN}$$

Problem 6-18

Determine the force in each member of the truss and state if the members are in tension or compression. *Hint:* The horizontal force component at *A* must be zero. Why?

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Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

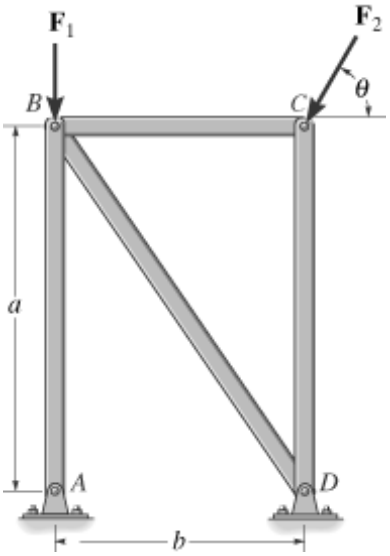
$F_1 = 600 \text{ lb}$

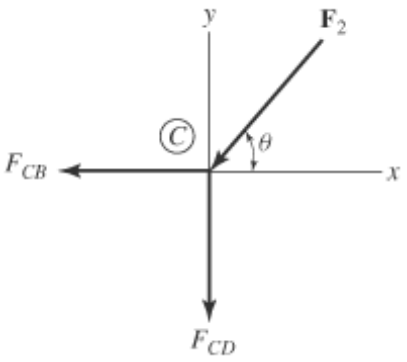
$F_2 = 800 \text{ lb}$

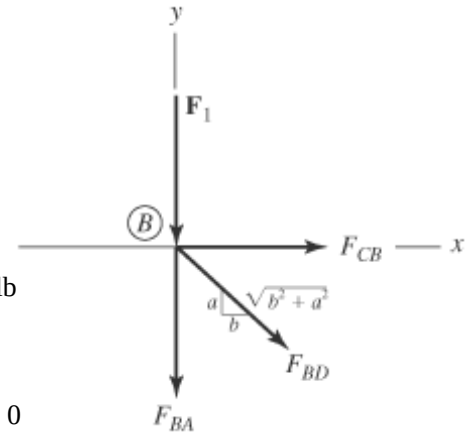
$a = 4 \text{ ft}$

$b = 3 \text{ ft}$

$\theta = 60 \text{ deg}$







Solution:

Initial Guesses

$F_{BA} = 1 \text{ lb} \quad F_{BD} = 1 \text{ lb} \quad F_{CB} = 1 \text{ lb} \quad F_{CD} = 1 \text{ lb}$

Given

Joint C $\quad -F_{CB} - F_2\cos(\theta) = 0 \qquad -F_{CD} - F_2\sin(\theta) = 0$

Joint B $\quad F_{CB} + F_{BD}\dfrac{b}{\sqrt{a^2 + b^2}} = 0 \qquad -F_{BA} - F_{BD}\dfrac{a}{\sqrt{a^2 + b^2}} - F_1 = 0$

$$\begin{pmatrix} F_{BA} \\ F_{BD} \\ F_{CB} \\ F_{CD} \end{pmatrix} = \text{Find}\left(F_{BA}, F_{BD}, F_{CB}, F_{CD}\right) \qquad \begin{pmatrix} F_{BA} \\ F_{BD} \\ F_{CB} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} -1.133 \times 10^3 \\ 666.667 \\ -400 \\ -692.82 \end{pmatrix} \text{ lb}$$

Positive means Tension
Negative means Compression

Problem 6-19

Determine the force in each member of the truss and state if the members are in tension or compression. *Hint:* The resultant force at the pin E acts along member ED. Why?

Units Used:

$\text{kN} = 10^3 \text{ N}$

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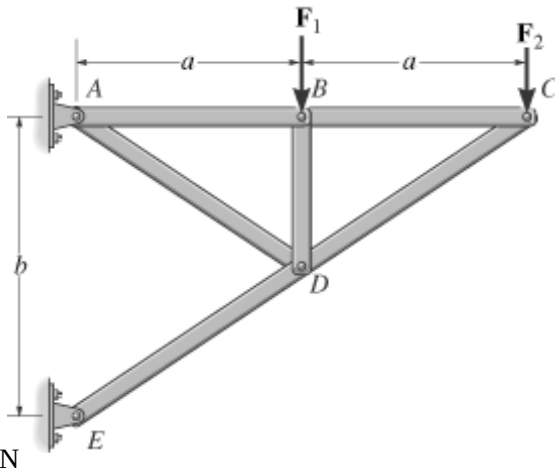
Given:

$F_1 = 3$ kN

$F_2 = 2$ kN

$a = 3$ m

$b = 4$ m



Solution:

Initial Guesses:

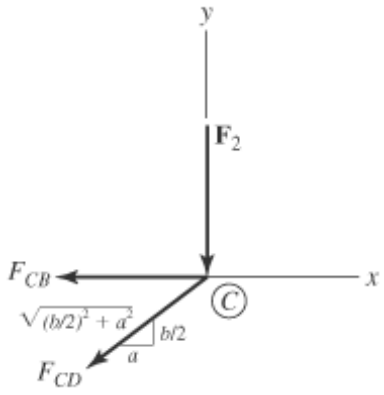
$F_{CB} = 1$ kN $\quad F_{CD} = 1$ kN $\quad F_{BA} = 1$ kN

$F_{BD} = 1$ kN $\quad F_{DA} = 1$ kN $\quad F_{DE} = 1$ kN

Given

Joint C $\quad -F_{CB} - F_{CD}\dfrac{2\,a}{\sqrt{(2\,a)^2 + b^2}} = 0$

$$-F_2 - F_{CD}\frac{b}{\sqrt{(2\,a)^2 + b^2}} = 0$$

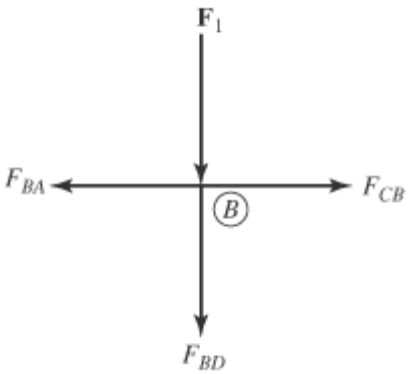


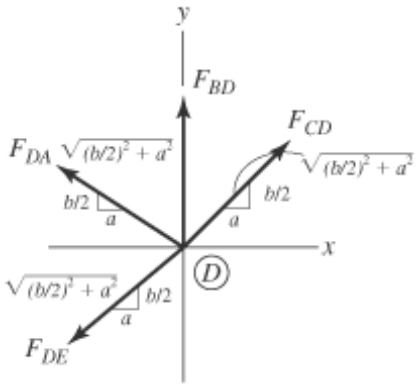
Joint B $\quad -F_{BA} + F_{CB} = 0$

$$-F_1 - F_{BD} = 0$$

Joint D $\quad \left(F_{CD} - F_{DA} - F_{DE}\right)\dfrac{2\,a}{\sqrt{(2\,a)^2 + b^2}} = 0$

$$F_{BD} + \left(F_{CD} + F_{DA} - F_{DE}\right)\frac{b}{\sqrt{(2\,a)^2 + b^2}} = 0$$





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$$\begin{pmatrix} F_{CB} \\ F_{CD} \\ F_{BA} \\ F_{BD} \\ F_{DA} \\ F_{DE} \end{pmatrix} = \text{Find}\left(F_{CB}, F_{CD}, F_{BA}, F_{BD}, F_{DA}, F_{DE}\right)$$

$$\begin{pmatrix} F_{CB} \\ F_{CD} \\ F_{BA} \\ F_{BD} \\ F_{DA} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} 3 \\ -3.606 \\ 3 \\ -3 \\ 2.704 \\ -6.31 \end{pmatrix} \text{kN}$$

Positive means Tension,
Negative means Compression

Problem 6-20

Each member of the truss is uniform and has a mass density ρ. Determine the approximate force in each member due to the weight of the truss. State if the members are in tension or compression. Solve the problem by *assuming* the weight of each member can be represented as a vertical force, half of which is applied at each end of the member.

Given:

$\rho = 8 \ \frac{\text{kg}}{\text{m}}$

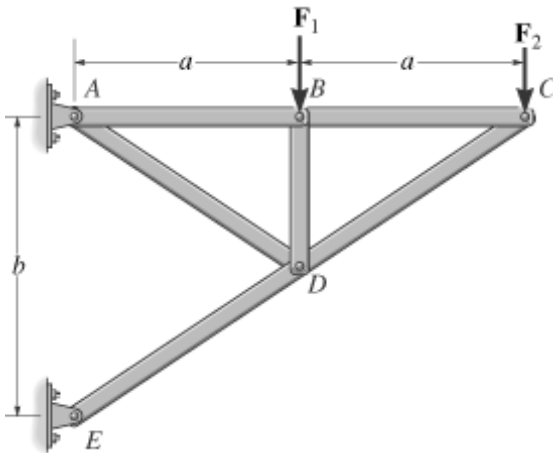
$g = 9.81 \ \frac{\text{m}}{\text{s}^2}$

$F_1 = 0 \text{ N}$

$F_2 = 0 \text{ N}$

$a = 3 \text{ m}$

$b = 4 \text{ m}$



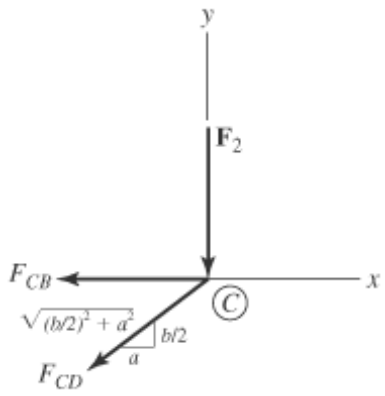
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Solution:

Initial Guesses:

$F_{CB} = 1\ \text{N}$ $F_{CD} = 1\ \text{N}$ $F_{BA} = 1\ \text{N}$

$F_{BD} = 1\ \text{N}$ $F_{DA} = 1\ \text{N}$ $F_{DE} = 1\ \text{N}$



Given

Joint C $-F_{CB} - F_{CD}\dfrac{2\,a}{\sqrt{(2\,a)^2 + b^2}} = 0$

$$-F_2 - F_{CD}\frac{b}{\sqrt{(2\,a)^2 + b^2}} - \rho g\left[\frac{a}{2} + \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{b}{4}\right)^2}\right] = 0$$

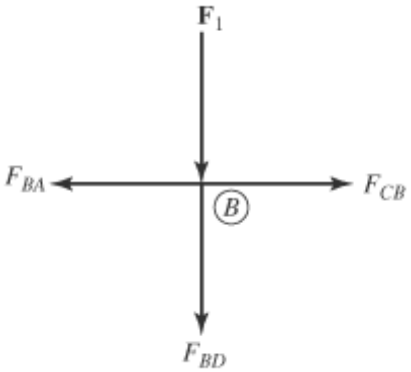
Joint B $-F_{BA} + F_{CB} = 0$

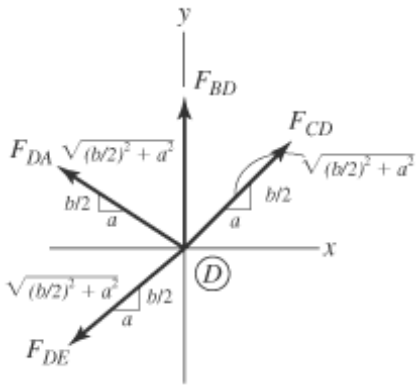
$$-F_1 - F_{BD} - \rho g\left(a + \frac{b}{4}\right) = 0$$

Joint D $F_{BD} + \left(F_{CD} + F_{DA} - F_{DE}\right)\dfrac{b}{\sqrt{(2\,a)^2 + b^2}} - \rho g\left[\dfrac{b}{4} + 3\sqrt{\left(\dfrac{a}{2}\right)^2 + \left(\dfrac{b}{4}\right)^2}\right] = 0$

$$\left(F_{CD} - F_{DA} - F_{DE}\right)\frac{2\,a}{\sqrt{(2\,a)^2 + b^2}} = 0$$

$$\begin{pmatrix} F_{CB} \\ F_{CD} \\ F_{BA} \\ F_{BD} \\ F_{DA} \\ F_{DE} \end{pmatrix} = \text{Find}\left(F_{CB}, F_{CD}, F_{BA}, F_{BD}, F_{DA}, F_{DE}\right)$$





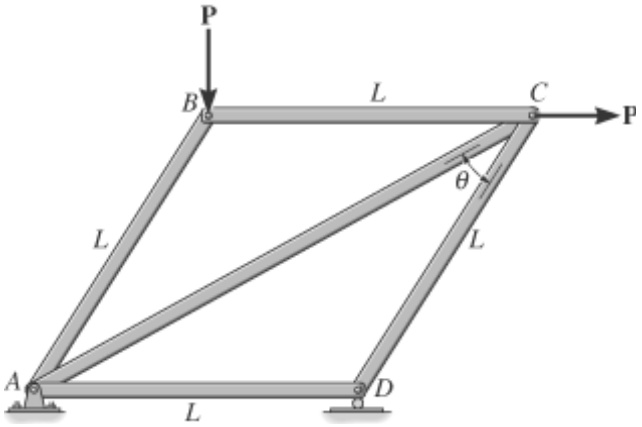
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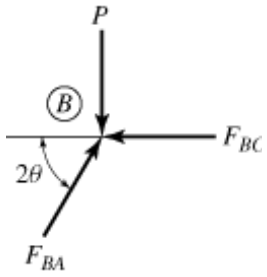
$$\begin{pmatrix} F_{CB} \\ F_{CD} \\ F_{BA} \\ F_{BD} \\ F_{DA} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} 389 \\ -467 \\ 389 \\ -314 \\ 736 \\ -1204 \end{pmatrix} \text{N}$$

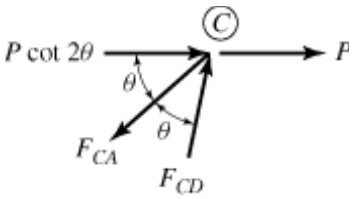
Positive means Tension,
Negative means Compression

Problem 6-21

Determine the force in each member of the truss in terms of the external loading and state if the members are in tension or compression.







Solution:

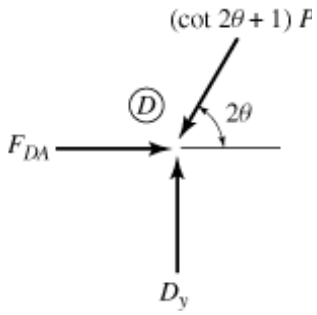
Joint *B*:

$$+\uparrow \Sigma F_y = 0; \quad F_{BA}\sin(2\theta) - P = 0$$

$$F_{BA} = P\csc(2\theta) \quad \text{(C)}$$

$$\rightarrow^{+} \Sigma F_x = 0; \quad F_{BA}\cos(2\theta) - F_{BC} = 0$$

$$F_{BC} = P\cot(2\theta) \quad \text{(C)}$$



Joint *C*:

$$\rightarrow^{+} \Sigma F_x = 0; \quad P\cot(2\theta) + P + F_{CD}\cos(2\theta) - F_{CA}\cos(\theta) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad F_{CD}\sin(2\theta) - F_{CA}\sin(\theta) = 0$$

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$$F_{CA} = \frac{\cot(2\ \theta) + 1}{\cos(\theta) - \sin(\theta)\cot(2\ \theta)}\mathrm{P}$$

$$F_{CA} = (\cot(\theta)\csc(\theta) - \sin(\theta) + 2\cos(\theta))P \quad \text{(T)}$$

$$F_{CD} = (\cot(2\ \theta) + 1)\ P \quad \text{(C)}$$

Joint *D*:

$$\xrightarrow{+} \Sigma F_x = 0; \quad F_{DA} - [\cot(2\ \theta) + 1]\ [\cos(2\ \theta)]\ P = 0$$

$$F_{DA} = [\cot(2\ \theta) + 1]\ [\cos(2\ \theta)]P \quad \text{(C)}$$

Problem 6-22

The maximum allowable tensile force in the members of the truss is T_{max}, and the maximum allowable compressive force is C_{max}. Determine the maximum magnitude *P* of the two loads that can be applied to the truss.

Units Used:

$\mathrm{kN} = 10^3\ \mathrm{N}$

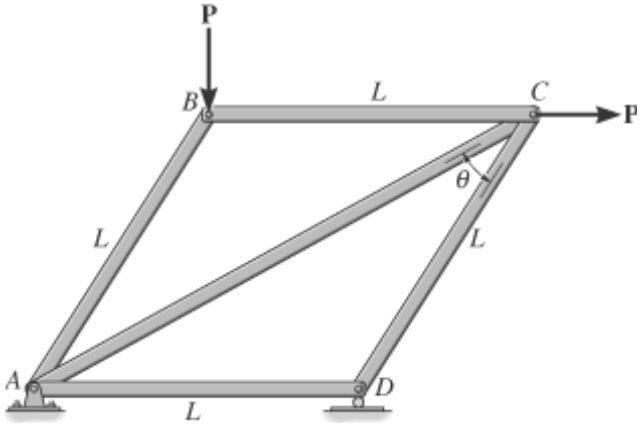
Given:

$T_{max} = 2\ \mathrm{kN}$

$C_{max} = 1.2\ \mathrm{kN}$

$L = 2\ \mathrm{m}$

$\theta = 30\ \mathrm{deg}$

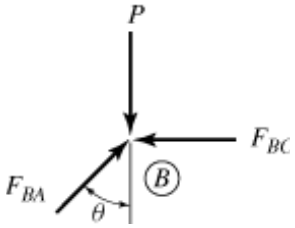


Solution:

Initial guesses (assume all bars are in tension). Use a unit load for *P* and then scale the answer later.

$F_{BA} = 1\ \mathrm{kN}$ $F_{BC} = 1\ \mathrm{kN}$ $F_{CA} = 1\ \mathrm{kN}$

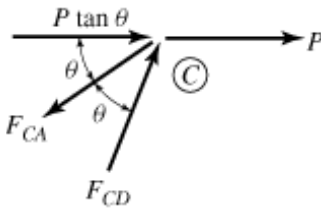
$F_{CD} = 1\ \mathrm{kN}$ $F_{DA} = 1\ \mathrm{kN}$ $P = 1\ \mathrm{kN}$



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Given

Joint B

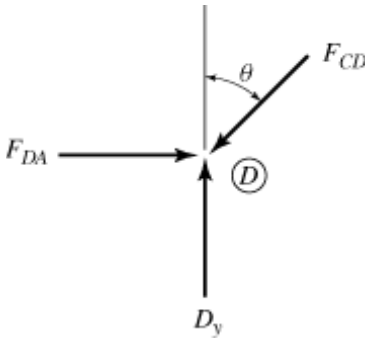


$+\uparrow \Sigma F_y = 0; \quad -F_{BA}\sin(2\ \theta) - P = 0$

$\xrightarrow{+} \Sigma F_x = 0; \quad -F_{BA}\cos(2\ \theta) + F_{BC} = 0$

Joint C

$+\uparrow \Sigma F_y = 0; \quad -F_{CA}\sin(\theta) - F_{CD}\sin(2\ \theta) = 0$



$\xrightarrow{+} \Sigma F_x = 0;$

Joint D $\quad -F_{BC} + P - F_{CD}\cos(2\ \theta) - F_{CA}\cos(\theta) = 0$

$\xrightarrow{+} \Sigma F_x = 0; \quad -F_{DA} + F_{CD}\cos(2\ \theta) = 0$

$$\begin{pmatrix} F_{BA} \\ F_{BC} \\ F_{CA} \\ F_{CD} \\ F_{DA} \end{pmatrix} = \text{Find}(F_{BA}, F_{BC}, F_{CA}, F_{CD}, F_{DA}) \qquad \text{ans} = \begin{pmatrix} F_{BA} \\ F_{BC} \\ F_{CA} \\ F_{CD} \\ F_{DA} \end{pmatrix} \qquad \text{ans} = \begin{pmatrix} -1.155 \\ -0.577 \\ 2.732 \\ -1.577 \\ -0.789 \end{pmatrix} \text{kN}$$

Now find the biggest tension and the biggest compression.

$T = \max(\text{ans}) \qquad T = 2.732\,\text{kN}$

$C = \min(\text{ans}) \qquad C = -1.577\,\text{kN}$

Decide which is more important and scale the answer

$$P = \min\left[\begin{pmatrix} \frac{T_{max}}{T} \\ \frac{-C_{max}}{C} \end{pmatrix} P\right] \qquad P = 732.051\,\text{N}$$

Problem 6-23

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The *Fink truss* supports the loads shown. Determine the force in each member and state if the members are in tension or compression. Approximate each joint as a pin.

Units Used:

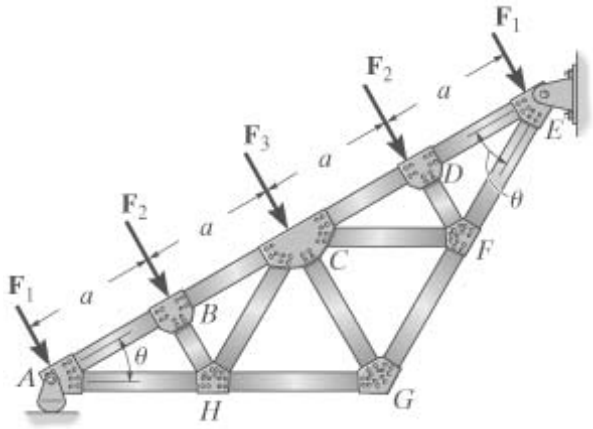
$\text{kip} = 10^3 \text{ lb}$

Given:

$F_1 = 500 \text{ lb}$ $a = 2.5 \text{ ft}$

$F_2 = 1 \text{ kip}$ $\theta = 30 \text{ deg}$

$F_3 = 1 \text{ kip}$



Solution:

Entire truss:

$\Sigma F_x = 0;$ $E_x = \left(F_1 + F_2 + F_3 + F_2 + F_1\right)\sin(\theta)$ $E_x = 2000 \text{ lb}$

$\Sigma M_E = 0;$ $-A_y 4a\cos(\theta) + F_1 4a + F_2 3a + F_3 2a + F_2 a = 0$

$$A_y = \frac{2\,F_1 + 2\,F_2 + F_3}{2\,\cos(\theta)} \qquad A_y = 2309.4 \text{ lb}$$

$\Sigma F_y = 0;$ $E_y = -A_y + 2\,\cos(\theta)\,F_1 + 2\,\cos(\theta)\,F_2 + \cos(\theta)\,F_3$ $E_y = 1154.7 \text{ lb}$

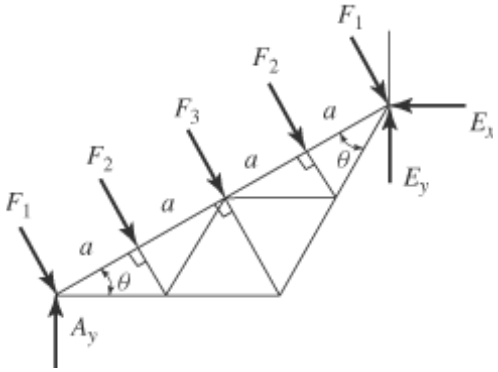
Joint *A*:

$\Sigma F_y = 0;$ $$F_{AB} = \frac{-\cos(\theta)\,F_1 + A_y}{\sin(\theta)}$$

$F_{AB} = 3.75 \text{ kip}$ (C)

$\Sigma F_x = 0;$ $F_{AH} = -\sin(\theta)\,F_1 + F_{AB}\cos(\theta)$

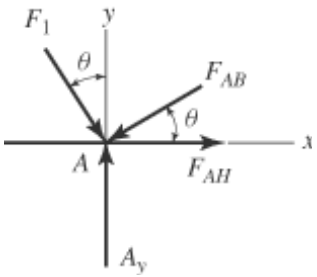
$F_{AH} = 3 \text{ kip}$ (T)



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Joint *B*:

$\Sigma F_x = 0;\quad F_{BC} = F_{AB} \quad F_{BC} = 3.75\,\text{kip}$ (C)



$\Sigma F_y = 0;\quad F_{BH} = F_2 \quad F_{BH} = 1\,\text{kip}$ (C)

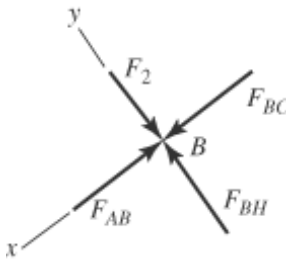
Joint *H*:

$\Sigma F_y = 0;\quad F_{HC} = F_2 \quad F_{HC} = 1\,\text{kip}$ (T)

$\Sigma F_x = 0;\quad F_{GH} = -F_2\cos(-90\text{ deg} + \theta) - F_{HC}\cos(-90\text{ deg} + \theta) + F_{AH}$

$F_{GH} = 2\,\text{kip}$ (T)

Joint *E*:

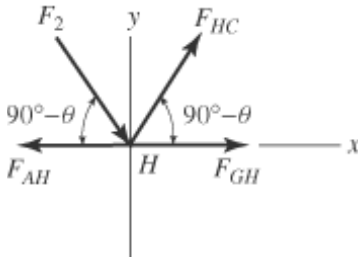


$\Sigma F_y = 0;\quad F_{EF} = \dfrac{-\left(F_1 - E_x\sin(\theta) - E_y\cos(\theta)\right)}{\sin(\theta)}$

$F_{EF} = 3\,\text{kip}$ (T)

$\Sigma F_x = 0;\quad F_{ED} = -E_y\sin(\theta) + E_x\cos(\theta) + F_{EF}\cos(\theta)$

$F_{ED} = 3.75\,\text{kip}$ (C)



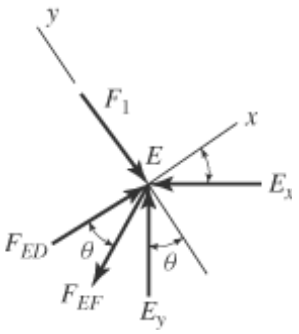
Joint *D*:

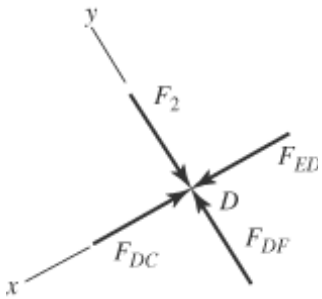
$\Sigma F_x = 0;\quad F_{DC} = F_{ED}$

$F_{DC} = 3.75\,\text{kip}$ (C)

$\Sigma F_y = 0;\quad F_{DF} = F_2$

$F_{DF} = 1\,\text{kip}$ (C)





Joint *C*:

$\Sigma F_x = 0;\quad F_{CF} = F_{HC}$

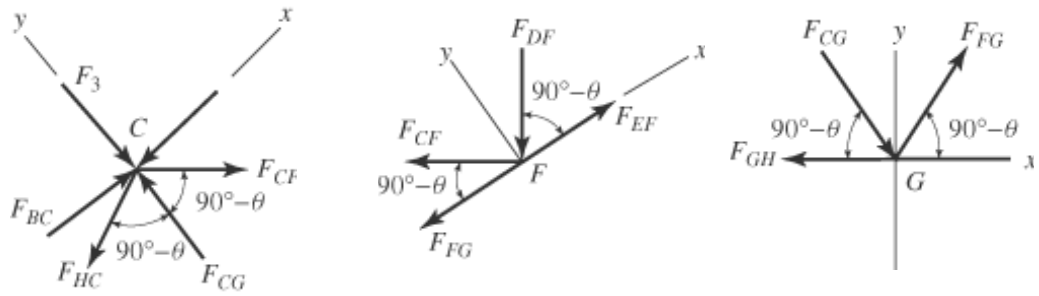
$F_{CF} = 1\,\text{kip}$ (T)

$\Sigma F_y = 0;\quad F_{CG} = F_3 + F_{HC}\cos(90\text{ deg} - \theta)(2) \quad F_{CG} = 2\,\text{kip}$ (C)

Joint *F*:

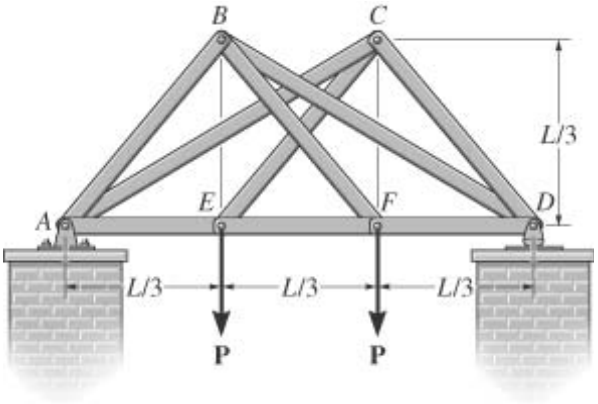
$\Sigma F_x = 0;\quad F_{FG} = F_{EF} - F_{CF}\cos(90\text{ deg} - \theta)(2) \quad F_{FG} = 2\,\text{kip}$ (T)

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Problem 6-24

Determine the force in each member of the double scissors truss in terms of the load P and state if the members are in tension or compression.



Solution:

$$\overset{+}{\circlearrowleft} \Sigma M_A = 0; \qquad P\frac{L}{3} + P\frac{2\,L}{3} - D_y\,L = 0$$

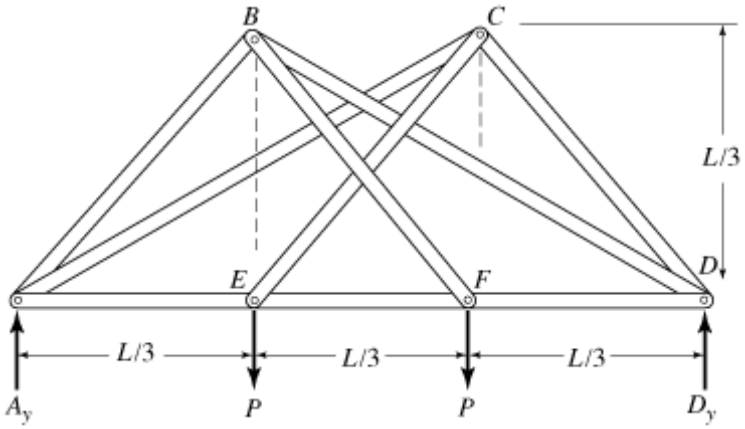
$$+\uparrow \Sigma F_y = 0; \qquad A_y + D_y - 2\,P = 0$$

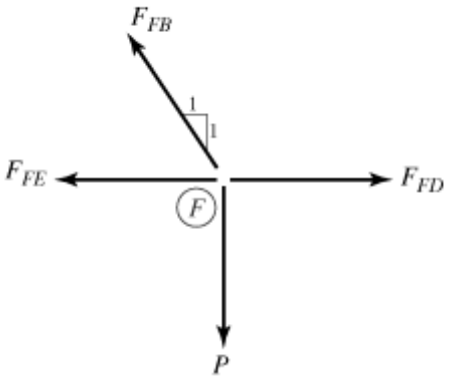
Joint F:

$$+\uparrow \Sigma F_y = 0; \qquad F_{FB}\frac{1}{\sqrt{2}} - P = 0$$

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad F_{FD} - F_{FE} - F_{FB}\frac{1}{\sqrt{2}} = 0$$

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Joint E:

$$+\uparrow \Sigma F_y = 0; \qquad F_{EC}\frac{1}{\sqrt{2}} - P = 0$$

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{EF} - F_{EA} + F_{EC}\frac{1}{\sqrt{2}} = 0$$

Joint B:

$$+\uparrow \Sigma F_y = 0; \qquad F_{BA}\frac{1}{\sqrt{2}} + F_{BD}\frac{1}{\sqrt{5}} - F_{FB}\frac{1}{\sqrt{2}} = 0$$

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{BA}\frac{1}{\sqrt{2}} + F_{FB}\frac{1}{\sqrt{2}} - F_{BD}\left(\frac{2}{\sqrt{5}}\right) = 0$$

Joint C:

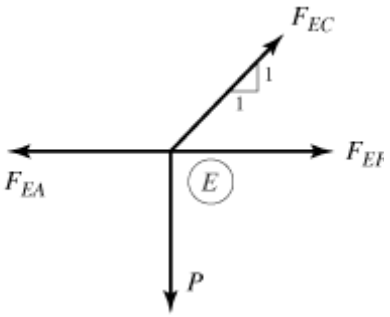
$$+\uparrow \Sigma F_y = 0; \qquad F_{CA}\frac{1}{\sqrt{5}} + F_{CD}\frac{1}{\sqrt{2}} - F_{EC}\frac{1}{\sqrt{2}} = 0$$

$$\xrightarrow{+} \Sigma F_x = 0; \qquad F_{CA}\frac{2}{\sqrt{5}} - F_{EC}\frac{1}{\sqrt{2}} - F_{CD}\frac{1}{\sqrt{2}} = 0$$

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Joint *A*:

$$\xrightarrow{+} \Sigma F_x = 0; \quad F_{AE} - F_{BA}\frac{1}{\sqrt{2}} - F_{CA}\frac{2}{\sqrt{5}} = 0$$



Solving we find

$$F_{EF} = 0.667\,P\,(\mathrm{T})$$

$$F_{FD} = 1.67\,P\,(\mathrm{T})$$

$$F_{AB} = 0.471\,P\,(\mathrm{C})$$

$$F_{AE} = 1.67\,P\,(\mathrm{T})$$

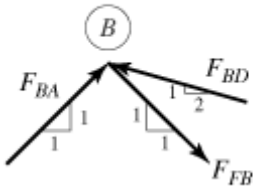
$$F_{AC} = 1.49\,P\,(\mathrm{C})$$

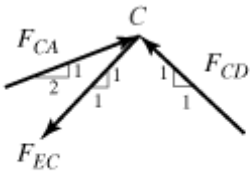
$$F_{BF} = 1.41\,P\,(\mathrm{T})$$

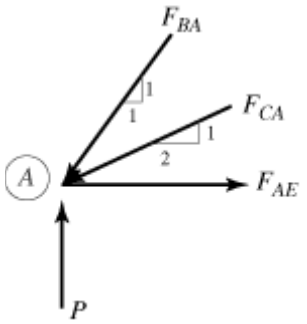
$$F_{BD} = 1.49\,P\,(\mathrm{C})$$

$$F_{EC} = 1.41\,P\,(\mathrm{T})$$

$$F_{CD} = 0.471\,P\,(\mathrm{C})$$







Problem 6-25

Determine the force in each member of the truss and state if the members are in tension or compression. *Hint*: The vertical component of force at *C* must equal zero. Why?

Units Used:

$\mathrm{kN} = 10^3\ \mathrm{N}$

Given:

$F_1 = 6\ \mathrm{kN}$

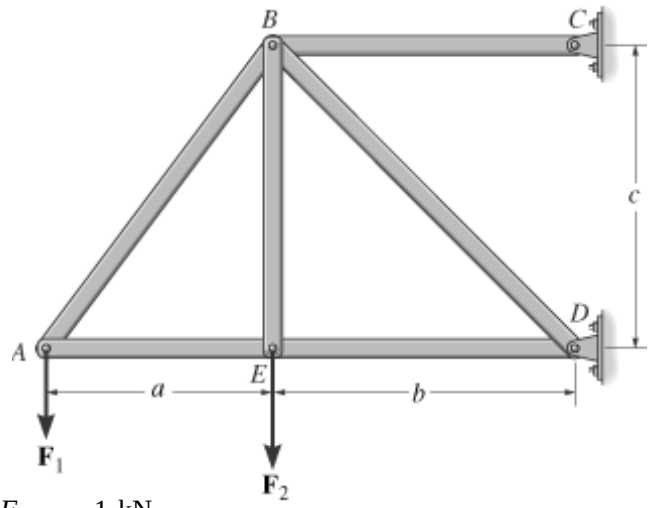
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$F_2 = 8$ kN

$a = 1.5$ m

$b = 2$ m

$c = 2$ m



Solution:

Initial Guesses:

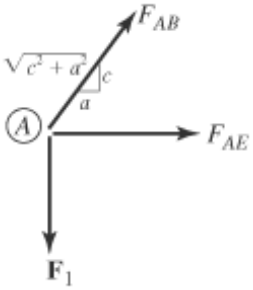
$F_{AB} = 1$ kN $F_{AE} = 1$ kN $F_{EB} = 1$ kN

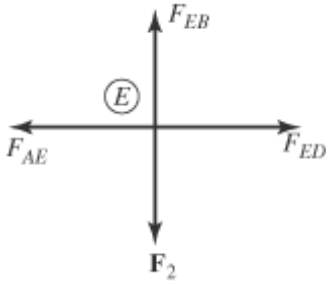
$F_{BC} = 1$ kN $F_{BD} = 1$ kN $F_{ED} = 1$ kN

Given

Joint A $F_{AB}\dfrac{a}{\sqrt{a^2 + c^2}} + F_{AE} = 0$

$F_{AB}\dfrac{c}{\sqrt{a^2 + c^2}} - F_1 = 0$



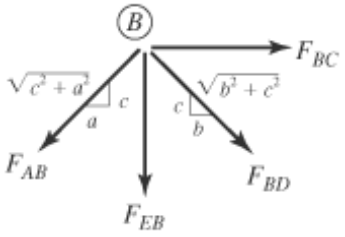


Joint E $F_{ED} - F_{AE} = 0$

$F_{EB} - F_2 = 0$

Joint B $F_{BC} + F_{BD}\dfrac{b}{\sqrt{b^2 + c^2}} - F_{AB}\dfrac{a}{\sqrt{a^2 + c^2}} = 0$

$-F_{EB} - F_{BD}\dfrac{c}{\sqrt{b^2 + c^2}} - F_{AB}\dfrac{c}{\sqrt{a^2 + c^2}} = 0$



$$\begin{pmatrix} F_{AB} \\ F_{AE} \\ F_{EB} \\ F_{BC} \\ F_{BD} \\ F_{ED} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AE}, F_{EB}, F_{BC}, F_{BD}, F_{ED}\right)$$

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$$\begin{pmatrix} F_{AB} \\ F_{AE} \\ F_{EB} \\ F_{BC} \\ F_{BD} \\ F_{ED} \end{pmatrix} = \begin{pmatrix} 7.5 \\ -4.5 \\ 8 \\ 18.5 \\ -19.799 \\ -4.5 \end{pmatrix} \text{kN}$$

Positive means Tension,
Negative means Compresson.

Problem 6-26

Each member of the truss is uniform and has a mass density ρ. Remove the external loads F_1 and F_2 and determine the approximate force in each member due to the weight of the truss. State if the members are in tension or compression. Solve the problem by *assuming* the weight of each member can be represented as a vertical force, half of which is applied at each end of the member.

Given:

$F_1 = 0$

$F_2 = 0$

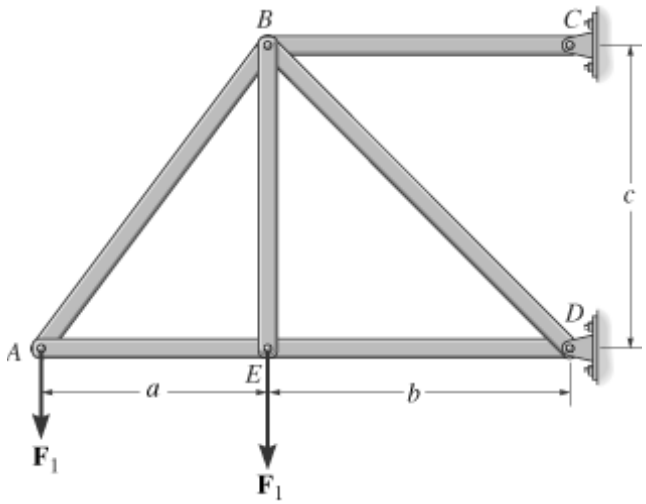
$\rho = 8 \dfrac{\text{kg}}{\text{m}}$

$a = 1.5 \text{ m}$

$b = 2 \text{ m}$

$c = 2 \text{ m}$

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

Find the weights of each bar.

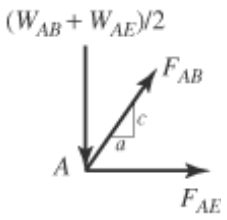
$W_{AB} = \rho g \sqrt{a^2 + c^2}$ $W_{BC} = \rho g b$ $W_{BE} = \rho g c$

$W_{AE} = \rho g a$ $W_{BD} = \rho g \sqrt{b^2 + c^2}$ $W_{DE} = \rho g b$

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Guesses $F_{AB} = 1\ \text{N}$ $F_{AE} = 1\ \text{N}$ $F_{BE} = 1\ \text{N}$

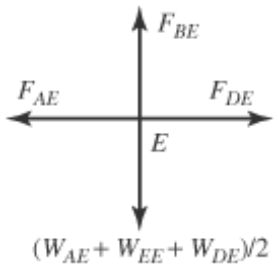
$F_{BC} = 1\ \text{N}$ $F_{BD} = 1\ \text{N}$ $F_{DE} = 1\ \text{N}$



Given

Joint A $F_{AE} + \dfrac{a}{\sqrt{a^2 + c^2}} F_{AB} = 0$

$$\frac{c}{\sqrt{a^2 + c^2}} F_{AB} - \frac{W_{AB} + W_{AE}}{2} = 0$$



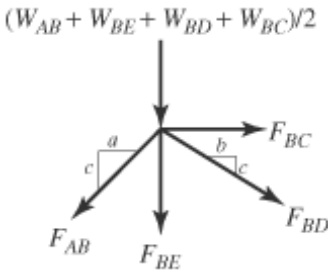
Joint E $F_{DE} - F_{AE} = 0$

$$F_{BE} - \frac{W_{AE} + W_{BE} + W_{DE}}{2} = 0$$

Joint B $F_{BC} + \dfrac{b}{\sqrt{b^2 + c^2}} F_{BD} - \dfrac{a}{\sqrt{a^2 + c^2}} F_{AB} = 0$

$$\frac{-c}{\sqrt{a^2 + c^2}} F_{AB} - F_{BE} - \frac{c}{\sqrt{b^2 + c^2}} F_{BD} - \frac{W_{AB} + W_{BE} + W_{BD} + W_{BC}}{2} = 0$$

$$\begin{pmatrix} F_{AB} \\ F_{AE} \\ F_{BC} \\ F_{BD} \\ F_{BE} \\ F_{DE} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AE}, F_{BC}, F_{BD}, F_{BE}, F_{DE}\right)$$



$$\begin{pmatrix} F_{AB} \\ F_{AE} \\ F_{BC} \\ F_{BD} \\ F_{BE} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} 196 \\ -118 \\ 857 \\ -1045 \\ 216 \\ -118 \end{pmatrix} \text{N}$$

Positive means tension,
Negative means Compression.

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Problem 6-27

Determine the force in each member of the truss and state if the members are in tension or compression.

Units Used:

$kN = 10^3 N$

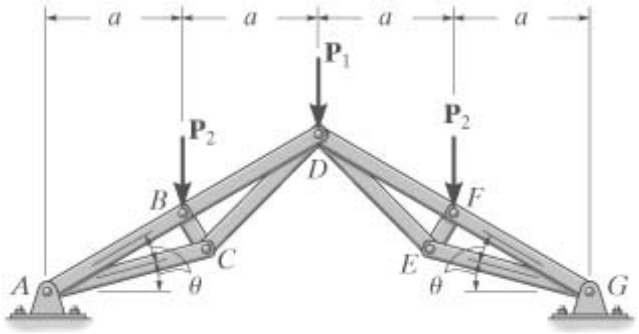
Given:

$P_1 = 4$ kN

$P_2 = 0$ kN

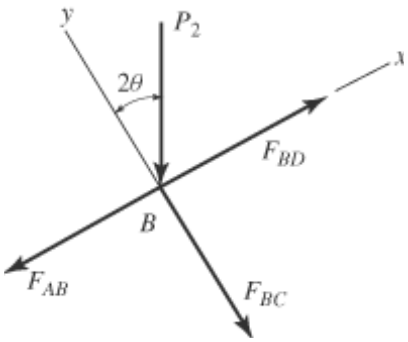
$a = 2$ m

$\theta = 15$ deg



Solution:

Take advantage of the symetry.



Initial Guesses:

$F_{BD} = 1$ kN $F_{CD} = 1$ kN $F_{AB} = 1$ kN

$F_{CA} = 1$ kN $F_{BC} = 1$ kN

Given

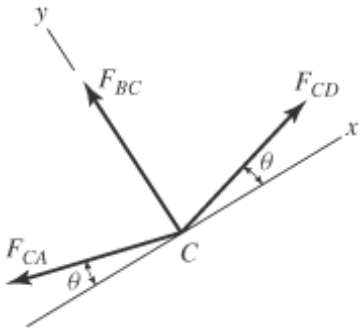
Joint D $\dfrac{-P_1}{2} - F_{BD}\sin(2\ \theta) - F_{CD}\sin(3\ \theta) = 0$

Joint B $-P_2\cos(2\ \theta) - F_{BC} = 0$

$F_{BD} - F_{AB} - P_2\sin(2\ \theta) = 0$

Joint C $F_{CD}\cos(\theta) - F_{CA}\cos(\theta) = 0$

$(F_{CD} + F_{CA})\sin(\theta) + F_{BC} = 0$



$$\begin{pmatrix} F_{BD} \\ F_{CD} \\ F_{AB} \\ F_{CA} \\ F_{BC} \end{pmatrix} = \text{Find}(F_{BD}, F_{CD}, F_{AB}, F_{CA}, F_{BC}) \qquad \begin{pmatrix} F_{FD} \\ F_{ED} \\ F_{GF} \\ F_{EG} \\ F_{FE} \end{pmatrix} = \begin{pmatrix} F_{BD} \\ F_{CD} \\ F_{AB} \\ F_{CA} \\ F_{BC} \end{pmatrix}$$

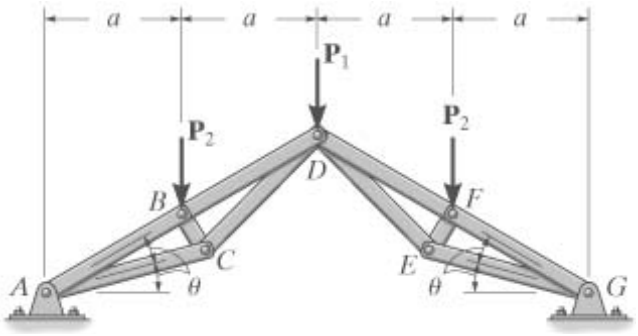
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$$\begin{pmatrix} F_{BD} \\ F_{CD} \\ F_{AB} \\ F_{CA} \\ F_{BC} \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \\ -4 \\ 0 \\ 0 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{FD} \\ F_{ED} \\ F_{GF} \\ F_{EG} \\ F_{FE} \end{pmatrix} = \begin{pmatrix} -4 \\ 0 \\ -4 \\ 0 \\ 0 \end{pmatrix} \text{kN}$$

Positvive means Tension,
Negative means Compression

Problem 6-28

Determine the force in each member of the truss and state if the members are in tension or compression.



Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

$P_1 = 2 \text{ kN}$

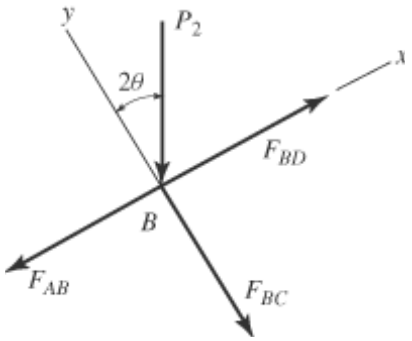
$P_2 = 4 \text{ kN}$

$a = 2 \text{ m}$

$\theta = 15 \text{ deg}$

Solution:

Take advantage of the symmetry.



Initial Guesses:

$F_{BD} = 1 \text{ kN} \quad F_{CD} = 1 \text{ kN} \quad F_{AB} = 1 \text{ kN}$

$F_{CA} = 1 \text{ kN} \quad F_{BC} = 1 \text{ kN}$

Given

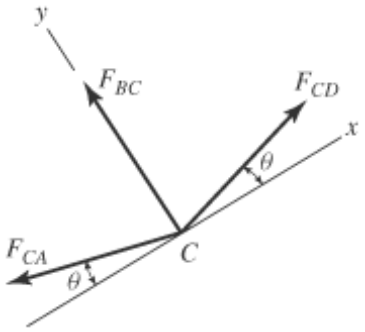
Joint D $\quad \dfrac{-P_1}{2} - F_{BD}\sin(2\ \theta) - F_{CD}\sin(3\ \theta) = 0$

Joint B $\quad -P_2\cos(2\ \theta) - F_{BC} = 0$

$F_{BD} - F_{AB} - P_2\sin(2\theta) = 0$

Joint C $\quad F_{CD}\cos(\theta) - F_{CA}\cos(\theta) = 0$

$(F_{CD} + F_{CA})\sin(\theta) + F_{BC} = 0$



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$$\begin{pmatrix} F_{BD} \\ F_{CD} \\ F_{AB} \\ F_{CA} \\ F_{BC} \end{pmatrix} = \text{Find}\left(F_{BD}, F_{CD}, F_{AB}, F_{CA}, F_{BC}\right) \qquad \begin{pmatrix} F_{FD} \\ F_{ED} \\ F_{GF} \\ F_{EG} \\ F_{FE} \end{pmatrix} = \begin{pmatrix} F_{BD} \\ F_{CD} \\ F_{AB} \\ F_{CA} \\ F_{BC} \end{pmatrix}$$

$$\begin{pmatrix} F_{BD} \\ F_{CD} \\ F_{AB} \\ F_{CA} \\ F_{BC} \end{pmatrix} = \begin{pmatrix} -11.46 \\ 6.69 \\ -13.46 \\ 6.69 \\ -3.46 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{FD} \\ F_{ED} \\ F_{GF} \\ F_{EG} \\ F_{FE} \end{pmatrix} = \begin{pmatrix} -11.46 \\ 6.69 \\ -13.46 \\ 6.69 \\ -3.46 \end{pmatrix} \text{kN}$$

Positvive means Tension,
Negative means Compression

Problem 6-29

Determine the force in each member of the truss and state if the members are in tension or compression.

Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$F_1 = 2 \text{ kip}$

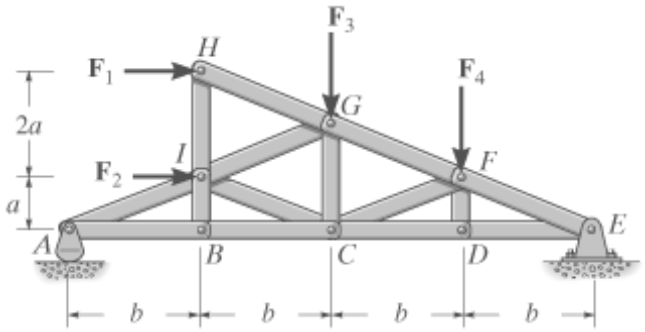
$F_2 = 1.5 \text{ kip}$

$F_3 = 3 \text{ kip}$

$F_4 = 3 \text{ kip}$

$a = 4 \text{ ft}$

$b = 10 \text{ ft}$



Solution: $\theta = \text{atan}\left(\frac{a}{b}\right)$

Initial Guesses

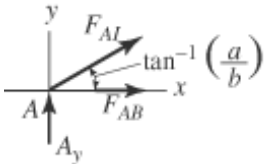
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$F_{AB} = 1$ lb $F_{BC} = 1$ lb $F_{CD} = 1$ lb $F_{DE} = 1$ lb

$F_{AI} = 1$ lb $F_{BI} = 1$ lb $F_{CI} = 1$ lb $F_{CG} = 1$ lb

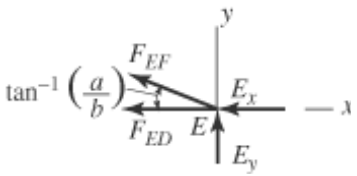
$F_{CF} = 1$ lb $F_{DF} = 1$ lb $F_{EF} = 1$ lb $F_{HI} = 1$ lb

$F_{GI} = 1$ lb $F_{GH} = 1$ lb $F_{FG} = 1$ lb



Given

Joint A $F_{AI}\cos(\theta) + F_{AB} = 0$



Joint B $F_{BC} - F_{AB} = 0$

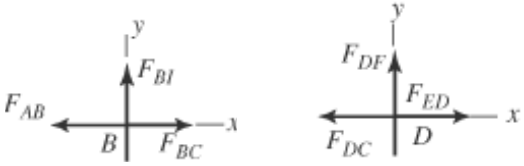
$F_{BI} = 0$

Joint C $F_{CD} - F_{BC} + \left(F_{CF} - F_{CI}\right)\cos(\theta) = 0$

$F_{CG} + \left(F_{CF} + F_{CI}\right)\sin(\theta) = 0$

Joint D $F_{DE} - F_{CD} = 0$

$F_{DF} = 0$

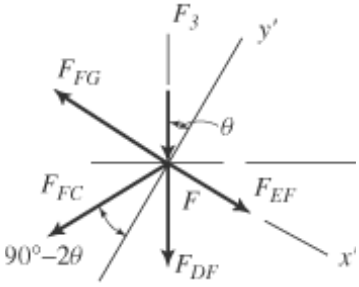


Joint I $F_2 + \left(F_{GI} + F_{CI} - F_{AI}\right)\cos(\theta) = 0$

$F_{HI} - F_{BI} + \left(F_{GI} - F_{AI} - F_{CI}\right)\sin(\theta) = 0$

Joint H $F_{GH}\cos(\theta) + F_1 = 0$

$-F_{GH}\sin(\theta) - F_{HI} = 0$

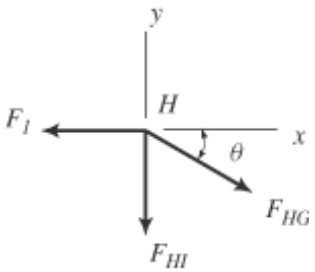


Joint G $\left(F_{FG} - F_{GH} - F_{GI}\right)\cos(\theta) = 0$

$-F_3 - F_{CG} + \left(F_{GH} - F_{FG} - F_{GI}\right)\sin(\theta) = 0$

Joint F $\left(F_{EF} - F_{FG} - F_{CF}\right)\cos(\theta) = 0$

$\left(F_{FG} - F_{CF} - F_{EF}\right)\sin(\theta) - F_4 - F_{DF} = 0$



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$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CD} \\ F_{DE} \\ F_{AI} \\ F_{BI} \\ F_{CI} \\ F_{CG} \\ F_{CF} \\ F_{DF} \\ F_{EF} \\ F_{HI} \\ F_{GI} \\ F_{GH} \\ F_{FG} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{BC}, F_{CD}, F_{DE}, F_{AI}, F_{BI}, F_{CI}, F_{CG}, F_{CF}, F_{DF}, F_{EF}, F_{HI}, F_{GI}, F_{GH}, F\right.$$

$$\begin{pmatrix} F_{AB} \\ F_{BC} \\ F_{CD} \\ F_{DE} \\ F_{AI} \end{pmatrix} = \begin{pmatrix} 3.75 \\ 3.75 \\ 7.75 \\ 7.75 \\ -4.04 \end{pmatrix} \text{kip} \qquad \begin{pmatrix} F_{BI} \\ F_{CI} \\ F_{CG} \\ F_{CF} \\ F_{DF} \end{pmatrix} = \begin{pmatrix} 0 \\ 0.27 \\ 1.4 \\ -4.04 \\ 0 \end{pmatrix} \text{kip} \qquad \begin{pmatrix} F_{EF} \\ F_{HI} \\ F_{GI} \\ F_{GH} \\ F_{FG} \end{pmatrix} = \begin{pmatrix} -12.12 \\ 0.8 \\ -5.92 \\ -2.15 \\ -8.08 \end{pmatrix} \text{kip}$$

Positive means Tension, Negative means Compression

Problem 6-30

The *Howe bridge truss* is subjected to the loading shown. Determine the force in members *DE*, *EH*, and *HG*, and state if the members are in tension or compression.

Units Used:

$$\text{kN} = 10^3 \text{ N}$$

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Given:

$F_1 = 30$ kN

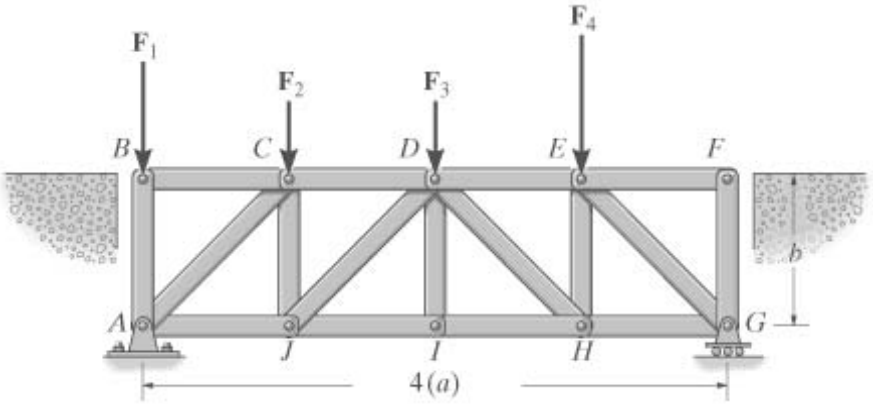
$F_2 = 20$ kN

$F_3 = 20$ kN

$F_4 = 40$ kN

$a = 4$ m

$b = 4$ m

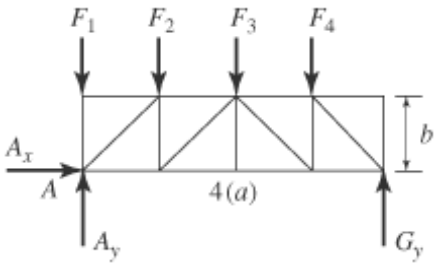


Solution:

$$-F_2\,a - F_3(2a) - F_4(3a) + G_y(4a) = 0$$

$$G_y = \frac{F_2 + 2F_3 + 3F_4}{4}$$

$$G_y = 45\,\text{kN}$$

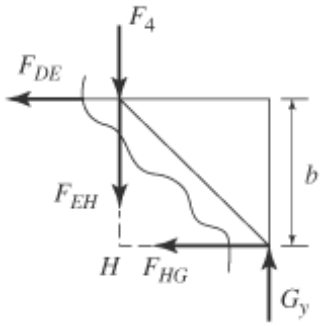


Guesses $F_{DE} = 1$ kN $F_{EH} = 1$ kN $F_{HG} = 1$ kN

Given

$$-F_{DE} - F_{HG} = 0 \qquad G_y - F_4 - F_{EH} = 0$$

$$F_{DE}\,b + G_y\,a = 0$$



$$\begin{pmatrix} F_{DE} \\ F_{EH} \\ F_{HG} \end{pmatrix} = \text{Find}\left(F_{DE}, F_{EH}, F_{HG}\right) \qquad \begin{pmatrix} F_{DE} \\ F_{EH} \\ F_{HG} \end{pmatrix} = \begin{pmatrix} -45 \\ 5 \\ 45 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-31

The *Pratt bridge truss* is subjected to the loading shown. Determine the force in members *LD*, *LK*, *CD*, and *KD*, and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3$ N

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Given:

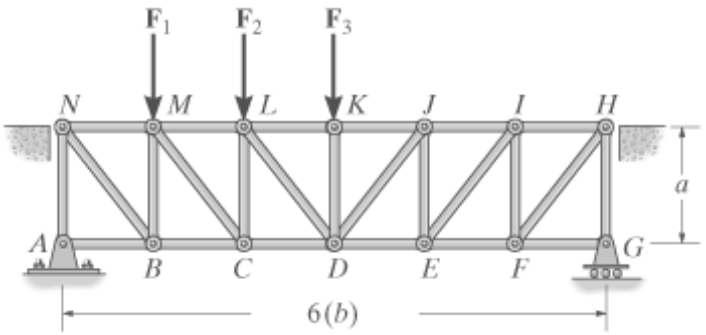
$F_1 = 50$ kN

$F_2 = 50$ kN

$F_3 = 50$ kN

$a = 4$ m

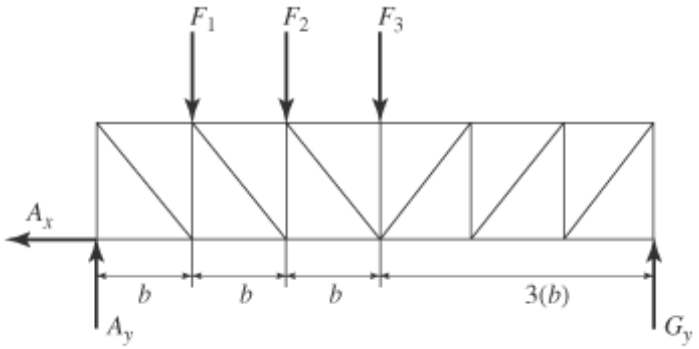
$b = 3$ m



Solution:

$A_x = 0$

$A_y = \dfrac{3F_3 + 4F_2 + 5F_1}{6}$



Guesses

$F_{LD} = 1$ kN $\quad F_{LK} = 1$ kN

$F_{CD} = 1$ kN $\quad F_{KD} = 1$ kN

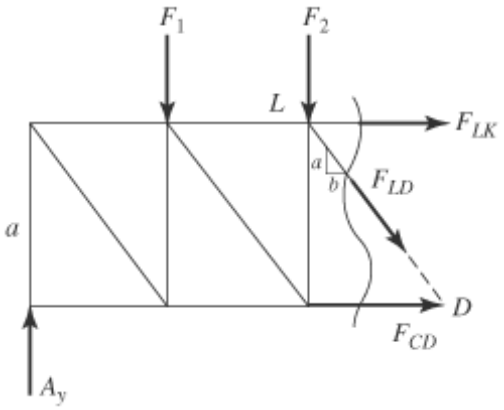
Given

$F_2\, b + F_1(2b) - A_y(3b) - F_{LK}\, a = 0$

$F_{CD} a + F_1 b - A_y(2b) = 0$

$A_y - F_1 - F_2 - \left(\dfrac{a}{\sqrt{a^2 + b^2}}\right) F_{LD} = 0$

$-F_3 - F_{KD} = 0$

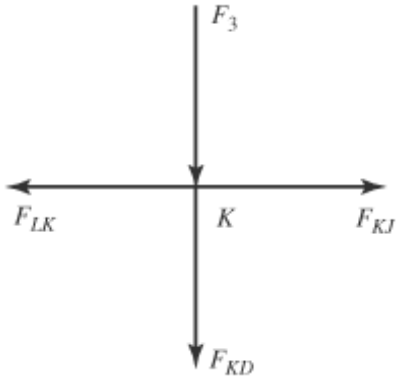


$$\begin{pmatrix} F_{LD} \\ F_{LK} \\ F_{CD} \\ F_{KD} \end{pmatrix} = \text{Find}\left(F_{LD}, F_{LK}, F_{CD}, F_{KD}\right)$$

$$\begin{pmatrix} F_{LD} \\ F_{LK} \\ F_{CD} \\ F_{KD} \end{pmatrix} = \begin{pmatrix} 0 \\ -112.5 \\ 112.5 \\ -50 \end{pmatrix} \text{kN}$$

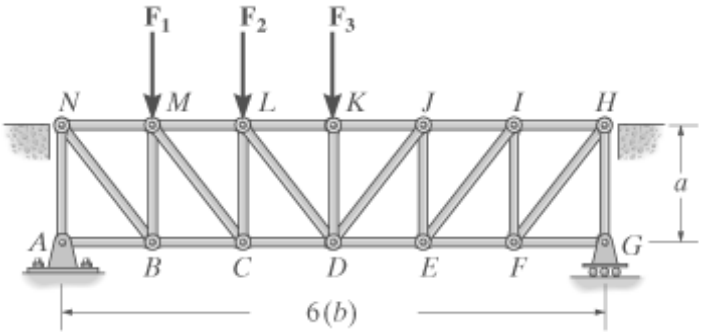
Positive (T)
Negative (C)

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Problem 6-32

The *Pratt bridge truss* is subjected to the loading shown. Determine the force in members *JI*, *JE*, and *DE*, and state if the members are in tension or compression.



Units Used:

$kN = 10^3\ N$

Given:

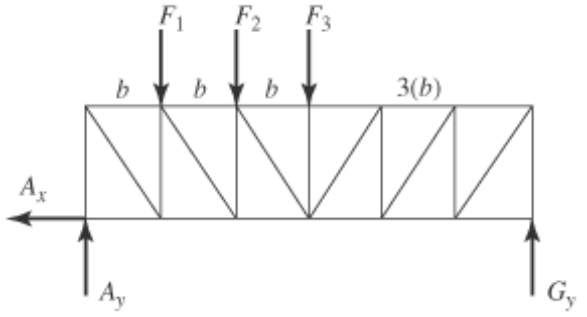
$F_1 = 50\ kN$

$F_2 = 50\ kN$

$F_3 = 50\ kN$

$a = 4\ m$

$b = 3\ m$



Solution:

Initial Guesses

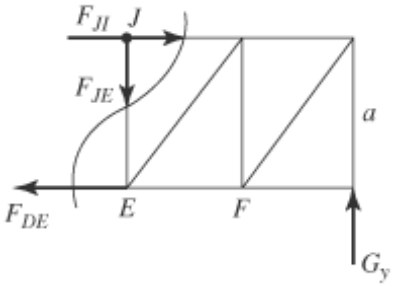
$G_y = 1\ kN \quad F_{JI} = 1\ kN$

$F_{JE} = 1\ kN \quad F_{DE} = 1\ kN$

Given

Entire Truss

$$-F_1 b - F_2(2b) - F_3(3b) + G_y(6b) = 0$$



Section

$$-F_{DE} - F_{JI} = 0 \qquad F_{JE} + G_y = 0$$

$$G_y(2b) - F_{DE}\, a = 0$$

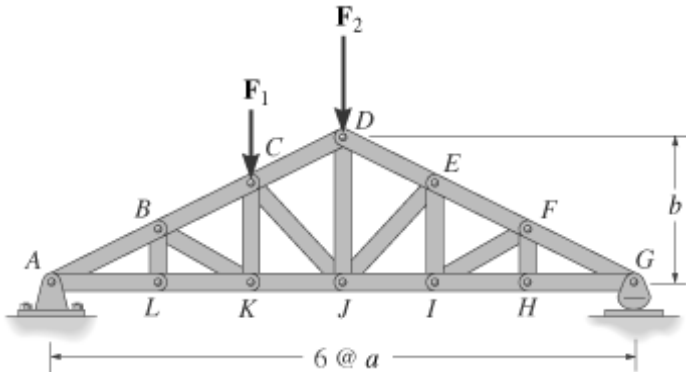
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$$\begin{pmatrix} G_y \\ F_{JI} \\ F_{JE} \\ F_{DE} \end{pmatrix} = \text{Find}\left(G_y, F_{JI}, F_{JE}, F_{DE}\right) \quad G_y = 50\,\text{kN} \qquad \begin{pmatrix} F_{JI} \\ F_{JE} \\ F_{DE} \end{pmatrix} = \begin{pmatrix} -75 \\ -50 \\ 75 \end{pmatrix} \text{kN}$$

Positive means Tension,
Negative means Compression

Problem 6-33

The roof truss supports the vertical loading shown. Determine the force in members *BC*, *CK*, and *KJ* and state if these members are in tension or compression.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 4\ \text{kN}$

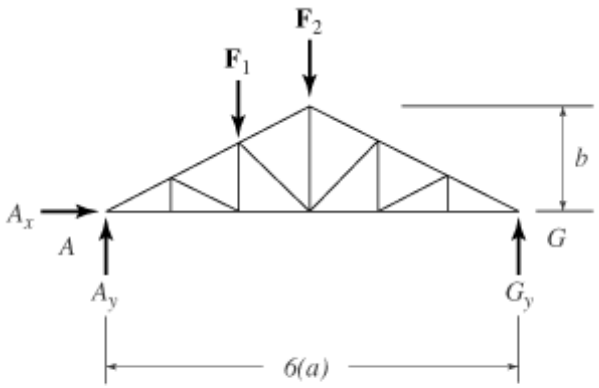
$F_2 = 8\ \text{kN}$

$a = 2\ \text{m}$

$b = 3\ \text{m}$

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Solution:



Initial Guesses

$A_x = 1$ kN $\quad A_y = 1$ kN

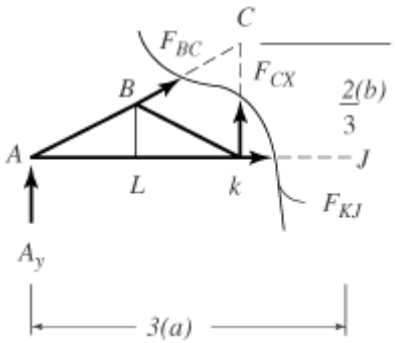
$F_{BC} = 1$ kN $\quad F_{CK} = 1$ kN

$F_{KJ} = 1$ kN

Given

$A_x = 0$

$F_2(3a) + F_1(4a) - A_y(6a) = 0$



$$F_{KJ}\left(\frac{2b}{3}\right) + A_x\left(\frac{2b}{3}\right) - A_y(2a) = 0$$

$$F_{KJ} + A_x + \left(\frac{3a}{\sqrt{b^2 + 9a^2}}\right)F_{BC} = 0$$

$$F_{CK} + A_y + \left(\frac{b}{\sqrt{b^2 + 9a^2}}\right)F_{BC} = 0$$

$$\begin{pmatrix} A_x \\ A_y \\ F_{KJ} \\ F_{CK} \\ F_{BC} \end{pmatrix} = \text{Find}\left(A_x, A_y, F_{KJ}, F_{CK}, F_{BC}\right)$$

$$\begin{pmatrix} A_x \\ A_y \\ F_{KJ} \\ F_{CK} \\ F_{BC} \end{pmatrix} = \begin{pmatrix} 0 \\ 6.667 \\ 13.333 \\ 0 \\ -14.907 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-34

Determine the force in members *CD, CJ, KJ,* and *DJ* of the truss which serves to support the deck of a bridge. State if these members are in tension or compression.

Units Used:

kip $= 10^3$ lb

Given:

$F_1 = 4000$ lb

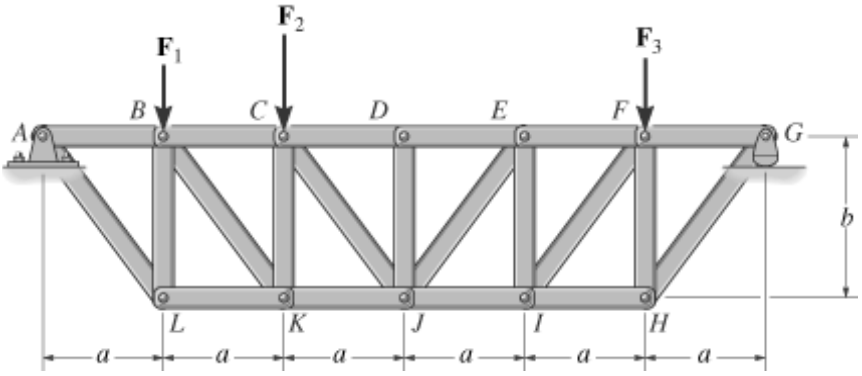
$F_2 = 8000$ lb

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$F_3 = 5000$ lb

$a = 9$ ft

$b = 12$ ft



Solution

Initial Guesses: $F_{DJ} = 1$ kip

$A_y = 1$ kip $F_{CD} = 1$ kip

$F_{CJ} = 1$ kip $F_{KJ} = 1$ kip

Given

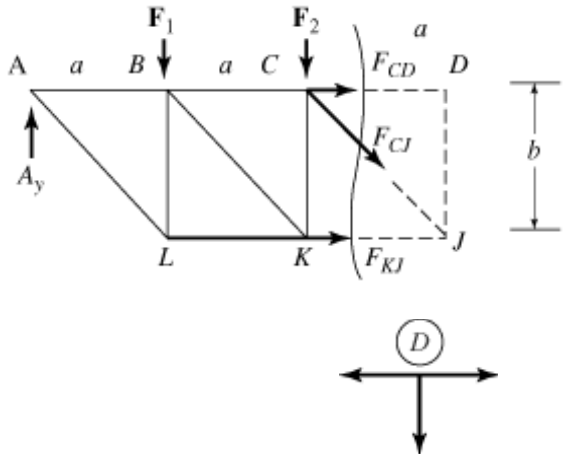
$$F_3\, a + F_2(4a) + F_1(5a) - A_y(6a) = 0$$

$$-A_y(2a) + F_1\, a + F_{KJ}\, b = 0$$

$$F_{CD} + F_{KJ} + \left(\frac{a}{\sqrt{a^2 + b^2}}\right) F_{CJ} = 0$$

$$A_y - F_1 - F_2 - \left(\frac{b}{\sqrt{a^2 + b^2}}\right) F_{CJ} = 0$$

$$-F_{DJ} = 0$$



$$\begin{pmatrix} A_y \\ F_{KJ} \\ F_{CJ} \\ F_{DJ} \\ F_{CD} \end{pmatrix} = \text{Find}\left(A_y, F_{KJ}, F_{CJ}, F_{DJ}, F_{CD}\right)$$

$$\begin{pmatrix} A_y \\ F_{KJ} \\ F_{CJ} \\ F_{DJ} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} 9.5 \\ 11.25 \\ -3.125 \\ 0 \\ -9.375 \end{pmatrix} \text{kip}$$

Positive (T)
Negative (C)

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Problem 6-35

Determine the force in members *EI* and *JI* of the truss which serves to support the deck of a bridge. State if these members are in tension or compression.

Units Used:

$\text{kip} = 10^3\ \text{lb}$

Given:

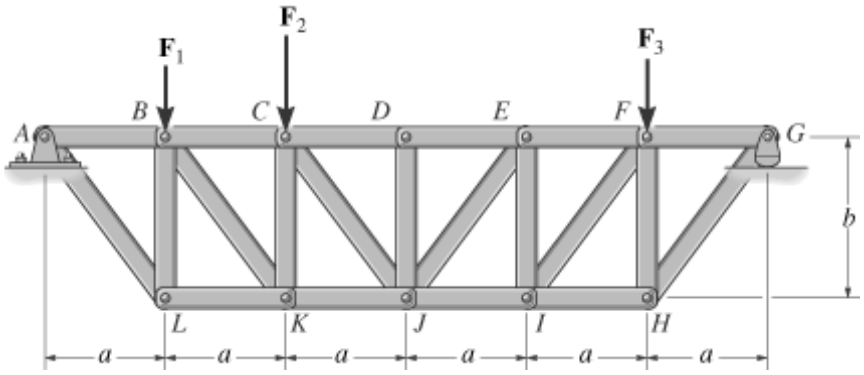
$F_1 = 4000\ \text{lb}$

$F_2 = 8000\ \text{lb}$

$F_3 = 5000\ \text{lb}$

$a = 9\ \text{ft}$

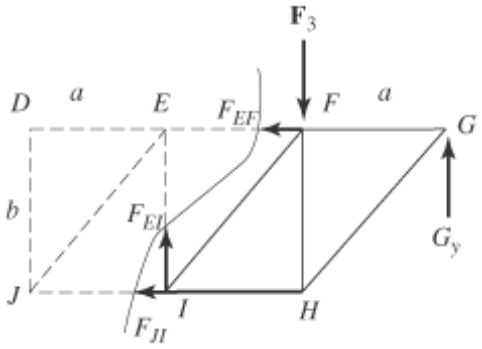
$b = 12\ \text{ft}$



Solution:

Initial Guesses:

$G_y = 1\ \text{kip} \quad F_{EI} = 1\ \text{kip} \quad F_{JI} = 1\ \text{kip}$



Given

$-F_1 a - F_2 2a - F_3 5a + G_y 6a = 0$

$G_y 2a - F_3 a - F_{JI} b = 0$

$F_{EI} - F_3 + G_y = 0$

$\begin{pmatrix} G_y \\ F_{JI} \\ F_{EI} \end{pmatrix} = \text{Find}\left(G_y, F_{JI}, F_{EI}\right)$

$\begin{pmatrix} G_y \\ F_{JI} \\ F_{EI} \end{pmatrix} = \begin{pmatrix} 7.5 \\ 7.5 \\ -2.5 \end{pmatrix} \text{kip}$

Positive (T)
Negative (C)

Problem 6-36

Determine the force in members *BE*, *EF*, and *CB*, and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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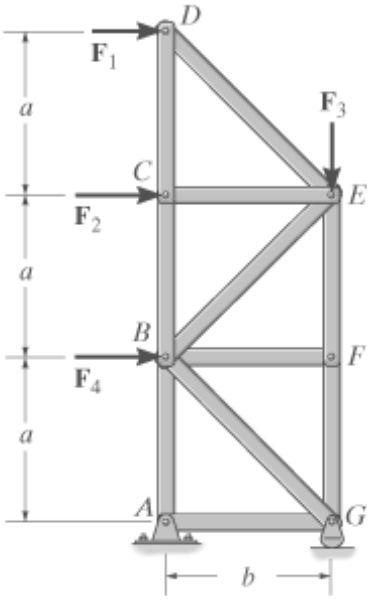
Given:

$F_1 = 5\ \text{kN} \quad F_4 = 10\ \text{kN}$

$F_2 = 10\ \text{kN} \quad a = 4\ \text{m}$

$F_3 = 5\ \text{kN} \quad b = 4\ \text{m}$

Solution: $\theta = \text{atan}\left(\frac{a}{b}\right)$



Inital Guesses

$F_{CB} = 1\ \text{kN} \quad F_{BE} = 1\ \text{kN} \quad F_{EF} = 1\ \text{kN}$

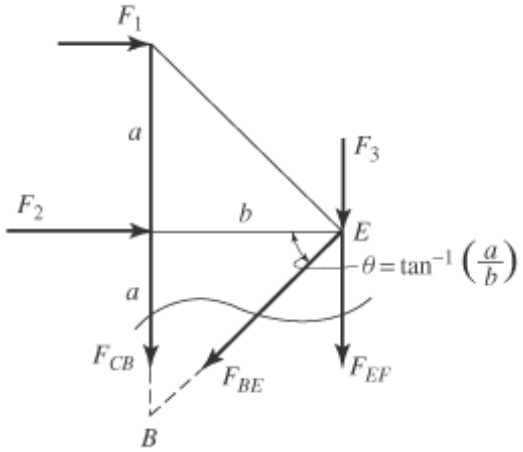
Given

$F_1 + F_2 - F_{BE}\cos(\theta) = 0$

$-F_{CB} - F_{EF} - F_{BE}\sin(\theta) - F_3 = 0$

$-F_1\, a + F_{CB}\, b = 0$

$\begin{pmatrix} F_{CB} \\ F_{BE} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{CB}, F_{BE}, F_{EF}\right)$



$\begin{pmatrix} F_{CB} \\ F_{BE} \\ F_{EF} \end{pmatrix} = \begin{pmatrix} 5 \\ 21.2 \\ -25 \end{pmatrix}\ \text{kN}$

Positive (T)
Negative (C)

Problem 6-37

Determine the force in members *BF*, *BG*, and A*B*, and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 5\ \text{kN} \quad F_4 = 10\ \text{kN}$

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$F_2 = 10\ \text{kN} \quad a = 4\ \text{m}$

$F_3 = 5\ \text{kN} \quad b = 4\ \text{m}$

Solution: $\theta = \text{atan}\left(\frac{a}{b}\right)$

Inital Guesses

$F_{AB} = 1\ \text{kN} \quad F_{BG} = 1\ \text{kN} \quad F_{BF} = 1\ \text{kN}$

Given

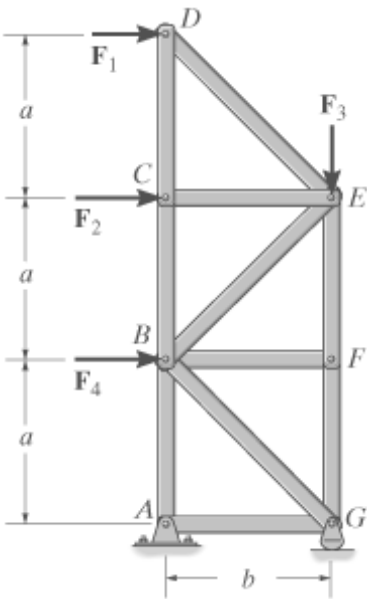
$F_1 + F_2 + F_4 + F_{BG}\cos(\theta) = 0$

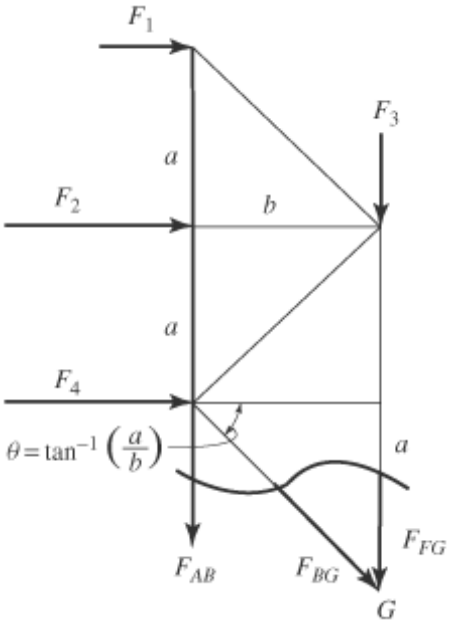
$-F_1 3a - F_2 2a - F_4 a + F_{AB} b = 0$

$-F_{BF} = 0$

$\begin{pmatrix} F_{AB} \\ F_{BG} \\ F_{BF} \end{pmatrix} = \text{Find}(F_{AB}, F_{BG}, F_{BF})$

$\begin{pmatrix} F_{AB} \\ F_{BG} \\ F_{BF} \end{pmatrix} = \begin{pmatrix} 45 \\ -35.4 \\ 0 \end{pmatrix} \text{kN}$ Positive (T) Negative (C)





Problem 6-38

Determine the force developed in members *GB* and *GF* of the bridge truss and state if these members are in tension or compression.

Given:

$F_1 = 600\ \text{lb}$

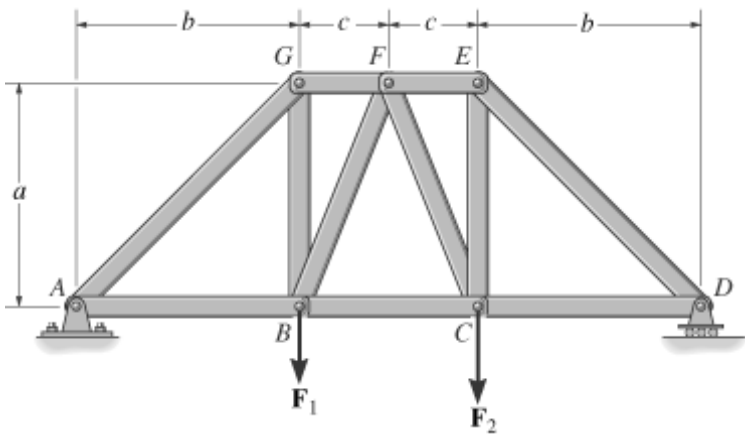
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$F_2 = 800$ lb

$a = 10$ ft

$b = 10$ ft

$c = 4$ ft



Solution:

Initial Guesses

$A_x = 1$ lb $\quad A_y = 1$ lb

$F_{GB} = 1$ lb $\quad F_{GF} = 1$ lb

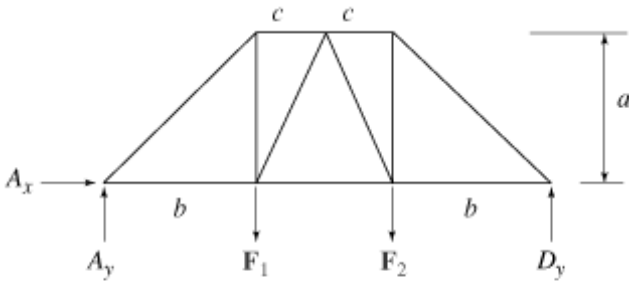
Given

$F_2 b + F_1(b + 2c) - A_y 2(b + c) = 0$

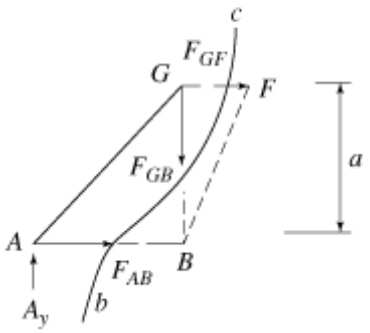
$A_x = 0$

$A_y - F_{GB} = 0$

$-A_y b - F_{GF} a = 0$



$$\begin{pmatrix} A_x \\ A_y \\ F_{GB} \\ F_{GF} \end{pmatrix} = \text{Find}\left(A_x, A_y, F_{GB}, F_{GF}\right)$$



$$\begin{pmatrix} A_x \\ A_y \\ F_{GB} \\ F_{GF} \end{pmatrix} = \begin{pmatrix} 0 \\ 671.429 \\ 671.429 \\ -671.429 \end{pmatrix} \text{lb}$$

Positive (T)
Negative (C)

Problem 6-39

Determine the force members *BC*, *FC*, and *FE*, and state if the members are in tension or compression.

Units Used:

kN $= 10^3$ N

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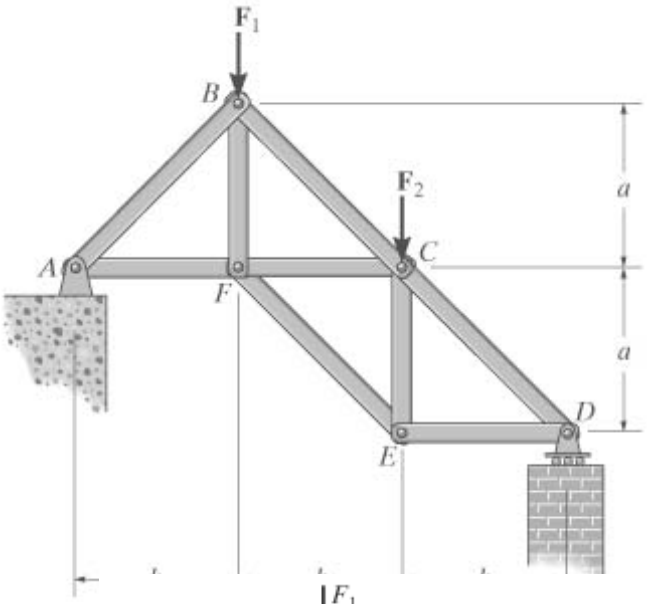
Given:

$F_1 = 6\ \text{kN}$

$F_2 = 6\ \text{kN}$

$a = 3\ \text{m}$

$b = 3\ \text{m}$



Solution: $\theta = \text{atan}\left(\frac{a}{b}\right)$

Initial Guesses

$D_y = 1\ \text{kN}$ $F_{BC} = 1\ \text{kN}$

$F_{FC} = 1\ \text{kN}$ $F_{FE} = 1\ \text{kN}$

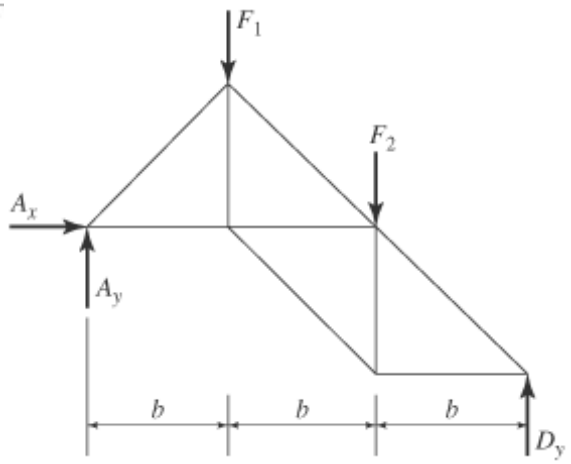
Given

$-F_1 b - F_2(2b) + D_y(3b) = 0$

$D_y b - F_{FE}\cos(\theta)a = 0$

$-F_{FC} - \left(F_{BC} + F_{FE}\right)\cos(\theta) = 0$

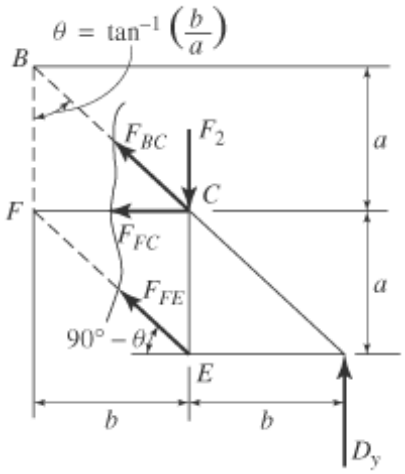
$-F_2 + D_y + \left(F_{FE} + F_{BC}\right)\sin(\theta) = 0$



$$\begin{pmatrix} D_y \\ F_{BC} \\ F_{FC} \\ F_{FE} \end{pmatrix} = \text{Find}\left(D_y, F_{BC}, F_{FC}, F_{FE}\right)$$

$$\begin{pmatrix} D_y \\ F_{BC} \\ F_{FC} \\ F_{FE} \end{pmatrix} = \begin{pmatrix} 6 \\ -8.49 \\ 0 \\ 8.49 \end{pmatrix} \text{kN}$$

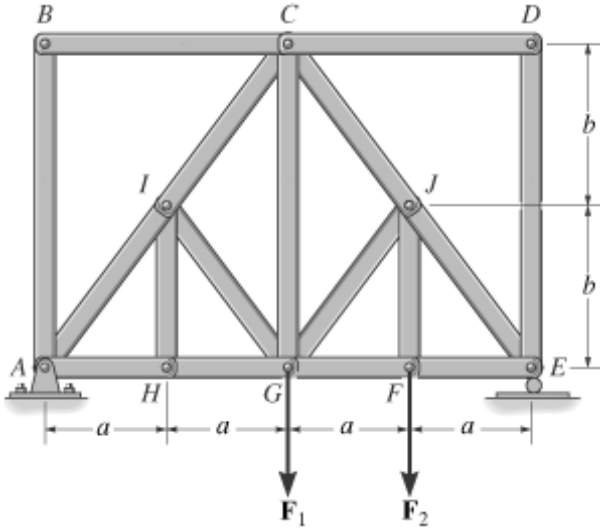
Positive (T)
Negative (C)



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Problem 6-40

Determine the force in members *IC* and *CG* of the truss and state if these members are in tension or compression. Also, indicate all zero-force members.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 6\ \text{kN}$

$F_2 = 6\ \text{kN}$

$a = 1.5\ \text{m}$

$b = 2\ \text{m}$

Solution:

By inspection of joints *B*, *D*, *H* and *I*.

AB, *BC*, *CD*, *DE*, *HI*, and *GI* are all zero-force members.

Guesses $A_y = 1\ \text{kN}$ $F_{IC} = 1\ \text{kN}$ $F_{CG} = 1\ \text{kN}$ $F_{CJ} = 1\ \text{kN}$

Given $-A_y(4a) + F_1(2a) + F_2 a = 0$

$$-A_y(2a) - \frac{b}{\sqrt{a^2 + b^2}} F_{IC}\, a - \frac{a}{\sqrt{a^2 + b^2}} F_{IC}\, b = 0$$

$$\frac{-a}{\sqrt{a^2 + b^2}} F_{IC} + \frac{a}{\sqrt{a^2 + b^2}} F_{CJ} = 0$$

$$\frac{-b}{\sqrt{a^2 + b^2}} F_{IC} - \frac{b}{\sqrt{a^2 + b^2}} F_{CJ} - F_{CG} = 0$$

$$\begin{pmatrix} A_y \\ F_{IC} \\ F_{CG} \\ F_{CJ} \end{pmatrix} = \text{Find}\left(A_y, F_{IC}, F_{CG}, F_{CJ}\right) \qquad \begin{pmatrix} A_y \\ F_{IC} \\ F_{CG} \\ F_{CJ} \end{pmatrix} = \begin{pmatrix} 4.5 \\ -5.625 \\ 9 \\ -5.625 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-41

Determine the force in members *JE* and *GF* of the truss and state if these members are in tension or compression. Also, indicate all zero-force members.

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Units Used:

$kN = 10^3 N$

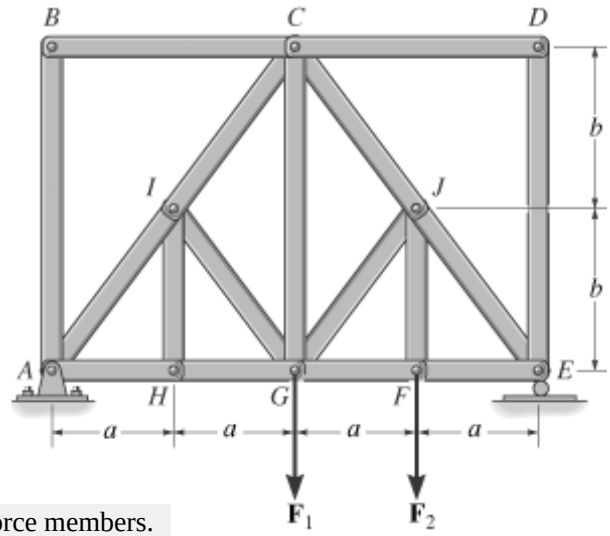
Given:

$F_1 = 6\ kN$

$F_2 = 6\ kN$

$a = 1.5\ m$

$b = 2\ m$



Solution:

By inspection of joints *B*, *D*, *H* and *I*.

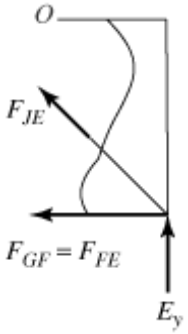
AB, *BC*, *CD*, *DE*, *HI*, and *GI* are all zero-force members.

Guesses $E_y = 1\ kN$ $F_{JE} = 1\ kN$ $F_{GF} = 1\ kN$

Given $-F_1(2a) - F_2(3a) + E_y(4a) = 0$

$$E_y + \frac{b}{\sqrt{a^2 + b^2}} F_{JE} = 0$$

$$\frac{-a}{\sqrt{a^2 + b^2}} F_{JE} - F_{GF} = 0$$



$$\begin{pmatrix} E_y \\ F_{JE} \\ F_{GF} \end{pmatrix} = \text{Find}\left(E_y, F_{JE}, F_{GF}\right)$$

$$\begin{pmatrix} E_y \\ F_{JE} \\ F_{GF} \end{pmatrix} = \begin{pmatrix} 7.5 \\ -9.375 \\ 5.625 \end{pmatrix} kN$$

Positive (T)
Negative (C)

Problem 6-42

Determine the force in members *BC*, *HC*, and *HG*. After the truss is sectioned use a single equation of equilibrium for the calculation of each force. State if these members are in tension or compression.

Units Used:

$kN = 10^3 N$

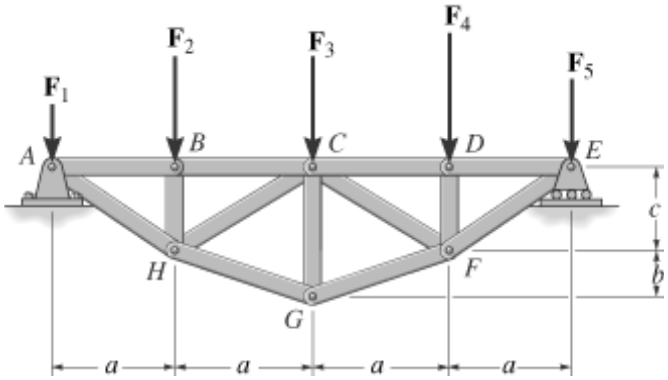
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Given:

$F_1 = 2$ kN $F_4 = 5$ kN $a = 5$ m

$F_2 = 4$ kN $F_5 = 3$ kN $b = 2$ m

$F_3 = 4$ kN $c = 3$ m



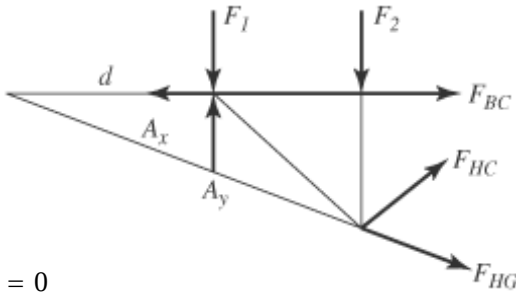
Solution:

Guesses

$A_x = 1$ kN $A_y = 1$ kN

$F_{BC} = 1$ kN $F_{HC} = 1$ kN

$F_{HG} = 1$ kN $d = 1$ m



Given

$$\frac{c}{a+d} = \frac{b}{a} \qquad -A_x = 0$$

$$(F_1 - A_y)(4a) + F_2(3a) + F_3(2a) + F_4(a) = 0$$

$$(F_1 - A_y)(a) + A_x(c) - F_{BC}(c) = 0$$

$$(F_1 - A_y)(2a) + F_2(a) + \frac{a}{\sqrt{a^2 + b^2}} F_{HG}(c) + \frac{b}{\sqrt{a^2 + b^2}} F_{HG}(a) = 0$$

$$(A_y - F_1)(d) - F_2(a + d) + \frac{c}{\sqrt{a^2 + c^2}} F_{HC}(a + d) + \frac{a}{\sqrt{a^2 + c^2}} F_{HC}(c) = 0$$

$$\begin{pmatrix} A_y \\ A_x \\ F_{BC} \\ F_{HC} \\ F_{HG} \\ d \end{pmatrix} = \text{Find}\left(A_y, A_x, F_{BC}, F_{HC}, F_{HG}, d\right) \quad \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 0 \\ 8.25 \end{pmatrix} \text{kN} \quad \begin{pmatrix} F_{BC} \\ F_{HC} \\ F_{HG} \end{pmatrix} = \begin{pmatrix} -10.417 \\ 2.235 \\ 9.155 \end{pmatrix} \text{kN}$$

$d = 2.5$ m

Positive (T)
Negative (C)

Problem 6-43

Determine the force in members *CD, CF,* and *CG* and state if these members are in tension or

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compression.

Units Used:

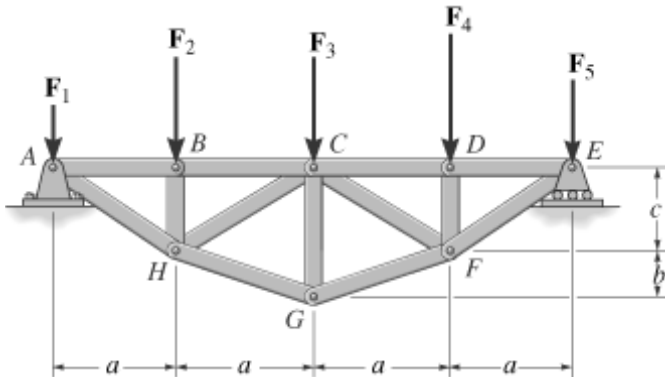
$kN = 10^3 N$

Given:

$F_1 = 2\ kN \quad F_4 = 5\ kN \quad a = 5\ m$

$F_2 = 4\ kN \quad F_5 = 3\ kN \quad b = 2\ m$

$F_3 = 4\ kN \quad c = 3\ m$



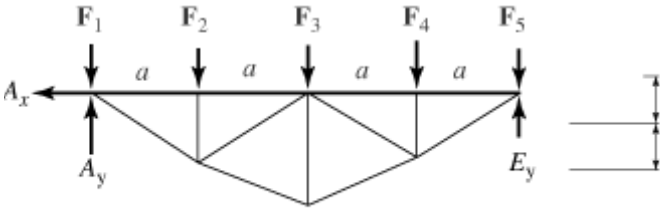
Solution:

Guesses

$E_y = 1\ kN \quad F_{CD} = 1\ kN$

$F_{CF} = 1\ kN \quad F_{CG} = 1\ kN$

$F_{FG} = 1\ kN \quad F_{GH} = 1\ kN$

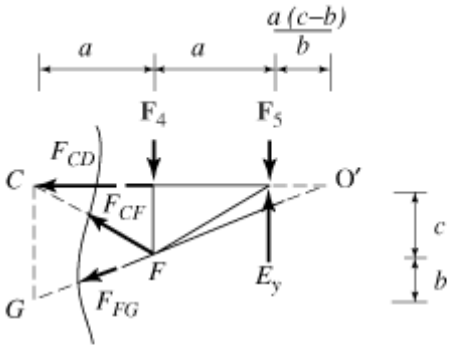


Given

$$-F_2(a) - F_3(2a) - F_4(3a) + (E_y - F_5)(4a) = 0$$

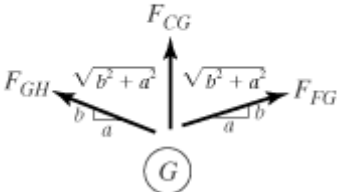
$$F_{CD}(c) + (E_y - F_5)(a) = 0$$

$$-F_4(a) - (F_5 - E_y)(2a) - \frac{a}{\sqrt{a^2 + b^2}} F_{FG}(b + c) = 0$$



$$\frac{a}{\sqrt{a^2 + b^2}} F_{FG} - \frac{a}{\sqrt{a^2 + b^2}} F_{GH} = 0$$

$$\frac{b}{\sqrt{a^2 + b^2}} (F_{FG} + F_{GH}) + F_{CG} = 0$$



$$(F_5 - E_y)\frac{a(c - b)}{b} + F_4\left[a + \frac{a(c - b)}{b}\right] - \frac{c}{\sqrt{a^2 + c^2}} F_{CF}\left[2\ a + \frac{a(c - b)}{b}\right] = 0$$

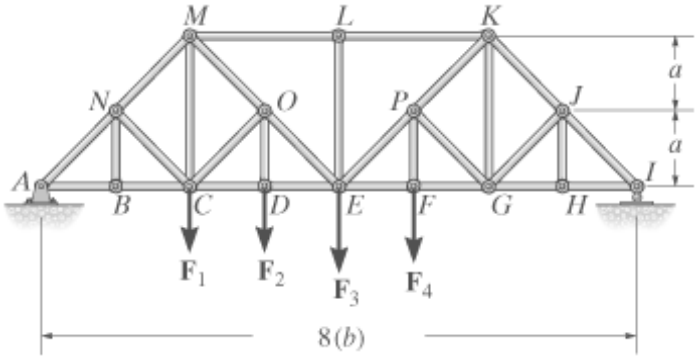
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$$\begin{pmatrix} E_y \\ F_{CD} \\ F_{CF} \\ F_{CG} \\ F_{FG} \\ F_{GH} \end{pmatrix} = \text{Find}\left(E_y, F_{CD}, F_{CF}, F_{CG}, F_{FG}, F_{GH}\right) \qquad \begin{pmatrix} E_y \\ F_{CD} \\ F_{CF} \\ F_{CG} \\ F_{FG} \\ F_{GH} \end{pmatrix} = \begin{pmatrix} 9.75 \\ -11.25 \\ 3.207 \\ -6.8 \\ 9.155 \\ 9.155 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-44

Determine the force in members *OE, LE,* and *LK* of the *Baltimore truss* and state if the members are in tension or compression.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 2\ \text{kN} \quad a = 2\ \text{m}$

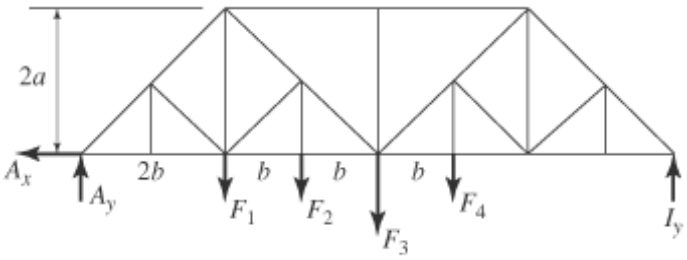
$F_2 = 2\ \text{kN} \quad b = 2\ \text{m}$

$F_3 = 5\ \text{kN}$

$F_4 = 3\ \text{kN}$

Solution:

$A_x = 0\ \text{kN}$



Initial Guesses

$A_y = 1\ \text{kN} \quad F_{OE} = 1\ \text{kN}$

$F_{DE} = 1\ \text{kN} \quad F_{LK} = 1\ \text{kN}$

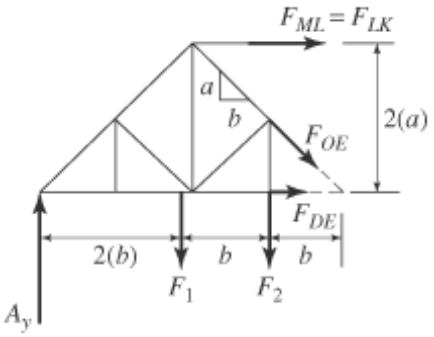
$F_{LE} = 1\ \text{kN}$

Given

$F_{LE} = 0$

$F_4(3b) + F_3(4b) + F_2(5b) + F_1(6b) - A_y(8b) = 0$

$$F_{LK} + F_{DE} + F_{OE}\frac{b}{\sqrt{a^2 + b^2}} = 0$$

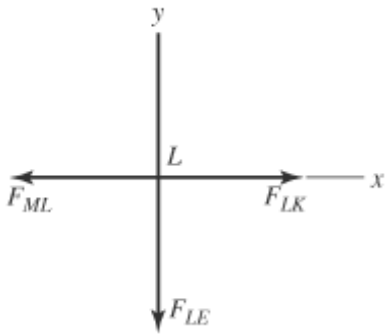


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$$A_y - F_1 - F_2 - F_{OE}\frac{a}{\sqrt{a^2 + b^2}} = 0$$

$$-F_{LK}(2a) + F_2(b) + F_1(2b) - A_y(4b) = 0$$

$$\begin{pmatrix} A_y \\ F_{OE} \\ F_{DE} \\ F_{LK} \\ F_{LE} \end{pmatrix} = \text{Find}\left(A_y, F_{OE}, F_{DE}, F_{LK}, F_{LE}\right)$$



$$\begin{pmatrix} A_y \\ F_{DE} \end{pmatrix} = \begin{pmatrix} 6.375 \\ 7.375 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} F_{OE} \\ F_{LE} \\ F_{LK} \end{pmatrix} = \begin{pmatrix} 3.36 \\ 0 \\ -9.75 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-45

Determine the force in member *GJ* of the truss and state if this member is in tension or compression.

Units Used:

kip = 10^3 lb

Given:

F_1 = 1000 lb

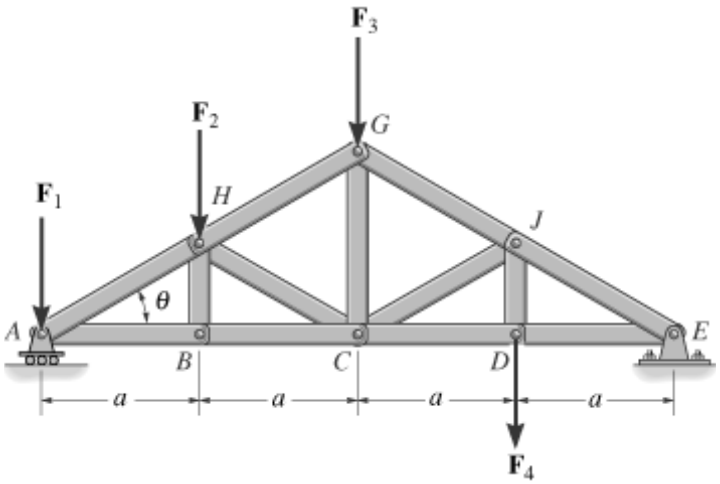
F_2 = 1000 lb

F_3 = 1000 lb

F_4 = 1000 lb

a = 10 ft

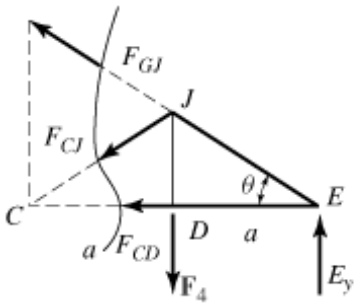
θ = 30 deg



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Solution:

Guess $E_y = 1$ lb $F_{GJ} = 1$ lb



Given

$$-F_2(a) - F_3(2a) - F_4(3a) + E_y(4a) = 0$$

$$-F_4(a) + E_y(2a) + F_{GJ}\sin(\theta)(2a) = 0$$

$$\begin{pmatrix} E_y \\ F_{GJ} \end{pmatrix} = \text{Find}(E_y, F_{GJ}) \qquad \begin{pmatrix} E_y \\ F_{GJ} \end{pmatrix} = \begin{pmatrix} 1.5 \\ -2 \end{pmatrix} \text{kip}$$

Positive (T)
Negative (C)

Problem 6-46

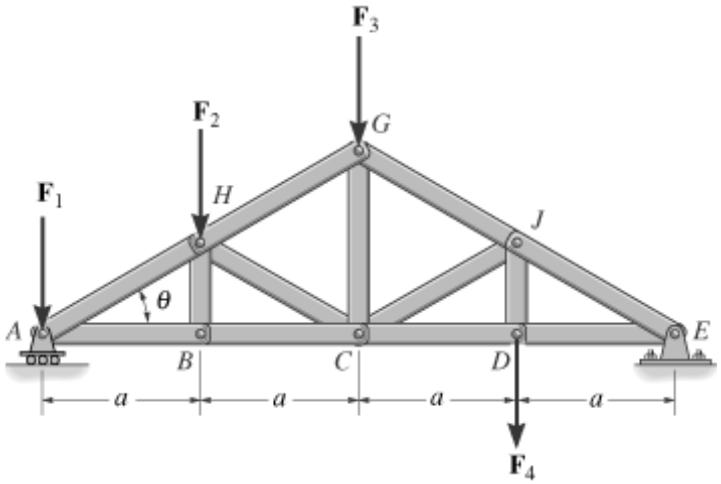
Determine the force in member *GC* of the truss and state if this member is in tension or compression.

Units Used:

$\text{kip} = 10^3$ lb

Given:

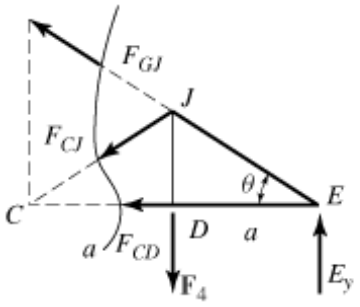
$F_1 = 1000$ lb
$F_2 = 1000$ lb
$F_3 = 1000$ lb
$F_4 = 1000$ lb
$a = 10$ ft
$\theta = 30$ deg



Solution:

Guess $E_y = 1$ lb $F_{GJ} = 1$ lb

$F_{HG} = 1$ lb $F_{GC} = 1$ lb



Given

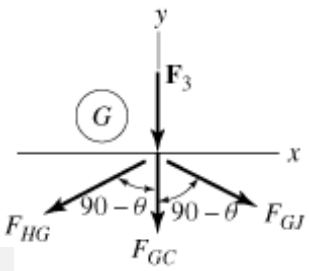
$$-F_2(a) - F_3(2a) - F_4(3a) + E_y(4a) = 0$$

$$-F_4(a) + E_y(2a) + F_{GJ}\sin(\theta)(2a) = 0$$

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$$-F_{HG}\cos(\theta) + F_{GJ}\cos(\theta) = 0$$

$$-F_3 - F_{GC} - (F_{HG} + F_{GJ})\sin(\theta) = 0$$



$$\begin{pmatrix} E_y \\ F_{GJ} \\ F_{GC} \\ F_{HG} \end{pmatrix} = \text{Find}(E_y, F_{GJ}, F_{GC}, F_{HG}) \qquad \begin{pmatrix} E_y \\ F_{GJ} \\ F_{GC} \\ F_{HG} \end{pmatrix} = \begin{pmatrix} 1.5 \\ -2 \\ 1 \\ -2 \end{pmatrix} \text{kip}$$

Positive (T)
Negative (C)

Problem 6-47

Determine the force in members *KJ*, *JN*, and *CD*, and state if the members are in tension or compression. Also indicate all zero-force members.

Units Used:

$\text{kip} = 10^3 \text{ lb}$

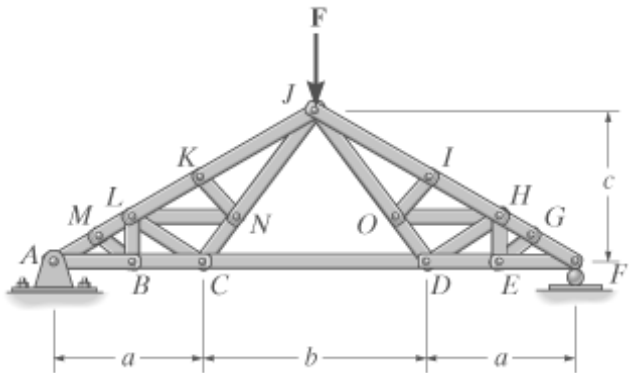
Given:

$F = 3 \text{ kip}$

$a = 20 \text{ ft}$

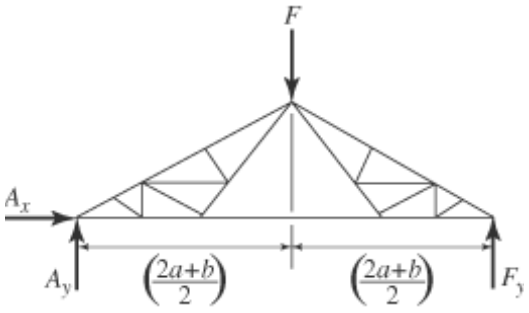
$b = 30 \text{ ft}$

$c = 20 \text{ ft}$



Solution: $A_x = 0$

$$\theta = \text{atan}\left(\frac{2c}{2a+b}\right) \qquad \phi = \text{atan}\left(\frac{2c}{b}\right)$$



Initial Guesses:

$A_y = 1 \text{ lb} \qquad F_{CD} = 1 \text{ lb}$

$F_{KJ} = 1 \text{ lb} \qquad F_{JN} = 1 \text{ lb}$

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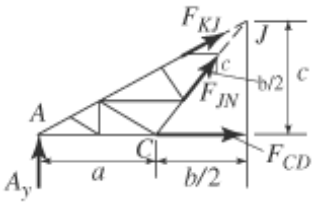
Given

$$F\left(a+\frac{b}{2}\right)-A_y(2a+b)=0$$

$$F_{CD}c-A_y\left(a+\frac{b}{2}\right)=0$$

$$F_{CD}+F_{JN}\cos(\phi)+F_{KJ}\cos(\theta)=0$$

$$A_y+F_{JN}\sin(\phi)+F_{KJ}\sin(\theta)=0$$



$$\begin{pmatrix} A_y \\ F_{CD} \\ F_{JN} \\ F_{KJ} \end{pmatrix} = \text{Find}\left(A_y, F_{CD}, F_{JN}, F_{KJ}\right) \qquad A_y = 1.5\,\text{kip}$$

$$\begin{pmatrix} F_{CD} \\ F_{JN} \\ F_{KJ} \end{pmatrix} = \begin{pmatrix} 2.625 \\ 0 \\ -3.023 \end{pmatrix} \text{kip}$$

Positive (T), Negative (C)

Problem 6-48

Determine the force in members *BG*, *HG*, and *BC* of the truss and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 6\ \text{kN}$

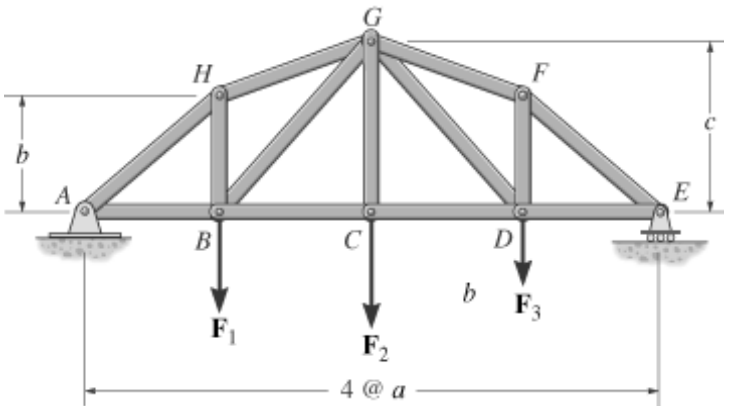
$F_2 = 7\ \text{kN}$

$F_3 = 4\ \text{kN}$

$a = 3\ \text{m}$

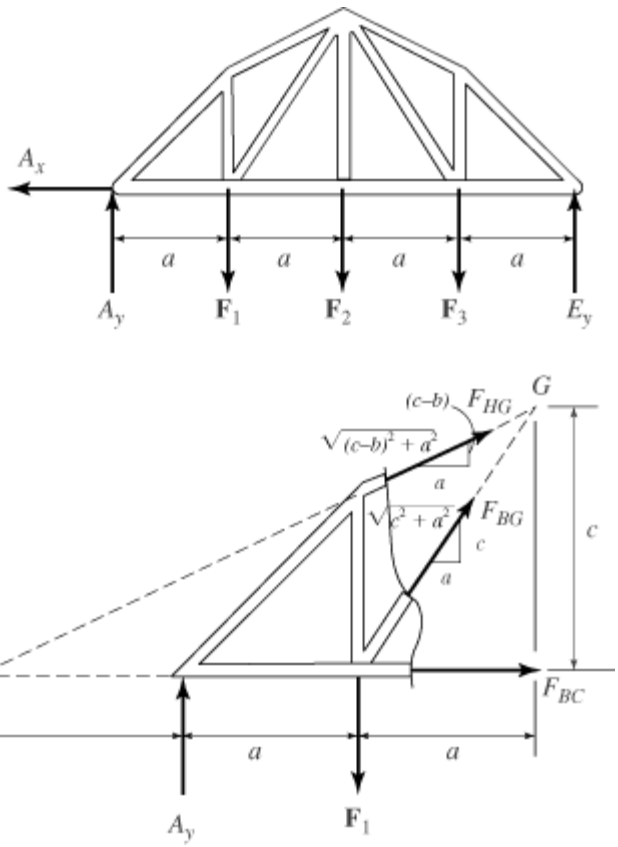
$b = 3\ \text{m}$

$c = 4.5\ \text{m}$



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Solution:



Initial Guesses $F_{BG} = 1$ kN

$A_x = 1$ kN $F_{HG} = 1$ kN

$A_y = 1$ kN $F_{BC} = 1$ kN

Given

$$-A_x = 0$$

$$-A_y(a) - \left[\frac{a}{\sqrt{(c-b)^2 + a^2}}\right]F_{HG}(b) = 0$$

$$F_3(a) + F_2(2a) + F_1(3a) - A_y(4a) = 0$$

$$F_{BC} + \left[\frac{a}{\sqrt{(c-b)^2 + a^2}}\right]F_{HG} + \left(\frac{a}{\sqrt{a^2 + c^2}}\right)F_{BG} - A_x = 0$$

$$A_y - F_1 + \left[\frac{c-b}{\sqrt{(c-b)^2 + a^2}}\right]F_{HG} + \left(\frac{c}{\sqrt{a^2 + c^2}}\right)F_{BG} = 0$$

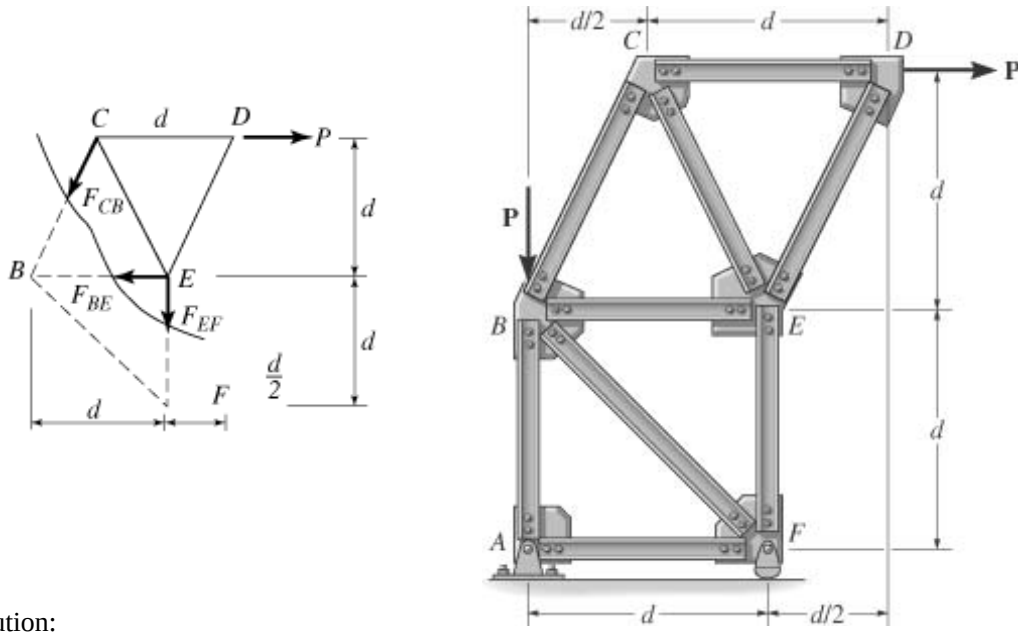
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$$\begin{pmatrix} A_x \\ A_y \\ F_{HG} \\ F_{BG} \\ F_{BC} \end{pmatrix} = \text{Find}\left(A_x, A_y, F_{HG}, F_{BG}, F_{BC}\right) \qquad \begin{pmatrix} A_x \\ A_y \\ F_{HG} \\ F_{BG} \\ F_{BC} \end{pmatrix} = \begin{pmatrix} 0 \\ 9 \\ -10.062 \\ 1.803 \\ 8 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-49

The skewed truss carries the load shown. Determine the force in members *CB*, *BE*, and *EF* and state if these members are in tension or compression. Assume that all joints are pinned.



Solution:

$$\circlearrowleft + \Sigma M_B = 0; \quad -P\,d - F_{EF}\,d = 0 \qquad F_{EF} = -P \qquad F_{EF} = P \quad \text{(C)}$$

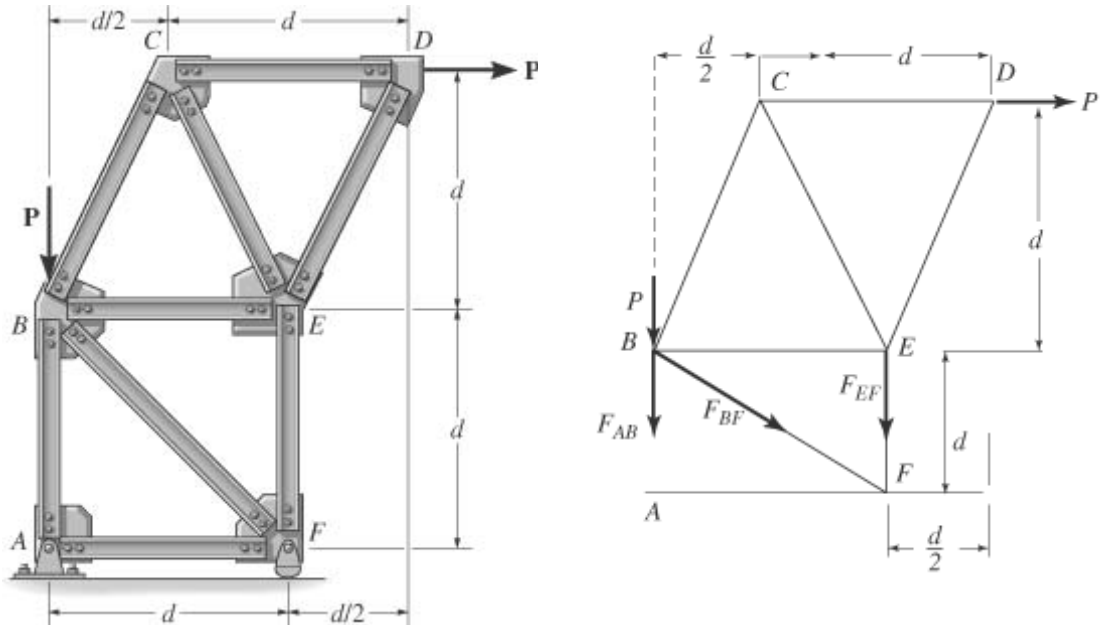
$$\circlearrowleft + \Sigma M_E = 0; \quad -P\,d + \frac{2}{\sqrt{5}} F_{CB}\,d = 0 \qquad F_{CB} = \frac{\sqrt{5}}{2} P \qquad F_{CB} = 1.12P \quad \text{(T)}$$

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad P - \frac{1}{\sqrt{5}} F_{CB} - F_{BE} = 0 \qquad P_{BE} = \frac{P}{2} \qquad F_{BE} = 0.5P \quad \text{(T)}$$

Problem 6-50

The skewed truss carries the load shown. Determine the force in members *AB*, *BF*, and *EF* and state if these members are in tension or compression. Assume that all joints are pinned.

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Solution:

$\circlearrowleft + \Sigma M_F = 0;$ $\quad -P\,2d + P\,d + F_{AB}\,d = 0$ $\quad F_{AB} = P$ $\quad \boxed{F_{AB} = P}$ (T)

$\circlearrowleft + \Sigma M_B = 0;$ $\quad -P\,d - F_{EF}\,d = 0$ $\quad F_{EF} = -P$ $\quad \boxed{F_{EF} = P}$ (C)

$\overset{+}{\rightarrow} \Sigma F_x = 0;$ $\quad P + F_{BF}\dfrac{1}{\sqrt{2}} = 0$ $\quad F_{BE} = -\sqrt{2}\,P$ $\quad \boxed{F_{BF} = 1.41P}$ (C)

Problem 6-51

Determine the force developed in members *BC* and *CH* of the roof truss and state if the members are in tension or compression.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 1.5\ \text{kN}$

$F_2 = 2\ \text{kN}$

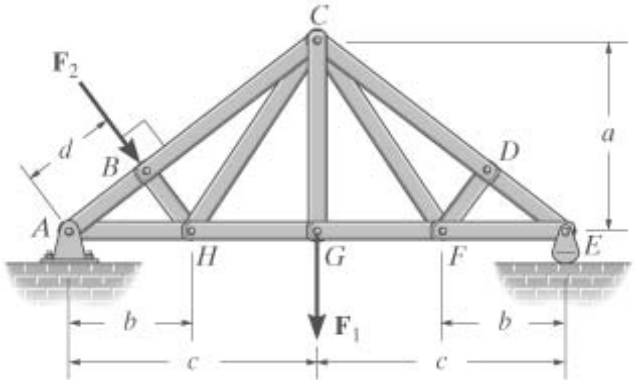
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$a = 1.5$ m

$b = 1$ m

$c = 2$ m

$d = 0.8$ m

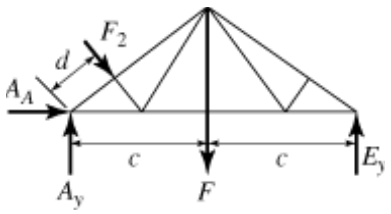


Solution:

$\theta = \text{atan}\left(\frac{a}{c}\right)$ $\phi = \text{atan}\left(\frac{a}{c-b}\right)$

Initial Guesses:

$E_y = 1$ kN $F_{BC} = 1$ kN $F_{CH} = 1$ kN

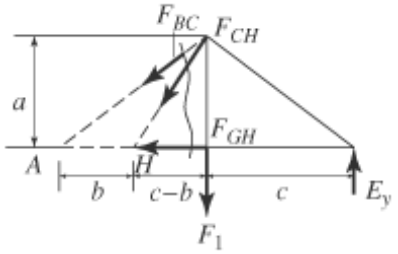


Given

$-F_2(d) - F_1(c) + E_y(2c) = 0$

$F_{BC}\sin(\theta)(c) + F_{CH}\sin(\phi)(c-b) + E_y(c) = 0$

$-F_{BC}\sin(\theta) - F_{CH}\sin(\phi) - F_1 + E_y = 0$

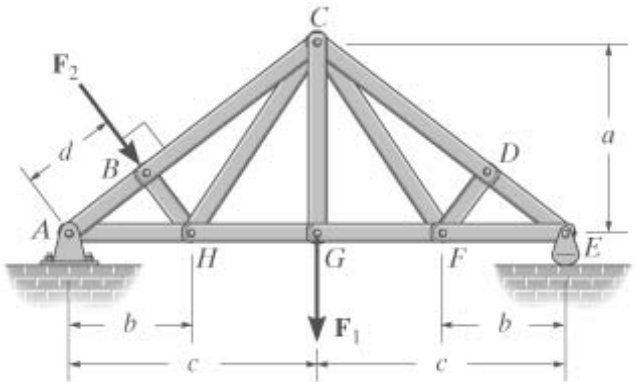


$\begin{pmatrix} E_y \\ F_{BC} \\ F_{CH} \end{pmatrix} = \text{Find}(E_y, F_{BC}, F_{CH})$ $E_y = 1.15\,\text{kN}$ $\begin{pmatrix} F_{BC} \\ F_{CH} \end{pmatrix} = \begin{pmatrix} -3.25 \\ 1.923 \end{pmatrix}$ kN

Positive (T)
Negative (C)

Problem 6-52

Determine the force in members *CD* and *GF* of the truss and state if the members are in tension or compression. Also indicate all zero-force members.



Units Used:

$\text{kN} = 10^3$ N

Given:

$F_1 = 1.5$ kN

$F_2 = 2$ kN

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$a = 1.5$ m

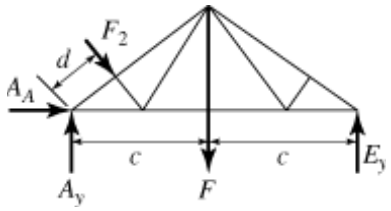
$b = 1$ m

$c = 2$ m

$d = 0.8$ m

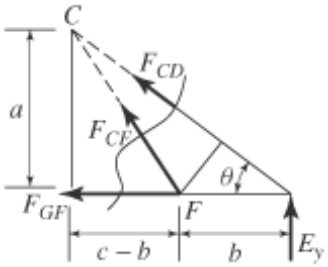
Solution:

$$\theta = \operatorname{atan}\left(\frac{a}{c}\right) \qquad \phi = \operatorname{atan}\left(\frac{a}{c-b}\right)$$



Initial Guesses:

$E_y = 1$ kN $\quad F_{CD} = 1$ kN $\quad F_{GF} = 1$ kN



Given

$$-F_2(d) - F_1(c) + E_y(2c) = 0$$

$$E_y(b) + F_{CD}\sin(\theta)(b) = 0$$

$$E_y(c) - F_{GF}(a) = 0$$

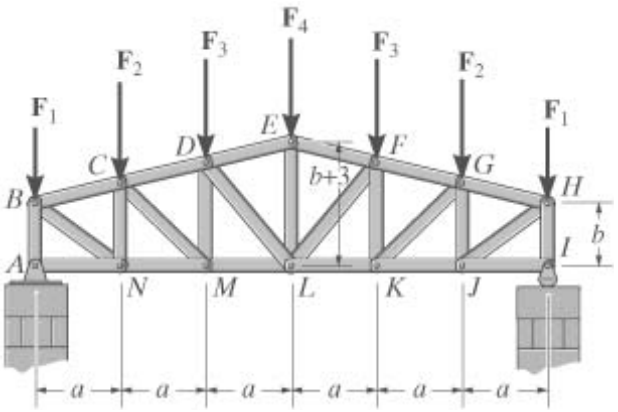
$$\begin{pmatrix} E_y \\ F_{CD} \\ F_{GF} \end{pmatrix} = \text{Find}\left(E_y, F_{CD}, F_{GF}\right) \quad E_y = 1.15\,\text{kN} \qquad \begin{pmatrix} F_{CD} \\ F_{GF} \end{pmatrix} = \begin{pmatrix} -1.917 \\ 1.533 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

DF and *CF* are zero force members.

Problem 6-53

Determine the force in members *DE*, *DL*, and *ML* of the roof truss and state if the members are in tension or compression.



Units Used:

$\text{kN} = 10^3$ N

Given:

$F_1 = 6$ kN

$F_2 = 12$ kN

$F_3 = 12$ kN

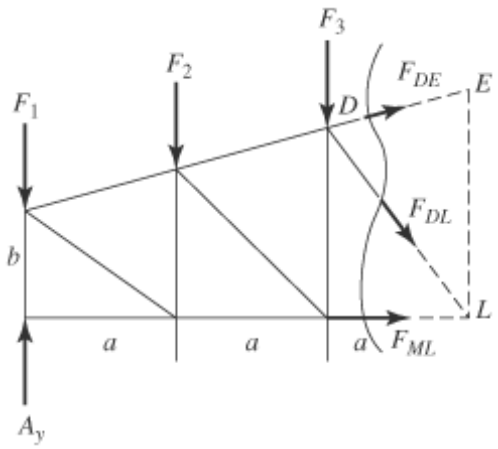
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$F_4 = 12$ kN

$a = 4$ m

$b = 3$ m

$c = 6$ m



Solution:

$$\theta = \operatorname{atan}\left(\frac{c-b}{3a}\right)$$

$$\phi = \operatorname{atan}\left[\frac{b+\frac{2}{3}(c-b)}{a}\right]$$

Initial Guesses:

$A_y = 1$ kN $\quad F_{ML} = 1$ kN $\quad F_{DL} = 1$ kN $\quad F_{DE} = 1$ kN

Given

$$F_2(a) + F_3(2a) + F_4(3a) + F_3(4a) + F_2(5a) + F_1(6a) - A_y(6a) = 0$$

$$F_1(2a) + F_2(a) - A_y(2a) + F_{ML}\left[b+\frac{2}{3}(c-b)\right] = 0$$

$$A_y - F_1 - F_2 - F_3 + F_{DE}\sin(\theta) - F_{DL}\sin(\phi) = 0$$

$$F_{ML} + F_{DL}\cos(\phi) + F_{DE}\cos(\theta) = 0$$

$$\begin{pmatrix} A_y \\ F_{ML} \\ F_{DE} \\ F_{DL} \end{pmatrix} = \text{Find}\left(A_y, F_{ML}, F_{DE}, F_{DL}\right) \qquad A_y = 36\,\text{kN} \qquad \begin{pmatrix} F_{ML} \\ F_{DE} \\ F_{DL} \end{pmatrix} = \begin{pmatrix} 38.4 \\ -37.1 \\ -3.8 \end{pmatrix} \text{kN}$$

Positive (T), Negative (C)

Problem 6-54

Determine the force in members *EF* and *EL* of the roof truss and state if the members are in

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tension or compression.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 6\ \text{kN}$

$F_2 = 12\ \text{kN}$

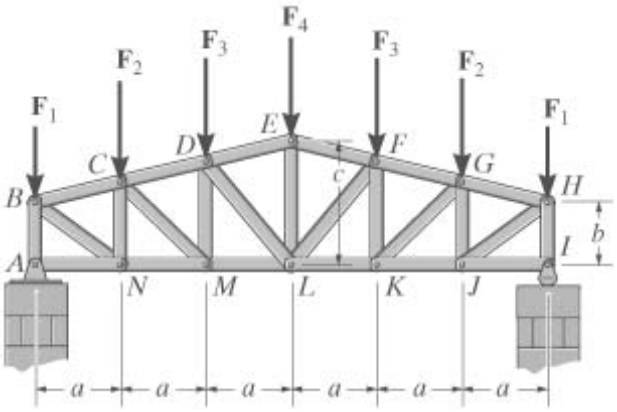
$F_3 = 12\ \text{kN}$

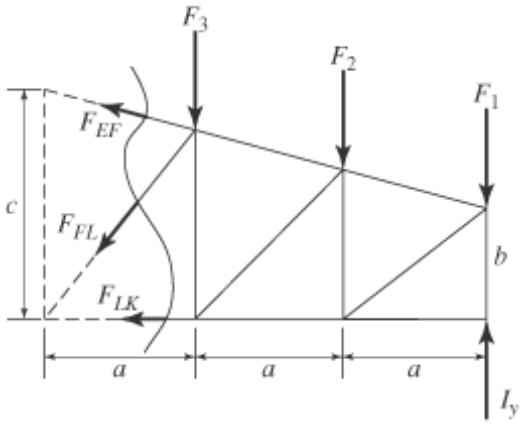
$F_4 = 12\ \text{kN}$

$a = 4\ \text{m}$

$b = 3\ \text{m}$

$c = 6\ \text{m}$





Solution:

$$\theta = \text{atan}\left(\frac{c-b}{3a}\right)$$

Initial Guesses:

$I_y = 1\ \text{kN} \qquad F_{EF} = 1\ \text{kN}$

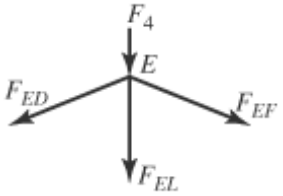
$F_{EL} = 1\ \text{kN}$

Given

$$-F_2(a) - F_3(2a) - F_4(3a) - F_3(4a) - F_2(5a) - F_1(6a) + I_y(6a) = 0$$

$$-F_3(a) - F_2(2a) - F_1(3a) + I_y(3a) + F_{EF}\cos(\theta)(c) = 0$$

$$-F_4 - F_{EL} - 2F_{EF}\sin(\theta) = 0$$



$$\begin{pmatrix} I_y \\ F_{EF} \\ F_{EL} \end{pmatrix} = \text{Find}\left(I_y, F_{EF}, F_{EL}\right) \qquad I_y = 36\,\text{kN} \qquad \begin{pmatrix} F_{EF} \\ F_{EL} \end{pmatrix} = \begin{pmatrix} -37.108 \\ 6 \end{pmatrix}\text{kN}$$

Positive (T)
Negative (C)

Problem 6-55

Two space trusses are used to equally support the uniform sign of mass *M*. Determine the force developed in members *AB, AC,* and *BC* of truss *ABCD* and state if the members are in tension or compression. Horizontal short links support the truss at joints *B* and *D* and there is a ball-and-

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socket joint at *C*.

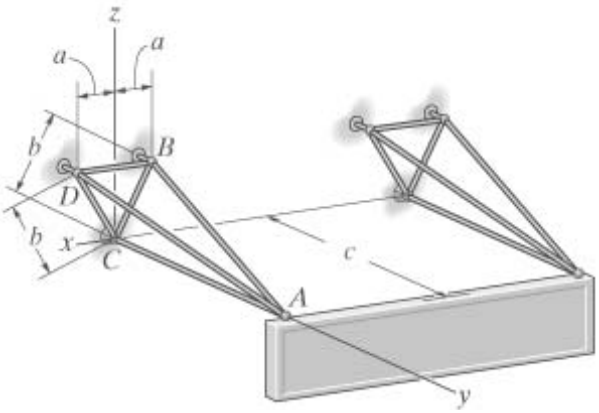
Given:

$M = 50\ \text{kg} \quad g = 9.81\ \frac{\text{m}}{\text{s}^2}$

$a = 0.25\ \text{m}$

$b = 0.5\ \text{m}$

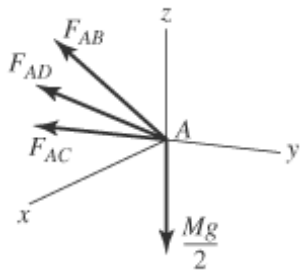
$c = 2\ \text{m}$



Solution: $h = \sqrt{b^2 - a^2}$

$$\mathbf{AB} = \begin{pmatrix} -a \\ -c \\ h \end{pmatrix} \quad \mathbf{AD} = \begin{pmatrix} a \\ -c \\ h \end{pmatrix} \quad \mathbf{AC} = \begin{pmatrix} 0 \\ -c \\ 0 \end{pmatrix}$$

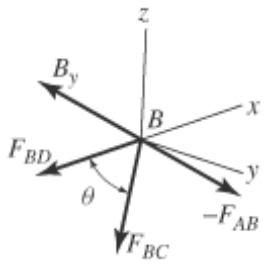
$$\mathbf{BD} = \begin{pmatrix} 2a \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{BC} = \begin{pmatrix} a \\ 0 \\ -h \end{pmatrix}$$



Guesses

$F_{AB} = 1\ \text{N} \quad F_{AD} = 1\ \text{N} \quad F_{AC} = 1\ \text{N}$

$F_{BC} = 1\ \text{N} \quad F_{BD} = 1\ \text{N} \quad B_y = 1\ \text{N}$



Given

$$F_{AB}\frac{\mathbf{AB}}{|\mathbf{AB}|} + F_{AD}\frac{\mathbf{AD}}{|\mathbf{AD}|} + F_{AC}\frac{\mathbf{AC}}{|\mathbf{AC}|} + \begin{pmatrix} 0 \\ 0 \\ \frac{-Mg}{2} \end{pmatrix} = 0$$

$$F_{AB}\frac{-\mathbf{AB}}{|\mathbf{AB}|} + F_{BD}\frac{\mathbf{BD}}{|\mathbf{BD}|} + F_{BC}\frac{\mathbf{BC}}{|\mathbf{BC}|} + \begin{pmatrix} 0 \\ -B_y \\ 0 \end{pmatrix} = 0$$

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$$\begin{pmatrix} F_{AB} \\ F_{AD} \\ F_{AC} \\ F_{BC} \\ F_{BD} \\ B_y \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AD}, F_{AC}, F_{BC}, F_{BD}, B_y\right)$$

$$\begin{pmatrix} B_y \\ F_{AD} \\ F_{BD} \end{pmatrix} = \begin{pmatrix} 566 \\ 584 \\ 0 \end{pmatrix} \text{ N}$$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{BC} \end{pmatrix} = \begin{pmatrix} 584 \\ -1133 \\ -142 \end{pmatrix} \text{ N}$$

Positive (T), Negative (C)

Problem 6-56

Determine the force in each member of the space truss and state if the members are in tension or compression.The truss is supported by short links at *B*, *C*, and *D*.

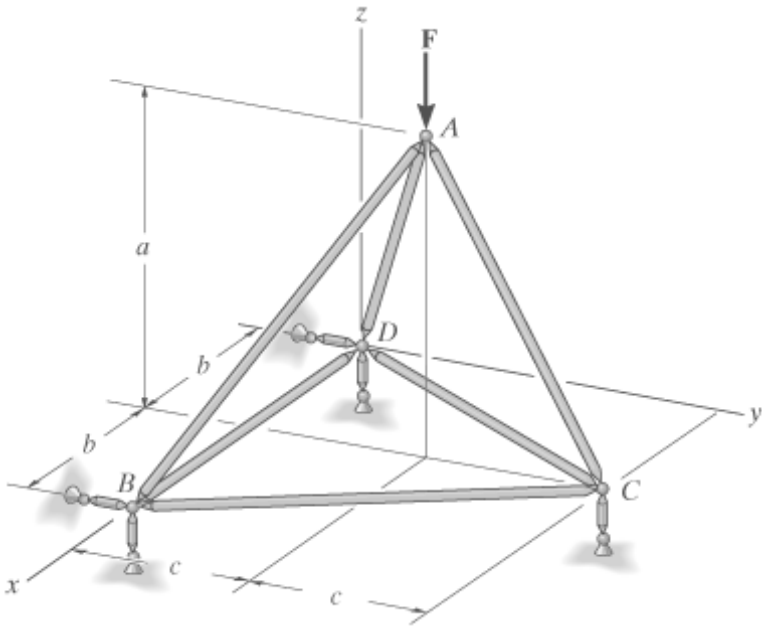
Given:

$F = 600$ N

$a = 3$ m

$b = 1$ m

$c = 1.5$ m



Solution:

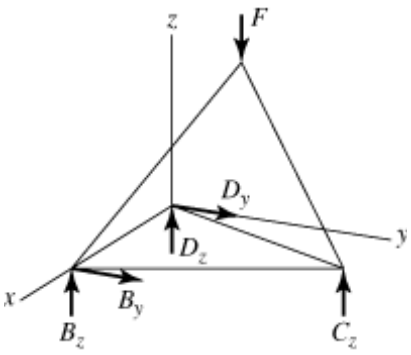
$$\mathbf{AB} = \begin{pmatrix} b \\ -c \\ -a \end{pmatrix}$$

$$\mathbf{AC} = \begin{pmatrix} 0 \\ c \\ -a \end{pmatrix}$$

$$\mathbf{AD} = \begin{pmatrix} -b \\ -c \\ -a \end{pmatrix} \qquad \mathbf{CD} = \begin{pmatrix} -b \\ -2c \\ 0 \end{pmatrix}$$

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$$\mathbf{CB} = \begin{pmatrix} b \\ -2c \\ 0 \end{pmatrix} \qquad \mathbf{BD} = \begin{pmatrix} -2b \\ 0 \\ 0 \end{pmatrix}$$



Guesses

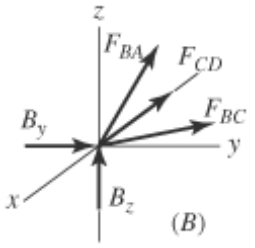
$F_{BA} = 1\text{ N}$ $F_{BC} = 1\text{ N}$ $F_{CA} = 1\text{ N}$

$F_{DA} = 1\text{ N}$ $F_{BD} = 1\text{ N}$ $F_{DC} = 1\text{ N}$

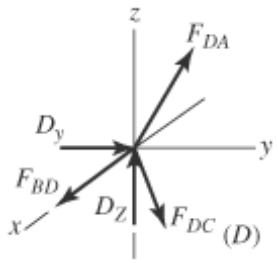
$B_y = 1\text{ N}$ $B_z = 1\text{ N}$ $C_z = 1\text{ N}$

Given

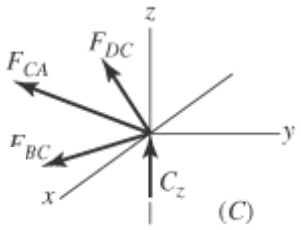
$$F_{BA}\frac{\mathbf{AB}}{|\mathbf{AB}|} + F_{CA}\frac{\mathbf{AC}}{|\mathbf{AC}|} + F_{DA}\frac{\mathbf{AD}}{|\mathbf{AD}|} + \begin{pmatrix} 0 \\ 0 \\ -F \end{pmatrix} = 0$$



$$F_{CA}\frac{-\mathbf{AC}}{|\mathbf{AC}|} + F_{DC}\frac{\mathbf{CD}}{|\mathbf{CD}|} + F_{BC}\frac{\mathbf{CB}}{|\mathbf{CB}|} + \begin{pmatrix} 0 \\ 0 \\ C_z \end{pmatrix} = 0$$



$$F_{BC}\frac{-\mathbf{CB}}{|\mathbf{CB}|} + F_{BD}\frac{\mathbf{BD}}{|\mathbf{BD}|} + F_{BA}\frac{-\mathbf{AB}}{|\mathbf{AB}|} + \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} = 0$$



$$\begin{pmatrix} F_{BA} \\ F_{BC} \\ F_{CA} \\ F_{DA} \\ F_{BD} \\ F_{DC} \\ B_y \\ B_z \\ C_z \end{pmatrix} = \text{Find}\left(F_{BA}, F_{BC}, F_{CA}, F_{DA}, F_{BD}, F_{DC}, B_y, B_z, C_z\right)$$

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$$\begin{pmatrix} B_y \\ B_z \\ C_z \end{pmatrix} = \begin{pmatrix} 1.421 \times 10^{-14} \\ 150 \\ 300 \end{pmatrix} \text{ N} \qquad \begin{pmatrix} F_{BA} \\ F_{BC} \\ F_{CA} \\ F_{DA} \\ F_{BD} \\ F_{DC} \end{pmatrix} = \begin{pmatrix} -175 \\ 79.1 \\ -335.4 \\ -175 \\ 25 \\ 79.1 \end{pmatrix} \text{ N}$$

Positive (T),
Negative (C)

Problem 6-57

Determine the force in each member of the space truss and state if the members are in tension or compression.The truss is supported by short links at *A*, *B*, and *C*.

Given:

$a = 4$ ft

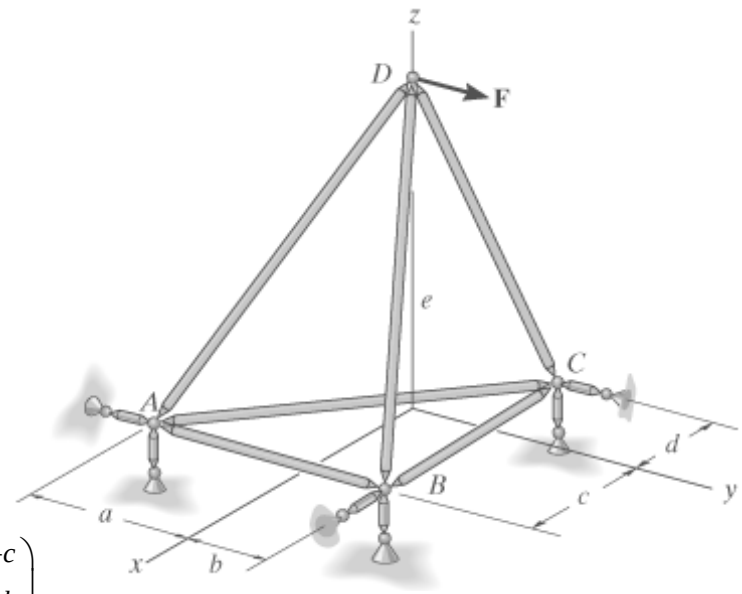
$b = 2$ ft

$c = 3$ ft

$d = 2$ ft

$e = 8$ ft

$$\mathbf{F} = \begin{pmatrix} 0 \\ 500 \\ 0 \end{pmatrix} \text{ lb}$$

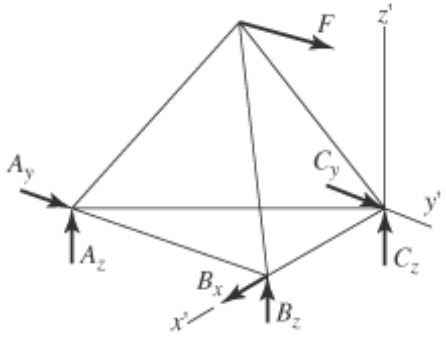


Solution:

$$\mathbf{AD} = \begin{pmatrix} -c \\ a \\ e \end{pmatrix} \qquad \mathbf{BD} = \begin{pmatrix} -c \\ -b \\ e \end{pmatrix}$$

$$\mathbf{CD} = \begin{pmatrix} d \\ -b \\ e \end{pmatrix} \qquad \mathbf{AB} = \begin{pmatrix} 0 \\ a+b \\ 0 \end{pmatrix}$$

$$\mathbf{AC} = \begin{pmatrix} -c-d \\ a+b \\ 0 \end{pmatrix} \qquad \mathbf{BC} = \begin{pmatrix} -c-d \\ 0 \\ 0 \end{pmatrix}$$



Guesses

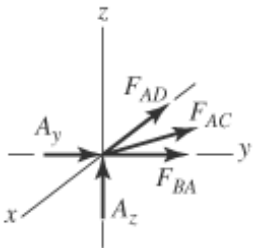
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$F_{BA} = 1$ lb $F_{BC} = 1$ lb $F_{BD} = 1$ lb

$F_{AD} = 1$ lb $F_{AC} = 1$ lb $F_{CD} = 1$ lb

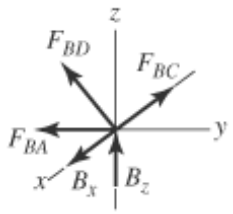
$A_y = 1$ lb $A_z = 1$ lb $B_x = 1$ lb

$B_z = 1$ lb $C_y = 1$ lb $C_z = 1$ lb



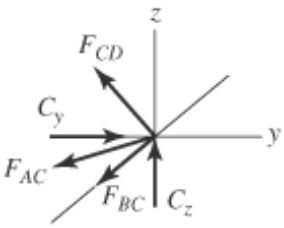
Given

$$\mathbf{F} + F_{AD}\frac{-\mathbf{AD}}{|\mathbf{AD}|} + F_{BD}\frac{-\mathbf{BD}}{|\mathbf{BD}|} + F_{CD}\frac{-\mathbf{CD}}{|\mathbf{CD}|} = \mathbf{0}$$



$$F_{AD}\frac{\mathbf{AD}}{|\mathbf{AD}|} + F_{BA}\frac{\mathbf{AB}}{|\mathbf{AB}|} + F_{AC}\frac{\mathbf{AC}}{|\mathbf{AC}|} + \begin{pmatrix} 0 \\ A_y \\ A_z \end{pmatrix} = \mathbf{0}$$

$$F_{BA}\frac{-\mathbf{AB}}{|\mathbf{AB}|} + F_{BC}\frac{\mathbf{BC}}{|\mathbf{BC}|} + F_{BD}\frac{\mathbf{BD}}{|\mathbf{BD}|} + \begin{pmatrix} B_x \\ 0 \\ B_z \end{pmatrix} = \mathbf{0}$$



$$F_{CD}\frac{\mathbf{CD}}{|\mathbf{CD}|} + F_{AC}\frac{-\mathbf{AC}}{|\mathbf{AC}|} + F_{BC}\frac{-\mathbf{BC}}{|\mathbf{BC}|} + \begin{pmatrix} 0 \\ C_y \\ C_z \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} F_{BA} \\ F_{BC} \\ F_{BD} \\ F_{AD} \\ F_{AC} \\ F_{CD} \\ A_y \\ A_z \\ B_x \\ B_z \\ C_y \\ C_z \end{pmatrix} = \text{Find}\left(F_{BA}, F_{BC}, F_{BD}, F_{AD}, F_{AC}, F_{CD}, A_y, A_z, B_x, B_z, C_y, C_z\right)$$

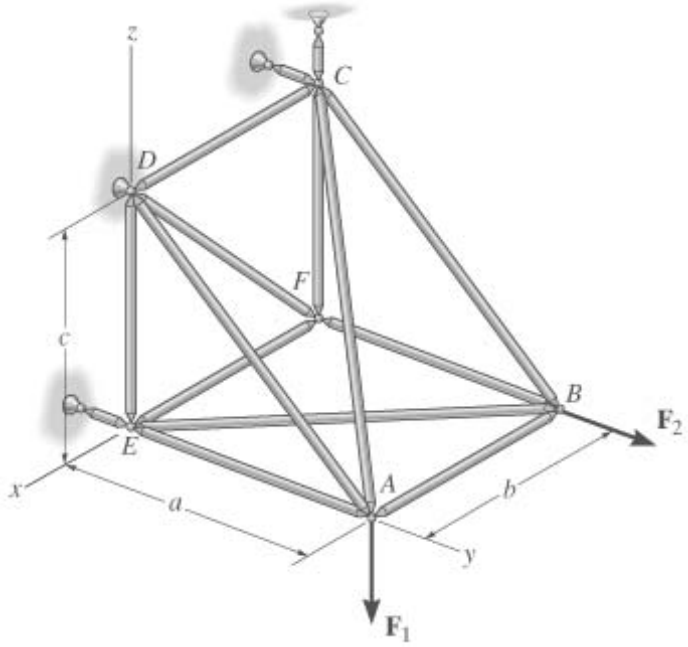
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$$\begin{pmatrix} A_y \\ A_z \\ B_x \\ B_z \\ C_y \\ C_z \end{pmatrix} = \begin{pmatrix} -200 \\ -667 \\ 0 \\ 667 \\ -300 \\ 0 \end{pmatrix} \text{lb} \qquad \begin{pmatrix} F_{BA} \\ F_{BC} \\ F_{BD} \\ F_{AD} \\ F_{AC} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} 167 \\ 250 \\ -731 \\ 786 \\ -391 \\ 0 \end{pmatrix} \text{lb}$$

Positive (T)
Negative (C)

Problem 6-58

The space truss is supported by a ball-and-socket joint at *D* and short links at *C* and *E*. Determine the force in each member and state if the members are in tension or compression.



Given:

$$\mathbf{F_1} = \begin{pmatrix} 0 \\ 0 \\ -500 \end{pmatrix} \text{lb}$$

$$\mathbf{F_2} = \begin{pmatrix} 0 \\ 400 \\ 0 \end{pmatrix} \text{lb}$$

$a = 4$ ft

$b = 3$ ft

$c = 3$ ft

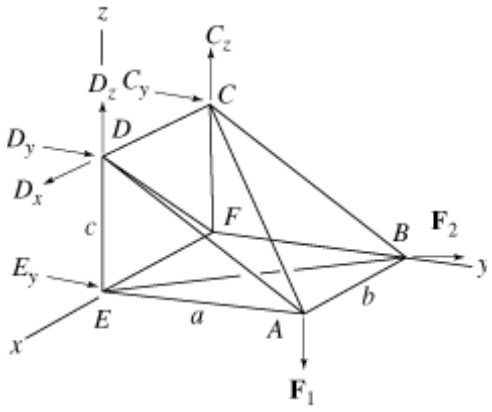
Solution:

Find the external reactions

Guesses

$E_y = 1$ lb $\quad C_y = 1$ lb $\quad C_z = 1$ lb

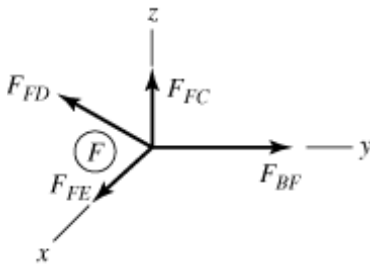
$D_x = 1$ lb $\quad D_y = 1$ lb $\quad D_z = 1$ lb



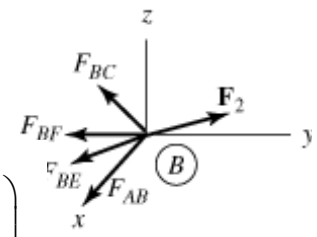
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Given

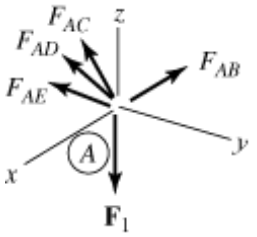
$$\begin{pmatrix} D_x \\ D_y \\ D_z \end{pmatrix} + \begin{pmatrix} 0 \\ E_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ C_y \\ C_z \end{pmatrix} + \mathbf{F_1} + \mathbf{F_2} = 0$$



$$\begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \mathbf{F_1} + \begin{pmatrix} -b \\ a \\ 0 \end{pmatrix} \times \mathbf{F_2} + \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} \times \begin{pmatrix} D_x \\ D_y \\ D_z \end{pmatrix} + \begin{pmatrix} -b \\ 0 \\ c \end{pmatrix} \times \begin{pmatrix} 0 \\ C_y \\ C_z \end{pmatrix} = 0$$



$$\begin{pmatrix} E_y \\ C_y \\ C_z \\ D_x \\ D_y \\ D_z \end{pmatrix} = \text{Find}\left(E_y, C_y, C_z, D_x, D_y, D_z\right) \qquad \begin{pmatrix} E_y \\ C_y \\ C_z \\ D_x \\ D_y \\ D_z \end{pmatrix} = \begin{pmatrix} 266.667 \\ -400 \\ 0 \\ 0 \\ -266.667 \\ 500 \end{pmatrix} \text{lb}$$

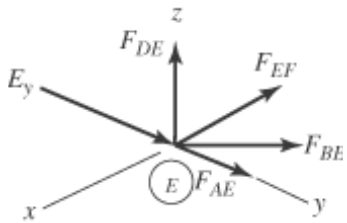


Now find the force in each member.

$$\mathbf{AB} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{AC} = \begin{pmatrix} -b \\ -a \\ c \end{pmatrix} \quad \mathbf{AD} = \begin{pmatrix} 0 \\ -a \\ c \end{pmatrix} \quad \mathbf{AE} = \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix}$$

$$\mathbf{BC} = \begin{pmatrix} 0 \\ -a \\ c \end{pmatrix} \quad \mathbf{BE} = \begin{pmatrix} b \\ -a \\ 0 \end{pmatrix} \quad \mathbf{BF} = \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix} \quad \mathbf{CD} = \begin{pmatrix} b \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{CF} = \begin{pmatrix} 0 \\ 0 \\ -c \end{pmatrix} \quad \mathbf{DE} = \begin{pmatrix} 0 \\ 0 \\ -c \end{pmatrix} \quad \mathbf{DF} = \begin{pmatrix} -b \\ 0 \\ -c \end{pmatrix} \quad \mathbf{EF} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix}$$



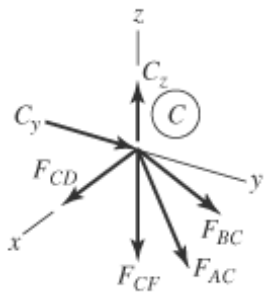
Guesses

$F_{AB} = 1$ lb $\quad F_{AC} = 1$ lb $\quad F_{AD} = 1$ lb $\quad F_{AE} = 1$ lb

$F_{BC} = 1$ lb $\quad F_{BE} = 1$ lb $\quad F_{BF} = 1$ lb $\quad F_{CD} = 1$ lb

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$F_{CF} = 1$ lb $F_{DE} = 1$lb $F_{DF} = 1$ lb $F_{EF} = 1$ lb

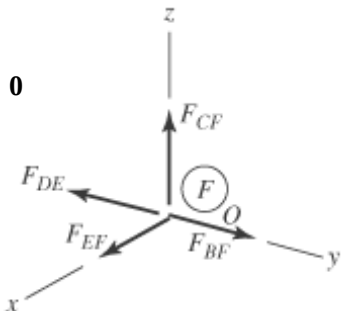


Given

$$\mathbf{F_1} + F_{AB}\frac{\mathbf{AB}}{|\mathbf{AB}|} + F_{AC}\frac{\mathbf{AC}}{|\mathbf{AC}|} + F_{AD}\frac{\mathbf{AD}}{|\mathbf{AD}|} + F_{AE}\frac{\mathbf{AE}}{|\mathbf{AE}|} = \mathbf{0}$$

$$\mathbf{F_2} + F_{BC}\frac{\mathbf{BC}}{|\mathbf{BC}|} + F_{BF}\frac{\mathbf{BF}}{|\mathbf{BF}|} + F_{BE}\frac{\mathbf{BE}}{|\mathbf{BE}|} + F_{AB}\frac{-\mathbf{AB}}{|\mathbf{AB}|} = \mathbf{0}$$

$$\begin{pmatrix} 0 \\ E_y \\ 0 \end{pmatrix} + F_{AE}\frac{-\mathbf{AE}}{|\mathbf{AE}|} + F_{BE}\frac{-\mathbf{BE}}{|\mathbf{BE}|} + F_{EF}\frac{\mathbf{EF}}{|\mathbf{EF}|} + F_{DE}\frac{-\mathbf{DE}}{|\mathbf{DE}|} = \mathbf{0}$$



$$F_{BF}\frac{-\mathbf{BF}}{|\mathbf{BF}|} + F_{CF}\frac{-\mathbf{CF}}{|\mathbf{CF}|} + F_{DF}\frac{-\mathbf{DF}}{|\mathbf{DF}|} + F_{EF}\frac{-\mathbf{EF}}{|\mathbf{EF}|} = \mathbf{0}$$

$$\begin{pmatrix} 0 \\ C_y \\ C_z \end{pmatrix} + F_{BC}\frac{-\mathbf{BC}}{|\mathbf{BC}|} + F_{AC}\frac{-\mathbf{AC}}{|\mathbf{AC}|} + F_{CD}\frac{\mathbf{CD}}{|\mathbf{CD}|} + F_{CF}\frac{\mathbf{CF}}{|\mathbf{CF}|} = \mathbf{0}$$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \\ F_{AE} \\ F_{BC} \\ F_{BE} \\ F_{BF} \\ F_{CD} \\ F_{CF} \\ F_{DE} \\ F_{DF} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AC}, F_{AD}, F_{AE}, F_{BC}, F_{BE}, F_{BF}, F_{CD}, F_{CF}, F_{DE}, F_{DF}, F_{EF}\right)$$

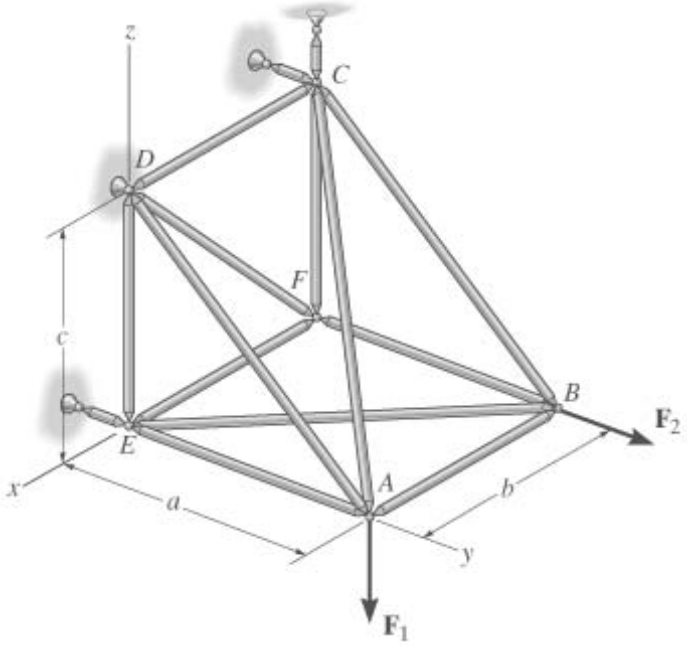
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$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \\ F_{AE} \\ F_{BC} \\ F_{BE} \end{pmatrix} = \begin{pmatrix} -300 \\ 583.095 \\ 333.333 \\ -666.667 \\ 0 \\ 500 \end{pmatrix} \text{lb} \qquad \begin{pmatrix} F_{BF} \\ F_{CD} \\ F_{CF} \\ F_{DE} \\ F_{DF} \\ F_{EF} \end{pmatrix} = \begin{pmatrix} 0 \\ -300 \\ -300 \\ 0 \\ 424.264 \\ -300 \end{pmatrix} \text{lb}$$

Positive (T)
Negative (C)

Problem 6-59

The space truss is supported by a ball-and-socket joint at *D* and short links at *C* and *E*. Determine the force in each member and state if the members are in tension or compression.



Given:

$$\mathbf{F_1} = \begin{pmatrix} 200 \\ 300 \\ -500 \end{pmatrix} \text{lb}$$

$$\mathbf{F_2} = \begin{pmatrix} 0 \\ 400 \\ 0 \end{pmatrix} \text{lb}$$

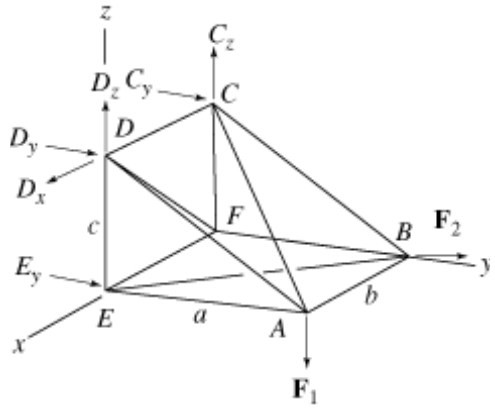
$a = 4$ ft

$b = 3$ ft

$c = 3$ ft

Solution:

Find the external reactions



Guesses

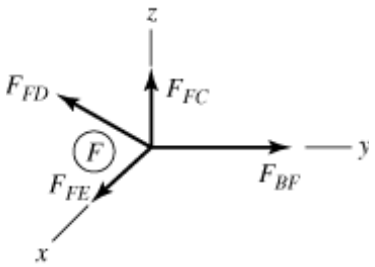
$E_y = 1$ lb $\quad C_y = 1$ lb $\quad C_z = 1$ lb

$D_x = 1$ lb $\quad D_y = 1$ lb $\quad D_z = 1$ lb

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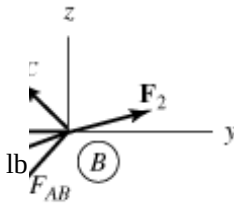
Given

$$\begin{pmatrix} D_x \\ D_y \\ D_z \end{pmatrix} + \begin{pmatrix} 0 \\ E_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ C_y \\ C_z \end{pmatrix} + \mathbf{F_1} + \mathbf{F_2} = \mathbf{0}$$

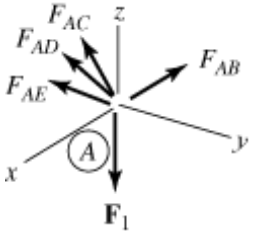


$$\begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \mathbf{F_1} + \begin{pmatrix} -b \\ a \\ 0 \end{pmatrix} \times \mathbf{F_2} + \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} \times \begin{pmatrix} D_x \\ D_y \\ D_z \end{pmatrix} + \begin{pmatrix} -b \\ 0 \\ c \end{pmatrix} \times \begin{pmatrix} 0 \\ C_y \\ C_z \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} E_y \\ C_y \\ C_z \\ D_x \\ D_y \\ D_z \end{pmatrix} = \text{Find}\left(E_y, C_y, C_z, D_x, D_y, D_z\right) \qquad \begin{pmatrix} E_y \\ C_y \\ C_z \\ D_x \\ D_y \\ D_z \end{pmatrix} = \begin{pmatrix} -33.333 \\ -666.667 \\ 200 \\ -200 \\ -1.253 \times 10^{-13} \\ 300 \end{pmatrix} \text{lb}$$



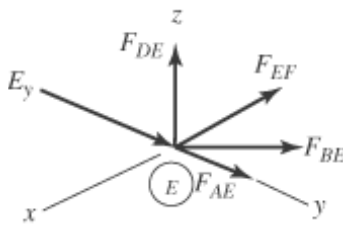
Now find the force in each member.



$$\mathbf{AB} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{AC} = \begin{pmatrix} -b \\ -a \\ c \end{pmatrix} \quad \mathbf{AD} = \begin{pmatrix} 0 \\ -a \\ c \end{pmatrix} \quad \mathbf{AE} = \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix}$$

$$\mathbf{BC} = \begin{pmatrix} 0 \\ -a \\ c \end{pmatrix} \quad \mathbf{BE} = \begin{pmatrix} b \\ -a \\ 0 \end{pmatrix} \quad \mathbf{BF} = \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix} \quad \mathbf{CD} = \begin{pmatrix} b \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{CF} = \begin{pmatrix} 0 \\ 0 \\ -c \end{pmatrix} \quad \mathbf{DE} = \begin{pmatrix} 0 \\ 0 \\ -c \end{pmatrix} \quad \mathbf{DF} = \begin{pmatrix} -b \\ 0 \\ -c \end{pmatrix} \quad \mathbf{EF} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix}$$

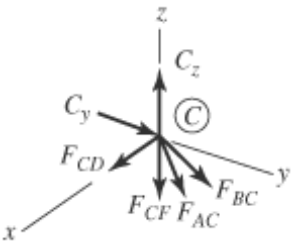


Guesses

$F_{AB} = 1$ lb $F_{AC} = 1$ lb $F_{AD} = 1$ lb $F_{AE} = 1$ lb

$F_{BC} = 1$ lb $F_{BE} = 1$ lb $F_{BF} = 1$ lb $F_{CD} = 1$ lb

$F_{CF} = 1$ lb $F_{DE} = 1$lb $F_{DF} = 1$ lb $F_{EF} = 1$ lb

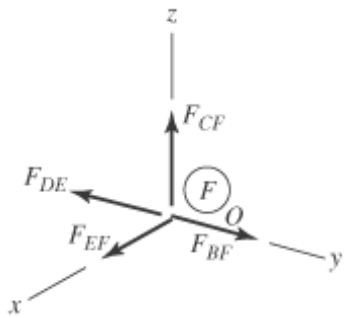


Given

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$$\mathbf{F_1} + F_{AB}\frac{\mathbf{AB}}{|\mathbf{AB}|} + F_{AC}\frac{\mathbf{AC}}{|\mathbf{AC}|} + F_{AD}\frac{\mathbf{AD}}{|\mathbf{AD}|} + F_{AE}\frac{\mathbf{AE}}{|\mathbf{AE}|} = \mathbf{0}$$

$$\mathbf{F_2} + F_{BC}\frac{\mathbf{BC}}{|\mathbf{BC}|} + F_{BF}\frac{\mathbf{BF}}{|\mathbf{BF}|} + F_{BE}\frac{\mathbf{BE}}{|\mathbf{BE}|} + F_{AB}\frac{-\mathbf{AB}}{|\mathbf{AB}|} = \mathbf{0}$$



$$\begin{pmatrix} 0 \\ E_y \\ 0 \end{pmatrix} + F_{AE}\frac{-\mathbf{AE}}{|\mathbf{AE}|} + F_{BE}\frac{-\mathbf{BE}}{|\mathbf{BE}|} + F_{EF}\frac{\mathbf{EF}}{|\mathbf{EF}|} + F_{DE}\frac{-\mathbf{DE}}{|\mathbf{DE}|} = \mathbf{0}$$

$$F_{BF}\frac{-\mathbf{BF}}{|\mathbf{BF}|} + F_{CF}\frac{-\mathbf{CF}}{|\mathbf{CF}|} + F_{DF}\frac{-\mathbf{DF}}{|\mathbf{DF}|} + F_{EF}\frac{-\mathbf{EF}}{|\mathbf{EF}|} = \mathbf{0}$$

$$\begin{pmatrix} 0 \\ C_y \\ C_z \end{pmatrix} + F_{BC}\frac{-\mathbf{BC}}{|\mathbf{BC}|} + F_{AC}\frac{-\mathbf{AC}}{|\mathbf{AC}|} + F_{CD}\frac{\mathbf{CD}}{|\mathbf{CD}|} + F_{CF}\frac{\mathbf{CF}}{|\mathbf{CF}|} = \mathbf{0}$$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \\ F_{AE} \\ F_{BC} \\ F_{BE} \\ F_{BF} \\ F_{CD} \\ F_{CF} \\ F_{DE} \\ F_{DF} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AC}, F_{AD}, F_{AE}, F_{BC}, F_{BE}, F_{BF}, F_{CD}, F_{CF}, F_{DE}, F_{DF}, F_{EF}\right)$$

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$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \\ F_{AE} \\ F_{BC} \\ F_{BE} \end{pmatrix} = \begin{pmatrix} -300 \\ 971.825 \\ 1.121 \times 10^{-11} \\ -366.667 \\ 0 \\ 500 \end{pmatrix} \begin{pmatrix} F_{BF} \\ F_{CD} \\ F_{CF} \\ F_{DE} \\ F_{DF} \\ F_{EF} \end{pmatrix} = \begin{pmatrix} 0 \\ -500 \\ -300 \\ 0 \\ 424.264 \\ -300 \end{pmatrix} \text{lb}$$

Positive (T)
Negative (C)

Problem 6-60

Determine the force in each member of the space truss and state if the members are in tension or compression. The truss is supported by a ball-and-socket joints at *A*, *B*, and *E*. *Hint:* The support reaction at *E* acts along member *EC*. Why?

Given:

$$\mathbf{F} = \begin{pmatrix} -200 \\ 400 \\ 0 \end{pmatrix} \text{N}$$

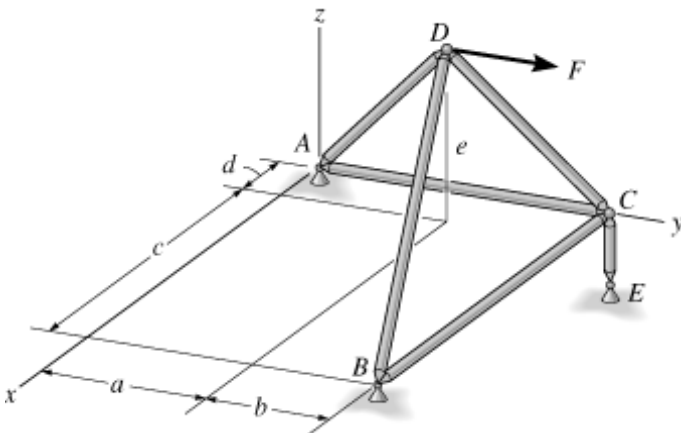
$a = 2$ m

$b = 1.5$ m

$c = 5$ m

$d = 1$ m

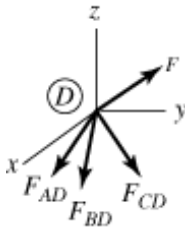
$e = 2$ m



Solution:

$$\mathbf{AC} = \begin{pmatrix} 0 \\ a+b \\ 0 \end{pmatrix} \quad \mathbf{AD} = \begin{pmatrix} d \\ a \\ e \end{pmatrix}$$

$$\mathbf{BC} = \begin{pmatrix} -c-d \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{BD} = \begin{pmatrix} -c \\ -b \\ e \end{pmatrix} \quad \mathbf{CD} = \begin{pmatrix} d \\ -b \\ e \end{pmatrix}$$

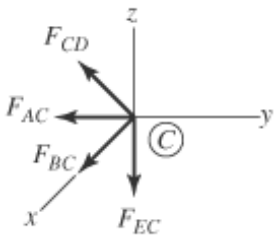


Guesses $F_{AC} = 1$ N $F_{AD} = 1$ N $F_{BC} = 1$ N

$F_{CD} = 1$ N $F_{EC} = 1$ N $F_{BD} = 1$ N

Given

$$\mathbf{F} + F_{AD}\frac{-\mathbf{AD}}{|\mathbf{AD}|} + F_{BD}\frac{-\mathbf{BD}}{|\mathbf{BD}|} + F_{CD}\frac{-\mathbf{CD}}{|\mathbf{CD}|} = \mathbf{0}$$



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$$F_{CD}\frac{\mathbf{CD}}{|\mathbf{CD}|}+F_{BC}\frac{-\mathbf{BC}}{|\mathbf{BC}|}+F_{AC}\frac{-\mathbf{AC}}{|\mathbf{AC}|}+\begin{pmatrix}0\\0\\-F_{EC}\end{pmatrix}=\mathbf{0}$$

$$\begin{pmatrix}F_{AC}\\F_{AD}\\F_{BC}\\F_{BD}\\F_{CD}\\F_{EC}\end{pmatrix}=\text{Find}\left(F_{AC},F_{AD},F_{BC},F_{BD},F_{CD},F_{EC}\right)\qquad \begin{pmatrix}F_{AC}\\F_{AD}\\F_{BC}\\F_{BD}\\F_{CD}\\F_{EC}\end{pmatrix}=\begin{pmatrix}221\\343\\148\\186\\-397\\-295\end{pmatrix}\text{ N}$$

Positive (T)
Negative (C)

Problem 6-61

Determine the force in each member of the space truss and state if the members are in tension or compression. The truss is supported by ball-and-socket joints at *C*, *D*, *E*, and *G*.

Units Used:

$$\text{kN}=10^3\text{ N}$$

Given:

$F = 3$ kN

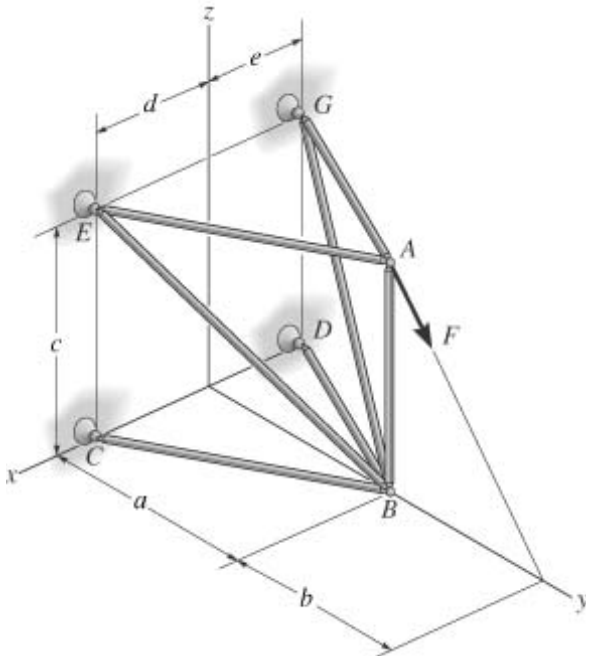
$a = 2$ m

$b = 1.5$ m

$c = 2$ m

$d = 1$ m

$e = 1$ m



Solution:

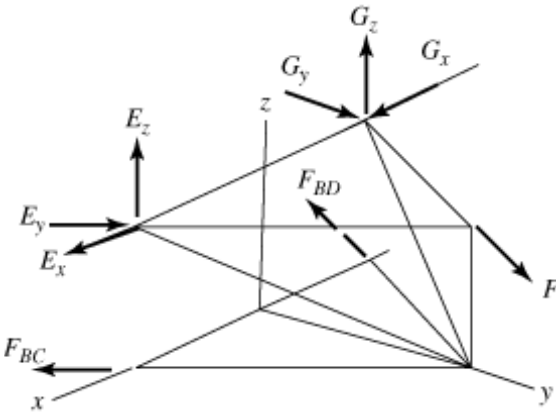
$$\mathbf{F_V}=\frac{F}{\sqrt{b^2+c^2}}\begin{pmatrix}0\\b\\-c\end{pmatrix}$$

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$$\mathbf{u_{AG}} = \frac{1}{\sqrt{a^2 + e^2}}\begin{pmatrix} -e \\ -a \\ 0 \end{pmatrix}$$

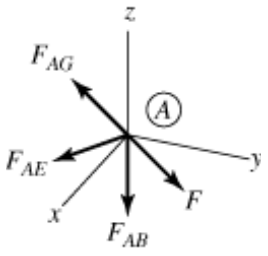
$$\mathbf{u_{AE}} = \frac{1}{\sqrt{a^2 + d^2}}\begin{pmatrix} d \\ -a \\ 0 \end{pmatrix}$$

$$\mathbf{u_{AB}} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \qquad \mathbf{u_{BC}} = \mathbf{u_{AE}} \qquad \mathbf{u_{BD}} = \mathbf{u_{AG}}$$



$$\mathbf{u_{BE}} = \frac{1}{\sqrt{a^2 + c^2 + d^2}}\begin{pmatrix} d \\ -a \\ c \end{pmatrix}$$

$$\mathbf{u_{BG}} = \frac{1}{\sqrt{a^2 + e^2 + c^2}}\begin{pmatrix} -e \\ -a \\ c \end{pmatrix}$$

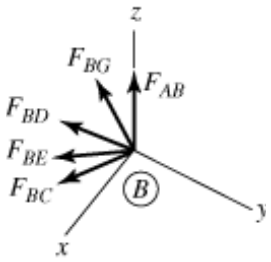


Guesses

$F_{AB} = 1$ kN $\quad F_{AE} = 1$ kN $\quad F_{AG} = 1$ kN

$F_{BC} = 1$ kN $\quad F_{BD} = 1$ kN

$F_{BE} = 1$ kN $\quad F_{BG} = 1$ kN



Given

$$\frac{-c}{\sqrt{c^2 + b^2}}F(a) - \frac{a}{\sqrt{a^2 + d^2}}F_{BC}(c) - \frac{a}{\sqrt{a^2 + e^2}}F_{BD}(c) = 0$$

$$\frac{e}{\sqrt{a^2 + e^2}}F_{BD}(a) - \frac{d}{\sqrt{a^2 + d^2}}F_{BC}(a) = 0$$

$$\mathbf{F_v} + F_{AE}\mathbf{u_{AE}} + F_{AG}\mathbf{u_{AG}} + F_{AB}\mathbf{u_{AB}} = \mathbf{0}$$

$$-F_{AB}\mathbf{u_{AB}} + F_{BG}\mathbf{u_{BG}} + F_{BE}\mathbf{u_{BE}} + F_{BC}\mathbf{u_{BC}} + F_{BD}\mathbf{u_{BD}} = \mathbf{0}$$

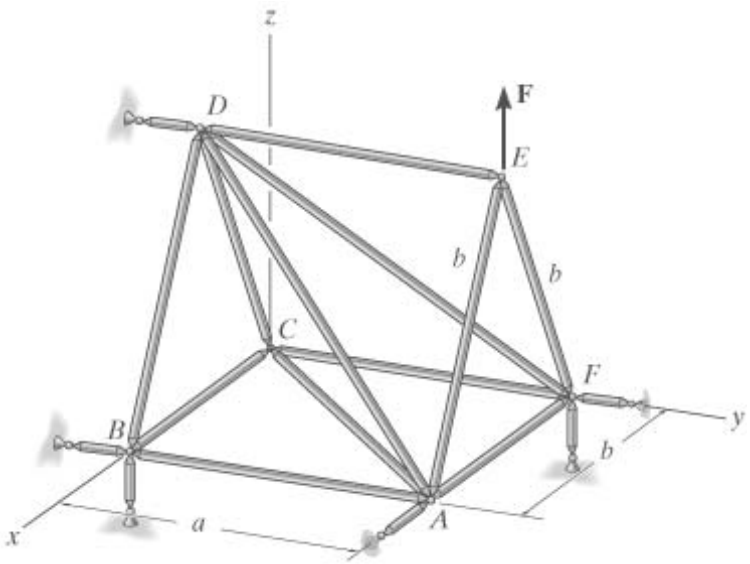
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$$\begin{pmatrix} F_{AB} \\ F_{AE} \\ F_{AG} \\ F_{BC} \\ F_{BD} \\ F_{BE} \\ F_{BG} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AE}, F_{AG}, F_{BC}, F_{BD}, F_{BE}, F_{BG}\right) \qquad \begin{pmatrix} F_{AB} \\ F_{AE} \\ F_{AG} \\ F_{BC} \\ F_{BD} \\ F_{BE} \\ F_{BG} \end{pmatrix} = \begin{pmatrix} -2.4 \\ 1.006 \\ 1.006 \\ -1.342 \\ -1.342 \\ 1.8 \\ 1.8 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-62

Determine the force in members *BD*, *AD*, and *AF* of the space truss and state if the members are in tension or compression. The truss is supported by short links at *A*, *B*, *D*, and *F*.



Given:

$$\mathbf{F} = \begin{pmatrix} 0 \\ 250 \\ -250 \end{pmatrix} \text{lb}$$

$a = 6$ ft

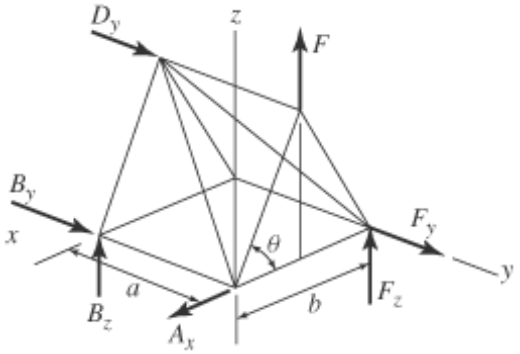
$b = 6$ ft

$\theta = 60$ deg

Solution:

Find the external reactions

$h = b\sin(\theta)$



Guesses

$A_x = 1$ lb $\quad B_y = 1$ lb $\quad B_z = 1$ lb

$D_y = 1$ lb $\quad F_y = 1$ lb $\quad F_z = 1$ lb

Given

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$$\mathbf{F} + \begin{pmatrix} A_x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} + \begin{pmatrix} 0 \\ D_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ F_y \\ F_z \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} 0.5b \\ a \\ h \end{pmatrix} \times \mathbf{F} + \begin{pmatrix} b \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} A_x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} b \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} + \begin{pmatrix} 0.5b \\ 0 \\ h \end{pmatrix} \times \begin{pmatrix} 0 \\ D_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ F_y \\ F_z \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} A_x \\ B_y \\ B_z \\ D_y \\ F_y \\ F_z \end{pmatrix} = \text{Find}\left(A_x, B_y, B_z, D_y, F_y, F_z\right) \qquad \begin{pmatrix} A_x \\ B_y \\ B_z \\ D_y \\ F_y \\ F_z \end{pmatrix} = \begin{pmatrix} 0 \\ 72 \\ 125 \\ -394 \\ 72 \\ 125 \end{pmatrix} \text{lb}$$

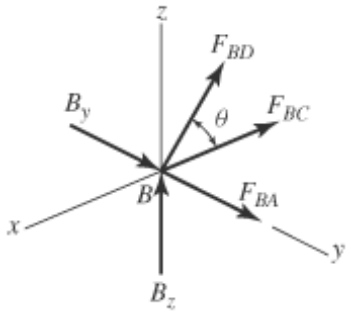
Now find the forces in the members

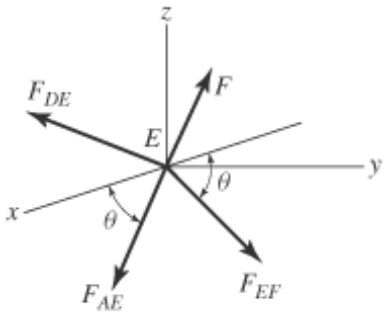
$$\mathbf{AB} = \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix} \quad \mathbf{AC} = \begin{pmatrix} -b \\ -a \\ 0 \end{pmatrix} \quad \mathbf{AD} = \begin{pmatrix} -0.5b \\ -a \\ h \end{pmatrix}$$

$$\mathbf{AE} = \begin{pmatrix} -0.5b \\ 0 \\ h \end{pmatrix} \quad \mathbf{AF} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{BC} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{BD} = \begin{pmatrix} -0.5b \\ 0 \\ h \end{pmatrix} \quad \mathbf{CD} = \begin{pmatrix} 0.5b \\ 0 \\ h \end{pmatrix} \quad \mathbf{CF} = \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix}$$

$$\mathbf{DE} = \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \quad \mathbf{DF} = \begin{pmatrix} -0.5b \\ a \\ -h \end{pmatrix} \quad \mathbf{EF} = \begin{pmatrix} -0.5b \\ 0 \\ -h \end{pmatrix}$$





Guesses

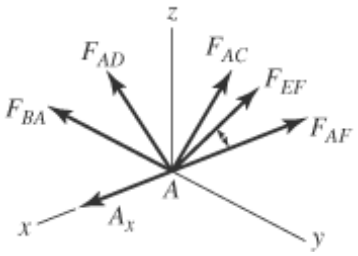
$F_{AB} = 1$ lb $F_{AC} = 1$ lb $F_{AD} = 1$ lb

$F_{AE} = 1$ lb $F_{AF} = 1$ lb $F_{BC} = 1$ lb

$F_{BD} = 1$ lb $F_{CD} = 1$ lb $F_{CF} = 1$ lb

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$F_{DE} = 1 \text{ lb} \quad F_{DF} = 1 \text{ lb} \quad F_{EF} = 1 \text{ lb}$



Given

$$\mathbf{F} + F_{DE}\frac{-\mathbf{DE}}{|\mathbf{DE}|} + F_{AE}\frac{-\mathbf{AE}}{|\mathbf{AE}|} + F_{EF}\frac{\mathbf{EF}}{|\mathbf{EF}|} = \mathbf{0}$$

$$F_{CF}\frac{\mathbf{CF}}{|\mathbf{CF}|} + F_{CD}\frac{\mathbf{CD}}{|\mathbf{CD}|} + F_{BC}\frac{-\mathbf{BC}}{|\mathbf{BC}|} + F_{AC}\frac{-\mathbf{AC}}{|\mathbf{AC}|} = \mathbf{0}$$

$$F_{DE}\frac{\mathbf{DE}}{|\mathbf{DE}|} + F_{DF}\frac{\mathbf{DF}}{|\mathbf{DF}|} + F_{AD}\frac{-\mathbf{AD}}{|\mathbf{AD}|} + F_{BD}\frac{-\mathbf{BD}}{|\mathbf{BD}|} + F_{CD}\frac{-\mathbf{CD}}{|\mathbf{CD}|} + \begin{pmatrix} 0 \\ D_y \\ 0 \end{pmatrix} = \mathbf{0}$$

$$F_{AB}\frac{-\mathbf{AB}}{|\mathbf{AB}|} + F_{BC}\frac{\mathbf{BC}}{|\mathbf{BC}|} + F_{BD}\frac{\mathbf{BD}}{|\mathbf{BD}|} + \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} = \mathbf{0}$$

$$F_{AB}\frac{\mathbf{AB}}{|\mathbf{AB}|} + F_{AC}\frac{\mathbf{AC}}{|\mathbf{AC}|} + F_{AF}\frac{\mathbf{AF}}{|\mathbf{AF}|} + F_{AD}\frac{\mathbf{AD}}{|\mathbf{AD}|} + F_{AE}\frac{\mathbf{AE}}{|\mathbf{AE}|} + \begin{pmatrix} A_x \\ 0 \\ 0 \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \\ F_{AE} \\ F_{AF} \\ F_{BC} \\ F_{BD} \\ F_{CD} \\ F_{CF} \\ F_{DE} \\ F_{DF} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AC}, F_{AD}, F_{AE}, F_{AF}, F_{BC}, F_{BD}, F_{CD}, F_{CF}, F_{DE}, F_{DF}, F_{EF}\right)$$

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$$\begin{pmatrix} F_{BD} \\ F_{AD} \\ F_{AF} \end{pmatrix} = \begin{pmatrix} -144.3 \\ 204.1 \\ 72.2 \end{pmatrix} \text{lb}$$

Positive (T)
Negative (C)

Problem 6-63

Determine the force in members *CF*, *EF*, and *DF* of the space truss and state if the members are in tension or compression. The truss is supported by short links at *A*, *B*, *D*, and *F*.

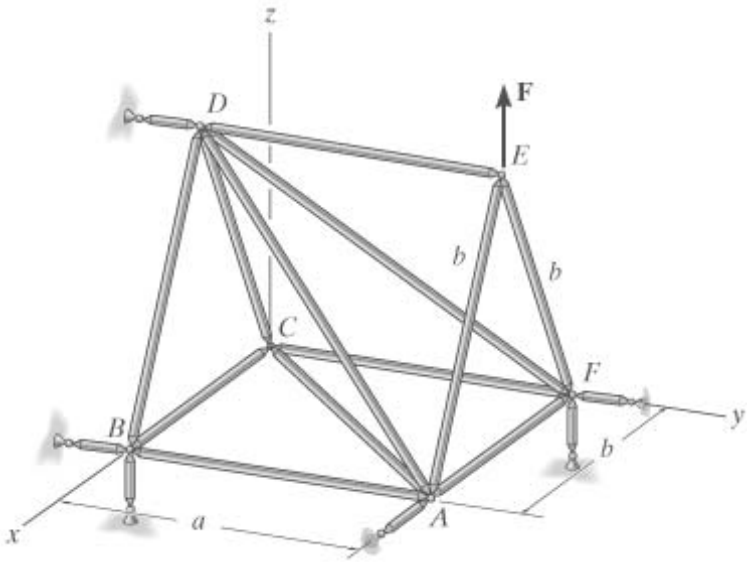
Given:

$$\mathbf{F} = \begin{pmatrix} 0 \\ 250 \\ -250 \end{pmatrix} \text{lb}$$

$a = 6$ ft

$b = 6$ ft

$\theta = 60$ deg



Solution:

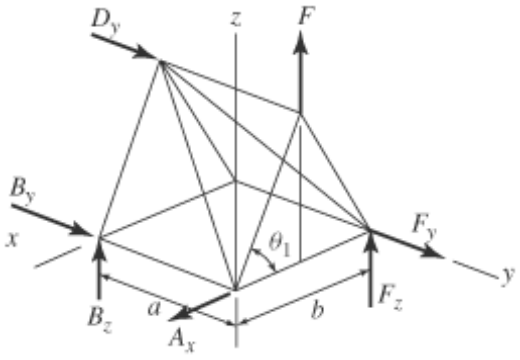
Find the external reactions

$h = b\sin(\theta)$

Guesses

$A_x = 1$ lb $\quad B_y = 1$ lb $\quad B_z = 1$ lb

$D_y = 1$ lb $\quad F_y = 1$ lb $\quad F_z = 1$ lb



Given

$$\mathbf{F} + \begin{pmatrix} A_x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} + \begin{pmatrix} 0 \\ D_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ F_y \\ F_z \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} 0.5b \\ a \\ h \end{pmatrix} \times \mathbf{F} + \begin{pmatrix} b \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} A_x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} b \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ B_y \\ B_z \end{pmatrix} + \begin{pmatrix} 0.5b \\ 0 \\ h \end{pmatrix} \times \begin{pmatrix} 0 \\ D_y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ F_y \\ F_z \end{pmatrix} = \mathbf{0}$$

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$$\begin{pmatrix} A_x \\ B_y \\ B_z \\ D_y \\ F_y \\ F_z \end{pmatrix} = \text{Find}\left(A_x, B_y, B_z, D_y, F_y, F_z\right) \qquad \begin{pmatrix} A_x \\ B_y \\ B_z \\ D_y \\ F_y \\ F_z \end{pmatrix} = \begin{pmatrix} 0 \\ 72 \\ 125 \\ -394 \\ 72 \\ 125 \end{pmatrix} \text{lb}$$

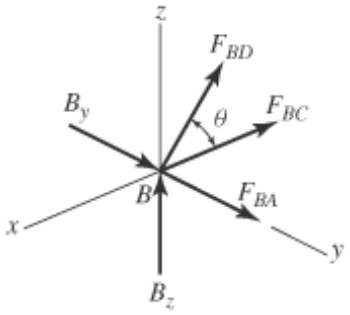
Now find the forces in the members

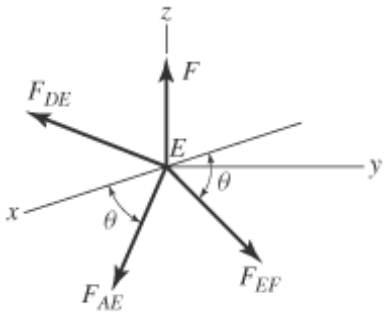
$$\mathbf{AB} = \begin{pmatrix} 0 \\ -a \\ 0 \end{pmatrix} \quad \mathbf{AC} = \begin{pmatrix} -b \\ -a \\ 0 \end{pmatrix} \quad \mathbf{AD} = \begin{pmatrix} -0.5b \\ -a \\ h \end{pmatrix}$$

$$\mathbf{AE} = \begin{pmatrix} -0.5b \\ 0 \\ h \end{pmatrix} \quad \mathbf{AF} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix} \quad \mathbf{BC} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{BD} = \begin{pmatrix} -0.5b \\ 0 \\ h \end{pmatrix} \quad \mathbf{CD} = \begin{pmatrix} 0.5b \\ 0 \\ h \end{pmatrix} \quad \mathbf{CF} = \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix}$$

$$\mathbf{DE} = \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \quad \mathbf{DF} = \begin{pmatrix} -0.5b \\ a \\ -h \end{pmatrix} \quad \mathbf{EF} = \begin{pmatrix} -0.5b \\ 0 \\ -h \end{pmatrix}$$





Guesses

$F_{AB} = 1$ lb $F_{AC} = 1$ lb $F_{AD} = 1$ lb

$F_{AE} = 1$ lb $F_{AF} = 1$ lb $F_{BC} = 1$ lb

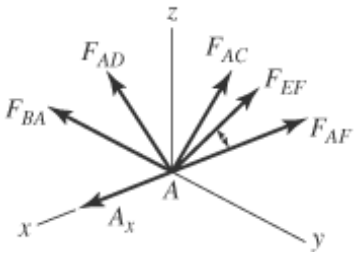
$F_{BD} = 1$ lb $F_{CD} = 1$ lb $F_{CF} = 1$ lb

$F_{DE} = 1$ lb $F_{DF} = 1$ lb $F_{EF} = 1$ lb

Given

$$\mathbf{F} + F_{DE}\frac{-\mathbf{DE}}{|\mathbf{DE}|} + F_{AE}\frac{-\mathbf{AE}}{|\mathbf{AE}|} + F_{EF}\frac{\mathbf{EF}}{|\mathbf{EF}|} = \mathbf{0}$$

$$F_{CF}\frac{\mathbf{CF}}{|\mathbf{CF}|} + F_{CD}\frac{\mathbf{CD}}{|\mathbf{CD}|} + F_{BC}\frac{-\mathbf{BC}}{|\mathbf{BC}|} + F_{AC}\frac{-\mathbf{AC}}{|\mathbf{AC}|} = \mathbf{0}$$



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$$F_{DE}\frac{\mathbf{DE}}{|\mathbf{DE}|}+F_{DF}\frac{\mathbf{DF}}{|\mathbf{DF}|}+F_{AD}\frac{-\mathbf{AD}}{|\mathbf{AD}|}+F_{BD}\frac{-\mathbf{BD}}{|\mathbf{BD}|}+F_{CD}\frac{-\mathbf{CD}}{|\mathbf{CD}|}+\begin{pmatrix}0\\D_y\\0\end{pmatrix}=\mathbf{0}$$

$$F_{AB}\frac{-\mathbf{AB}}{|\mathbf{AB}|}+F_{BC}\frac{\mathbf{BC}}{|\mathbf{BC}|}+F_{BD}\frac{\mathbf{BD}}{|\mathbf{BD}|}+\begin{pmatrix}0\\B_y\\B_z\end{pmatrix}=\mathbf{0}$$

$$F_{AB}\frac{\mathbf{AB}}{|\mathbf{AB}|}+F_{AC}\frac{\mathbf{AC}}{|\mathbf{AC}|}+F_{AF}\frac{\mathbf{AF}}{|\mathbf{AF}|}+F_{AD}\frac{\mathbf{AD}}{|\mathbf{AD}|}+F_{AE}\frac{\mathbf{AE}}{|\mathbf{AE}|}+\begin{pmatrix}A_x\\0\\0\end{pmatrix}=\mathbf{0}$$

$$\begin{pmatrix}F_{AB}\\F_{AC}\\F_{AD}\\F_{AE}\\F_{AF}\\F_{BC}\\F_{BD}\\F_{CD}\\F_{CF}\\F_{DE}\\F_{DF}\\F_{EF}\end{pmatrix}=\text{Find}\left(F_{AB},F_{AC},F_{AD},F_{AE},F_{AF},F_{BC},F_{BD},F_{CD},F_{CF},F_{DE},F_{DF},F_{EF}\right)$$

$$\begin{pmatrix}F_{CF}\\F_{EF}\\F_{DF}\end{pmatrix}=\begin{pmatrix}72.2\\-144.3\\0\end{pmatrix}\text{ lb}$$

Positive (T)
Negative (C)

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Problem 6-64

Determine the force developed in each member of the space truss and state if the members are in tension or compression. The crate has weight W.

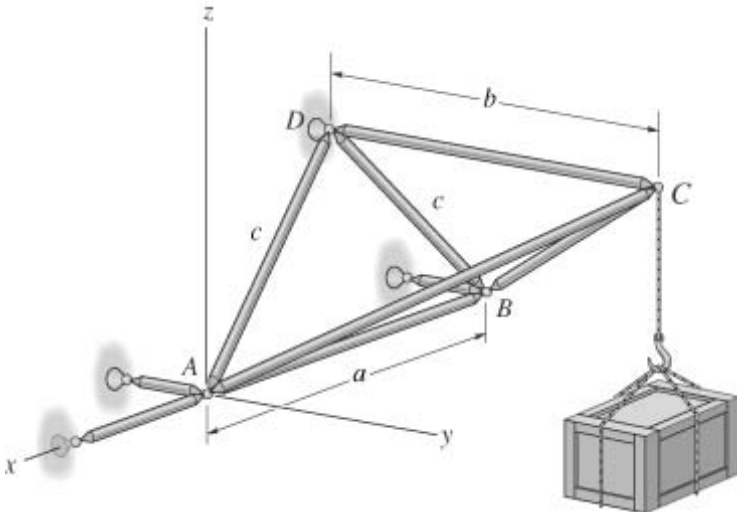
Given:

$W = 150$ lb

$a = 6$ ft

$b = 6$ ft

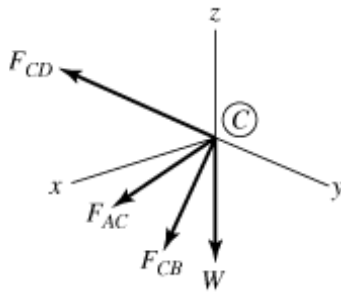
$c = 6$ ft



Solution: Unit Vectors

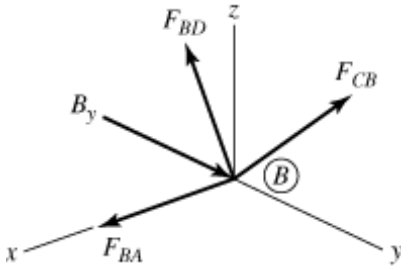
$$h = \sqrt{c^2 - \left(\frac{a}{2}\right)^2}$$

$$\mathbf{u_{AD}} = \frac{1}{\sqrt{h^2 + \left(\frac{a}{2}\right)^2}} \begin{pmatrix} \frac{-a}{2} \\ 0 \\ h \end{pmatrix}$$



$$\mathbf{u_{BD}} = \frac{1}{\sqrt{h^2 + \left(\frac{a}{2}\right)^2}} \begin{pmatrix} \frac{a}{2} \\ 0 \\ h \end{pmatrix}$$

$$\mathbf{u_{AC}} = \frac{1}{\sqrt{h^2 + b^2 + \left(\frac{a}{2}\right)^2}} \begin{pmatrix} -\frac{a}{2} \\ b \\ h \end{pmatrix}$$



$$\mathbf{u_{BC}} = \frac{1}{\sqrt{h^2 + b^2 + \left(\frac{a}{2}\right)^2}} \begin{pmatrix} \frac{a}{2} \\ b \\ h \end{pmatrix}$$

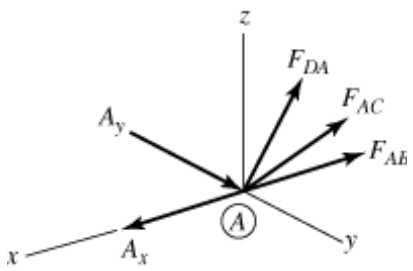
Guesses $B_y = 1$ lb $A_x = 1$ lb $A_y = 1$ lb

$F_{AB} = 1$ lb $F_{AC} = 1$ lb $F_{AD} = 1$ lb $F_{BC} = 1$ lb $F_{BD} = 1$ lb $F_{CD} = 1$ lb

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Given

$$\begin{pmatrix} 0 \\ -F_{CD} \\ -W \end{pmatrix} - F_{AC}\mathbf{u_{AC}} - F_{BC}\mathbf{u_{BC}} = \mathbf{0}$$



$$\begin{pmatrix} F_{AB} \\ B_y \\ 0 \end{pmatrix} + F_{BC}\mathbf{u_{BC}} + F_{BD}\mathbf{u_{BD}} = \mathbf{0}$$

$$\begin{pmatrix} A_x - F_{AB} \\ A_y \\ 0 \end{pmatrix} + F_{AC}\mathbf{u_{AC}} + F_{AD}\mathbf{u_{AD}} = \mathbf{0}$$

$$\begin{pmatrix} A_x \\ A_y \\ B_y \\ F_{AB} \\ F_{AC} \\ F_{AD} \\ F_{BC} \\ F_{BD} \\ F_{CD} \end{pmatrix} = \text{Find}\left(A_x, A_y, B_y, F_{AB}, F_{AC}, F_{AD}, F_{BC}, F_{BD}, F_{CD}\right)$$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \\ F_{BC} \\ F_{BD} \\ F_{CD} \end{pmatrix} = \begin{pmatrix} 0.0 \\ -122.5 \\ 86.6 \\ -122.5 \\ 86.6 \\ 173.2 \end{pmatrix} \text{lb}$$

Positive (T)
Negative (C)

Problem 6-65

The space truss is used to support vertical forces at joints *B*, *C*, and *D*. Determine the force in each member and state if the members are in tension or compression.

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Units Used:

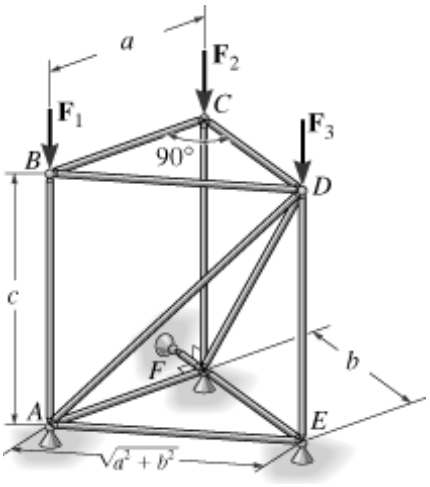
$$\text{kN} = 10^3 \text{ N}$$

Given:

$F_1 = 6$ kN $\quad a = 0.75$ m

$F_2 = 8$ kN $\quad b = 1.00$ m

$F_3 = 9$ kN $\quad c = 1.5$ m



Solution:

Assume that the connections at A, E, and F are rollers

Guesses

$F_{BC} = 1$ kN $\quad F_{CF} = 1$ kN

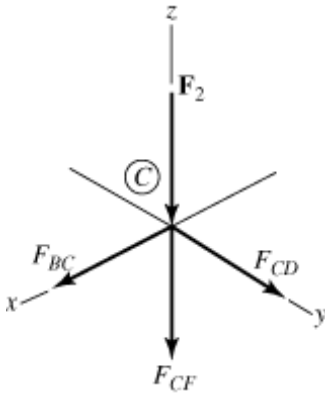
$F_{CD} = 1$ kN $\quad F_{AD} = 1$ kN

$F_{DF} = 1$ kN $\quad F_{DE} = 1$ kN

$F_{BD} = 1$ kN $\quad F_{BA} = 1$ kN

$F_{EF} = 1$ kN $\quad F_{AE} = 1$ kN

$F_{AF} = 1$ kN

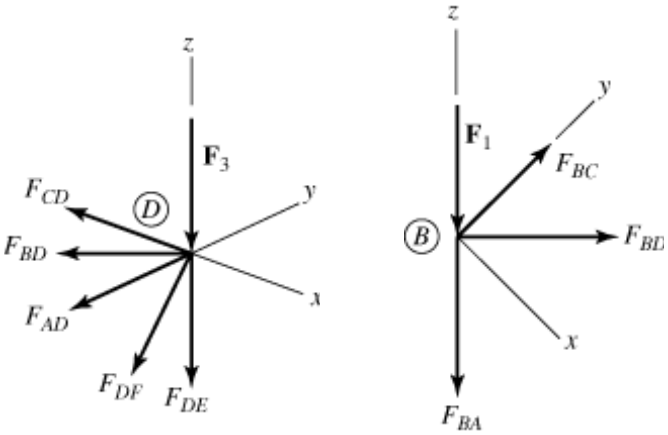


Given

Joint C

$F_{BC} = 0 \quad F_{CD} = 0$

$-F_2 - F_{CF} = 0$



Joint D

$$\frac{a}{\sqrt{a^2 + b^2}} F_{BD} + \frac{a}{\sqrt{a^2 + b^2 + c^2}} F_{AD} = 0$$

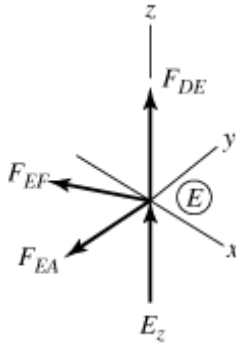
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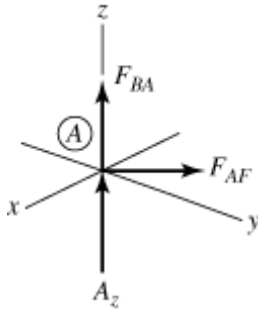
$$-F_{CD} - \frac{b}{\sqrt{a^2 + b^2}} F_{BD} \cdots = 0$$

$$+ \frac{-b}{\sqrt{a^2 + b^2 + c^2}} F_{AD} - \frac{b}{\sqrt{b^2 + c^2}} F_{DF}$$

$$-F_3 - F_{DE} - \frac{c}{\sqrt{b^2 + c^2}} F_{DF} \cdots = 0$$

$$+ \frac{-c}{\sqrt{a^2 + b^2 + c^2}} F_{AD}$$





Joint *B*

$$-F_{BC} - \frac{a}{\sqrt{a^2 + b^2}} F_{BD} = 0 \qquad \frac{b}{\sqrt{a^2 + b^2}} F_{BD} = 0 \qquad -F_1 - F_{BA} = 0$$

Joint *E*

$$\frac{a}{\sqrt{a^2 + b^2}} F_{AE} = 0 \qquad -F_{EF} - \frac{b}{\sqrt{a^2 + b^2}} F_{AE} = 0$$

$$\begin{pmatrix} F_{BC} \\ F_{CF} \\ F_{CD} \\ F_{AD} \\ F_{DF} \\ F_{DE} \\ F_{BD} \\ F_{BA} \\ F_{EF} \\ F_{AE} \\ F_{AF} \end{pmatrix} = \text{Find}\left(F_{BC}, F_{CF}, F_{CD}, F_{AD}, F_{DF}, F_{DE}, F_{BD}, F_{BA}, F_{EF}, F_{AE}, F_{AF}\right)$$

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$$\begin{pmatrix} F_{BC} \\ F_{CF} \\ F_{CD} \\ F_{AD} \\ F_{DF} \\ F_{DE} \\ F_{BD} \\ F_{BA} \\ F_{EF} \\ F_{AE} \\ F_{AF} \end{pmatrix} = \begin{pmatrix} 0.00 \\ -8.00 \\ 0.00 \\ 0.00 \\ 0.00 \\ -9.00 \\ 0.00 \\ -6.00 \\ 0.00 \\ 0.00 \\ 0.00 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-66

A force **P** is applied to the handles of the pliers. Determine the force developed on the smooth bolt *B* and the reaction that pin *A* exerts on its attached members.

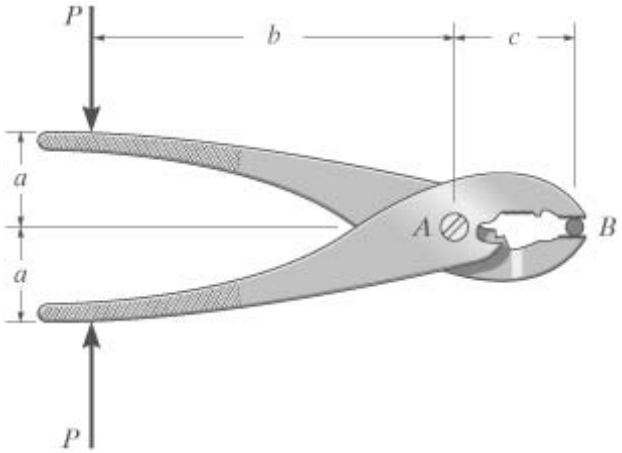
Given:

$P = 8$ lb

$a = 1.25$ in

$b = 5$ in

$c = 1.5$ in

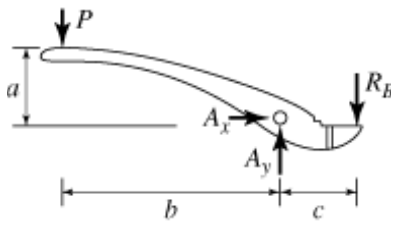


Solution:

$\Sigma M_A = 0; \quad -R_B c + P b = 0$

$$R_B = P \frac{b}{c}$$

$$R_B = 26.7 \text{ lb}$$



$\Sigma F_x = 0; \quad A_x = 0$

$\Sigma F_y = 0; \quad A_y - P - R_B = 0$

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$$A_y = P + R_B$$

$$A_y = 34.7\,\text{lb}$$

Problem 6-67

The eye hook has a positive locking latch when it supports the load because its two parts are pin-connected at A and they bear against one another along the smooth surface at B. Determine the resultant force at the pin and the normal force at B when the eye hook supports load $\mathbf{F}$.

Given:

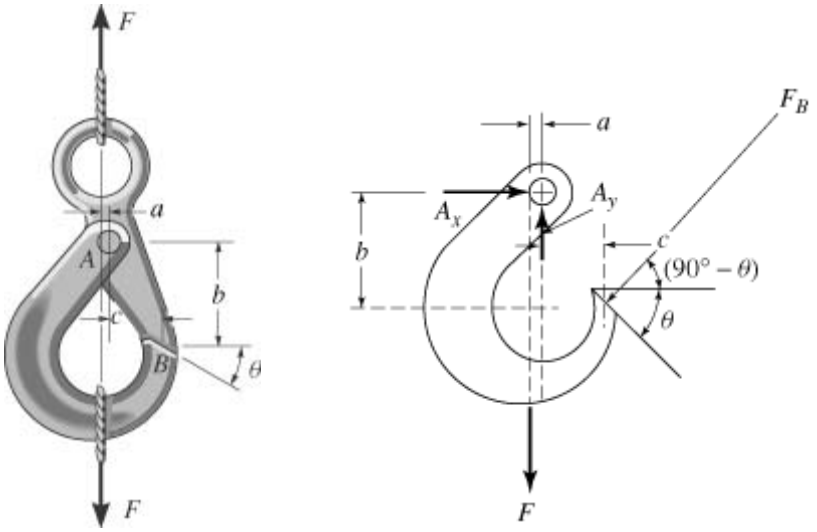
$F = 800$ lb

$a = 0.25$ in

$b = 3$ in

$c = 2$ in

$\theta = 30$ deg



Solution:

$$\curvearrowright + \Sigma M_A = 0; \qquad -F_B\cos(90\text{ deg} - \theta)(b) - F_B\sin(90\text{ deg} - \theta)(c) + F\,a = 0$$

$$F_B = F\frac{a}{\cos(90\text{ deg} - \theta)\,b + \sin(90\text{ deg} - \theta)\,c} \qquad F_B = 61.9\,\text{lb}$$

$$+\uparrow \Sigma F_y = 0; \qquad -F - F_B\sin(90\text{ deg} - \theta) + A_y = 0$$

$$A_y = F + F_B\sin(90\text{ deg} - \theta) \qquad A_y = 854\,\text{lb}$$

$$\overset{+}{\rightarrow} \Sigma F_x = 0; \qquad A_x - F_B\cos(90\text{ deg} - \theta) = 0$$

$$A_x = F_B\cos(90\text{ deg} - \theta) \qquad A_x = 30.9\,\text{lb}$$

$$F_A = \sqrt{A_x^2 + A_y^2} \qquad F_A = 854\,\text{lb}$$

Problem 6-68

Determine the force $\mathbf{P}$ needed to hold the block of mass F in equilibrium.

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Given:

$F = 20$ lb

Solution:

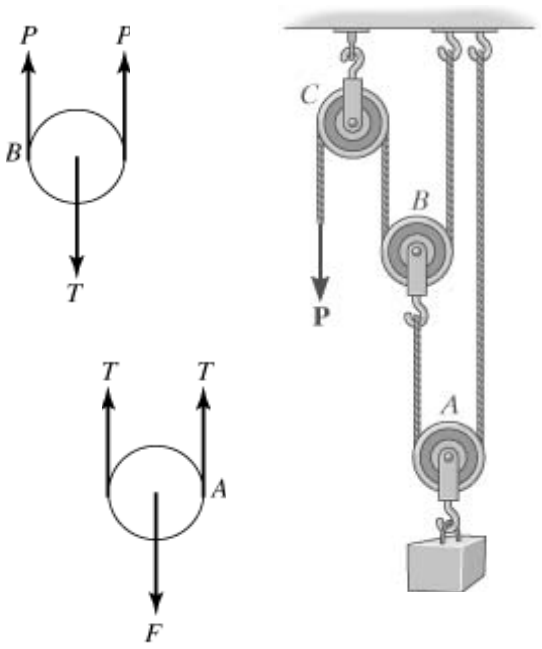
Pulley B: $\Sigma F_y = 0;$ $2P - T = 0$

Pulley A: $\Sigma F_y = 0;$ $2T - F = 0$

$T = \frac{1}{2}F$ $T = 10\,\text{lb}$

$2P = T$ $P = \frac{1}{2}T$

$P = 5\,\text{lb}$



Problem 6-69

The link is used to hold the rod in place. Determine the required axial force on the screw at E if the largest force to be exerted on the rod at B, C or D is to be $\mathbf{F}_{\mathbf{max}}$. Also, find the magnitude of the force reaction at pin A. Assume all surfaces of contact are smooth.

Given:

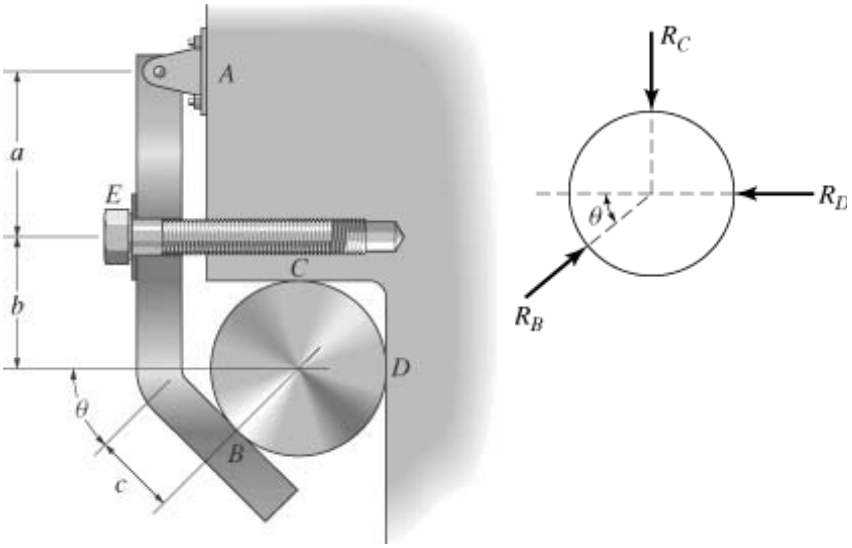
$F_{max} = 100$ lb

$a = 100$ mm

$b = 80$ mm

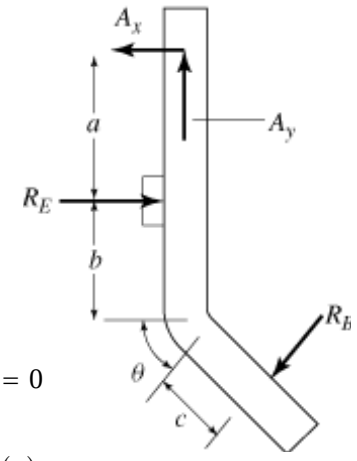
$c = 50$ mm

$\theta = 45$ deg



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Solution:



Assign an initial value for R_E. This will be scaled at the end of the problem.

Guesses $A_x = 1$ lb $A_y = 1$ lb $R_B = 1$ lb

$R_C = 1$ lb $R_D = 1$ lb $R_E = 1$ lb

Given $-A_x + R_E - R_B\cos(\theta) = 0$ $A_y - R_B\sin(\theta) = 0$

$R_E a - R_B\cos(\theta)(a + b + c\cos(\theta)) - R_B\sin(\theta)c\sin(\theta) = 0$

$R_B\cos(\theta) - R_D = 0$ $R_B\sin(\theta) - R_C = 0$

$$\begin{pmatrix} A_x \\ A_y \\ R_B \\ R_C \\ R_D \end{pmatrix} = \text{Find}(A_x, A_y, R_B, R_C, R_D) \qquad \begin{pmatrix} A_x \\ A_y \\ R_B \\ R_C \\ R_D \end{pmatrix} = \begin{pmatrix} 0.601 \\ 0.399 \\ 0.564 \\ 0.399 \\ 0.399 \end{pmatrix} \text{lb}$$

Now find the critical load and scale the problem

$$ans = \begin{pmatrix} R_B \\ R_C \\ R_D \end{pmatrix} \qquad F_{scale} = \frac{F_{max}}{\max(ans)} \qquad R_E = F_{scale} R_E \qquad R_E = 177.3\,\text{lb}$$

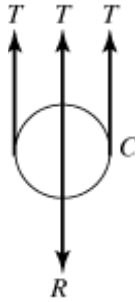
$$F_A = F_{scale}\sqrt{A_x^{\,2} + A_y^{\,2}} \qquad F_A = 127.9\,\text{lb}$$

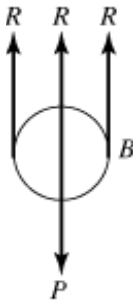
Problem 6-70

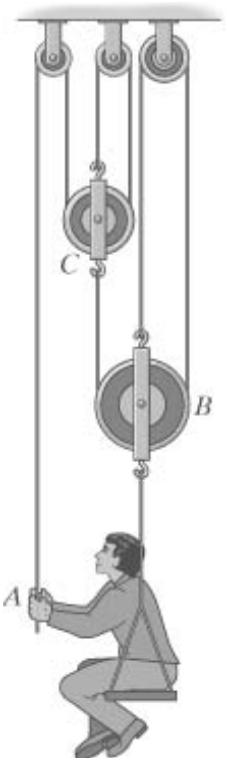
The man of weight W_1 attempts to lift himself and the seat of weight W_2 using the rope and pulley system shown. Determine the force at A needed to do so, and also find his reaction on the seat.

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Given:







$W_1 = 150$ lb

$W_2 = 10$ lb

Solution:

Pulley C:

$\Sigma F_y = 0; \quad 3T - R = 0$

Pulley B:

$\Sigma F_y = 0; \quad 3R - P = 0$

Thus, $P = 9T$



Man and seat:

$\Sigma F_y = 0; \quad T + P - W_1 - W_2 = 0$

$10T = W_1 + W_2$

$T = \dfrac{W_1 + W_2}{10}$ $\quad T = 16\,\text{lb}$

$P = 9T$ $\quad P = 144\,\text{lb}$

Seat:

$\Sigma F_y = 0; \quad P - N - W_2 = 0$

$N = P - W_2$ $\quad N = 134\,\text{lb}$

Problem 6-71

Determine the horizontal and vertical components of force that pins A and C exert on the frame.

Given:

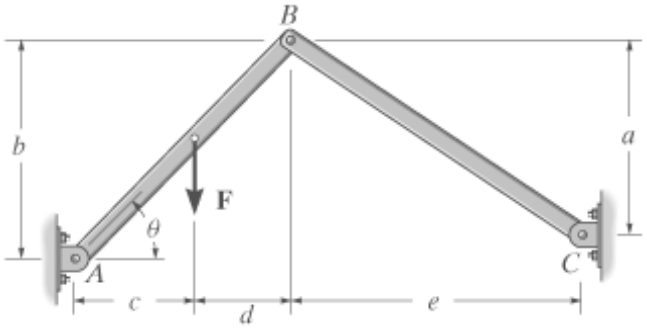
$F = 500$ N

$a = 0.8$ m $\quad d = 0.4$ m

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$b = 0.9$ m $\quad e = 1.2$ m

$c = 0.5$ m $\quad \theta = 45$deg



Solution:

BC is a two-force member

Member *AB* :

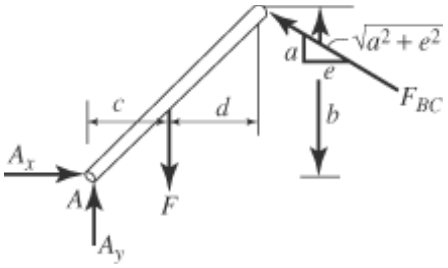
$$\Sigma M_A = 0; \quad -F\,c + F_{BC}\frac{e}{\sqrt{a^2+e^2}}b + F_{BC}\frac{a}{\sqrt{a^2+e^2}}(c+d) = 0$$

$$F_{BC} = F\,c\,\frac{\sqrt{a^2+e^2}}{e\,b + a\,c + a\,d} \qquad F_{BC} = 200.3 \text{ N}$$

Thus,

$$C_x = F_{BC}\frac{e}{\sqrt{a^2+e^2}} \qquad C_x = 167 \text{ N}$$

$$C_y = F_{BC}\frac{a}{\sqrt{a^2+e^2}} \qquad C_y = 111 \text{ N}$$



$$\Sigma F_x = 0; \quad A_x - F_{BC}\frac{e}{\sqrt{a^2+e^2}} = 0 \qquad A_x = F_{BC}\frac{e}{\sqrt{a^2+e^2}} \qquad A_x = 167 \text{ N}$$

$$\Sigma F_y = 0; \quad A_y - F + F_{BC}\frac{a}{\sqrt{a^2+e^2}} = 0 \qquad A_y = F - F_{BC}\frac{a}{\sqrt{a^2+e^2}} \qquad A_y = 389 \text{ N}$$

Problem 6-72

Determine the horizontal and vertical components of force that pins *A* and *C* exert on the frame.

Units Used:

$\text{kN} = 10^3$ N

Given:

$F_1 = 1$ kN

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$F_2 = 500$ N

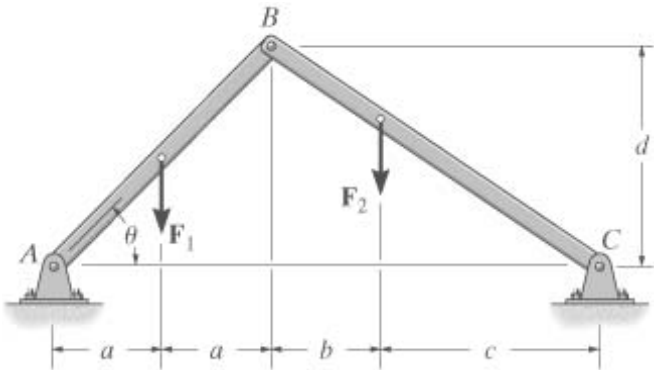
$\theta = 45$ deg

$a = 0.2$ m

$b = 0.2$ m

$c = 0.4$ m

$d = 0.4$ m

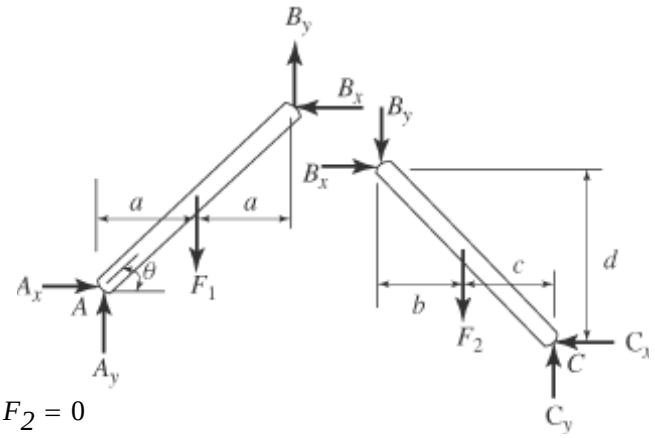


Solution:

Guesses

$A_x = 1$ N $\quad A_y = 1$ N

$C_x = 1$ N $\quad C_y = 1$ N



Given

$A_x - C_x = 0 \qquad A_y + C_y - F_1 - F_2 = 0$

$F_1 a - A_y\, 2\, a + A_x d = 0 \qquad -F_2 b + C_y(b + c) - C_x d = 0$

$$\begin{pmatrix} A_x \\ A_y \\ C_x \\ C_y \end{pmatrix} = \text{Find}\left(A_x, A_y, C_x, C_y\right) \qquad \begin{pmatrix} A_x \\ A_y \\ C_x \\ C_y \end{pmatrix} = \begin{pmatrix} 500 \\ 1000 \\ 500 \\ 500 \end{pmatrix} \text{N}$$

Problem 6-73

The truck exerts the three forces shown on the girders of the bridge. Determine the reactions at the supports when the truck is in the position shown. The girders are connected together by a short vertical link *DC*.

Units Used:

kip $= 10^3$ lb

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Given:

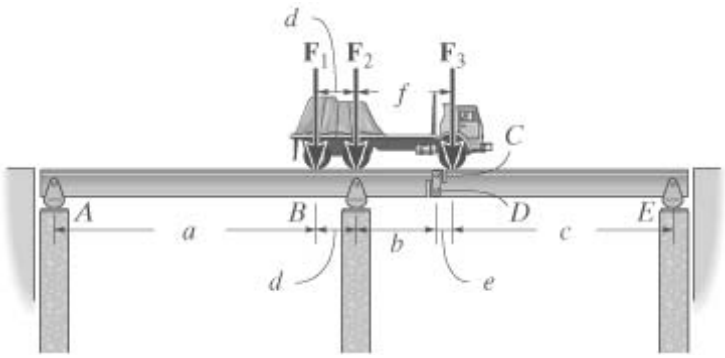
$a = 55$ ft $\quad f = 12$ ft

$b = 10$ ft $\quad F_1 = 5$ kip

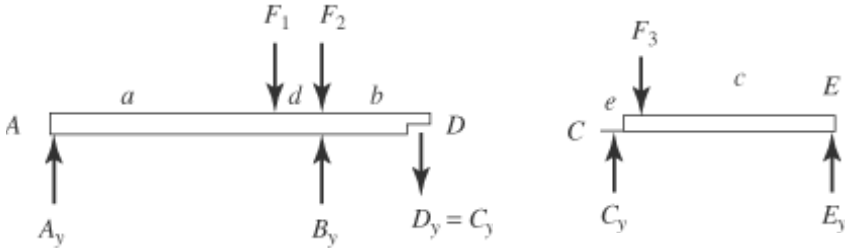
$c = 48$ ft $\quad F_2 = 4$ kip

$d = 5$ ft $\quad F_3 = 2$ kip

$e = 2$ ft



Solution:



Member *CE*:

$\Sigma M_C = 0; \quad -F_3 e + E_y(e + c) = 0 \quad E_y = F_3 \dfrac{e}{e + c} \quad E_y = 80\,\text{lb}$

$\Sigma F_y = 0; \quad C_y - F_3 + E_y = 0 \quad C_y = F_3 - E_y \quad C_y = 1920\,\text{lb}$

Member *ABD*:

$\Sigma M_A = 0; \quad -F_1 a - F_2(d + a) - C_y(a + d + b) + B_y(a + d) = 0$

$B_y = \dfrac{F_1 a + F_2(d + a) + C_y(a + d + b)}{d + a} \quad B_y = 10.8\,\text{kip}$

$\Sigma F_y = 0; \quad A_y - F_1 + B_y - F_2 - C_y = 0$

$A_y = C_y + F_1 - B_y + F_2 \quad A_y = 96.7\,\text{lb}$

Problem 6-74

Determine the greatest force **P** that can be applied to the frame if the largest force resultant acting at *A* can have a magnitude F_{max}.

Units Used:

$\text{kN} = 10^3\ \text{N}$

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Given:

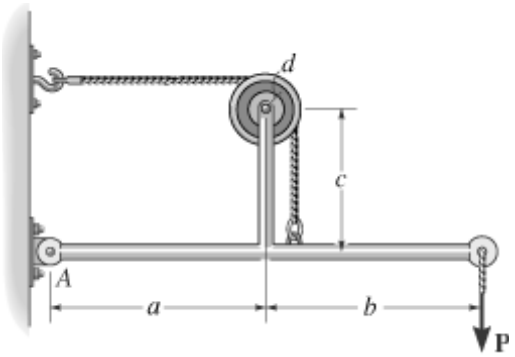
$F_{max} = 2$ kN

$a = 0.75$ m

$b = 0.75$ m

$c = 0.5$ m

$d = 0.1$ m



Solution:

$\circlearrowleft + \Sigma M_A = 0; \quad T(c + d) - P(a + b) = 0$

$\xrightarrow{+} \Sigma F_x = 0; \quad A_x - T = 0$

$+\uparrow \Sigma F_y = 0; \quad A_y - P = 0$

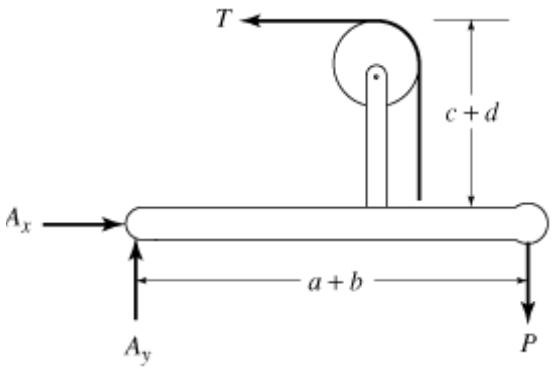
Thus, $T = \dfrac{a+b}{c+d}P \qquad A_y = P \qquad A_x = \dfrac{a+b}{c+d}P$

Require,

$$F_{max} = \sqrt{A_x^2 + A_y^2}$$

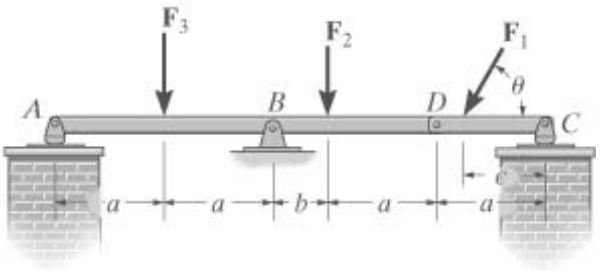
$$P = \frac{F_{max}}{\sqrt{\left(\dfrac{a+b}{c+d}\right)^2 + 1}}$$

$P = 743$ N



Problem 6-75

The compound beam is pin supported at *B* and supported by rockers at *A* and *C*. There is a hinge (pin) at *D*. Determine the reactions at the supports.



Units Used:

kN $= 10^3$ N

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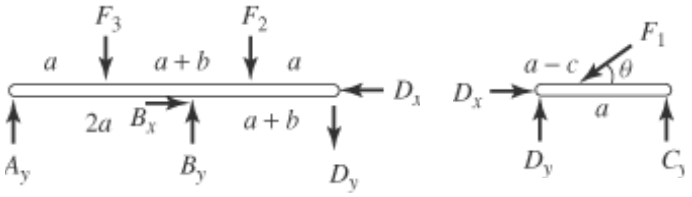
Given:

$F_1 = 7$ kN $a = 4$ m

$F_2 = 6$ kN $b = 2$ m

$F_3 = 16$ kN $c = 3$ m

$\theta = 60$ deg $d = 4$ m



Solution:

Member *DC* :

$\Sigma M_D = 0;$ $\quad -F_1 \sin(\theta)(a - c) + C_y a = 0$

$$C_y = F_1 \sin(\theta) \frac{a - c}{a} \qquad C_y = 1.52\,\text{kN}$$

$\Sigma F_y = 0;$ $\quad D_y - F_1 \sin(\theta) + C_y = 0$

$$D_y = F_1 \sin(\theta) - C_y \qquad D_y = 4.55\,\text{kN}$$

$\Sigma F_x = 0;$ $\quad D_x - F_1 \cos(\theta) = 0$

$$D_x = F_1 \cos(\theta) \qquad D_x = 3.5\,\text{kN}$$

Member *ABD* :

$\Sigma M_A = 0;$ $\quad -F_3 a - F_2(2\,a + b) - D_y(3\,a + b) + B_y 2\,a = 0$

$$B_y = \frac{F_3 a + F_2(2a + b) + D_y\,(3a + b)}{2\,a} \qquad B_y = 23.5\,\text{kN}$$

$\Sigma F_y = 0;$ $\quad A_y - F_3 + B_y - F_2 - D_y = 0$

$$A_y = D_y + F_3 - B_y + F_2 \qquad A_y = 3.09\,\text{kN}$$

$\Sigma F_x = 0;$ $\quad B_x - F_1 \cos(\theta) = 0$

$$B_x = F_1 \cos(\theta) \qquad B_x = 3.5\,\text{kN}$$

Problem 6-76

The compound beam is fixed supported at *A* and supported by rockers at *B* and *C*. If there are hinges (pins) at *D* and *E*, determine the reactions at the supports *A*, *B*, and *C*.

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Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

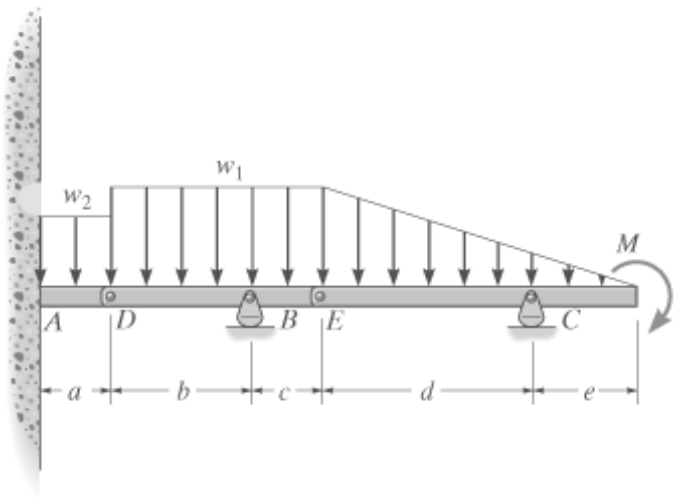
$a = 2\ \text{m}$ $M = 48\ \text{kN}\cdot\text{m}$

$b = 4\ \text{m}$ $w_1 = 8\ \frac{\text{kN}}{\text{m}}$

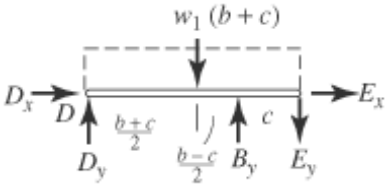
$c = 2\ \text{m}$

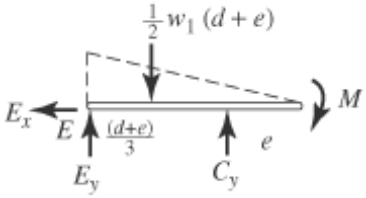
$d = 6\ \text{m}$ $w_2 = 6\ \frac{\text{kN}}{\text{m}}$

$e = 3\ \text{m}$



Solution:





Guesses

$A_x = 1\ \text{N}$ $A_y = 1\ \text{N}$ $M_A = 1\ \text{N m}$

$D_x = 1\ \text{N}$ $D_y = 1\ \text{N}$ $B_y = 1\ \text{N}$

$E_y = 1\ \text{N}$ $E_x = 1\ \text{N}$ $C_y = 1\ \text{N}$

Given

$A_y - w_2 a - D_y = 0$

$-A_x - D_x = 0$

$M_A - w_2 a \frac{a}{2} - D_y a = 0$

$D_y - w_1(b + c) + B_y - E_y = 0$

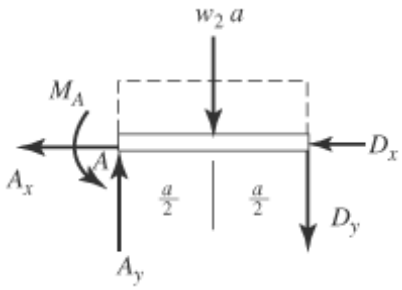
$-w_1 \frac{(b + c)^2}{2} + B_y b - E_y(b + c) = 0$

$D_x + E_x = 0$

$E_y - w_1 \frac{d + e}{2} + C_y = 0$

$-E_x = 0$

$-w_1\left(\frac{d + e}{2}\right)\left(\frac{d + e}{3}\right) + C_y d - M = 0$



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$$\begin{pmatrix} A_x \\ A_y \\ M_A \\ D_x \\ D_y \\ B_y \\ E_y \\ E_x \\ C_y \end{pmatrix} = \text{Find}\left(A_x, A_y, M_A, D_x, D_y, B_y, E_y, E_x, C_y\right)$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 0 \\ 19 \end{pmatrix} \text{kN}$$

$$M_A = 26\,\text{kN m}$$

$$B_y = 51\,\text{kN}$$

$$C_y = 26\,\text{kN}$$

Problem 6-77

Determine the reactions at supports *A* and *B*.

Units Used:

$\text{kip} = 10^3\ \text{lb}$

Given:

$w_1 = 500\ \frac{\text{lb}}{\text{ft}}$

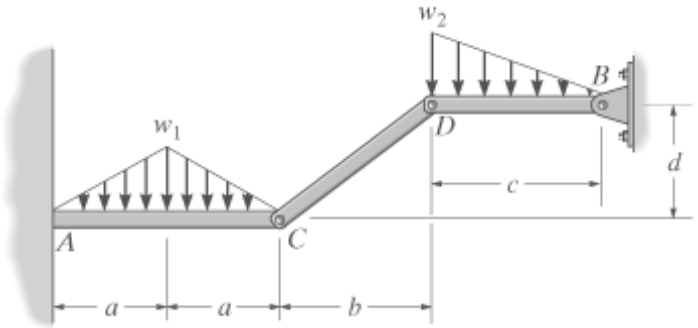
$w_2 = 700\ \frac{\text{lb}}{\text{ft}}$

$a = 6\ \text{ft}$

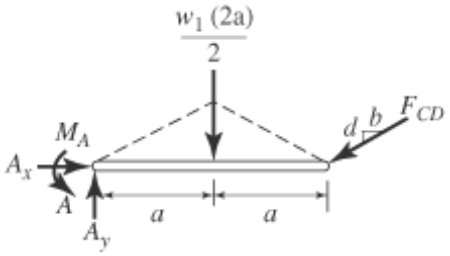
$b = 8\ \text{ft}$

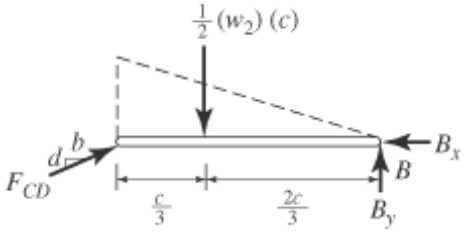
$c = 9\ \text{ft}$

$d = 6\ \text{ft}$



Solution:





Guesses

$A_x = 1\ \text{lb}$ $A_y = 1\ \text{lb}$ $B_x = 1\ \text{lb}$ $B_y = 1\ \text{lb}$ $F_{CD} = 1\ \text{lb}$ $M_A = 1\ \text{lb}\cdot\text{ft}$

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Given

$$A_y - w_1 a - \frac{d}{\sqrt{b^2 + d^2}} F_{CD} = 0 \qquad A_x - \frac{b}{\sqrt{b^2 + d^2}} F_{CD} = 0$$

$$M_A - w_1 a \, a - \frac{d}{\sqrt{d^2 + b^2}} F_{CD} 2\, a = 0 \qquad \frac{b}{\sqrt{b^2 + d^2}} F_{CD} - B_x = 0$$

$$\frac{d}{\sqrt{b^2 + d^2}} F_{CD} - w_2 \frac{c}{2} + B_y = 0 \qquad B_y c - w_2 \frac{c}{2} \frac{c}{3} = 0$$

$$\begin{pmatrix} A_x \\ A_y \\ B_x \\ B_y \\ F_{CD} \\ M_A \end{pmatrix} = \text{Find}\left(A_x, A_y, B_x, B_y, F_{CD}, M_A\right)$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 2.8 \\ 5.1 \end{pmatrix} \text{kip}$$

$$M_A = 43.2 \, \text{kip} \cdot \text{ft}$$

$$\begin{pmatrix} B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 2.8 \\ 1.05 \end{pmatrix} \text{kip}$$

Problem 6-78

Determine the horizontal and vertical components of force at C which member ABC exerts on member CEF.

Given:

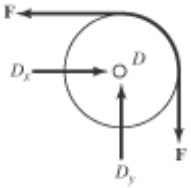
$F = 300$ lb

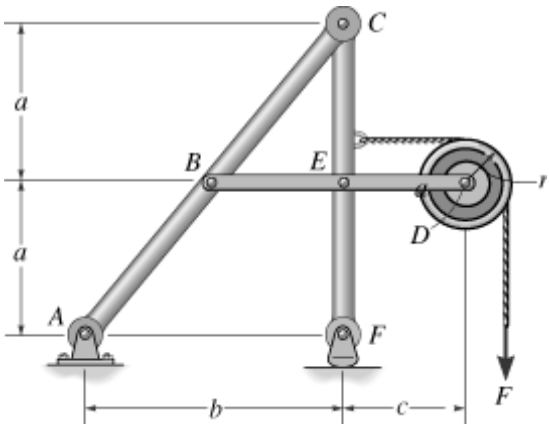
$a = 4$ ft

$b = 6$ ft

$c = 3$ ft

$r = 1$ ft





Solution:

Guesses

$A_x = 1$ lb $\quad A_y = 1$ lb $\quad F_y = 1$ lb

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$C_x = 1$ lb $\quad C_y = 1$ lb

Given

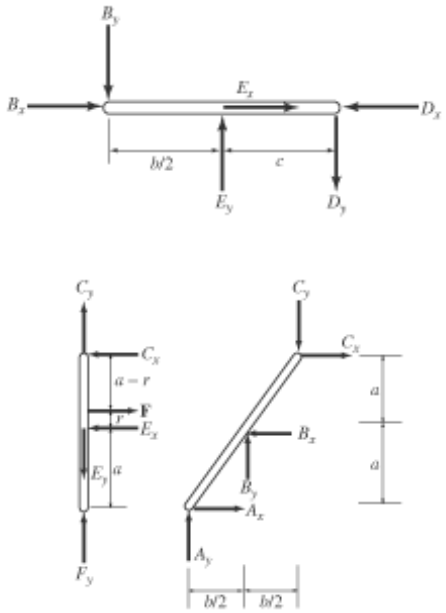
$$A_x a - A_y \frac{b}{2} - C_x a - C_y \frac{b}{2} = 0$$

$$C_x a - F r = 0$$

$$A_x = 0$$

$$A_y + F_y - F = 0$$

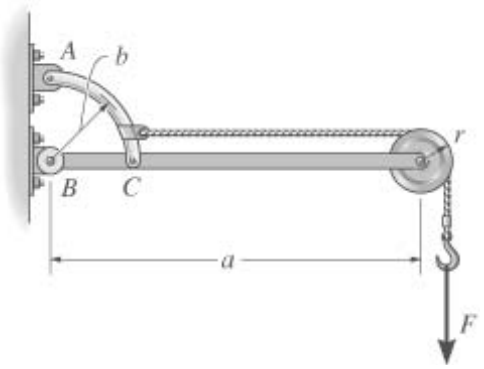
$$F_y b - F(b + c + r) = 0$$



$$\begin{pmatrix} A_x \\ A_y \\ F_y \\ C_x \\ C_y \end{pmatrix} = \text{Find}\left(A_x, A_y, F_y, C_x, C_y\right) \qquad \begin{pmatrix} A_x \\ A_y \\ F_y \end{pmatrix} = \begin{pmatrix} 0 \\ -200 \\ 500 \end{pmatrix} \text{lb} \qquad \begin{pmatrix} C_x \\ C_y \end{pmatrix} = \begin{pmatrix} 75 \\ 100 \end{pmatrix} \text{lb}$$

Problem 6-79

Determine the horizontal and vertical components of force that the pins at *A*, *B*, and *C* exert on their connecting members.



Units Used:

$\text{kN} = 10^3$ N

Given:

$F = 800$ N

$a = 1$ m

$r = 50$ mm

$b = 0.2$ m

Solution:

$-F(a + r) + A_x b = 0$

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$$A_x = F\frac{a+r}{b} \qquad A_x = 4.2\,\text{kN}$$

$$-A_x + B_x = 0$$

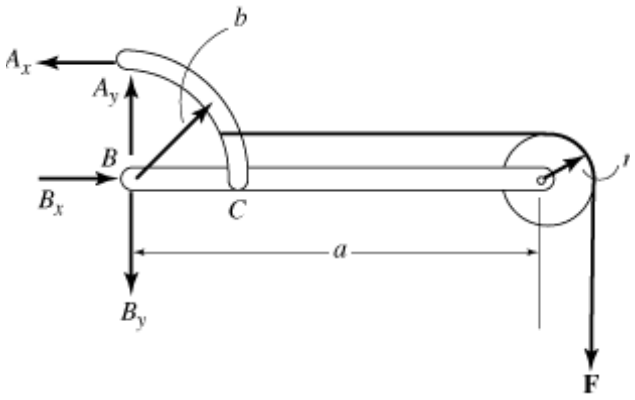
$$B_x = A_x \qquad B_x = 4.2\,\text{kN}$$

$$-F\,r - A_y b + A_x b = 0$$

$$A_y = \frac{-F\,r + A_x b}{b} \qquad A_y = 4\,\text{kN}$$

$$A_y - B_y - F = 0$$

$$B_y = A_y - F \qquad B_y = 3.2\,\text{kN}$$

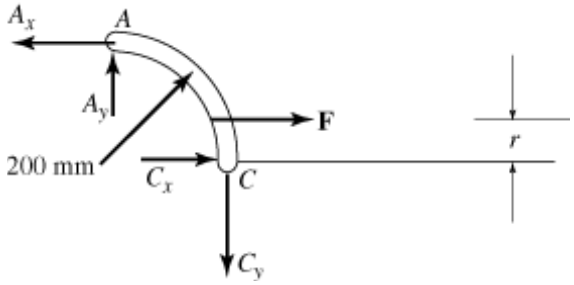


$$-A_x + F + C_x = 0$$

$$C_x = A_x - F \qquad C_x = 3.4\,\text{kN}$$

$$A_y - C_y = 0$$

$$C_y = A_y \qquad C_y = 4\,\text{kN}$$



Problem 6-80

Operation of exhaust and intake valves in an automobile engine consists of the cam *C*, push rod *DE*, rocker arm *EFG* which is pinned at *F*, and a spring and valve, *V*. If the spring is compressed a distance δ when the valve is open as shown, determine the normal force acting on the cam lobe at *C*. Assume the cam and bearings at *H*, *I*, and *J* are smooth.The spring has a stiffness *k*.

Given:

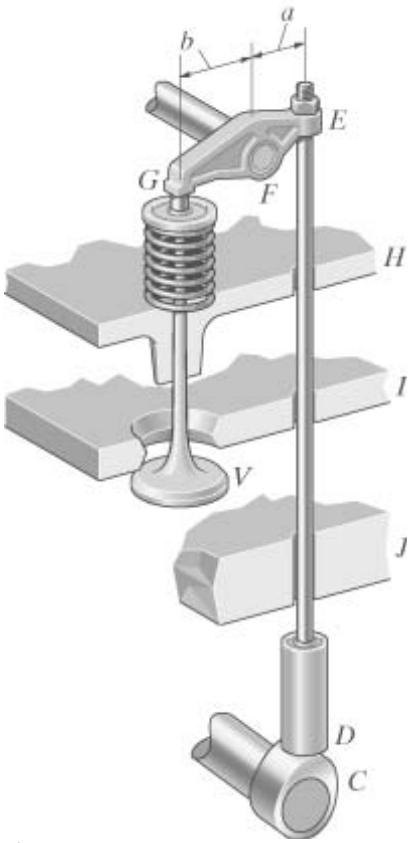
$a = 25\ \text{mm}$

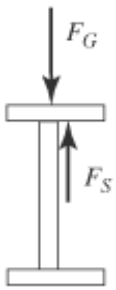
$b = 40\ \text{mm}$

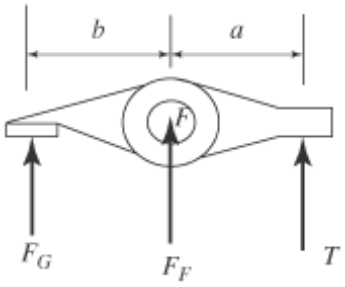
$\delta = 20\ \text{mm}$

$k = 300\ \frac{\text{N}}{\text{m}}$

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Solution:

$F_S = k\delta \qquad F_S = 6 \text{ N}$

$\Sigma F_y = 0;$

$-F_G + F_S = 0$

$F_G = F_S \qquad F_G = 6 \text{ N}$

$\Sigma M_F = 0;$

$F_G b + T a = 0$

$T = F_G \dfrac{b}{a} \qquad T = 9.60 \text{ N}$

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Problem 6-81

Determine the force **P** on the cord, and the angle θ that the pulley-supporting link *AB* makes with the vertical. Neglect the mass of the pulleys and the link. The block has weight *W* and the cord is attached to the pin at *B*. The pulleys have radii of r_1 and r_2.

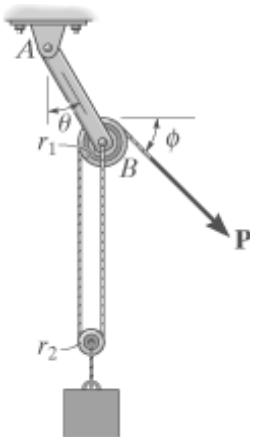
Given:

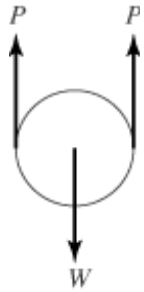
$W = 200$ lb

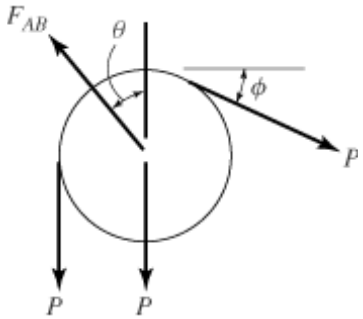
$r_1 = 2$ in

$r_2 = 1$ in

$\phi = 45$ deg







Solution:

The initial guesses are

$\theta = 30$ deg $\quad F_{AB} = 30$ lb

$+\uparrow \Sigma F_y = 0; \quad 2P - W = 0$

$P = \frac{1}{2}W \qquad P = 100\,\text{lb}$

Given

$\rightarrow^{+} \Sigma F_x = 0; \quad P\cos(\phi) - F_{AB}\sin(\theta) = 0$

$+\uparrow \Sigma F_y = 0; \quad F_{AB}\cos(\theta) - P - P - P\sin(\phi) = 0$

$\begin{pmatrix} \theta \\ F_{AB} \end{pmatrix} = \text{Find}(\theta, F_{AB}) \qquad F_{AB} = 280\,\text{lb} \qquad \theta = 14.6\,\text{deg}$

Problem 6-82

The nail cutter consists of the handle and the two cutting blades. Assuming the blades are pin connected at *B* and the surface at *D* is smooth, determine the normal force on the fingernail when a force **F** is applied to the handles as shown.The pin *AC* slides through a smooth hole at *A* and is attached to the bottom member at *C*.

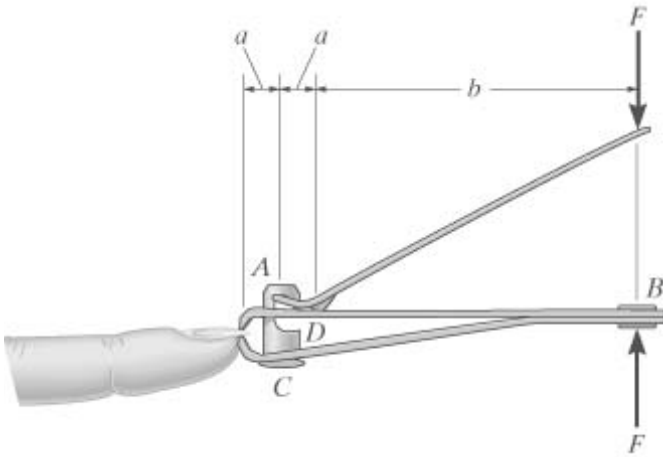
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Given:

$F = 1$ lb

$a = 0.25$ in

$b = 1.5$ in



Solution:

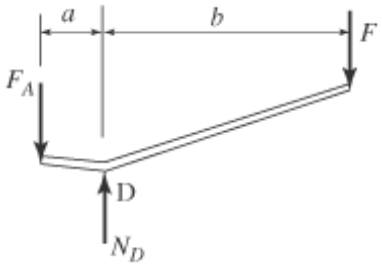
Handle :

$\Sigma M_D = 0;$ $F_A\, a - F\, b = 0$

$F_A = F\left(\frac{b}{a}\right)$ $F_A = 6\,\text{lb}$

$\Sigma F_y = 0;$ $N_D - F_A - F = 0$

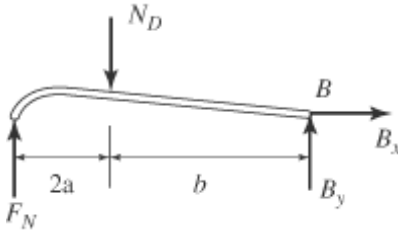
$N_D = F_A + F$ $N_D = 7\,\text{lb}$

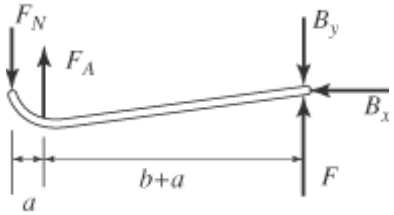


Top blade :

$\Sigma M_B = 0;$ $N_D\, b - F_N(2\, a + b) = 0$

$F_N = N_D\left(\frac{b}{2\, a + b}\right)$ $F_N = 5.25\,\text{lb}$





Problem 6-83

The wall crane supports load F. Determine the horizontal and vertical components of reaction at the pins A and D. Also, what is the force in the cable at the winch W?

Units Used: kip $= 10^3$ lb

Given:

$F = 700$ lb

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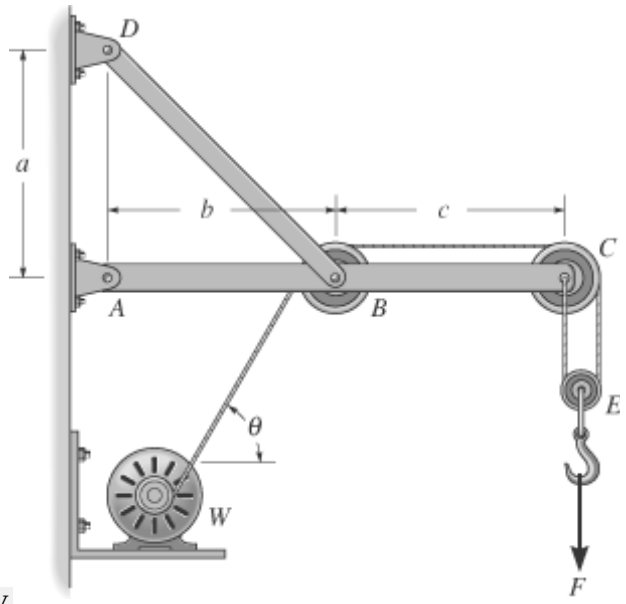
$a = 4$ ft

$b = 4$ ft

$c = 4$ ft

$\theta = 60$ deg





Solution:

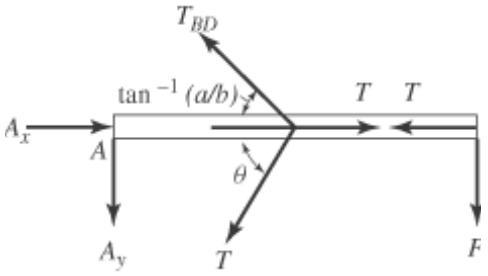
Pulley E:

$+\uparrow \Sigma F_y = 0; \qquad 2T - F = 0$

$T = \frac{1}{2}F \qquad T = 350\,\text{lb}$

This is the force in the cable at the winch W

Member ABC: $\phi = \operatorname{atan}\left(\frac{a}{b}\right)$

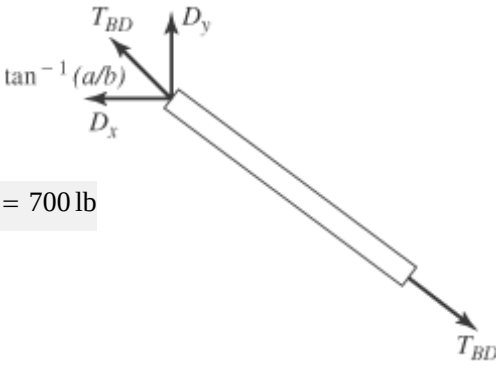


$\curvearrowleft + \Sigma M_A = 0;$

$-F(b + c) + \left(T_{BD}\sin(\phi) - T\sin(\theta)\right)b = 0$

$T_{BD} = \dfrac{F\left(\frac{b + c}{b}\right) + T\sin(\theta)}{\sin(\phi)}$

$T_{BD} = 2.409 \times 10^3\,\text{lb}$



$+\uparrow \Sigma F_y = 0;$

$-A_y + T_{BD}\sin(\phi) - T\sin(\theta) - F = 0$

$A_y = T_{BD}\sin(\phi) - T\sin(\theta) - F \qquad A_y = 700\,\text{lb}$

$\xrightarrow{+} \Sigma F_x = 0;$

$A_x - T_{BD}\cos(\phi) - T\cos(\theta) = 0$

$A_x = T_{BD}\cos(\phi) + T\cos(\theta) \qquad A_x = 1.878\,\text{kip}$

At D:

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$D_x = T_{BD}\cos(\phi)$ $D_x = 1.703\,\text{kip}$

$D_y = T_{BD}\sin(\phi)$ $D_y = 1.703\,\text{kip}$

Problem 6-84

Determine the force that the smooth roller *C* exerts on beam *AB*. Also, what are the horizontal and vertical components of reaction at pin *A*? Neglect the weight of the frame and roller.

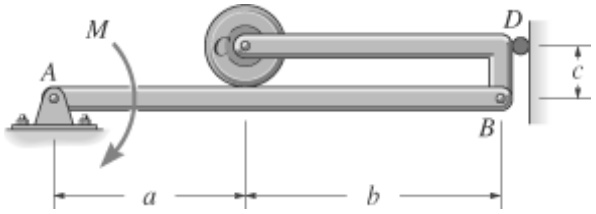
Given:

$M = 60\ \text{lb}\cdot\text{ft}$

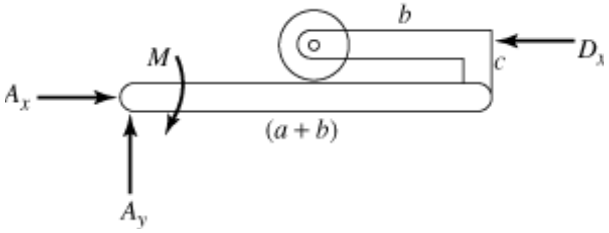
$a = 3\ \text{ft}$

$b = 4\ \text{ft}$

$c = 0.5\ \text{ft}$



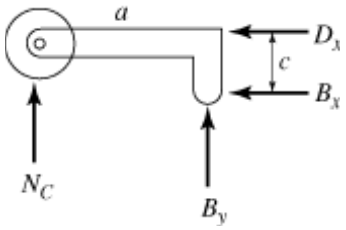
Solution:



$\curvearrowleft + \Sigma M_A = 0;$ $-M + D_x c = 0$

$D_x = \dfrac{M}{c}$

$D_x = 120\,\text{lb}$



$\overset{+}{\rightarrow} \Sigma F_x = 0;$ $A_x - D_x = 0$

$A_x = D_x$ $A_x = 120\,\text{lb}$

$+\uparrow \Sigma F_y = 0;$ $A_y = 0$ $A_y = 0\ \text{lb}$

$\curvearrowright + \Sigma M_B = 0;$ $-N_C b + D_x c = 0$

$N_C = D_x \dfrac{c}{b}$ $N_C = 15.0\,\text{lb}$

Problem 6-85

Determine the horizontal and vertical components of force which the pins exert on member *ABC*.

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Given:

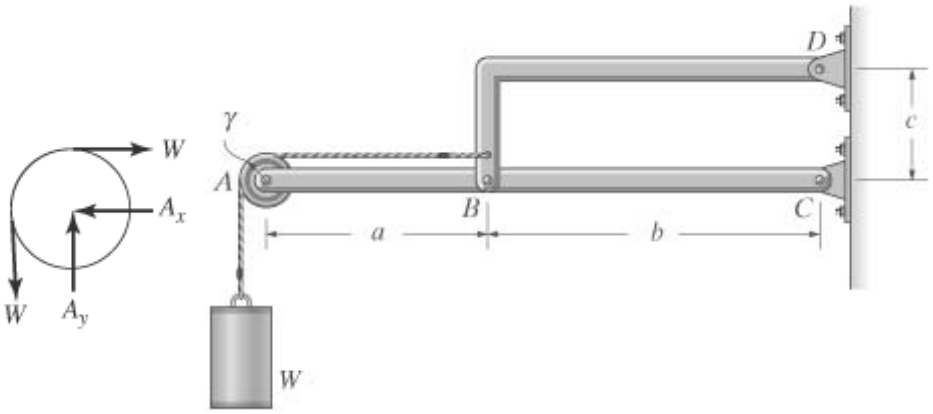
$W = 80$ lb

$a = 6$ ft

$b = 9$ ft

$c = 3$ ft

$r = 0.5$ ft



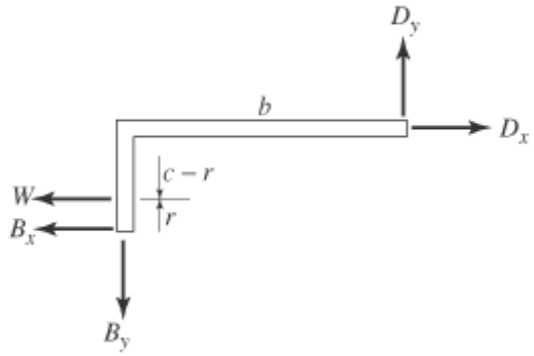
Solution:

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad -A_x + W = 0$

$A_x = W \qquad A_x = 80\,\text{lb}$

$+\uparrow \Sigma F_y = 0; \quad A_y - W = 0$

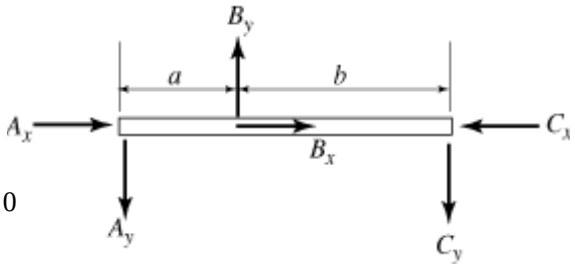
$A_y = W \qquad A_y = 80\,\text{lb}$



$\circlearrowleft + \Sigma M_C = 0; \quad A_y(a + b) - B_y b = 0$

$B_y = A_y \dfrac{a + b}{b}$

$B_y = 133\,\text{lb}$



$\circlearrowleft + \Sigma M_D = 0; \quad -W(c - r) + B_y b - B_x c = 0$

$B_x = \dfrac{B_y\, b - W\,(c - r)}{c} \qquad B_x = 333\,\text{lb}$

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad A_x + B_x - C_x = 0$

$C_x = A_x + B_x \qquad C_x = 413\,\text{lb}$

$+\uparrow \Sigma F_y = 0; \quad -A_y + B_y - C_y = 0$

$C_y = B_y - A_y \qquad C_y = 53.3\,\text{lb}$

Problem 6-86

The floor beams *AB* and *BC* are stiffened using the two tie rods *CD* and *AD*. Determine the force along each rod when the floor beams are subjected to a uniform load *w*. Assume the three contacting members at *B* are smooth and the joints at *A*, *C*, and *D* are pins. *Hint:*

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Members *AD*, *CD*, and *BD* are two-force members

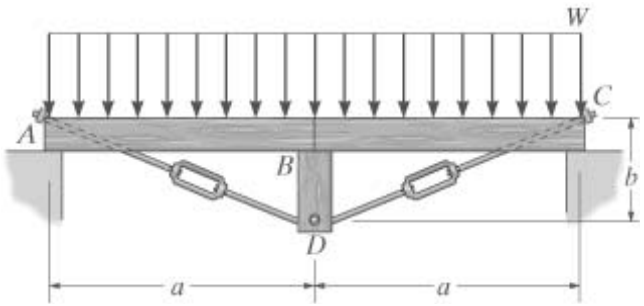
Units Used: $\text{kip} = 10^3\ \text{lb}$

Given:

$w = 80\ \frac{\text{lb}}{\text{ft}}$

$b = 5\ \text{ft}$

$a = 12\ \text{ft}$

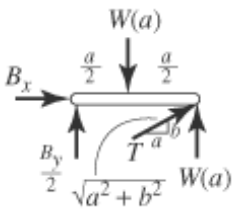


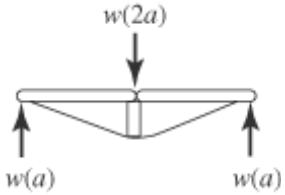
Solution:

Due to summetry:

$C_y = \frac{w(2a)}{2}$

$C_y = 960\,\text{lb}$





Member *BC* :

$\Sigma M_B = 0;\qquad C_y(a) - w a\left(\frac{a}{2}\right) - T\left(\frac{b}{\sqrt{a^2 + b^2}}\right)a = 0$

$T = \left(C_y - w\frac{a}{2}\right)\frac{\sqrt{a^2 + b^2}}{b}$ $\qquad T = 1.248\,\text{kip}$

Problem 6-87

Determine the horizontal and vertical components of force at pins *B* and *C*.

Given:

$F = 50\ \text{lb}$ $\quad c = 6\ \text{ft}$

$a = 4\ \text{ft}$ $\quad d = 1.5\ \text{ft}$

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$b = 4$ ft $\quad r = 0.5$ ft

Solution: Guesses

$C_x = 1$ lb $\quad C_y = 1$ lb

$B_x = 1$ lb $\quad B_y = 1$ lb

Given

$F(a - r) + C_x c - C_y(a + b) = 0$

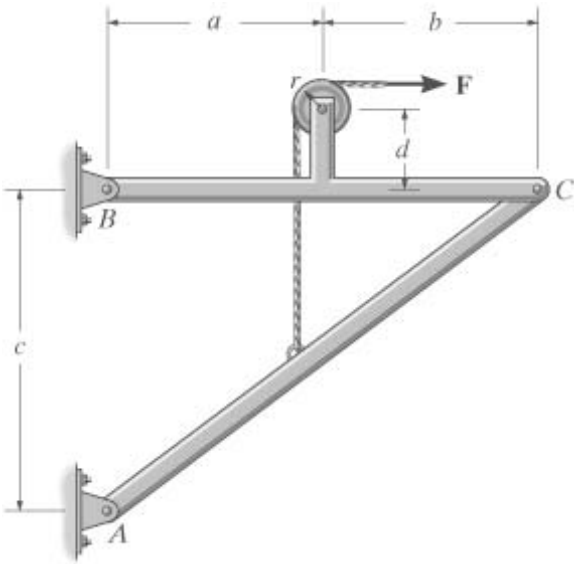
$-F(a - r) - F(d + r) + C_y(a + b) = 0$

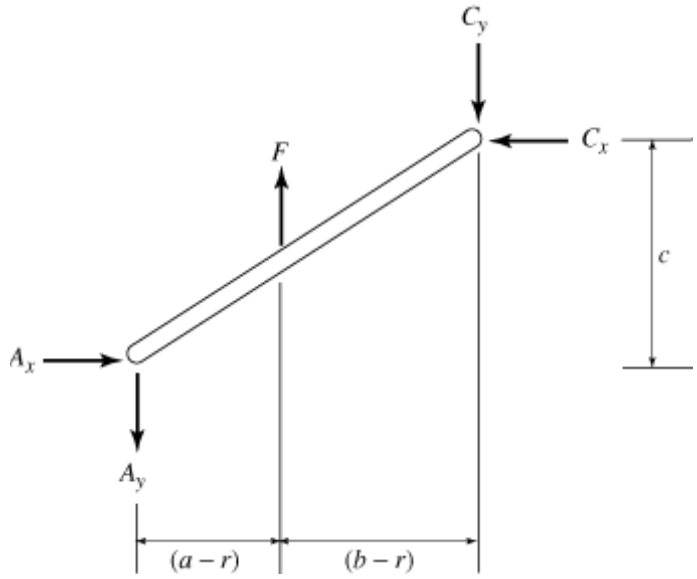
$-B_x + F + C_x = 0$

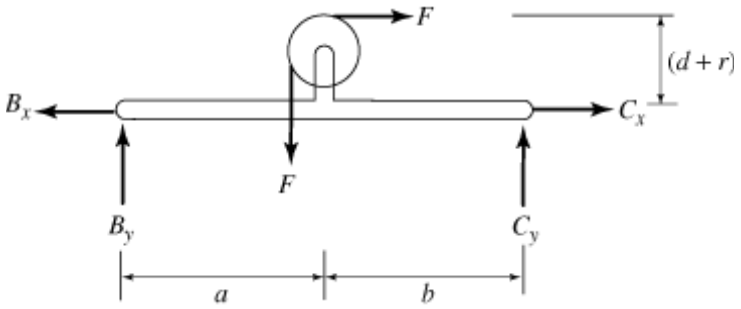
$B_y - F + C_y = 0$

$$\begin{pmatrix} B_x \\ B_y \\ C_x \\ C_y \end{pmatrix} = \text{Find}\left(B_x, B_y, C_x, C_y\right)$$

$$\begin{pmatrix} B_x \\ B_y \\ C_x \\ C_y \end{pmatrix} = \begin{pmatrix} 66.667 \\ 15.625 \\ 16.667 \\ 34.375 \end{pmatrix} \text{lb}$$







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Problem 6-88

The skid steer loader has a mass M_1, and in the position shown the center of mass is at G_1. If there is a stone of mass M_2 in the bucket, with center of mass at G_2 determine the reactions of each pair of wheels A and B on the ground and the force in the hydraulic cylinder CD and at the pin E. There is a similar linkage on each side of the loader.

Units Used:

$\text{Mg} = 10^3 \text{ kg}$

$\text{kN} = 10^3 \text{ N}$

Given:

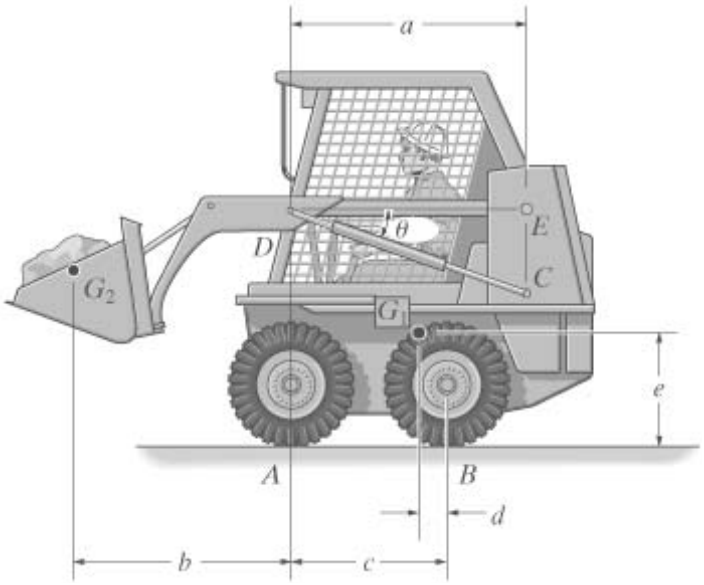
$M_1 = 1.18 \text{ Mg}$

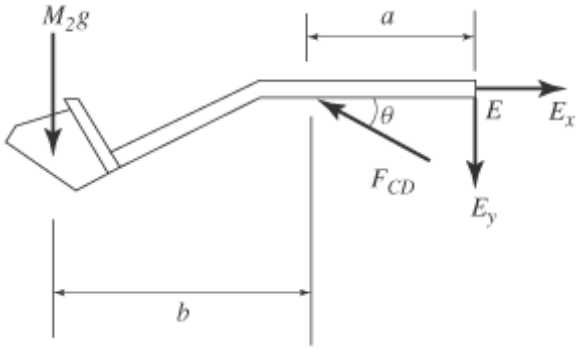
$M_2 = 300 \text{ kg}$

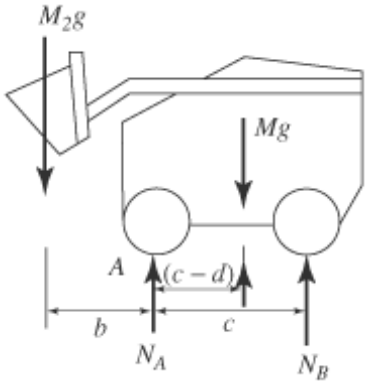
$a = 1.25 \text{ m}$ $d = 0.15 \text{ m}$

$b = 1.5 \text{ m}$ $e = 0.5 \text{ m}$

$c = 0.75 \text{ m}$ $\theta = 30 \text{ deg}$







Solution: Entire System:

$\Sigma M_A = 0;$ $M_2 g\, b - M_1 g(c - d) + N_B c = 0$

$$N_B = \frac{M_1 g(c - d) - M_2 g\, b}{c} \qquad N_B = 3.37 \text{ kN} \quad \text{(Both wheels)}$$

$\Sigma F_y = 0;$ $N_B - M_2 g - M_1 g + N_A = 0$

$$N_A = -N_B + M_2 g + M_1 g \qquad N_A = 11.1 \text{ kN} \quad \text{(Both wheels)}$$

Upper member:

$\Sigma M_E = 0;$ $M_2 g(a + b) - 2\, F_{CD} \sin(\theta) a = 0$

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$$F_{CD} = \frac{M_2 g(a+b)}{2\sin(\theta)a} \qquad F_{CD} = 6.5\,\text{kN}$$

$\Sigma F_x = 0;$ $\qquad E_x = F_{CD}(\cos(\theta))$ $\qquad E_x = 5607\,\text{N}$

$\Sigma F_y = 0;$ $\qquad E_y - \frac{M_2 g}{2} + F_{CD}\sin(\theta) = 0$

$$E_y = \frac{M_2 g}{2} - F_{CD}\sin(\theta) \qquad E_y = -1766\,\text{N}$$

$$F_R = \sqrt{E_x^2 + E_y^2} \qquad F_R = 5.879\,\text{kN}$$

Problem 6-89

Determine the horizontal and vertical components of force at each pin. The suspended cylinder has a weight W.

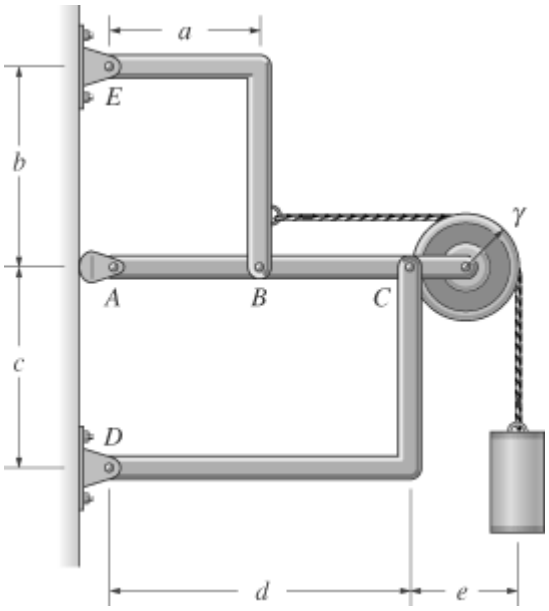
Given:

$W = 80$ lb $\qquad d = 6$ ft

$a = 3$ ft $\qquad e = 2$ ft

$b = 4$ ft $\qquad r = 1$ ft

$c = 4$ ft



Solution:

Guesses

$A_x = 1$ lb $\qquad B_x = 1$ lb

$B_y = 1$ lb $\qquad F_{CD} = 1$ lb

$E_x = 1$ lb $\qquad E_y = 1$ lb

Given

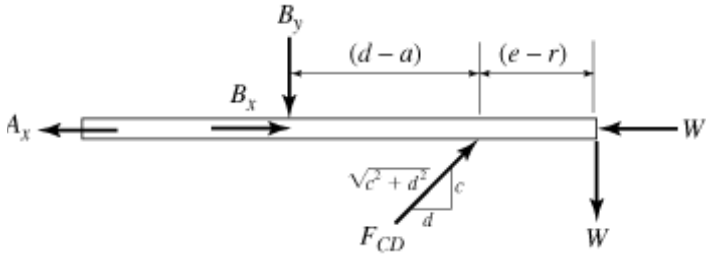
$E_x - B_x + W = 0$

$-E_y + B_y = 0$

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$$-Wr + E_y a - E_x d = 0$$

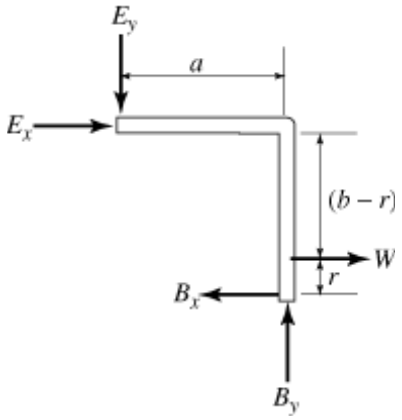
$$-B_y + \frac{c}{\sqrt{c^2 + d^2}} F_{CD} - W = 0$$



$$-B_y a + \frac{c}{\sqrt{c^2 + d^2}} F_{CD} d - W(d + e - r) = 0$$

$$-A_x + B_x + \frac{d}{\sqrt{c^2 + d^2}} F_{CD} - W = 0$$

$$\begin{pmatrix} A_x \\ B_x \\ B_y \\ F_{CD} \\ E_x \\ E_y \end{pmatrix} = \text{Find}\left(A_x, B_x, B_y, F_{CD}, E_x, E_y\right)$$



$$C_x = F_{CD} \frac{d}{\sqrt{c^2 + d^2}} \qquad C_y = F_{CD} \frac{c}{\sqrt{c^2 + d^2}} \qquad D_x = -C_x \qquad D_y = -C_y$$

$$\begin{pmatrix} A_x \\ B_x \\ B_y \\ C_x \\ C_y \\ D_x \\ D_y \\ E_x \\ E_y \end{pmatrix} = \begin{pmatrix} 160 \\ 80 \\ 26.667 \\ 160 \\ 106.667 \\ -160 \\ -106.667 \\ -8.694 \times 10^{-13} \\ 26.667 \end{pmatrix} \text{lb}$$

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Problem 6-90

The two-member frame is pin connected at *C*, *D*, and *E*. The cable is attached to *A*, passes over the smooth peg at *B*, and is attached to a load *W*. Determine the horizontal and vertical reactions at each pin.

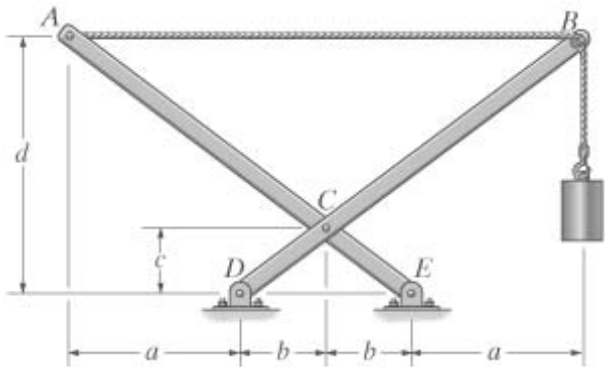
Given:

$a = 2$ ft

$b = 1$ ft

$c = 0.75$ ft

$W = 100$ lb



Solution:

$$d = \frac{c}{b}(a + 2\,b)$$

Initial guesses:

$C_x = 1$ lb $\quad C_y = 1$ lb $\quad D_x = 1$ lb $\quad D_y = 1$ lb $\quad E_x = 1$ lb $\quad E_y = 1$ lb

Given

$$-D_x + C_x - W = 0$$

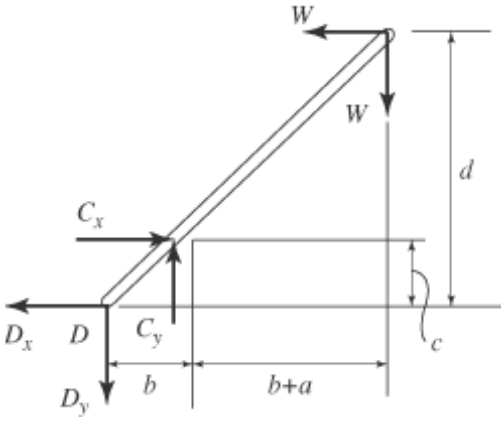
$$-D_y + C_y - W = 0$$

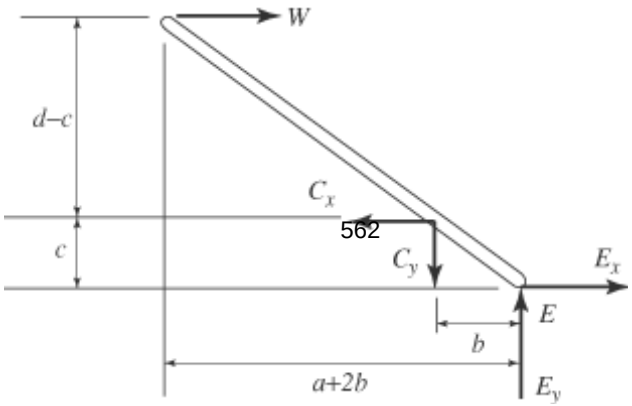
$$-C_x c + C_y b + W d - W(a + 2\,b) = 0$$

$$W - C_x + E_x = 0$$

$$E_y - C_y = 0$$

$$-W d + C_x c + C_y b = 0$$





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$$\begin{pmatrix} C_x \\ C_y \\ D_x \\ D_y \\ E_x \\ E_y \end{pmatrix} = \text{Find}\left(C_x, C_y, D_x, D_y, E_x, E_y\right) \qquad \begin{pmatrix} C_x \\ C_y \\ D_x \\ D_y \\ E_x \\ E_y \end{pmatrix} = \begin{pmatrix} 133 \\ 200 \\ 33 \\ 100 \\ 33 \\ 200 \end{pmatrix} \text{lb}$$

Problem 6-91

Determine the horizontal and vertical components of force which the pins at *A*, *B*, and *C* exert on member *ABC* of the frame.

Given:

$F_1 = 400$ N

$F_2 = 300$ N

$F_3 = 300$ N

$a = 1.5$ m

$b = 2$ m

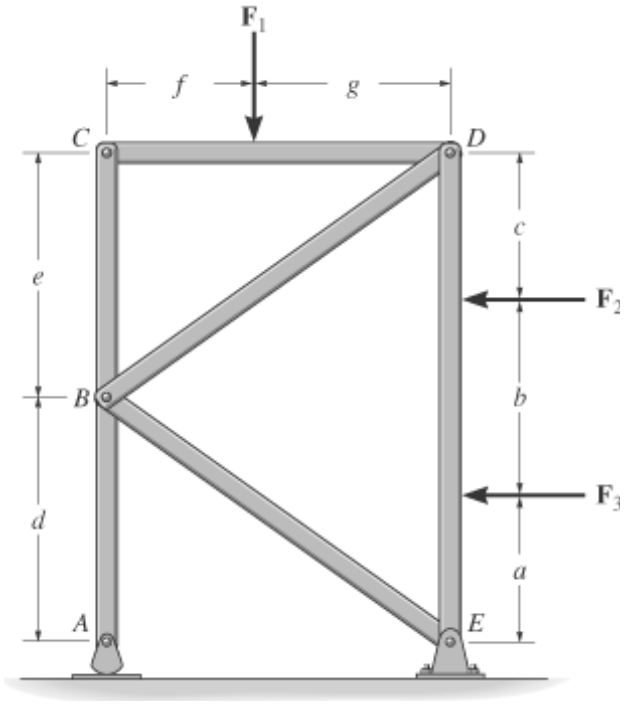
$c = 1.5$ m

$d = 2.5$ m

$f = 1.5$ m

$g = 2$ m

$e = a + b + c - d$



Solution:

Guesses

$A_y = 1$ N $\quad C_x = 1$ N $\quad C_y = 1$ N

$F_{BD} = 1$ N $\quad F_{BE} = 1$ N

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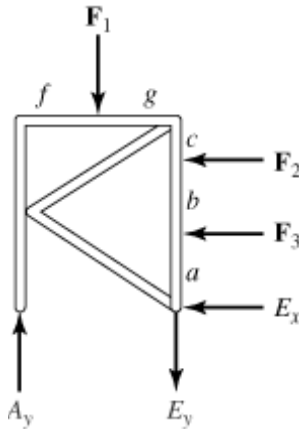
Given

$$F_1 g + F_2(a + b) + F_3 a - A_y(f + g) = 0$$

$$F_1 g - C_y(f + g) = 0$$

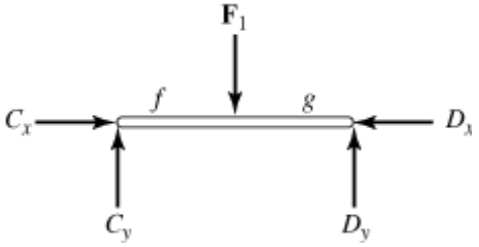
$$C_x e = 0$$

$$-C_x - \frac{f + g}{\sqrt{e^2 + (f + g)^2}} F_{BD} + \frac{f + g}{\sqrt{d^2 + (f + g)^2}} F_{BE} = 0$$



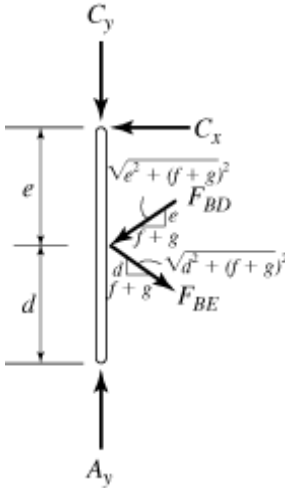
$$A_y - C_y - \frac{e}{\sqrt{e^2 + (f + g)^2}} F_{BD} - \frac{d}{\sqrt{d^2 + (f + g)^2}} F_{BE} = 0$$

$$\begin{pmatrix} A_y \\ C_x \\ C_y \\ F_{BD} \\ F_{BE} \end{pmatrix} = \text{Find}\left(A_y, C_x, C_y, F_{BD}, F_{BE}\right)$$



$$B_x = -\frac{f + g}{\sqrt{e^2 + (f + g)^2}} F_{BD} + \frac{f + g}{\sqrt{d^2 + (f + g)^2}} F_{BE}$$

$$B_y = \frac{e}{\sqrt{e^2 + (f + g)^2}} F_{BD} + \frac{d}{\sqrt{d^2 + (f + g)^2}} F_{BE}$$



$A_y = 657$ N

$$\begin{pmatrix} B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 0 \\ 429 \end{pmatrix} \text{N}$$

$$\begin{pmatrix} C_x \\ C_y \end{pmatrix} = \begin{pmatrix} 0 \\ 229 \end{pmatrix} \text{N}$$

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Problem 6-92

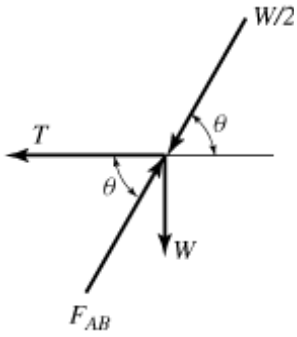
The derrick is pin-connected to the pivot at A. Determine the largest mass that can be supported by the derrick if the maximum force that can be sustained by the pin at A is F_{max}.

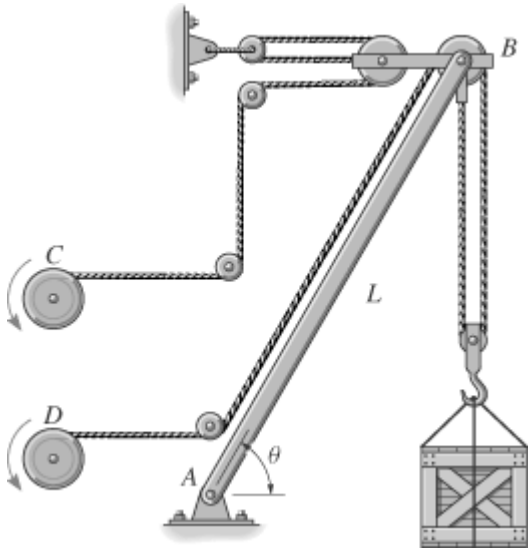
Units Used:

$kN = 10^3\ N$

$g = 9.81\ \frac{m}{s^2}$

$Mg = 10^3\ kg$





Given:

$F_{max} = 18\ kN$

$L = 5\ m$

$\theta = 60\ deg$

Solution:

AB is a two-force member.

Require $F_{AB} = F_{max}$

$$+\uparrow \Sigma F_y = 0; \quad F_{AB}\sin(\theta) - \frac{Mg}{2}\sin(\theta) - W = 0$$

$$M = 2\left(\frac{F_{AB}}{g}\right)\left(\frac{\sin(\theta)}{\sin(\theta) + 2}\right) \qquad M = 5.439\ \frac{1}{s^2}\,Mg$$

Problem 6-93

Determine the required mass of the suspended cylinder if the tension in the chain wrapped around the freely turning gear is T. Also, what is the magnitude of the resultant force on pin A?

Units Used:

$kN = 10^3\ N$

$g = 9.8\ \frac{m}{s^2}$

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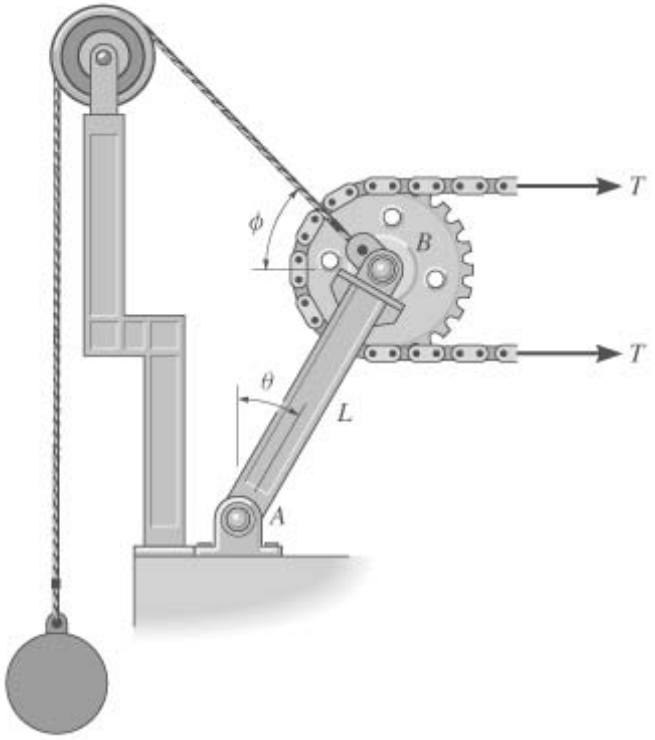
Given:

$T = 2$ kN

$L = 2$ ft

$\theta = 30$ deg

$\phi = 45$ deg

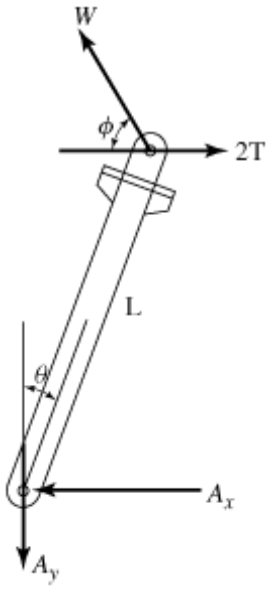


Solution:

$\curvearrowleft + \Sigma M_A = 0;$ $\quad -2\,TL\cos(\theta) + Mg\cos(\phi)L\cos(\theta) + Mg\sin(\phi)L\sin(\theta) = 0$

$$M = \frac{2\,T\cos(\theta)}{(\cos(\phi)\cos(\theta) + \sin(\phi)\sin(\theta))g}$$

$$M = 1793\ \frac{1}{s^2}\,\text{kg}$$



$\xrightarrow{+} \Sigma F_x = 0;$ $\quad 2\,T - Mg\cos(\phi) - A_x = 0$

$$A_x = 2\,T - Mg\cos(\phi)$$

$+\uparrow \Sigma F_y = 0;$ $\quad Mg\sin(\phi) - A_y = 0$

$$A_y = Mg\sin(\phi)$$

$$F_A = \sqrt{A_x^{\,2} + A_y^{\,2}}$$

$$F_A = 2.928\,\text{kN}$$

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Problem 6-94

The tongs consist of two jaws pinned to links at *A*, *B*, *C*, and *D*. Determine the horizontal and vertical components of force exerted on the stone of weight *W* at *F* and *G* in order to lift it.

Given:

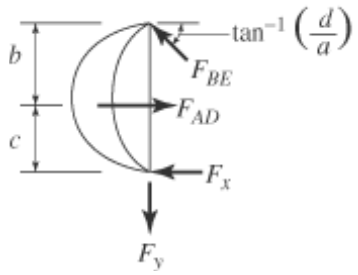
$a = 1$ ft

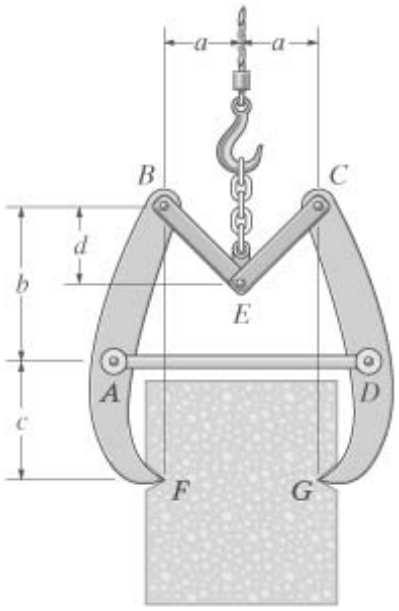
$b = 2$ ft

$c = 1.5$ ft

$d = 1$ ft

$W = 500$ lb





Solution:

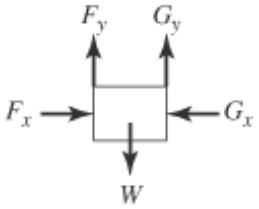
Guesses

$F_x = 1$ lb

$F_y = 1$ lb

$F_{AD} = 1$ lb

$F_{BE} = 1$ lb



Given $\quad 2\,F_y - W = 0 \qquad F_{AD}\,b - F_x(b + c) = 0$

$$F_{AD} - F_x - \left(\frac{a}{\sqrt{a^2 + d^2}}\right)F_{BE} = 0 \qquad -F_y + \left(\frac{d}{\sqrt{a^2 + d^2}}\right)F_{BE} = 0$$

$$\begin{pmatrix} F_x \\ F_y \\ F_{AD} \\ F_{BE} \end{pmatrix} = \text{Find}\left(F_x, F_y, F_{AD}, F_{BE}\right) \qquad \begin{pmatrix} G_x \\ G_y \end{pmatrix} = \begin{pmatrix} F_x \\ F_y \end{pmatrix} \qquad \begin{pmatrix} F_x \\ F_y \\ G_x \\ G_y \end{pmatrix} = \begin{pmatrix} 333 \\ 250 \\ 333 \\ 250 \end{pmatrix} \text{lb}$$

Problem 6-95

Determine the force **P** on the cable if the spring is compressed a distance δ when the mechanism is in the position shown. The spring has a stiffness *k*.

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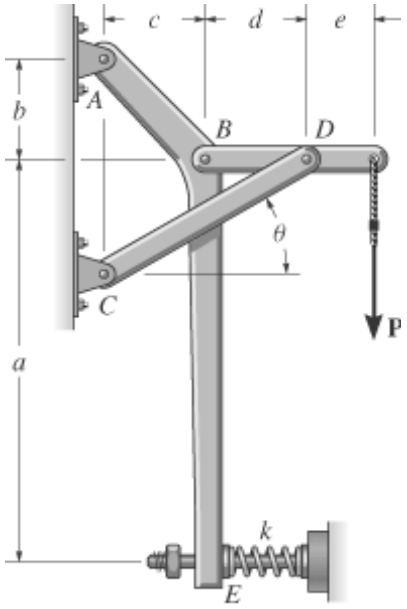
Given:

$\delta = 0.5$ in $\quad c = 6$ in

$k = 800 \dfrac{\text{lb}}{\text{ft}} \quad d = 6$ in

$a = 24$ in $\quad e = 4$ in

$b = 6$ in $\quad \theta = 30$ deg



Solution:

$F_E = k\delta \qquad F_E = 33.333\,\text{lb}$

The initial guesses are

$P = 20$ lb $\quad B_x = 11$ lb

$B_y = 34$ lb $\quad F_{CD} = 34$ lb

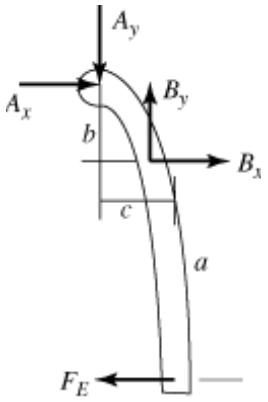
Given

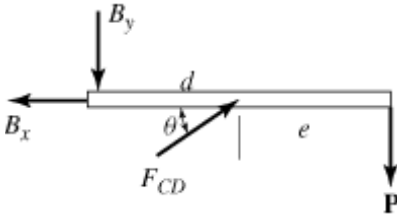
$\circlearrowleft + \quad \Sigma M_A = 0; \qquad B_x b + B_y c - F_E(a + b) = 0$

$\circlearrowleft + \quad \Sigma M_D = 0; \qquad B_y d - P e = 0$

$\overset{+}{\rightarrow} \quad \Sigma F_x = 0; \qquad -B_x + F_{CD}\cos(\theta) = 0$

$\circlearrowleft + \quad \Sigma M_B = 0; \qquad F_{CD}\sin(\theta) d - P(d + e) = 0$





$$\begin{pmatrix} F_{CD} \\ B_x \\ B_y \\ P \end{pmatrix} = \text{Find}\left(F_{CD}, B_x, B_y, P\right)$$

$B_x = 135.398\,\text{lb} \qquad B_y = 31.269\,\text{lb} \qquad F_{CD} = 156.344\,\text{lb} \qquad P = 46.903\,\text{lb}$

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Problem 6-96

The scale consists of five pin-connected members. Determine the load W on the pan EG if a weight F is suspended from the hook at A.

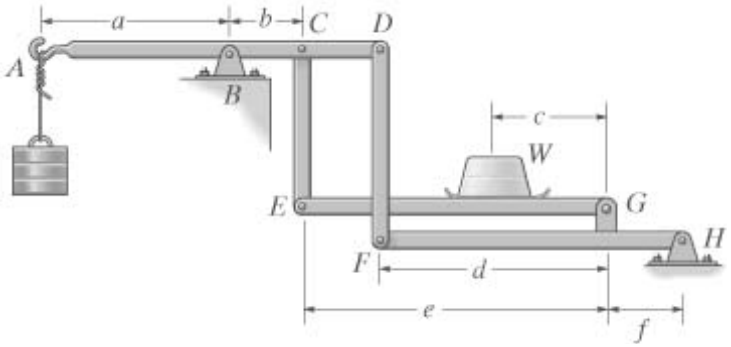
Given:

$F = 3$ lb $\quad b = 3$ in

$a = 5$ in $\quad c = 4$ in

$d = 6$ in $\quad f = 2$ in

$e = 8$ in



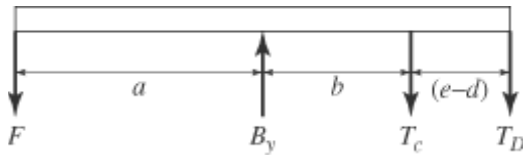
Solution:

Guesses $\quad T_C = 10$ lb $\quad T_D = 10$ lb $\quad T_G = 10$ lb $\quad W = 10$ lb

Given

Member $ABCD$: $\quad \Sigma M_B = 0;$

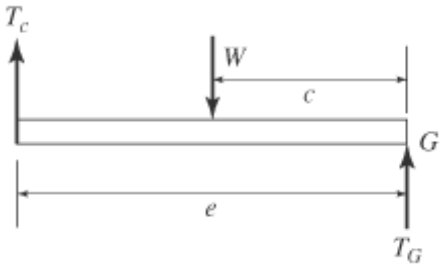
$$F\,a - T_C\,b - T_D(b + e - d) = 0$$



Member EG:

$\Sigma M_G = 0; \quad -T_C\,e + W\,c = 0$

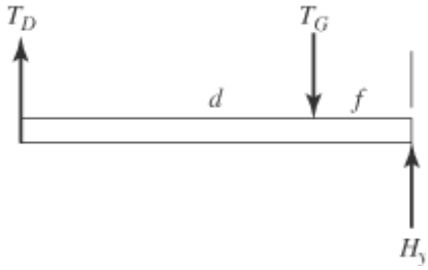
$\Sigma F_y = 0; \quad T_G - W + T_C = 0$



Member FH:

$\Sigma M_H = 0; \quad -T_D(d + f) + T_G\,f = 0$

$$\begin{pmatrix} T_C \\ T_D \\ T_G \\ W \end{pmatrix} = \text{Find}\left(T_C, T_D, T_G, W\right)$$



$W = 7.06$ lb

Problem 6-97

The machine shown is used for forming metal plates. It consists of two toggles ABC and DEF,

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which are operated by the hydraulic cylinder H. The toggles push the movable bar G forward, pressing the plate p into the cavity. If the force which the plate exerts on the head is P, determine the force F in the hydraulic cylinder for the given angle θ.

Units Used:

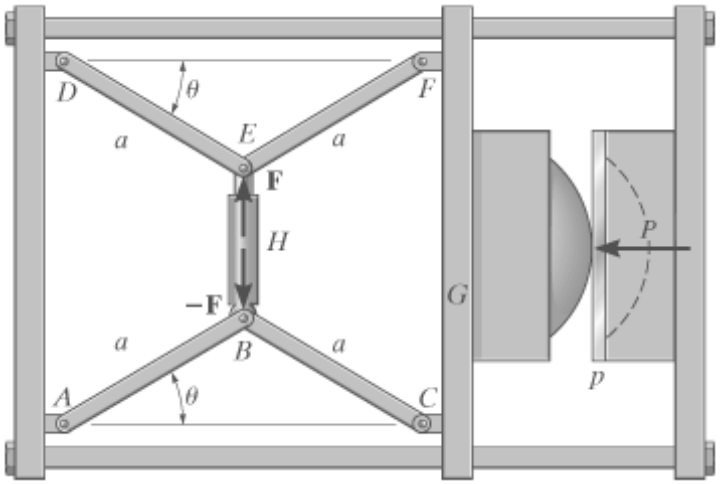
$kN = 10^3 N$

Given:

$P = 12\ kN$

$a = 200\ mm$

$\theta = 30\ deg$

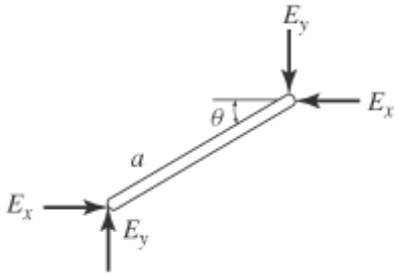


Solution:

Member EF:

$\Sigma M_E = 0; \quad -F_y a\cos(\theta) + \frac{P}{2} a\sin(\theta) = 0$

$F_y = \frac{P}{2}\tan(\theta) \qquad F_y = 3.464\,kN$

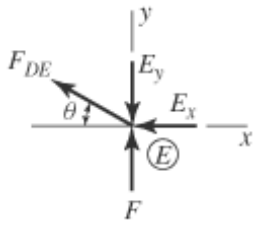


$\Sigma F_x = 0; \quad E_x - \frac{P}{2} = 0$

$E_x = \frac{P}{2} \qquad E_x = 6\,kN$

$\Sigma F_y = 0; \quad E_y - F_y = 0$

$E_y = F_y \qquad E_y = 3.464\,kN$



Joint E:

$\Sigma F_x = 0; \quad -F_{DE}\cos(\theta) - E_x = 0$

$F_{DE} = \frac{-E_x}{\cos(\theta)} \qquad F_{DE} = -6.928\,kN$

$\Sigma F_y = 0; \quad F - E_y + F_{DE}\sin(\theta) = 0$

$F = E_y - F_{DE}\sin(\theta) \qquad F = 6.93\,kN$

Problem 6-98

Determine the horizontal and vertical components of force at pins A and C of the two-member frame.

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Given:

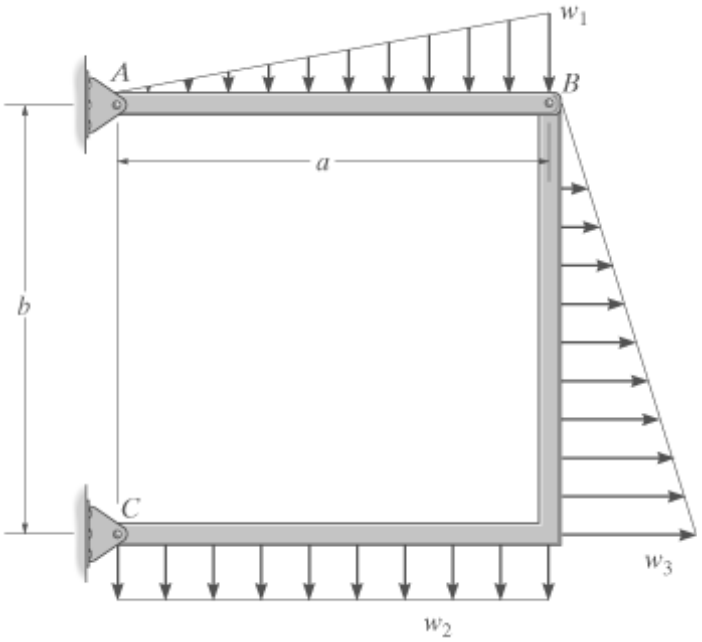
$w_1 = 500 \frac{\text{N}}{\text{m}}$

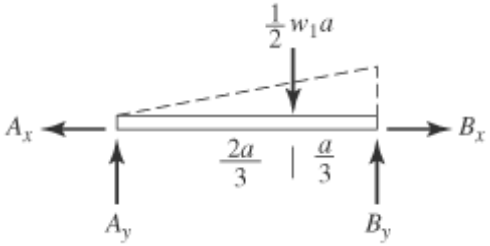
$w_2 = 400 \frac{\text{N}}{\text{m}}$

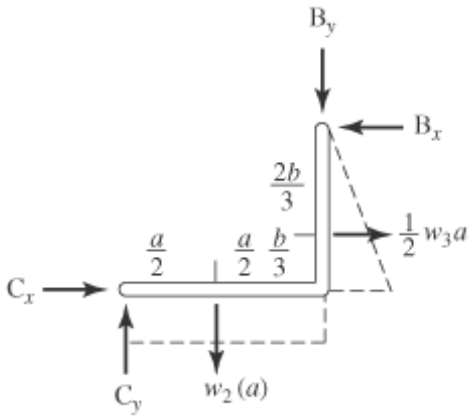
$w_3 = 600 \frac{\text{N}}{\text{m}}$

$a = 3$ m

$b = 3$ m







Solution:

Guesses

$A_x = 1$ N $\quad A_y = 1$ N $\quad C_x = 1$ N $\quad C_y = 1$ N

Given

$$A_y + C_y - \frac{1}{2} w_1 a - w_2 a = 0 \qquad A_x a - \frac{1}{2} w_1 a \frac{2}{3} \frac{a}{} - \frac{1}{2} w_3 b \frac{b}{3} - w_2 a \frac{a}{2} = 0$$

$$-A_x + C_x + \frac{1}{2} w_3 b = 0 \qquad -A_y a + \frac{1}{2} w_1 a \frac{a}{3} = 0$$

$$\begin{pmatrix} A_x \\ A_y \\ C_x \\ C_y \end{pmatrix} = \text{Find}\left(A_x, A_y, C_x, C_y\right) \qquad \begin{pmatrix} A_x \\ A_y \\ C_x \\ C_y \end{pmatrix} = \begin{pmatrix} 1400 \\ 250 \\ 500 \\ 1700 \end{pmatrix} \text{N}$$

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Problem 6-99

The truck rests on the scale, which consists of a series of compound levers. If a mass M_1 is placed on the pan P and it is required that the weight is located at a distance x to balance the "beam" ABC, determine the mass of the truck. There are pins at all lettered points. Is it necessary for the truck to be symmetrically placed on the scale? Explain.

Units Used:

$$\text{Mg} = 10^3 \text{ kg} \qquad \text{g} = 9.81 \frac{\text{m}}{\text{s}^2}$$

Given:

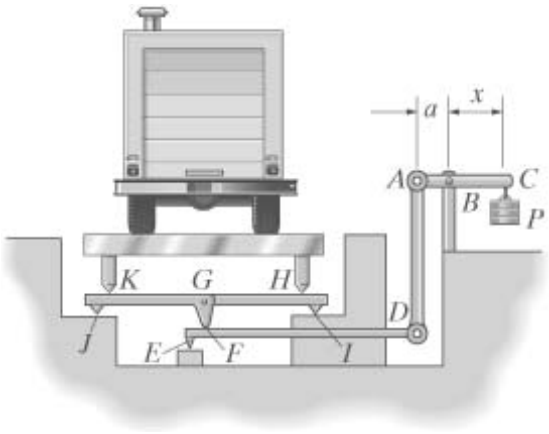
$M_1 = 15$ kg $\qquad FD = 3$ m

$x = 0.480$ m $\qquad EF = 0.2$ m

$a = 0.2$ m

$HI = 0.1$ m $\qquad GH = 2.5$ m

$KJ = HI \qquad KG = GH$

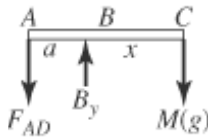


Solution:

Member ABC :

$$\Sigma M_B = 0; \qquad -M_1 g x + F_{AD} a = 0$$

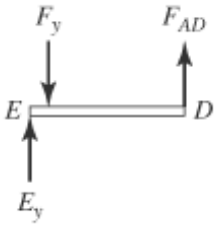
$$F_{AD} = M_1 g \frac{x}{a} \qquad F_{AD} = 72 \text{ s}^2 \text{ N}$$



Member EFD :

$$\Sigma M_E = 0; \qquad -F_y EF + F_{AD}(FD + EF) = 0$$

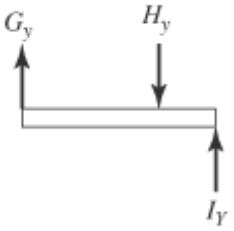
$$F_y = F_{AD}\left(\frac{FD + EF}{EF}\right) \qquad F_y = 1152 \text{ s}^2 \text{ N}$$



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Member *GHI* :

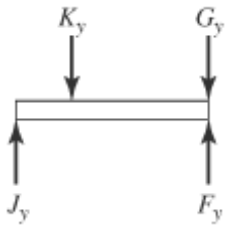
$\Sigma M_I = 0; \quad H_y HI - G_Y(GH + HI) = 0$



Member *JKG* :

$\Sigma M_J = 0; \quad \left(F_y - G_y\right)(KJ + GH) - K_y(KJ) = 0$

$$K_y + H_y = F_y \frac{KJ + KG}{HI}$$

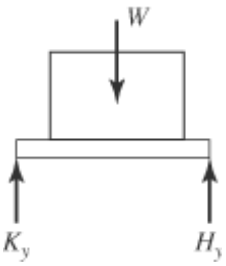


Scale Platform :

$\Sigma F_y = 0; \quad K_y + H_y = W$

$$W = F_y\left(\frac{KJ + KG}{HI}\right)$$

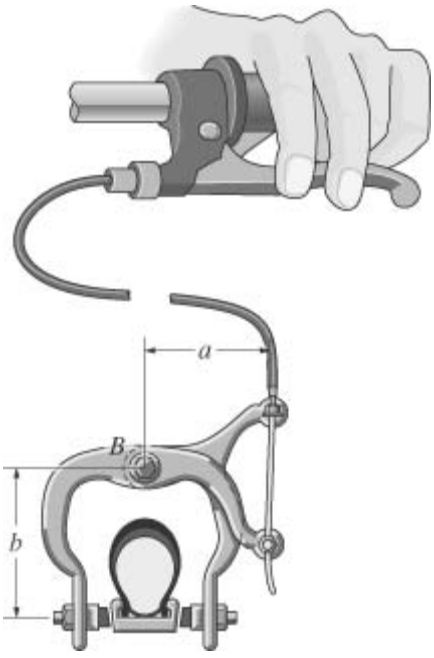
$$M = \frac{W}{g} \qquad M = 14.98\ \text{Mg}$$



Because $KJ = HI$ and $KG = GH$ it doesn't matter where the truck is on the scale.

Problem 6-100

By squeezing on the hand brake of the bicycle, the rider subjects the brake cable to a tension **T** If the caliper mechanism is pin-connected to the bicycle frame at *B*, determine the normal force each brake pad exerts on the rim of the wheel. Is this the force that stops the wheel from turning? Explain.



Given:

$T = 50$ lb

$a = 2.5$ in

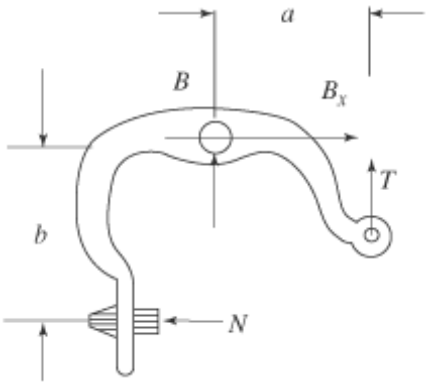
$b = 3$ in

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Solution:

$\circlearrowleft + \Sigma M_B = 0; \quad -Nb + Ta = 0$

$$N = T\frac{a}{b} \qquad N = 41.7\,\text{lb}$$



This normal force **does not** stop the wheel from turning. A frictional force (see Chapter 8), which acts along the wheel's rim stops the wheel.

Problem 6-101

If a force of magnitude P is applied perpendicular to the handle of the mechanism, determine the magnitude of force **F** for equilibrium. The members are pin-connected at A, B, C, and D.

Given:

$P = 6$ lb

$a = 25$ in

$b = 4$ in

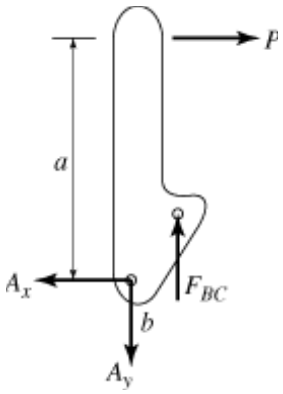
$c = 5$ in

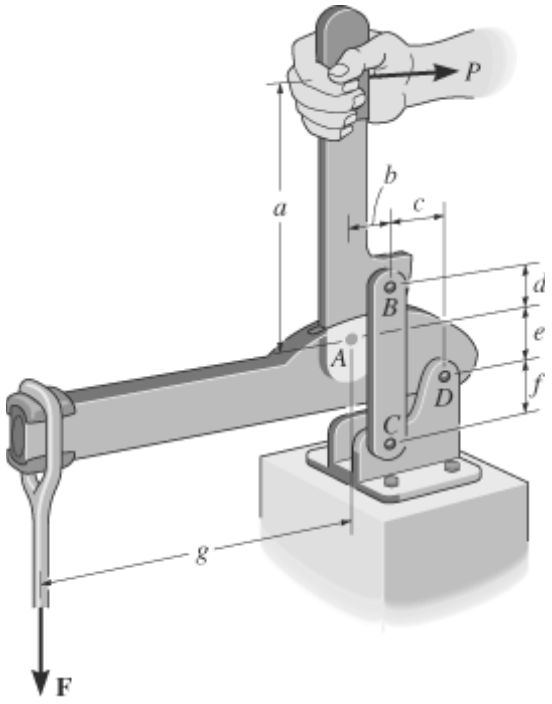
$d = 4$ in

$e = 5$ in

$f = 5$ in

$g = 30$ in





Solution:

$\circlearrowleft + \Sigma M_A = 0; \quad F_{BC}\,b - P\,a = 0$

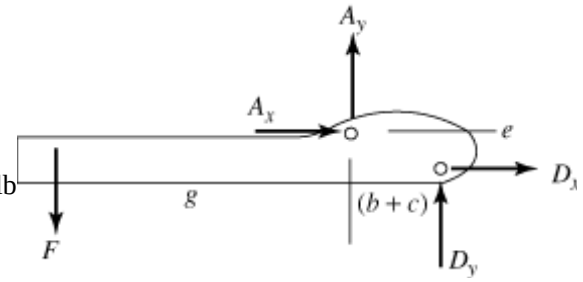
$$F_{BC} = \frac{P\,a}{b}$$

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$F_{BC} = 37.5\,\text{lb}$

$\overset{+}{\rightarrow} \Sigma F_x = 0; \quad -A_x + P = 0$

$A_x = P \qquad A_x = 6\,\text{lb}$



$+\uparrow \Sigma F_y = 0; \quad -A_y + F_{BC} = 0$

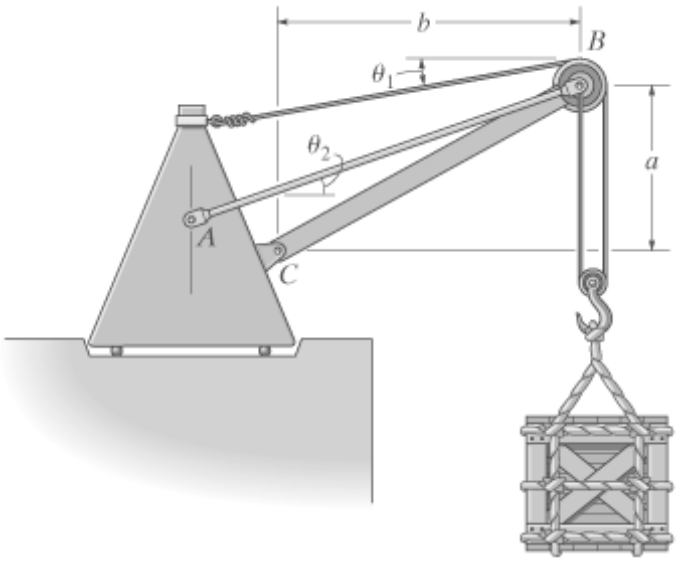
$A_y = F_{BC} \qquad A_y = 37.5\,\text{lb}$

$\circlearrowleft \Sigma M_D = 0; \quad -e A_x - A_y(b + c) + (g + b + c)F = 0$

$$F = \frac{e A_x + A_y(b + c)}{g + b + c} \qquad F = 9.423\,\text{lb}$$

Problem 6-102

The pillar crane is subjected to the load having a mass M. Determine the force developed in the tie rod AB and the horizontal and vertical reactions at the pin support C when the boom is tied in the position shown.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$M = 500\ \text{kg}$

$a = 1.8\ \text{m}$

$b = 2.4\ \text{m}$

$\theta_1 = 10\ \text{deg}$

$\theta_2 = 20\ \text{deg}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$

Solution:

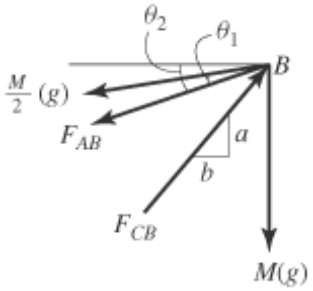
initial guesses: $F_{CB} = 10\ \text{kN} \qquad F_{AB} = 10\ \text{kN}$

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Given

$$\frac{-M}{2}g\cos(\theta_1) - F_{AB}\cos(\theta_2) + F_{CB}\frac{b}{\sqrt{a^2+b^2}} = 0$$

$$\frac{-M}{2}g\sin(\theta_1) - F_{AB}\sin(\theta_2) + F_{CB}\frac{a}{\sqrt{a^2+b^2}} - Mg = 0$$



$$\begin{pmatrix} F_{AB} \\ F_{CB} \end{pmatrix} = \text{Find}(F_{AB}, F_{CB}) \qquad \begin{pmatrix} C_x \\ C_y \end{pmatrix} = \frac{F_{CB}}{\sqrt{a^2+b^2}}\begin{pmatrix} b \\ a \end{pmatrix}$$

$$\begin{pmatrix} F_{AB} \\ C_x \\ C_y \end{pmatrix} = \begin{pmatrix} 9.7 \\ 11.53 \\ 8.65 \end{pmatrix} \text{kN}$$

Problem 6-103

The tower truss has a weight W and a center of gravity at G. The rope system is used to hoist it into the vertical position. If rope CB is attached to the top of the shear leg AC and a second rope CD is attached to the truss, determine the required tension in BC to hold the truss in the position shown. The base of the truss and the shear leg bears against the stake at A, which can be considered as a pin. Also, compute the compressive force acting along the shear leg.

Given:

$W = 575$ lb

$\theta = 40$ deg

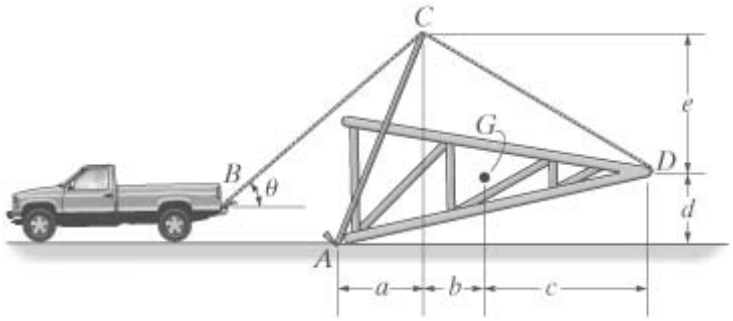
$a = 5$ ft

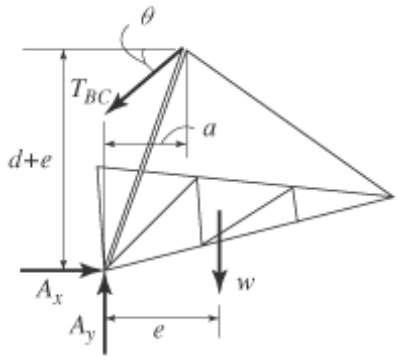
$b = 3$ ft

$c = 10$ ft

$d = 4$ ft

$e = 8$ ft





Solution: Entire system: $\Sigma M_A = 0$;

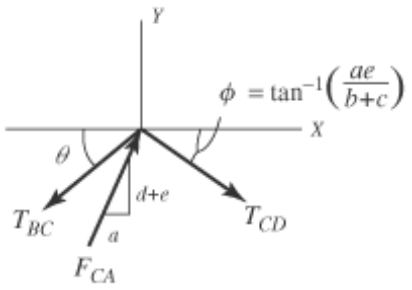
$$T_{BC}\cos(\theta)(d+e) - T_{BC}\sin(\theta)a - W(a+b) = 0$$

$$T_{BC} = \frac{W(a+b)}{\cos(\theta)(d+e) - \sin(\theta)a}$$

$$T_{BC} = 769\,\text{lb}$$

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CA is a two-force member. At *C*:



$$\phi = \operatorname{atan}\left(\frac{e}{b+c}\right)$$

initial guesses: $F_{CA} = 500\ \text{lb}$ $T_{CD} = 300\ \text{lb}$

Given

$$\Sigma F_x = 0; \quad F_{CA}\frac{a}{\sqrt{a^2 + (d+e)^2}} + T_{CD}\cos(\phi) - T_{BC}\cos(\theta) = 0$$

$$\Sigma F_y = 0; \quad F_{CA}\frac{d+e}{\sqrt{a^2 + (d+e)^2}} - T_{CD}\sin(\phi) - T_{BC}\sin(\theta) = 0$$

$$\begin{pmatrix} F_{CA} \\ T_{CD} \end{pmatrix} = \text{Find}(F_{CA}, T_{CD}) \qquad T_{CD} = 358\,\text{lb} \qquad F_{CA} = 739\,\text{lb}$$

Problem 6-104

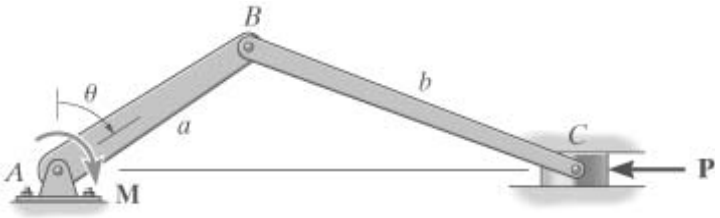
The constant moment *M* is applied to the crank shaft. Determine the compressive force *P* that is exerted on the piston for equilibrium as a function of θ. Plot the results of *P* (ordinate) versus θ (abscissa) for $0\ \text{deg} \le \theta \le 90\ \text{deg}$.

Given:

$a = 0.2\ \text{m}$

$b = 0.45\ \text{m}$

$M = 50\ \text{N}\cdot\text{m}$



Solution:

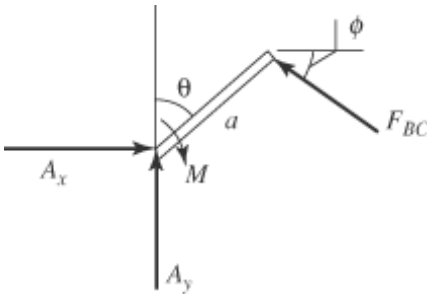
$$a\cos(\theta) = b\sin(\phi) \qquad \phi = \operatorname{asin}\left(\frac{a}{b}\cos(\theta)\right)$$

$$-M + F_{BC}\cos(\theta - \phi)a = 0$$

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$$F_{BC} = \frac{M}{a\cos(\theta - \phi)}$$

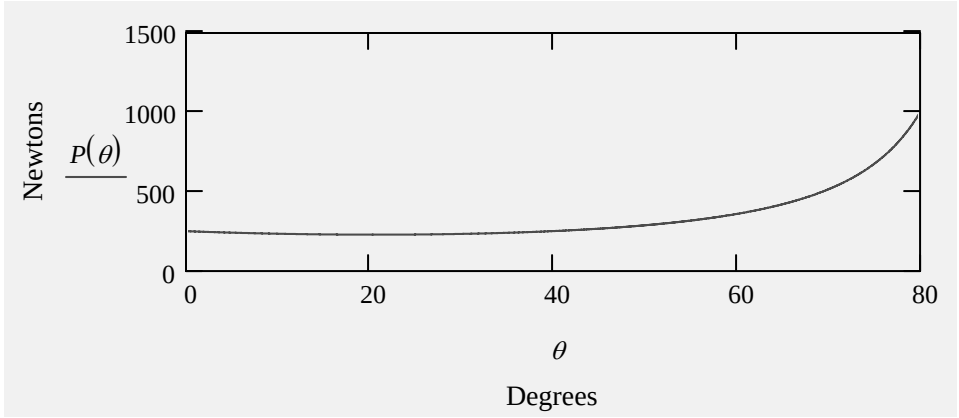
$$P = F_{BC}\cos(\phi) = \frac{M\cos(\phi)}{a\cos(\theta - \phi)}$$



This function goes to infinity at $\theta = 90$ deg, so we will only plot it to $\theta = 80$ deg.

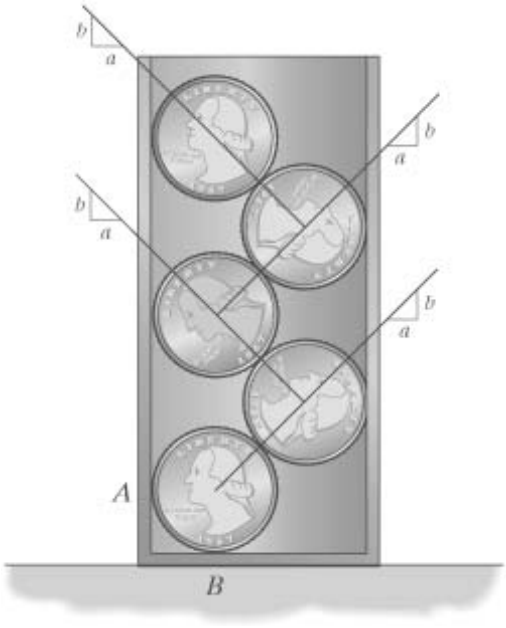
$\theta = 0, 0.1 .. 80$

$$\phi(\theta) = \operatorname{asin}\left(\frac{a}{b}\cos(\theta \text{ deg})\right) \qquad P(\theta) = \frac{M\cos(\phi(\theta))}{a\cos(\theta \text{ deg} - \phi(\theta))}$$



Problem 6-105

Five coins are stacked in the smooth plastic container shown. If each coin has weight *W*, determine the normal reactions of the bottom coin on the container at points *A* and *B*.



Given:

$W = 0.0235$ lb

$a = 3$

$b = 4$

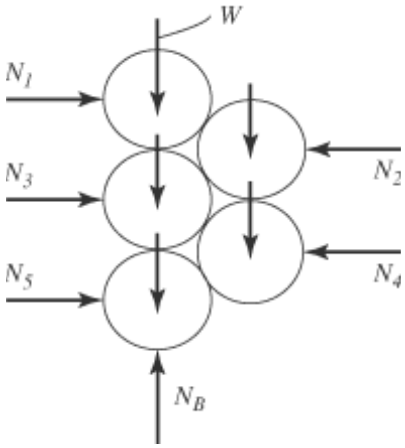
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Solution:

All coins :

$\Sigma F_y = 0; \quad N_B = 5W$

$N_B = 0.1175\,\text{lb}$

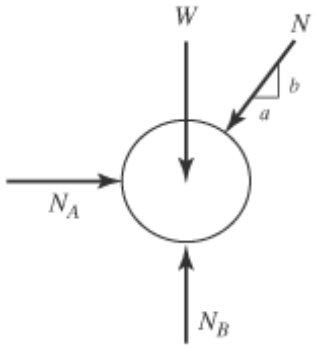


Bottom coin :

$\Sigma F_y = 0; \quad N_B - W - N\left(\frac{b}{\sqrt{a^2 + b^2}}\right) = 0$

$N = \left(N_B - W\right)\left(\frac{\sqrt{a^2 + b^2}}{b}\right)$

$N = 0.1175\,\text{lb}$



$\Sigma F_x = 0; \quad N_A = N\left(\frac{a}{\sqrt{a^2 + b^2}}\right)$

$N_A = 0.0705\,\text{lb}$

Problem 6-106

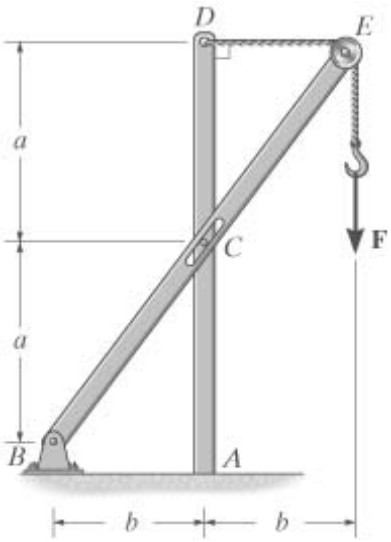
Determine the horizontal and vertical components of force at pin *B* and the normal force the pin at *C* exerts on the smooth slot. Also, determine the moment and horizontal and vertical reactions of force at *A*. There is a pulley at *E*.

Given:

$F = 50$ lb

$a = 4$ ft

$b = 3$ ft



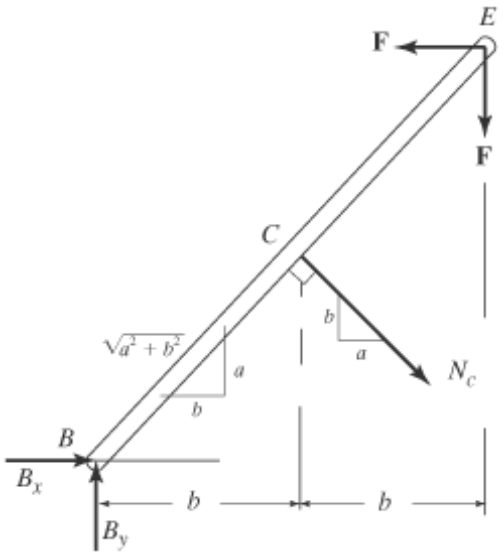
Solution:

Guesses

$B_x = 1$ lb

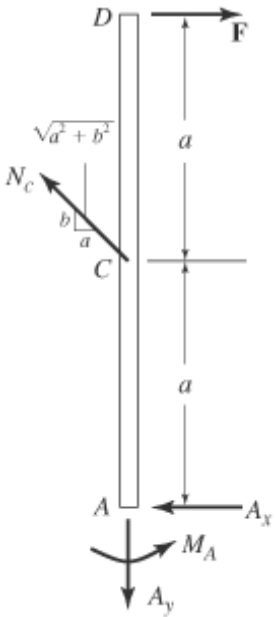
$B_y = 1$ lb

$N_C = 1$ lb



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$A_x = 1$ lb $\quad A_y = 1$ lb $\quad M_A = 1$ lb·ft



Given

$$B_x + \left(\frac{a}{\sqrt{a^2 + b^2}}\right) N_C - F = 0$$

$$B_y - \left(\frac{b}{\sqrt{a^2 + b^2}}\right) N_C - F = 0$$

$$\left(F + B_x\right) a - \left(F + B_y\right) b = 0$$

$$F - \left(\frac{a}{\sqrt{a^2 + b^2}}\right) N_C - A_x = 0$$

$$\left(\frac{b}{\sqrt{a^2 + b^2}}\right) N_C - A_y = 0$$

$$-F\ 2\ a + \left(\frac{a}{\sqrt{a^2 + b^2}}\right) N_C a + M_A = 0$$

$$\begin{pmatrix} B_x \\ B_y \\ N_C \\ A_x \\ A_y \\ M_A \end{pmatrix} = \text{Find}\left(B_x, B_y, N_C, A_x, A_y, M_A\right)$$

$N_C = 20\,\text{lb}$

$$\begin{pmatrix} B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 34 \\ 62 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 34 \\ 12 \end{pmatrix} \text{lb}$$

$M_A = 336\,\text{lb·ft}$

Problem 6-107

A force **F** is applied to the handles of the vise grip. Determine the compressive force developed on the smooth bolt shank *A* at the jaws.

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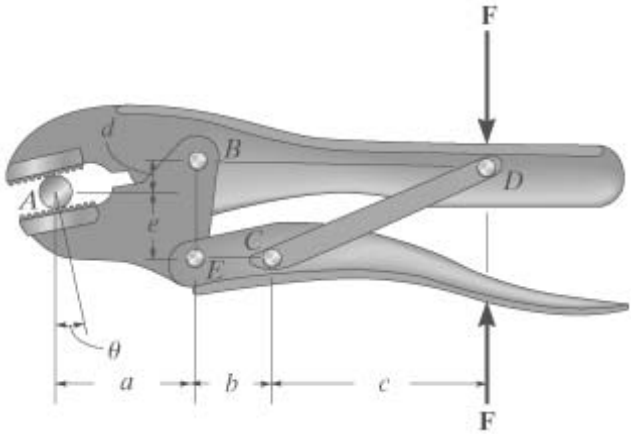
Given:

$F = 5$ lb $\quad b = 1$ in

$a = 1.5$ in $\quad c = 3$ in

$d = 0.75$ in $\quad e = 1$ in

$\theta = 20$ deg



Solution:

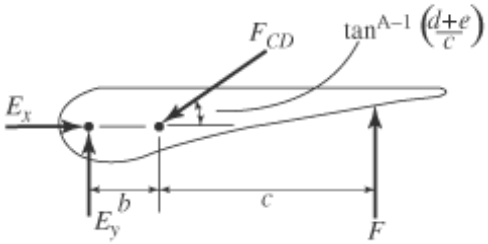
From *FBD* (a)

$$\Sigma M_E = 0; \quad F(b+c) - F_{CD}\left[\frac{d+e}{\sqrt{c^2+(d+e)^2}}\right]b = 0$$

$$F_{CD} = F(b+c)\left[\frac{\sqrt{c^2+(d+e)^2}}{b(d+e)}\right] \qquad F_{CD} = 39.693\,\text{lb}$$

$$\Sigma F_x = 0; \quad E_x = F_{CD}\left[\frac{c}{\sqrt{c^2+(d+e)^2}}\right]$$

$$E_x = 34.286\,\text{lb}$$

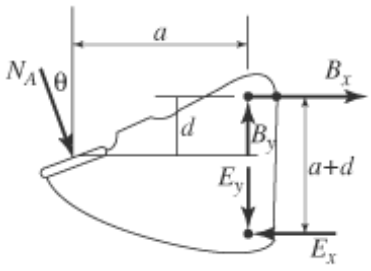


From *FBD* (b)

$$\Sigma M_B = 0; \quad N_A \sin(\theta) d + N_A \cos(\theta) a - E_x(d+e) = 0$$

$$N_A = E_x\left(\frac{d+e}{\sin(\theta)\, d + \cos(\theta)\, a}\right)$$

$$N_A = 36.0\,\text{lb}$$



Problem 6-108

If a force of magnitude P is applied to the grip of the clamp, determine the compressive force F that the wood block exerts on the clamp.

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Given:

$P = 10$ lb

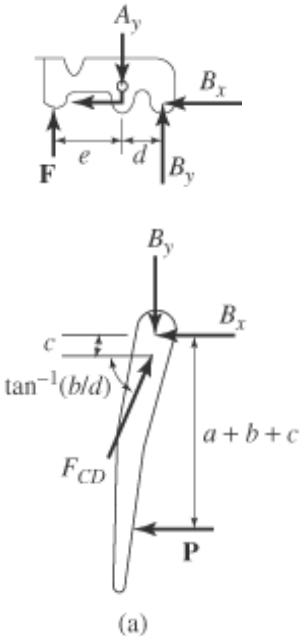
$a = 2$ in

$b = 2$ in

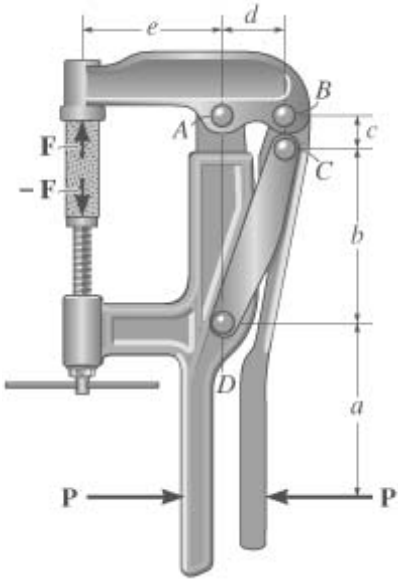
$c = 0.5$ in

$d = 0.75$ in

$e = 1.5$ in



(a)



Solution:

Define $\phi = \operatorname{atan}\left(\frac{b}{d}\right)$ $\phi = 69.444\,\text{deg}$

From *FBD* (a),

$\circlearrowleft + \Sigma M_B = 0;$ $F_{CD}\cos(\phi)c - P(a+b+c) = 0$

$F_{CD} = \frac{P(a+b+c)}{\cos(\phi)(c)}$ $F_{CD} = 256.32\,\text{lb}$

$+\uparrow \Sigma F_y = 0;$ $F_{CD}\sin(\phi) - B_y = 0$

$B_y = F_{CD}\sin(\phi)$ $B_y = 240\,\text{lb}$

From *FBD* (b),

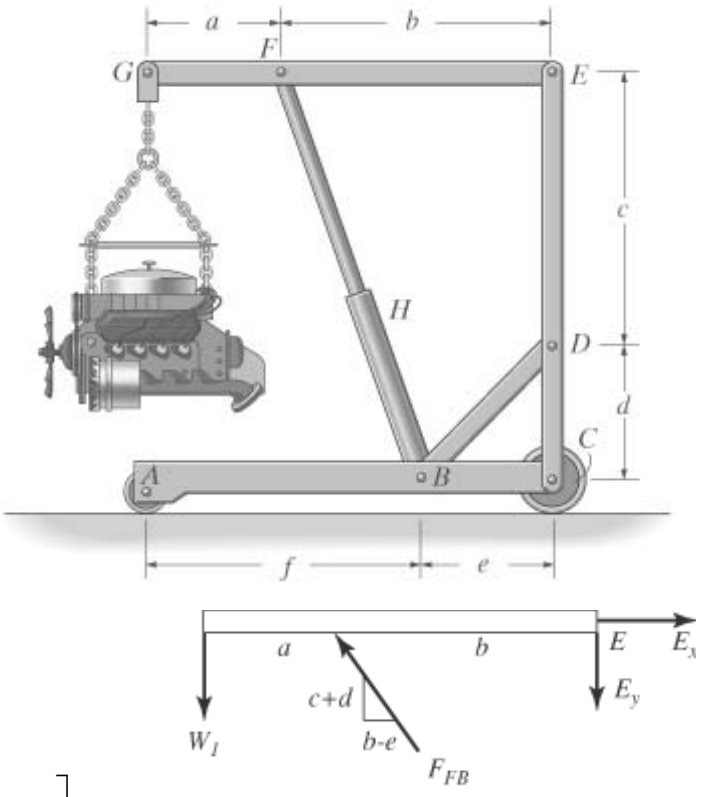
$\circlearrowleft + \Sigma M_A = 0;$ $B_y d - F e = 0$

$F = \frac{B_y d}{e}$ $F = 120\,\text{lb}$

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Problem 6-109

The hoist supports the engine of mass *M*. Determine the force in member *DB* and in the hydraulic cylinder *H* of member *FB*.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$M = 125\ \text{kg}$ $d = 1\ \text{m}$

$a = 1\ \text{m}$ $e = 1\ \text{m}$

$b = 2\ \text{m}$ $f = 2\ \text{m}$

$c = 2\ \text{m}$ $g = 9.81\ \frac{\text{m}}{\text{s}^2}$

Solution:

Member *GFE*:

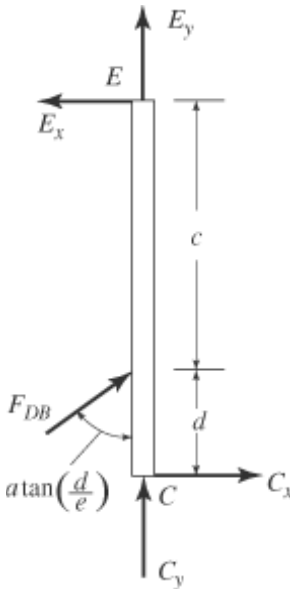
$$\Sigma M_E = 0;\ -F_{FB}\left[\frac{c+d}{\sqrt{(c+d)^2+(b-e)^2}}\right]b + Mg(a+b) = 0$$

$$F_{FB} = M\,g\left[\frac{a+b}{b(c+d)}\right]\sqrt{(c+d)^2+(b-e)^2}$$

$$F_{FB} = 1.94\,\text{kN}$$

$$\Sigma F_x = 0;\qquad E_x - F_{FB}\left[\frac{b-e}{\sqrt{(c+d)^2+(b-e)^2}}\right] = 0$$

$$E_x = F_{FB}\left[\frac{b-e}{\sqrt{(c+d)^2+(b-e)^2}}\right]$$



Member *EDC*:

$$\Sigma M_c = 0;\qquad E_x(c+d) - F_{DB}\left(\frac{e}{\sqrt{e^2+d^2}}\right)d = 0$$

$$F_{DB} = E_x\left(\frac{c+d}{e\,d}\right)\sqrt{e^2+d^2}\qquad F_{DB} = 2.601\,\text{kN}$$

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Problem 6-110

The flat-bed trailer has weight W_1 and center of gravity at G_T. It is pin-connected to the cab at D. The cab has a weight W_2 and center of gravity at G_C. Determine the range of values x for the position of the load L of weight W_3 so that no axle is subjected to a force greater than F_{Max}. The load has a center of gravity at G_L.

Given:

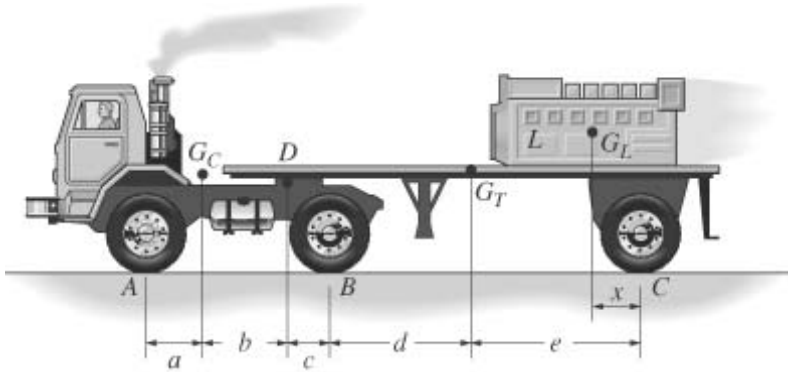
$W_1 = 7000$ lb $a = 4$ ft

$W_2 = 6000$ lb $b = 6$ ft

$W_3 = 2000$ lb $c = 3$ ft

$F_{max} = 5500$ lb $d = 10$ ft

$e = 12$ ft

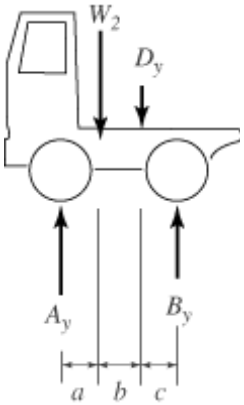


Solution:

Case 1: Assume $A_y = F_{max}$

Guesses $A_y = F_{max}$ $B_y = F_{max}$ $C_y = F_{max}$

$x = 1$ ft $D_y = F_{max}$



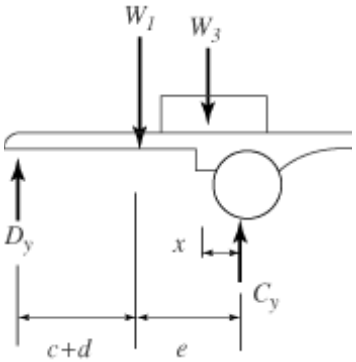
Given $A_y + B_y - W_2 - D_y = 0$

$-W_2 a - D_y(a + b) + B_y(a + b + c) = 0$

$D_y - W_1 - W_3 + C_y = 0$

$W_3 x + W_1 e - D_y(c + d + e) = 0$

$$\begin{pmatrix} B_y \\ C_y \\ D_y \\ x \end{pmatrix} = \text{Find}\left(B_y, C_y, D_y, x\right)$$



$$\begin{pmatrix} A_y \\ B_y \\ C_y \end{pmatrix} = \begin{pmatrix} 5.5 \times 10^3 \\ 6.333 \times 10^3 \\ 3.167 \times 10^3 \end{pmatrix} \text{lb} \quad x_1 = x \quad x_1 = 30.917 \text{ ft}$$

Since $B_y > F_{max}$ then this solution is no good.

Case 2: Assume $B_y = F_{max}$

Guesses $A_y = F_{max}$ $B_y = F_{max}$ $C_y = F_{max}$

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$x = 1\ \text{ft} \qquad D_y = F_{max}$

Given $A_y + B_y - W_2 - D_y = 0$

$-W_2 a - D_y(a + b) + B_y(a + b + c) = 0$

$D_y - W_1 - W_3 + C_y = 0$

$W_3 x + W_1 e - D_y(c + d + e) = 0$

$\begin{pmatrix} A_y \\ C_y \\ D_y \\ x \end{pmatrix} = \text{Find}\left(A_y, C_y, D_y, x\right) \qquad \begin{pmatrix} A_y \\ B_y \\ C_y \end{pmatrix} = \begin{pmatrix} 5.25 \times 10^3 \\ 5.5 \times 10^3 \\ 4.25 \times 10^3 \end{pmatrix} \text{lb} \quad x_2 = x \qquad x_2 = 17.375\ \text{ft}$

Since $A_y < F_{max}$ and $C_y < F_{max}$ then this solution is good.

Case 3: Assume $C_y = F_{max}$

Guesses $A_y = F_{max} \quad B_y = F_{max} \quad C_y = F_{max}$

$x = 1\ \text{ft} \qquad D_y = F_{max}$

Given $A_y + B_y - W_2 - D_y = 0$

$-W_2 a - D_y(a + b) + B_y(a + b + c) = 0$

$D_y - W_1 - W_3 + C_y = 0$

$W_3 x + W_1 e - D_y(c + d + e) = 0$

$\begin{pmatrix} A_y \\ B_y \\ D_y \\ x \end{pmatrix} = \text{Find}\left(A_y, B_y, D_y, x\right) \qquad \begin{pmatrix} A_y \\ B_y \\ C_y \end{pmatrix} = \begin{pmatrix} 4.962 \times 10^3 \\ 4.538 \times 10^3 \\ 5.5 \times 10^3 \end{pmatrix} \text{lb} \quad x_3 = x \qquad x_3 = 1.75\ \text{ft}$

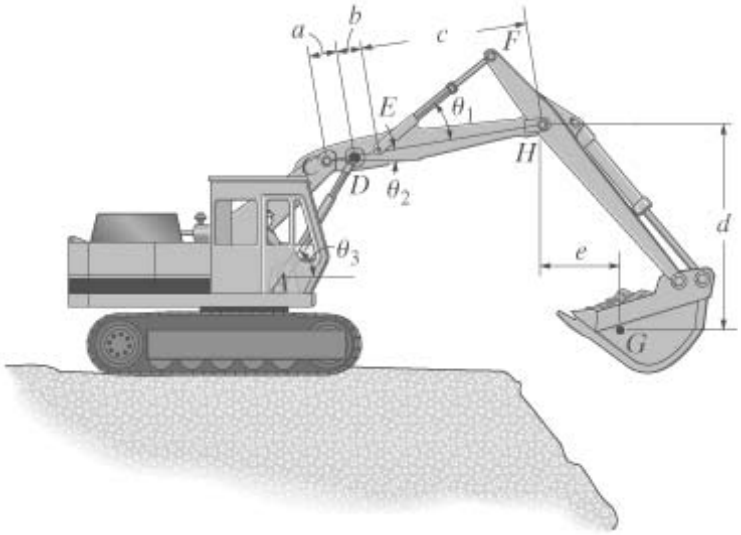
Since $A_y < F_{max}$ and $B_y < F_{max}$ then this solution is good.

We conclude that $x_3 = 1.75\ \text{ft} < x < x_2 = 17.375\ \text{ft}$

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Problem 6-111

Determine the force created in the hydraulic cylinders *EF* and *AD* in order to hold the shovel in equilibrium. The shovel load has a mass *W* and a center of gravity at *G*. All joints are pin connected.



Units Used:

$\text{Mg} = 10^3\ \text{kg}$

$\text{kN} = 10^3\ \text{N}$

Given:

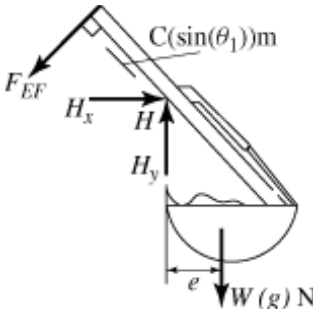
$a = 0.25\ \text{m} \quad \theta_1 = 30\ \text{deg}$

$b = 0.25\ \text{m} \quad \theta_2 = 10\ \text{deg}$

$c = 1.5\ \text{m} \quad \theta_3 = 60\ \text{deg}$

$d = 2\ \text{m} \quad W = 1.25\ \text{Mg}$

$e = 0.5\ \text{m}$

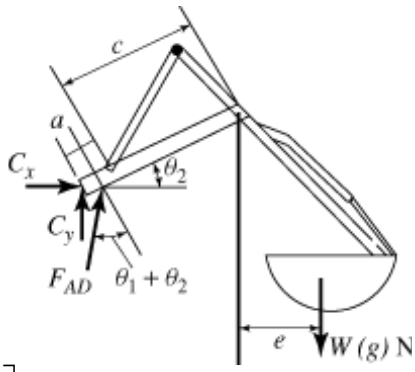


Solution:

Assembly *FHG* :

$\Sigma M_H = 0; \quad -[W g(e)] + F_{EF}\left(c \sin\left(\theta_1\right)\right) = 0$

$$F_{EF} = W\, g \left(\frac{e}{c \sin\left(\theta_1\right)} \right) \qquad F_{EF} = 8.175\,\text{k(T)}$$



Assembly *CEFHG*:

$\Sigma M_C = 0; \; F_{AD} \cos\left(\theta_1 + \theta_2\right) b - W g \left[(a + b + c) \cos\left(\theta_2\right) + e \right] = 0$

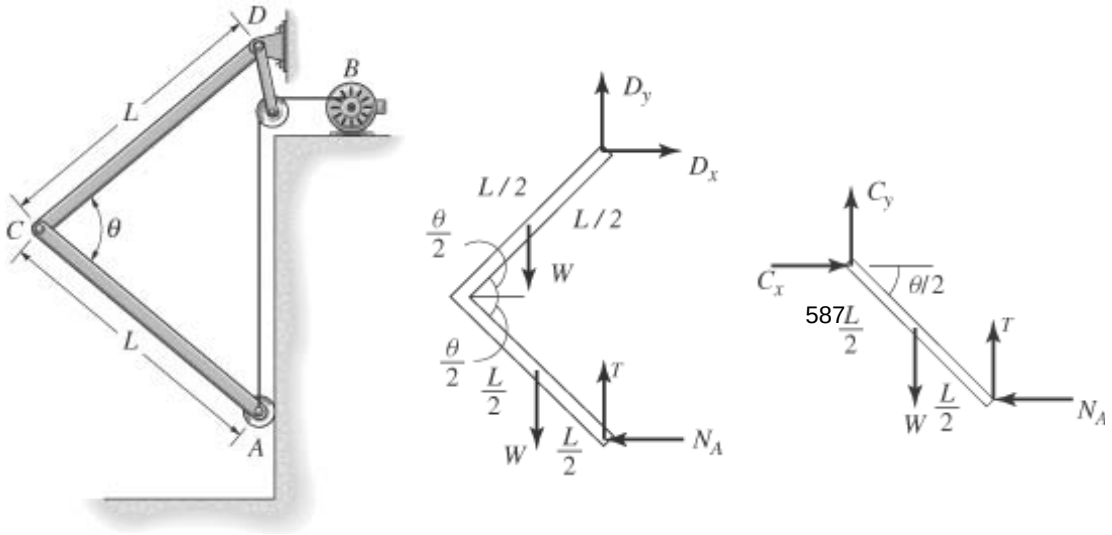
$$F_{AD} = W\, g \left(\frac{\cos\left(\theta_2\right)\, a + \cos\left(\theta_2\right)\, b + \cos\left(\theta_2\right)\, c + e}{\cos\left(\theta_1 + \theta_2\right)\, b} \right)$$

$F_{AD} = 158\,\text{kN}$ (C)

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Problem 6-112

The aircraft-hangar door opens and closes slowly by means of a motor which draws in the cable *AB*. If the door is made in two sections (bifold) and each section has a uniform weight *W* and length *L*, determine the force in the cable as a function of the door's position θ. The sections are pin-connected at *C* and *D* and the bottom is attached to a roller that travels along the vertical track.



Solution:

$$\circlearrowleft + \Sigma M_D = 0; \qquad 2W\frac{L}{2}\cos\left(\frac{\theta}{2}\right) - 2L\sin\left(\frac{\theta}{2}\right)N_A = 0 \qquad N_A = \frac{W}{2}\cot\left(\frac{\theta}{2}\right)$$

$$\circlearrowleft + \Sigma M_C = 0; \qquad TL\cos\left(\frac{\theta}{2}\right) - N_A L\sin\left(\frac{\theta}{2}\right) - W\frac{L}{2}\cos\left(\frac{\theta}{2}\right) = 0 \qquad T = W$$

Problem 6-113

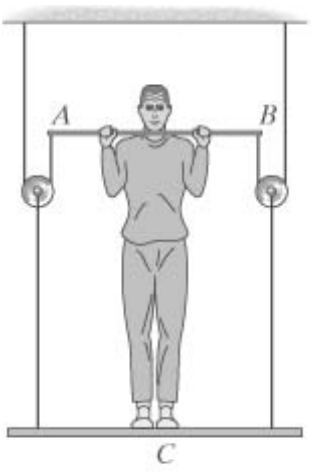
A man having weight *W* attempts to lift himself using one of the two methods shown. Determine the total force he must exert on bar *AB* in each case and the normal reaction he exerts on the platform at *C*. Neglect the weight of the platform.

Given:

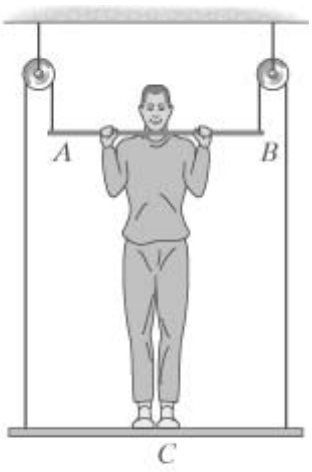
$W = 175$ lb

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Solution:



(a)



(b)

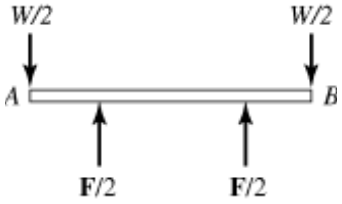


(a)

Bar:

$$+\uparrow \Sigma F_y = 0; \qquad 2\left(\frac{F}{2}\right) - 2\left(\frac{W}{2}\right) = 0$$

$$F = W \qquad F = 175\,\text{lb}$$



Man:

$$+\uparrow \Sigma F_y = 0; \qquad N_C - W - 2\left(\frac{F}{2}\right) = 0$$

$$N_C = W + F \qquad N_C = 350\,\text{lb}$$

(b)



Bar:

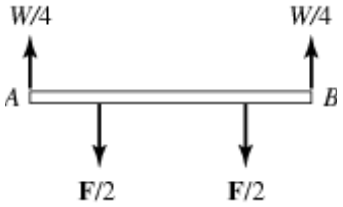
$$+\uparrow \Sigma F_y = 0; \qquad 2\left(\frac{W}{4}\right) - 2\,\frac{F}{2} = 0$$

$$F = \frac{W}{2} \qquad F = 87.5\,\text{lb}$$

Man:

$$+\uparrow \Sigma F_y = 0; \qquad N_C - W + 2\left(\frac{F}{2}\right) = 0$$

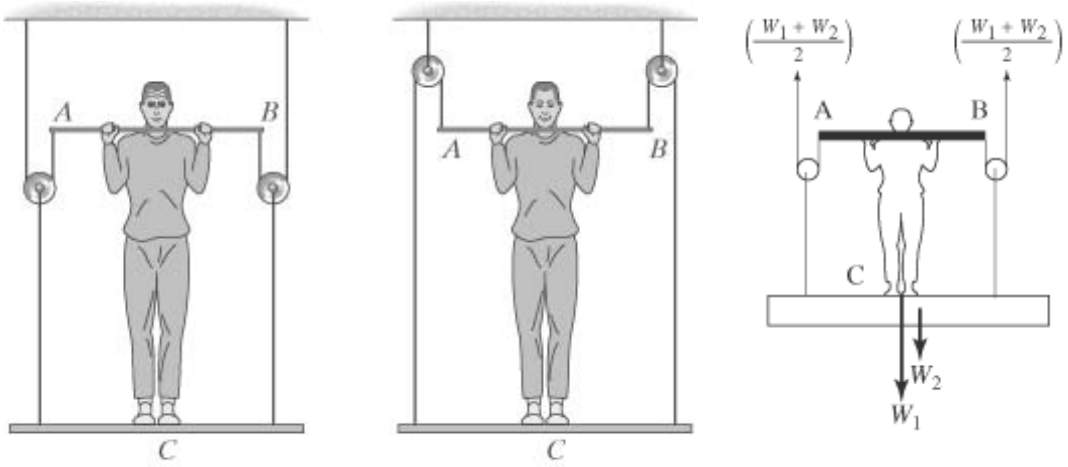
$$N_C = W - F \qquad N_C = 87.5\,\text{lb}$$



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Problem 6-114

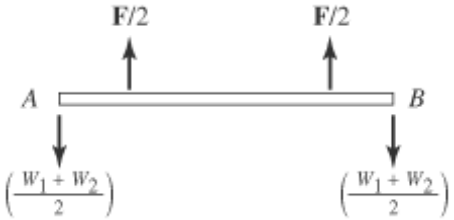
A man having weight W_1 attempts to lift himself using one of the two methods shown. Determine the total force he must exert on bar *AB* in each case and the normal reaction he exerts on the platform at *C*. The platform has weight W_2.



(a) (b)

Given:

$W_1 = 175$ lb $W_2 = 30$ lb



Solution:

(a)

Bar:

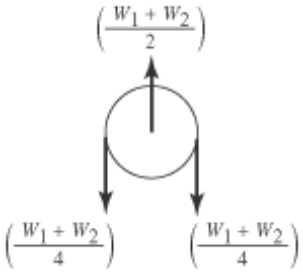
$$+\uparrow \Sigma F_y = 0; \qquad 2\frac{F}{2} - (W_1 + W_2) = 0$$

$$F = W_1 + W_2$$

$$F = 205\,\text{lb}$$

Man:





$$+\uparrow \Sigma F_y = 0; \qquad N_C - W_1 - 2\frac{F}{2} = 0$$

$$N_C = F + W_1$$

$$N_C = 380\,\text{lb}$$

(b)

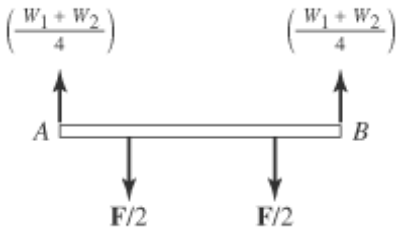
Bar:

$$+\uparrow \Sigma F_y = 0; \qquad -2\left(\frac{F}{2}\right) + 2\left(\frac{W_1 + W_2}{4}\right) = 0$$

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$$F = \frac{W_1 + W_2}{2}$$

$$F = 102\,\text{lb}$$



Man:

$$+\uparrow \Sigma F_y = 0; \quad N_C - W_1 + 2\left(\frac{F}{2}\right) = 0$$

$$N_C = W_1 - F$$

$$N_C = 72.5\,\text{lb}$$



Problem 6-115

The piston *C* moves vertically between the two smooth walls. If the spring has stiffness k and is unstretched when $\theta = 0$, determine the couple **M** that must be applied to *AB* to hold the mechanism in equilibrium.

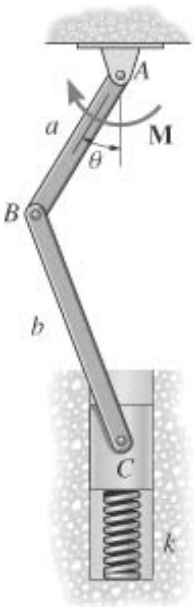
Given:

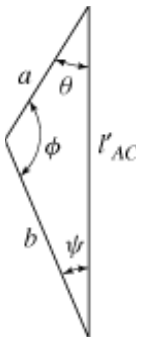
$k = 15\ \frac{\text{lb}}{\text{in}}$

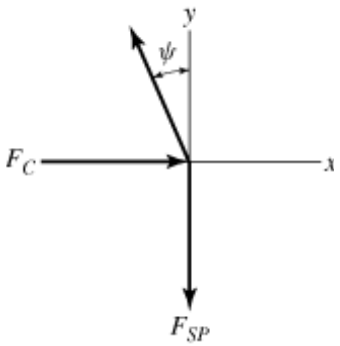
$\theta = 30\ \text{deg}$

$a = 8\ \text{in}$

$b = 12\ \text{in}$







Solution:

Geometry:

$b\sin(\psi) = a\sin(\theta)$

$\psi = \text{asin}\left(\sin(\theta)\,\frac{a}{b}\right)$ $\psi = 19.471\,\text{deg}$

$\phi = 180\ \text{deg} - \psi - \theta$ $\phi = 130.529\,\text{deg}$

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$$\frac{r_{AC}}{\sin(\phi)} = \frac{b}{\sin(\theta)} \qquad r_{AC} = b\left(\frac{\sin(\phi)}{\sin(\theta)}\right) \qquad r_{AC} = 18.242\,\text{in}$$

Free Body Diagram: The solution for this problem will be simplified if one realizes that member *CB* is a two force member. Since the spring stretches

$$x = (a + b) - r_{AC} \qquad x = 1.758\,\text{in}$$

the spring force is $F_{sp} = kx$ $F_{sp} = 26.371\,\text{lb}$

Equations of Equilibrium: Using the method of joints

$$+\uparrow \Sigma F_y = 0; \qquad F_{CB}\cos(\psi) - F_{sp} = 0 \qquad F_{CB} = \frac{F_{sp}}{\cos(\psi)} \qquad F_{CB} = 27.971\,\text{lb}$$

From *FBD* of bar *AB*

$$+\Sigma M_A = 0; \qquad F_{CB}\sin(\phi)a - M = 0 \qquad M = F_{CB}\sin(\phi)a \qquad M = 14.2\,\text{lb}\cdot\text{ft}$$

Problem 6-116

The compound shears are used to cut metal parts. Determine the vertical cutting force exerted on the rod *R* if a force **F** is applied at the grip *G*. The lobe *CDE* is in smooth contact with the head of the shear blade at *E*.

Given:

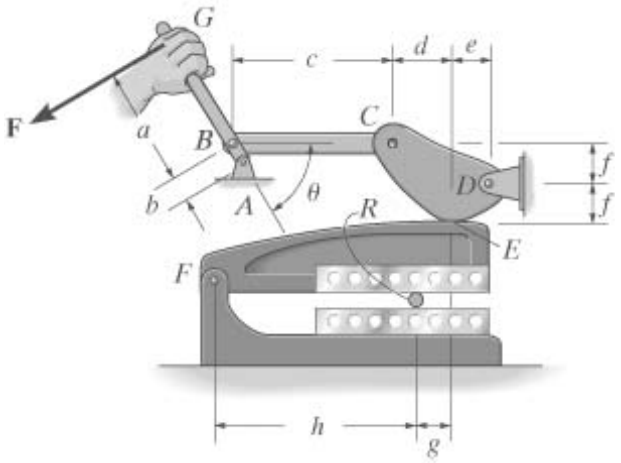
$F = 20$ lb $e = 0.5$ ft

$a = 1.4$ ft $f = 0.5$ ft

$b = 0.2$ ft $g = 0.5$ ft

$c = 2$ ft $h = 2.5$ ft

$d = 0.75$ ft $\theta = 60$ deg



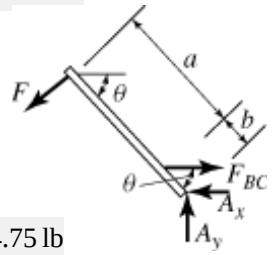
Solution:

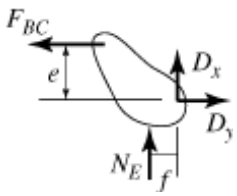
Member *AG*:

$$\Sigma M_A = 0; \qquad F(a + b) - F_{BC}\,b\sin(\theta) = 0$$

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$$F_{BC} = F\left(\frac{a+b}{b\,\sin(\theta)}\right) \qquad F_{BC} = 184.75\,\text{lb}$$





Lobe:

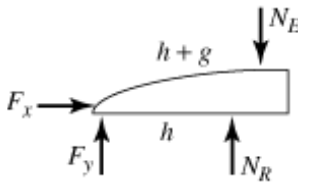
$\Sigma M_D = 0; \quad F_{BC}\,f - N_E\,e = 0$

$$N_E = F_{BC}\,\frac{f}{e} \qquad N_E = 184.75\,\text{lb}$$

Head:

$\Sigma M_F = 0; \quad -N_E(h+g) + h\,N_R = 0$

$$N_R = N_E\left(\frac{h+g}{h}\right) \qquad N_R = 222\,\text{lb}$$



Problem 6-117

The handle of the sector press is fixed to gear *G*, which in turn is in mesh with the sector gear *C*. Note that *AB* is pinned at its ends to gear *C* and the underside of the table *EF*, which is allowed to move vertically due to the smooth guides at *E* and *F*. If the gears exert tangential forces between them, determine the compressive force developed on the cylinder *S* when a vertical force **F** is applied to the handle of the press.

Given:

$F = 40$ N

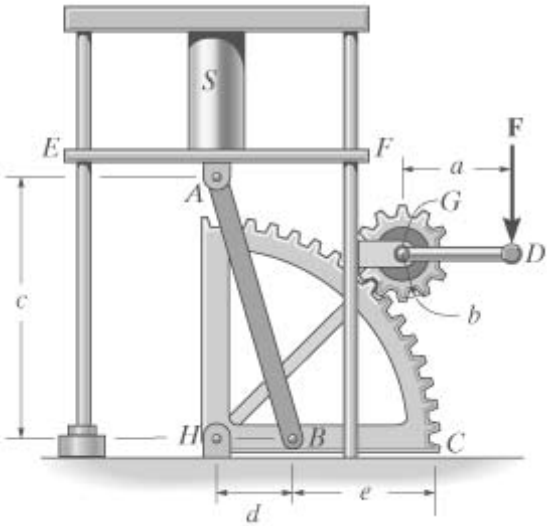
$a = 0.5$ m

$b = 0.2$ m

$c = 1.2$ m

$d = 0.35$ m

$e = 0.65$ m



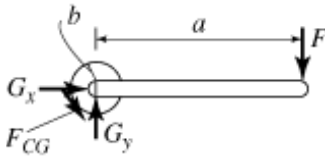
Solution:

Member *GD*:

$\Sigma M_G = 0; \quad -F\,a + F_{CG}\,b = 0$

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$$F_{CG} = F\,\frac{a}{b} \qquad F_{CG} = 100\ \text{N}$$



Sector gear :

$$\Sigma M_H = 0; \quad F_{CG}(d+e) - F_{AB}\left(\frac{c}{\sqrt{c^2+d^2}}\right)d = 0$$

$$F_{AB} = F_{CG}\,(d+e)\left(\frac{\sqrt{c^2+d^2}}{c\,d}\right) \qquad F_{AB} = 297.62\ \text{N}$$

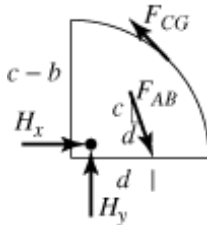
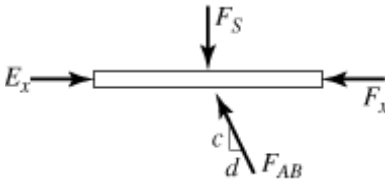


Table:

$$\Sigma F_y = 0; \quad F_{AB}\left(\frac{c}{\sqrt{c^2+d^2}}\right) - F_S = 0$$

$$F_S = F_{AB}\left(\frac{c}{\sqrt{c^2+d^2}}\right) \qquad F_S = 286\ \text{N}$$



Problem 6-118

The mechanism is used to hide kitchen appliances under a cabinet by allowing the shelf to rotate downward. If the mixer has weight *W*, is centered on the shelf, and has a mass center at *G*, determine the stretch in the spring necessary to hold the shelf in the equilibrium position shown. There is a similar mechanism on each side of the shelf, so that each mechanism supports half of the load *W*. The springs each have stiffness *k*.

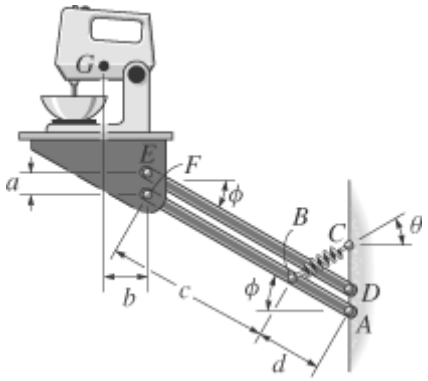
Given:

$W = 10\ \text{lb}$ $\quad a = 2\ \text{in}$

$k = 4\ \frac{\text{lb}}{\text{in}}$ $\quad b = 4\ \text{in}$

$\phi = 30\ \text{deg}$ $\quad c = 15\ \text{in}$

$\theta = 30\ \text{deg}$ $\quad d = 6\ \text{in}$

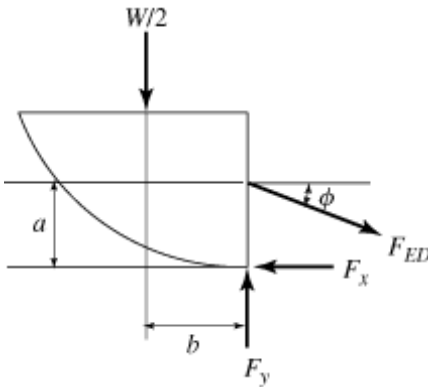


Solution:

$$\circlearrowleft + \Sigma M_F = 0; \qquad \frac{W}{2}b - a\,F_{ED}\cos(\phi) = 0$$

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$$F_{ED} = \frac{W b}{2\, a \cos(\phi)} \qquad F_{ED} = 11.547\,\text{lb}$$



$$\xrightarrow{+} \Sigma F_x = 0; \qquad -F_x + F_{ED}\cos(\phi) = 0$$

$$F_x = F_{ED}\cos(\phi) \qquad F_x = 10\,\text{lb}$$

$$+\uparrow \Sigma F_y = 0; \qquad \frac{-W}{2} + F_y - F_{ED}\sin(\phi) = 0$$

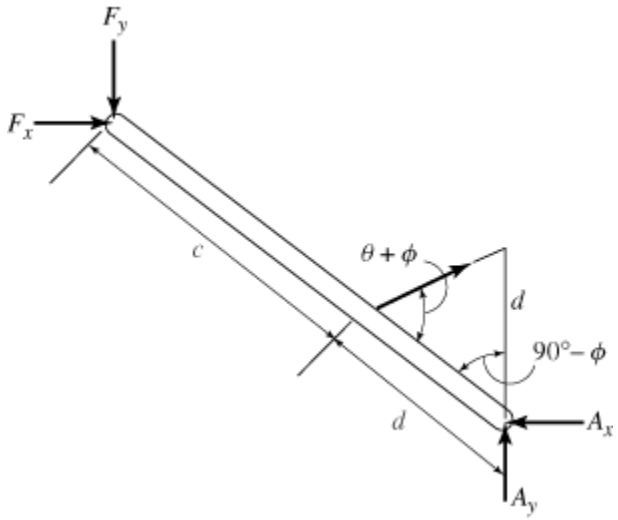
$$F_y = \frac{W}{2} + F_{ED}\cdot\sin(\phi) \qquad F_y = 10.774\,\text{lb}$$

Member *FBA*:

$$\circlearrowleft \Sigma M_A = 0; \qquad F_y(c+d)\cos(\phi) - F_x(c+d)\sin(\phi) - F_s\sin(\theta+\phi)d = 0$$

$$F_s = \frac{F_y(c+d)\cos(\phi) - F_x(c+d)\sin(\phi)}{d\sin(\theta+\phi)} \qquad F_s = 17.5\,\text{lb}$$

$$F_s = ks; \qquad F_s = kx \qquad x = \frac{F_s}{k} \qquad x = 4.375\,\text{in}$$



Problem 6-119

If each of the three links of the mechanism has a weight *W*, determine the angle θ for equilibrium.The spring, which always remains horizontal, is unstretched when $\theta = 0°$.

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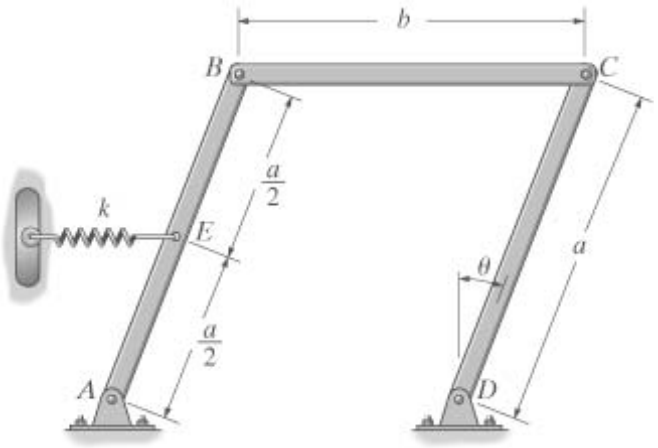
Given:

$W = 25$ lb

$k = 60 \dfrac{\text{lb}}{\text{ft}}$

$a = 4$ ft

$b = 4$ ft

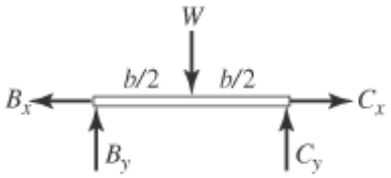


Solution:

Guesses $\theta = 30$ deg

$B_x = 10$ lb $B_y = 10$ lb

$C_x = 10$ lb $C_y = 10$ lb



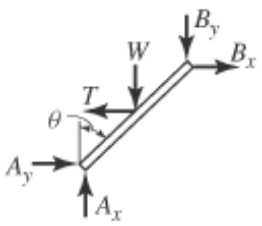
Given

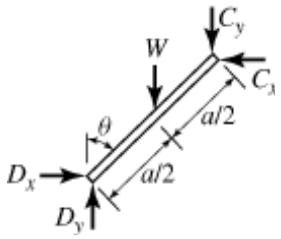
$$-W\left(\frac{a}{2}\right)\sin(\theta) - C_y a\sin(\theta) + C_x a\cos(\theta) = 0$$

$$-W\left(\frac{b}{2}\right) + C_y b = 0$$

$$B_y + C_y - W = 0$$

$$-B_x + C_x = 0$$





$$-B_x a\cos(\theta) - B_y a\sin(\theta) - W\left(\frac{a}{2}\right)\sin(\theta) + k\left(\frac{a}{2}\right)\sin(\theta)\left(\frac{a}{2}\right)\cos(\theta) = 0$$

$$\begin{pmatrix} B_x \\ B_y \\ C_x \\ C_y \\ \theta \end{pmatrix} = \text{Find}\left(B_x, B_y, C_x, C_y, \theta\right) \qquad \begin{pmatrix} B_x \\ B_y \\ C_x \\ C_y \end{pmatrix} = \begin{pmatrix} 16.583 \\ 12.5 \\ 16.583 \\ 12.5 \end{pmatrix} \text{lb} \qquad \theta = 33.6\,\text{deg}$$

Problem 6-120

Determine the required force **P** that must be applied at the blade of the pruning shears so that the blade exerts a normal force F on the twig at E.

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Given:

$F = 20$ lb

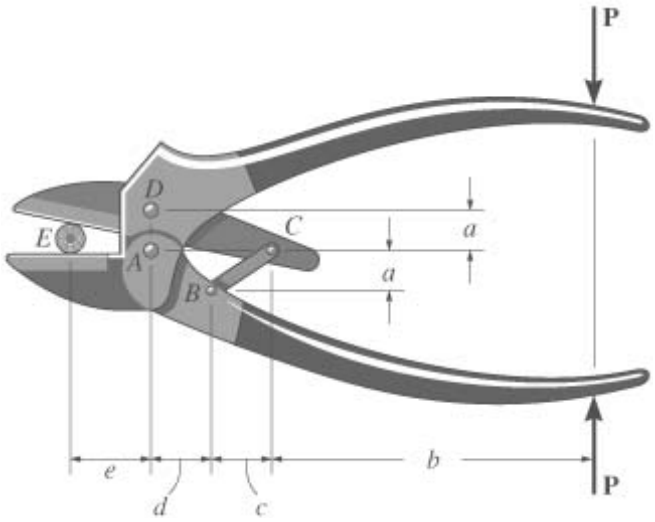
$a = 0.5$ in

$b = 4$ in

$c = 0.75$ in

$d = 0.75$ in

$e = 1$ in

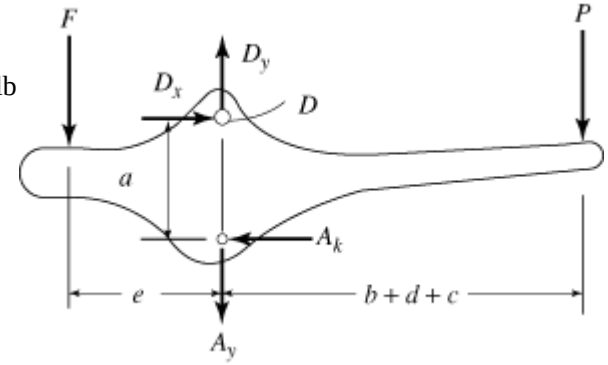


Solution:

initial guesses:

$A_x = 1$ lb $\quad A_y = 1$ lb $\quad D_x = 10$ lb

$D_y = 10$ lb $\quad P = 20$ lb $\quad F_{CB} = 20$ lb



Given

$$-P(b + c + d) - A_x a + F e = 0$$

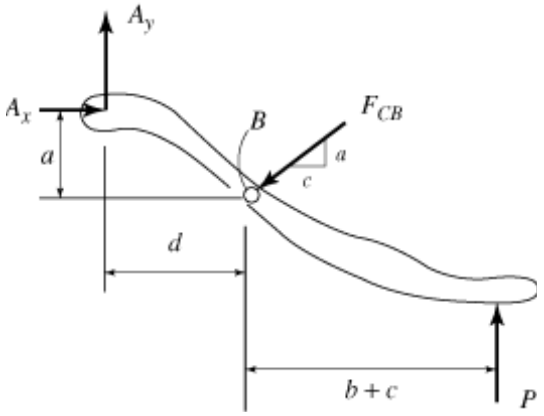
$$D_y - P - A_y - F = 0$$

$$D_x - A_x = 0$$

$$-A_y(d) - A_x(a) + (b + c)P = 0$$

$$A_x - F_{CB}\left(\frac{c}{\sqrt{c^2 + a^2}}\right) = 0$$

$$A_y + P - F_{CB}\left(\frac{a}{\sqrt{a^2 + c^2}}\right) = 0$$



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$$\begin{pmatrix} A_x \\ A_y \\ D_x \\ D_y \\ P \\ F_{CB} \end{pmatrix} = \text{Find}\left(A_x, A_y, D_x, D_y, P, F_{CB}\right) \qquad \begin{pmatrix} A_x \\ A_y \\ D_x \\ D_y \\ F_{CB} \end{pmatrix} = \begin{pmatrix} 13.333 \\ 6.465 \\ 13.333 \\ 28.889 \\ 16.025 \end{pmatrix} \text{lb} \qquad P = 2.424\,\text{lb}$$

Problem 6-121

The three power lines exert the forces shown on the truss joints, which in turn are pin-connected to the poles *AH* and *EG*. Determine the force in the guy cable *AI* and the pin reaction at the support *H*.

Units Used:

$\text{kip} = 10^3\ \text{lb}$

Given:

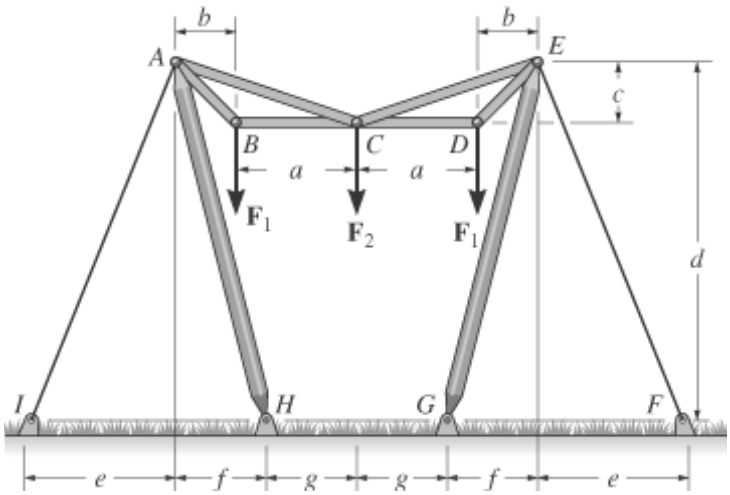
$F_1 = 800\ \text{lb}$ $d = 125\ \text{ft}$

$F_2 = 800\ \text{lb}$ $e = 50\ \text{ft}$

$a = 40\ \text{ft}$ $f = 30\ \text{ft}$

$b = 20\ \text{ft}$ $g = 30\ \text{ft}$

$c = 20\ \text{ft}$

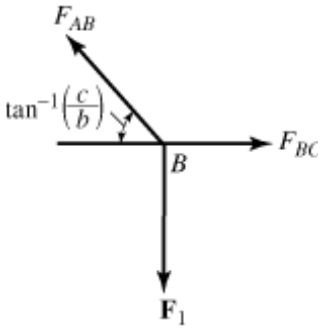


Solution:

AH is a two-force member.

$$\theta = \text{atan}\left(\frac{c}{b}\right) \qquad \phi = \text{atan}\left(\frac{c}{a+b}\right)$$

$$\beta = \text{atan}\left(\frac{d}{f}\right) \qquad \gamma = \text{atan}\left(\frac{e}{d}\right) \qquad \alpha = 90\ \text{deg} - \beta + \gamma$$



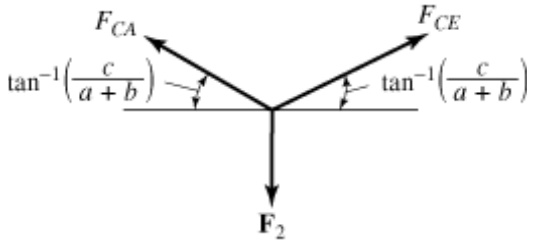
Guesses

$F_{AB} = 1\ \text{lb}$ $F_{BC} = 1\ \text{lb}$ $F_H = 1\ \text{lb}$

$F_{CA} = 1\ \text{lb}$ $F_{AI} = 1\ \text{lb}$

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Given



$$F_{BC} - F_{AB}\cos(\theta) = 0$$

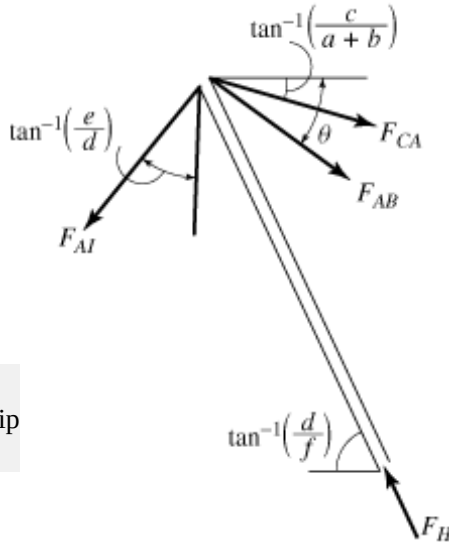
$$F_{AB}\sin(\theta) - F_1 = 0$$

$$2\,F_{CA}\sin(\phi) - F_2 = 0$$

$$-F_{AI}\sin(\alpha) + F_{AB}\sin(\beta - \theta) + F_{CA}\sin(\beta - \phi) = 0$$

$$-F_{AI}\cos(\alpha) - F_{AB}\cos(\beta - \theta) - F_{CA}\cos(\beta - \phi) + F_H = 0$$

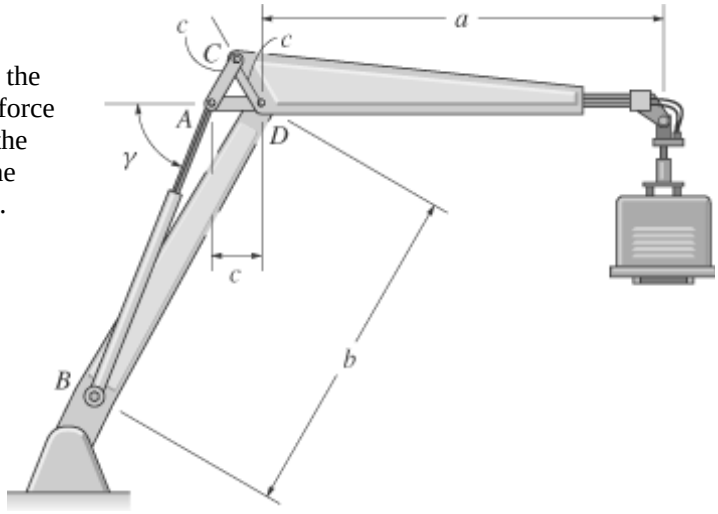
$$\begin{pmatrix} F_{AB} \\ F_{CA} \\ F_{BC} \\ F_{AI} \\ F_H \end{pmatrix} = \text{Find}\left(F_{AB}, F_{CA}, F_{BC}, F_{AI}, F_H\right)$$



$$\begin{pmatrix} F_{AB} \\ F_{CA} \\ F_{BC} \end{pmatrix} = \begin{pmatrix} 1.131 \\ 1.265 \\ 0.8 \end{pmatrix} \text{kip} \qquad \begin{pmatrix} F_{AI} \\ F_H \end{pmatrix} = \begin{pmatrix} 2.881 \\ 3.985 \end{pmatrix} \text{kip}$$

Problem 6-122

The hydraulic crane is used to lift the load of weight W. Determine the force in the hydraulic cylinder AB and the force in links AC and AD when the load is held in the position shown.



Units Used:

$$\text{kip} = 10^3 \text{ lb}$$

Given:

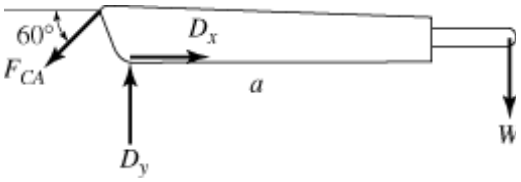
$W = 1400$ lb

$a = 8$ ft $\quad c = 1$ ft

$b = 7$ ft $\quad \gamma = 70$ deg

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Solution:

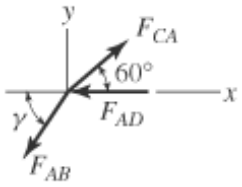


$$\circlearrowleft + \Sigma M_D = 0;$$

$$F_{CA}\sin(60\ \text{deg})\,c - W a = 0$$

$$F_{CA} = \frac{W a}{\sin(60\ \text{deg})\,c} \qquad F_{CA} = 12.9\,\text{kip}$$

$$+\uparrow \Sigma F_y = 0; \quad F_{CA}\sin(60\ \text{deg}) - F_{AB}\sin(\gamma) = 0$$



$$F_{AB} = F_{CA}\frac{\sin(60\ \text{deg})}{\sin(\gamma)} \qquad F_{AB} = 11.9\,\text{kip}$$

$$\rightarrow + \Sigma F_x = 0; \quad -F_{AB}\cos(\gamma) + F_{CA}\cos(60\ \text{deg}) - F_{AD} = 0$$

$$F_{AD} = -F_{AB}\cos(\gamma) + F_{CA}\cos(60\ \text{deg}) \qquad F_{AD} = 2.39\,\text{kip}$$

Problem 6-123

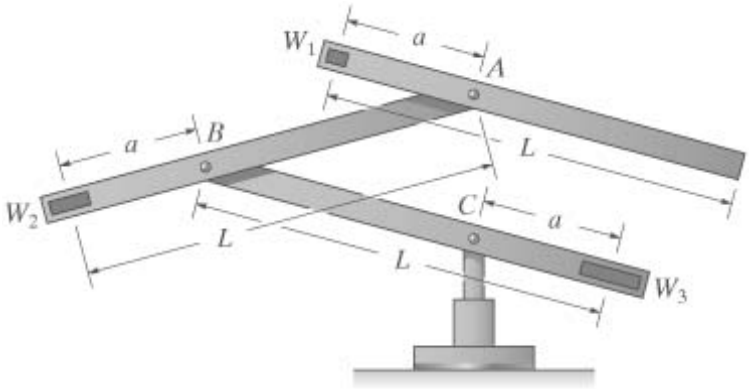
The kinetic sculpture requires that each of the three pinned beams be in perfect balance at all times during its slow motion. If each member has a uniform weight density γ and length L, determine the necessary counterweights W_1, W_2 and W_3 which must be added to the ends of each member to keep the system in balance for any position. Neglect the size of the counterweights.

Given:

$$\gamma = 2\ \frac{\text{lb}}{\text{ft}}$$

$$L = 3\ \text{ft}$$

$$a = 1\ \text{ft}$$

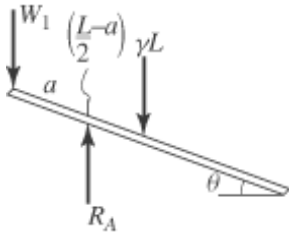


Solution:

$$\circlearrowleft + \Sigma M_A = 0; \qquad W_1 a\cos(\theta) - \gamma L\cos(\theta)\left(\frac{L}{2} - a\right) = 0$$

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$$W_1 = \frac{\gamma L\left(\frac{L}{2} - a\right)}{a} \qquad W_1 = 3\,\text{lb}$$

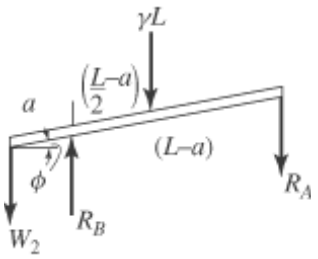


$$+\uparrow \Sigma F_y = 0; \qquad R_A - W_1 - \gamma L = 0$$

$$R_A = W_1 + \gamma L \qquad R_A = 9\,\text{lb}$$

$$\curvearrowleft + \Sigma M_B = 0;$$

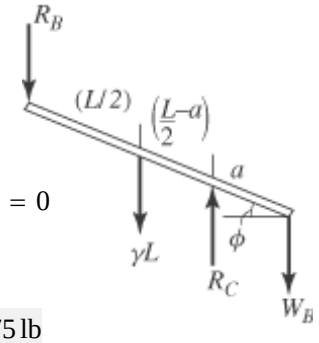
$$W_2\, a\cos(\phi) - \gamma L\cos(\phi)\left(\frac{L}{2} - a\right) - R_A(L - a)\cos(\phi) = 0$$



$$W_2 = \frac{\gamma L\left(\frac{L}{2} - a\right) + R_A(L - a)}{a} \qquad W_2 = 21\,\text{lb}$$

$$+\uparrow \Sigma F_y = 0; \quad R_B - W_2 - R_A - \gamma L = 0$$

$$R_B = W_2 + R_A + \gamma L \qquad R_B = 36\,\text{lb}$$



$$\curvearrowleft + \Sigma M_C = 0; \quad R_B(L - a)\cos(\phi) + \gamma L\left(\frac{L}{2} - a\right)\cos(\phi) - W_3\, a\cos(\phi) = 0$$

$$W_3 = \frac{R_B(L - a) + \gamma L\left(\frac{L}{2} - a\right)}{a} \qquad W_3 = 75\,\text{lb}$$

Problem 6-124

The three-member frame is connected at its ends using ball-and-socket joints. Determine the *x, y, z* components of reaction at *B* and the tension in member *ED*. The force acting at *D* is **F**.

Given:

$$\mathbf{F} = \begin{pmatrix} 135 \\ 200 \\ -180 \end{pmatrix} \text{lb}$$

$a = 6$ ft $\quad e = 3$ ft

$b = 4$ ft $\quad f = 1$ ft

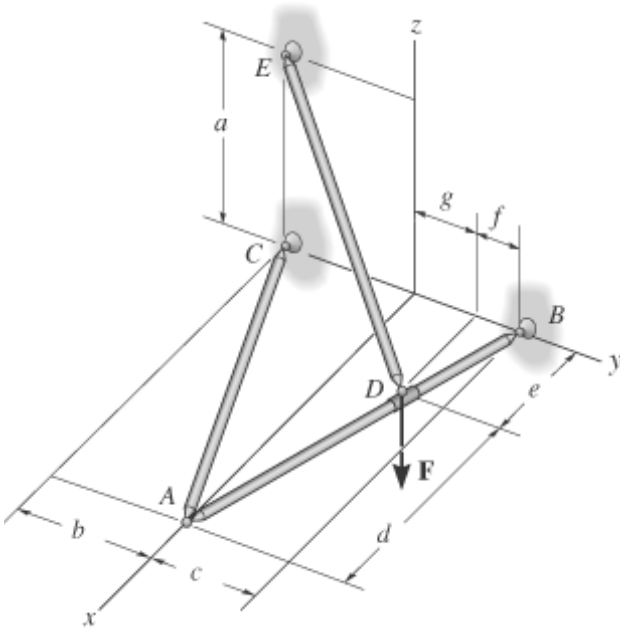
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$d = 6$ ft $\quad g = 2$ ft

$c = g + f$

Solution:

AC and *DE* are two-force members.

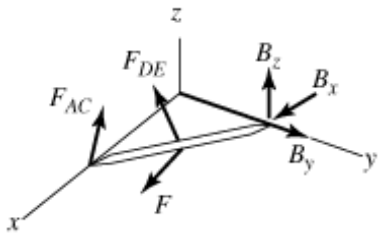


Define some vectors

$$\mathbf{r_{DE}} = \begin{pmatrix} -e \\ -b-g \\ a \end{pmatrix} \qquad \mathbf{u_{DE}} = \frac{\mathbf{r_{DE}}}{\left|\mathbf{r_{DE}}\right|}$$

$$\mathbf{r_{AC}} = \begin{pmatrix} -d-e \\ -b \\ 0 \end{pmatrix} \qquad \mathbf{u_{AC}} = \frac{\mathbf{r_{AC}}}{\left|\mathbf{r_{AC}}\right|}$$

$$\mathbf{r_{BD}} = \begin{pmatrix} e \\ -f \\ 0 \end{pmatrix} \qquad \mathbf{r_{BA}} = \begin{pmatrix} e+d \\ -c \\ 0 \end{pmatrix}$$



Guesses

$B_x = 1$ lb $\quad B_y = 1$ lb $\quad B_z = 1$ lb $\quad F_{DE} = 1$ lb $\quad F_{AC} = 1$ lb

Given

$$\begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} + F_{DE}\,\mathbf{u_{DE}} + F_{AC}\mathbf{u_{AC}} + \mathbf{F} = \mathbf{0} \qquad \mathbf{r_{BD}} \times \left(F_{DE}\mathbf{u_{DE}} + \mathbf{F}\right) + \mathbf{r_{BA}} \times \left(F_{AC}\mathbf{u_{AC}}\right) = \mathbf{0}$$

$$\begin{pmatrix} B_x \\ B_y \\ B_z \\ F_{DE} \\ F_{AC} \end{pmatrix} = \text{Find}\left(B_x, B_y, B_z, F_{DE}, F_{AC}\right) \qquad \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{pmatrix} -30 \\ -13.333 \\ 3.039 \times 10^{-12} \end{pmatrix} \begin{pmatrix} F_{DE} \\ F_{AC} \end{pmatrix} = \begin{pmatrix} 270 \\ 16.415 \end{pmatrix} \text{lb}$$

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Problem 6-125

The four-member "A" frame is supported at *A* and *E* by smooth collars and at *G* by a pin. All the other joints are ball-and-sockets. If the pin at *G* will fail when the resultant force there is F_{max}, determine the largest vertical force *P* that can be supported by the frame. Also, what are the *x, y, z* force components which member *BD* exerts on members *EDC* and *ABC*? The collars at *A* and *E* and the pon at *G* only exert force components on the frame.

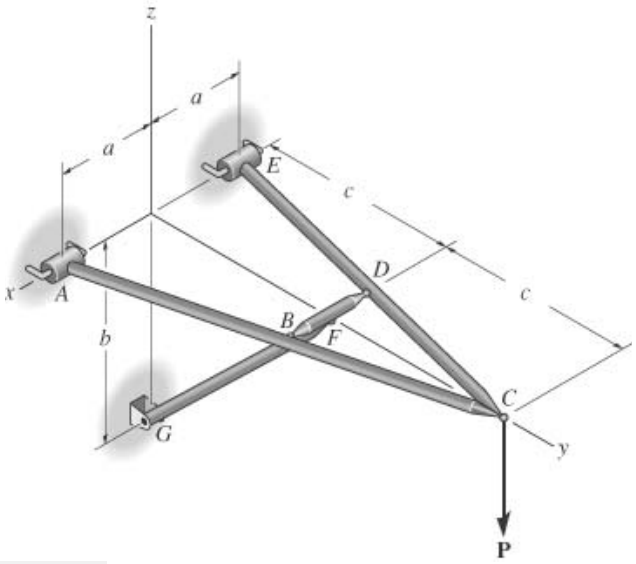
Given:

$F_{max} = 800$ N

$a = 300$ mm

$b = 600$ mm

$c = 600$ mm



Solution:

$\Sigma M_x = 0;$

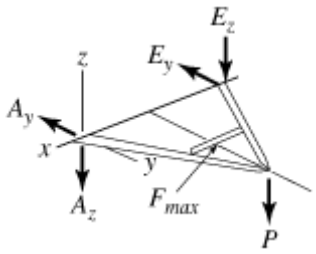
$$-P\,2\,c + \frac{b}{\sqrt{b^2 + c^2}} F_{max} c = 0$$

$$P = \frac{F_{max} b}{2\sqrt{b^2 + c^2}} \qquad P = 282.843 \text{ N}$$

$$B_z + D_z - F_{max}\frac{c}{\sqrt{b^2 + c^2}} = 0 \qquad D_z = B_z$$

$$B_z = \frac{F_{max} c}{2\sqrt{b^2 + c^2}} \qquad D_z = B_z \qquad B_z = 283 \text{ N}$$

$D_z = 283$ N



$$B_y + D_y - F_{max}\frac{b}{\sqrt{b^2 + c^2}} = 0 \qquad D_y = B_y$$

$$B_y = \frac{F_{max} b}{2\sqrt{b^2 + c^2}} \qquad D_y = B_y \qquad B_y = 283 \text{ N} \qquad D_y = 283 \text{ N}$$

$B_x = D_x = 0$

Problem 6-126

The structure is subjected to the loading shown. Member *AD* is supported by a cable *AB* and a roller at *C* and fits through a smooth circular hole at *D.* Member *ED* is supported by a roller at

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D and a pole that fits in a smooth snug circular hole at *E*. Determine the *x*, *y*, *z* components of reaction at *E* and the tension in cable *AB*.

Units Used:

$$\text{kN} = 10^3 \text{ N}$$

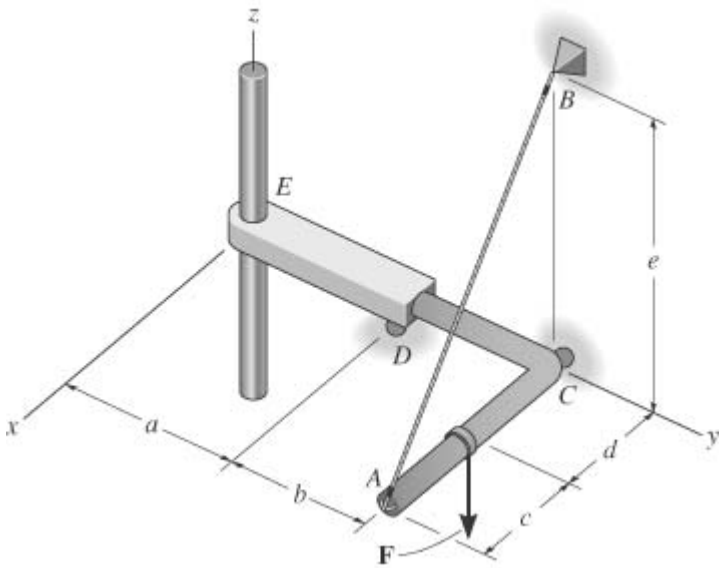
Given:

$$\mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ -2.5 \end{pmatrix} \text{kN}$$

$a = 0.5$ m $\quad d = 0.3$ m

$b = 0.4$ m $\quad e = 0.8$ m

$c = 0.3$ m



Solution: $AB = \begin{pmatrix} -c-d \\ 0 \\ e \end{pmatrix}$

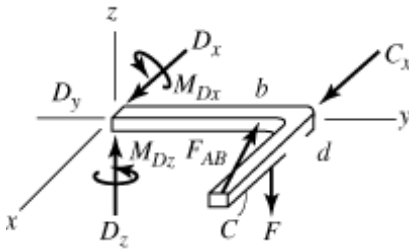
Guesses

$F_{AB} = 1$ kN $\quad D_x = 1$ kN $\quad D_z = 1$ kN

$D_{z2} = 1$ kN $\quad E_x = 1$ kN $\quad E_y = 1$ kN

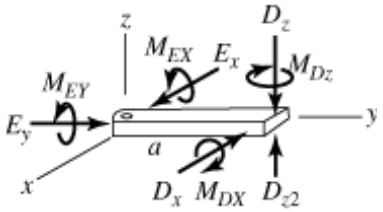
$M_{Dx} = 1$ kN·m $\quad M_{Dz} = 1$ kN·m $\quad C_x = 1$ kN

$M_{Ex} = 1$ kN·m $\quad M_{Ey} = 1$·kN m



Given

$$\mathbf{F} + F_{AB}\frac{AB}{|AB|} + \begin{pmatrix} C_x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} D_x \\ 0 \\ D_z \end{pmatrix} = \mathbf{0}$$



$$\begin{pmatrix} M_{Dx} \\ 0 \\ M_{Dz} \end{pmatrix} + \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} \times \begin{pmatrix} C_x \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} d \\ b \\ 0 \end{pmatrix} \times \mathbf{F} + \begin{pmatrix} c+d \\ b \\ 0 \end{pmatrix} \times \left(F_{AB}\frac{AB}{|AB|} \right) = \mathbf{0}$$

$$\begin{pmatrix} -D_x \\ 0 \\ D_{z2} - D_z \end{pmatrix} + \begin{pmatrix} E_x \\ E_y \\ 0 \end{pmatrix} = \mathbf{0} \qquad \begin{pmatrix} -M_{Dx} \\ 0 \\ -M_{Dz} \end{pmatrix} + \begin{pmatrix} M_{Ex} \\ M_{Ey} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} D_x \\ 0 \\ D_z - D_{z2} \end{pmatrix} = \mathbf{0}$$

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$$\begin{pmatrix} C_x \\ D_x \\ D_z \\ D_{z2} \\ E_x \\ E_y \\ F_{AB} \\ M_{Dx} \\ M_{Dz} \\ M_{Ex} \\ M_{Ey} \end{pmatrix} = \text{Find}\left(C_x, D_x, D_z, D_{z2}, E_x, E_y, F_{AB}, M_{Dx}, M_{Dz}, M_{Ex}, M_{Ey}\right)$$

$$\begin{pmatrix} C_x \\ D_x \\ D_z \\ D_{z2} \end{pmatrix} = \begin{pmatrix} 0.937 \\ 0 \\ 1.25 \\ 1.25 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} M_{Dx} \\ M_{Dz} \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} \text{kN}\cdot\text{m}$$

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{kN} \qquad \begin{pmatrix} M_{Ex} \\ M_{Ey} \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} \text{kN}\cdot\text{m} \qquad F_{AB} = 1.562\,\text{kN}$$

Problem 6-127

The structure is subjected to the loadings shown.Member *AB* is supported by a ball-and-socket at *A* and smooth collar at *B*. Member *CD* is supported by a pin at *C*. Determine the *x, y, z* components of reaction at *A* and *C*.

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Given:

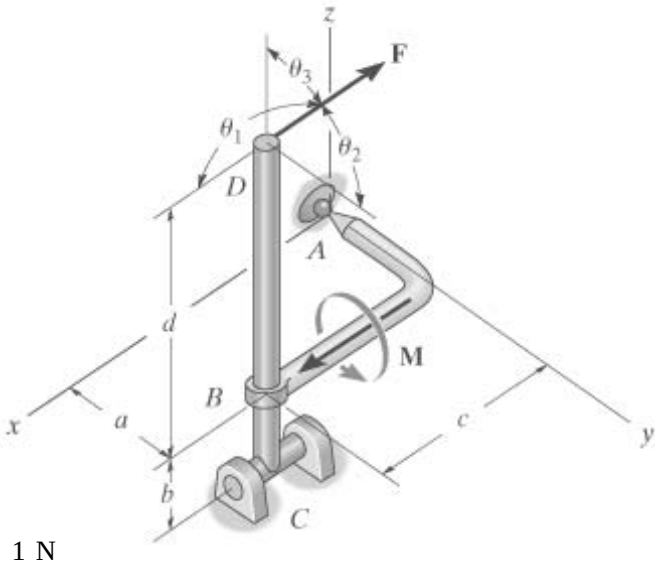
$a = 2$ m $\quad M = 800$ N·m

$b = 1.5$ m $\quad F = 250$ N

$c = 3$ m $\quad \theta_1 = 60$ deg

$d = 4$ m $\quad \theta_2 = 45$ deg

$\theta_3 = 60$ deg



Solution:

Guesses

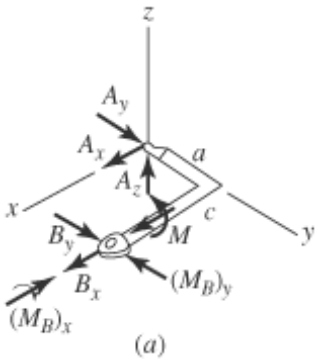
$B_x = 1$ N $\quad B_y = 1$ N

$A_x = 1$ N $\quad A_y = 1$ N $\quad A_z = 1$ N

$C_x = 1$ N $\quad C_y = 1$ N $\quad C_z = 1$ N

$M_{Bx} = 1$ N·m $\quad M_{By} = 1$ N·m

$M_{Cy} = 1$ N·m $\quad M_{Cz} = 1$ N·m



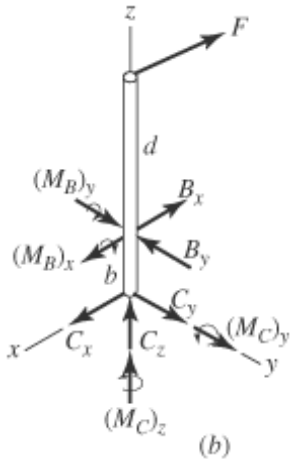
(a)

Given

$$\begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} + \begin{pmatrix} B_x \\ B_y \\ 0 \end{pmatrix} = \mathbf{0}$$

$$\begin{pmatrix} c \\ a \\ 0 \end{pmatrix} \times \begin{pmatrix} B_x \\ B_y \\ 0 \end{pmatrix} + \begin{pmatrix} M \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} -M_{Bx} \\ -M_{By} \\ 0 \end{pmatrix} = \mathbf{0}$$

$$F\begin{pmatrix} \cos(\theta_1) \\ \cos(\theta_2) \\ \cos(\theta_3) \end{pmatrix} + \begin{pmatrix} -B_x \\ -B_y \\ 0 \end{pmatrix} + \begin{pmatrix} C_x \\ C_y \\ C_z \end{pmatrix} = \mathbf{0}$$



(b)

$$\begin{pmatrix} 0 \\ 0 \\ b+d \end{pmatrix} \times \left[F\begin{pmatrix} \cos(\theta_1) \\ \cos(\theta_2) \\ \cos(\theta_3) \end{pmatrix} \right] + \begin{pmatrix} 0 \\ 0 \\ b \end{pmatrix} \times \begin{pmatrix} -B_x \\ -B_y \\ 0 \end{pmatrix} + \begin{pmatrix} M_{Bx} \\ M_{By} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ M_{Cy} \\ M_{Cz} \end{pmatrix} = \mathbf{0}$$

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$$\begin{pmatrix} A_x \\ A_y \\ A_z \\ C_x \\ C_y \\ C_z \\ B_x \\ B_y \\ M_{Bx} \\ M_{By} \\ M_{Cy} \\ M_{Cz} \end{pmatrix} = \text{Find}\left(A_x, A_y, A_z, C_x, C_y, C_z, B_x, B_y, M_{Bx}, M_{By}, M_{Cy}, M_{Cz}\right)$$

$$\begin{pmatrix} A_x \\ A_y \\ A_z \\ C_x \\ C_y \\ C_z \end{pmatrix} = \begin{pmatrix} -172.3 \\ -114.8 \\ 0 \\ 47.3 \\ -61.9 \\ -125 \end{pmatrix} \text{N} \qquad \begin{pmatrix} M_{Cy} \\ M_{Cz} \end{pmatrix} = \begin{pmatrix} -429 \\ 0 \end{pmatrix} \text{N}\cdot\text{m}$$

Problem 6-128

Determine the resultant forces at pins *B* and *C* on member *ABC* of the four-member frame.

Given:

$w = 150 \ \frac{\text{lb}}{\text{ft}}$

$a = 5 \text{ ft}$

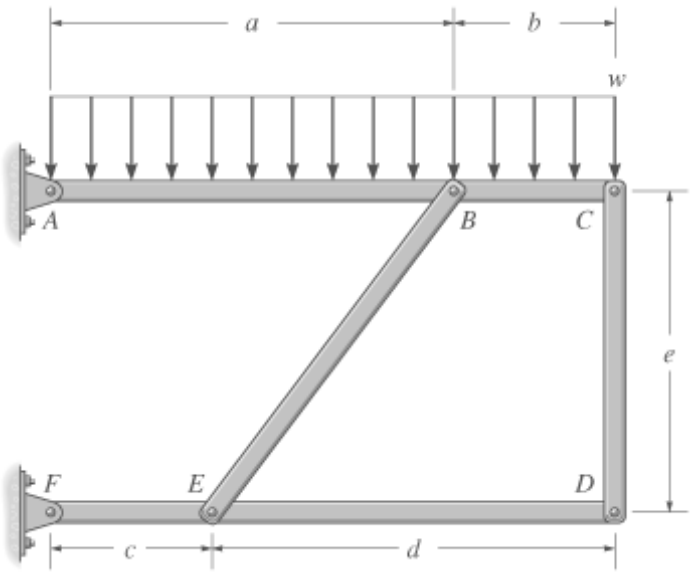
$b = 2 \text{ ft}$

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$c = 2$ ft

$e = 4$ ft

$d = a + b - c$



Solution:

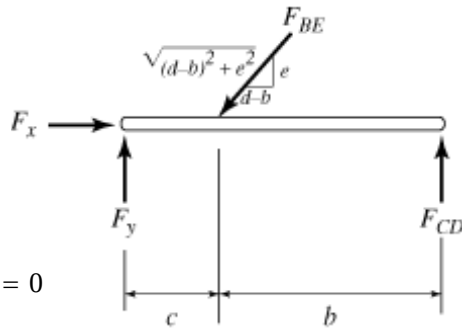
The initial guesses are

$F_{CD} = 20$ lb $\quad F_{BE} = 40$ lb

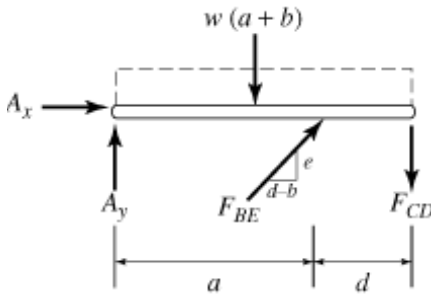
Given

$$F_{CD}(c + d) - F_{BE}\frac{e\,c}{\sqrt{(d - b)^2 + e^2}} = 0$$

$$-w(a + b)\left(\frac{a + b}{2}\right) + \frac{F_{BE}\,e}{\sqrt{(d - b)^2 + e^2}}a - F_{CD}(a + b) = 0$$



$$\begin{pmatrix} F_{CD} \\ F_{BE} \end{pmatrix} = \text{Find}(F_{CD}, F_{BE}) \qquad \begin{pmatrix} F_{CD} \\ F_{BE} \end{pmatrix} = \begin{pmatrix} 350 \\ 1531 \end{pmatrix} \text{lb}$$



Problem 6-129

The mechanism consists of identical meshed gears *A* and *B* and arms which are fixed to the gears. The spring attached to the ends of the arms has an unstretched length δ and a stiffness *k*. If a torque *M* is applied to gear *A*, determine the angle θ through which each arm rotates. The gears are each pinned to fixed supports at their centers.

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Given:

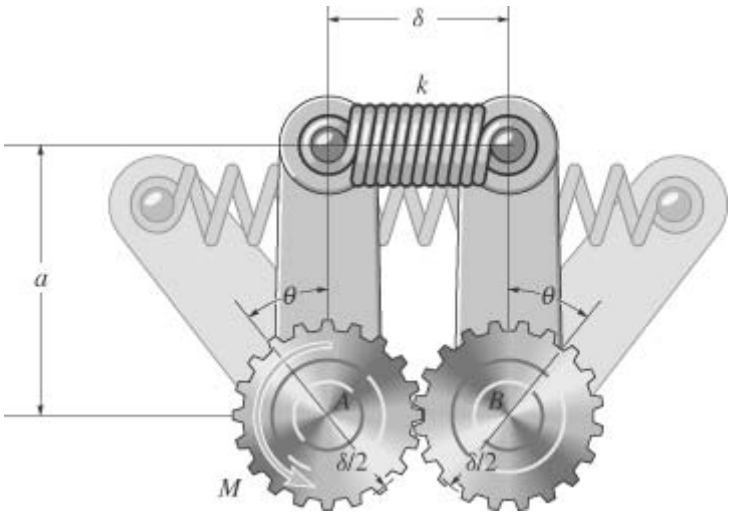
$\delta = 100\ \text{mm}$

$k = 250\ \frac{\text{N}}{\text{m}}$

$M = 6\ \text{N}\cdot\text{m}$

$r = \frac{\delta}{2}$

$a = 150\ \text{mm}$



Solution:

$\circlearrowleft + \quad \Sigma M_A = 0; \qquad -F\,r - P\,a\cos(\theta) + M = 0$

$\circlearrowright + \quad \Sigma M_B = 0; \qquad P\,a\cos(\theta) - F\,r = 0$

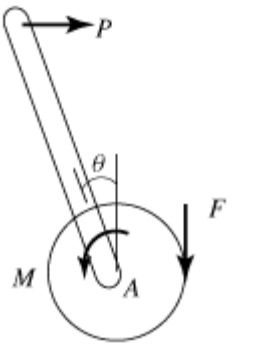
$2P\,a\cos(\theta) = M$

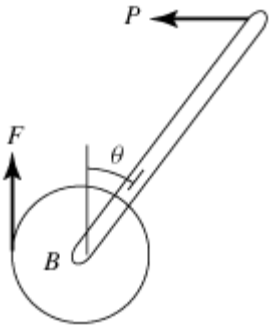
$2k(2a)\sin(\theta)a\cos(\theta) = M$

$2k\,a^2\sin(2\theta) = M$

$\theta = \frac{1}{2}\,\text{asin}\left(\frac{M}{2\,k\,a^2}\right)$

$\theta = 16.1\,\text{deg}$





Problem 6-130

Determine the force in each member of the truss and state if the members are in tension or compression.

Units Used:

kN = 1000 N

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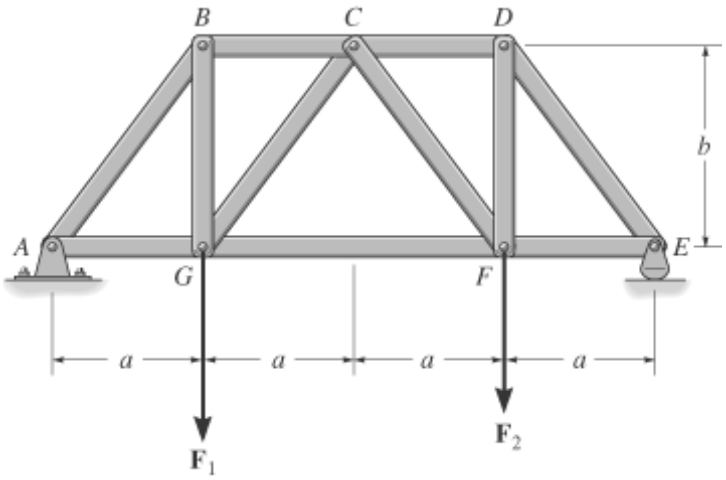
Given:

$F_1 = 20$ kN

$F_2 = 10$ kN

$a = 1.5$ m

$b = 2$ m



Solution:

$\theta = \text{atan}\left(\frac{b}{a}\right)$

Guesses $F_{AG} = 1$ kN $F_{BG} = 1$ kN

$F_{GC} = 1$ kN $F_{GF} = 1$ kN $F_{AB} = 1$ kN

$F_{BC} = 1$ kN $F_{CD} = 1$ kN $F_{CF} = 1$ kN

$F_{DF} = 1$ kN $F_{DE} = 1$ kN $F_{EF} = 1$ kN

Given

$-F_{AB}\cos(\theta) + F_{BC} = 0$

$-F_{AB}\sin(\theta) - F_{BG} = 0$

$F_{GC}\cos(\theta) + F_{GF} - F_{AG} = 0$

$F_{GC}\sin(\theta) + F_{BG} - F_1 = 0$

$-F_{BC} + F_{CD} - F_{GC}\cos(\theta) + F_{CF}\cos(\theta) = 0$

$-F_{GC}\sin(\theta) - F_{CF}\sin(\theta) = 0$

$-F_{CD} + F_{DE}\cos(\theta) = 0$

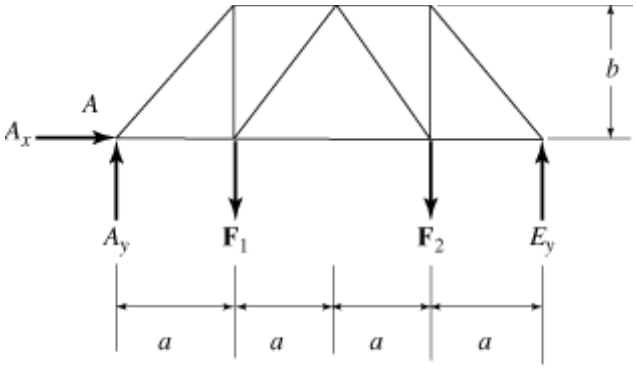
$-F_{DF} - F_{DE}\sin(\theta) = 0$

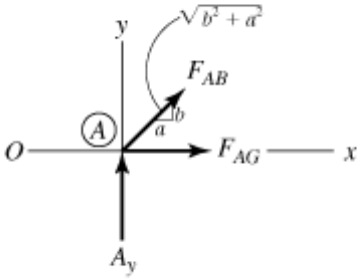
$-F_{GF} - F_{CF}\cos(\theta) + F_{EF} = 0$

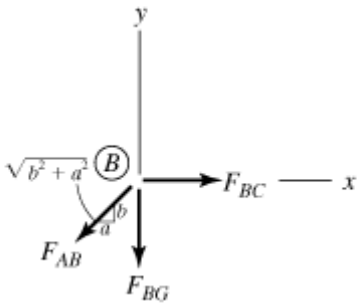
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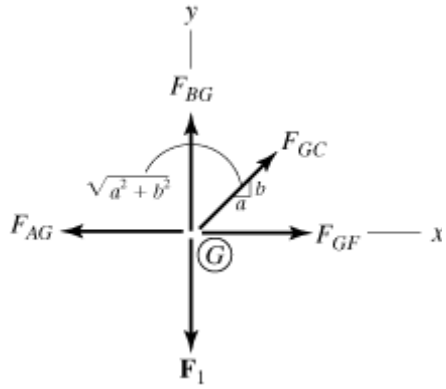
$$F_{DF} + F_{CF}\sin(\theta) - F_2 = 0$$

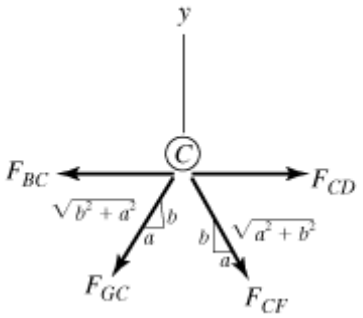
$$-F_{DE}\cos(\theta) - F_{EF} = 0$$

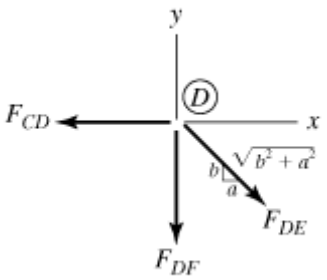


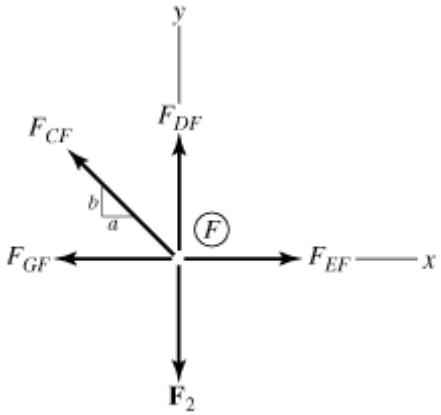












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$$\begin{pmatrix} F_{AG} \\ F_{BG} \\ F_{GC} \\ F_{GF} \\ F_{AB} \\ F_{BC} \\ F_{CD} \\ F_{CF} \\ F_{DF} \\ F_{DE} \\ F_{EF} \end{pmatrix} = \text{Find}\left(F_{AG}, F_{BG}, F_{GC}, F_{GF}, F_{AB}, F_{BC}, F_{CD}, F_{CF}, F_{DF}, F_{DE}, F_{EF}\right)$$

$$\begin{pmatrix} F_{AG} \\ F_{BG} \\ F_{GC} \\ F_{GF} \\ F_{AB} \\ F_{BC} \\ F_{CD} \\ F_{CF} \\ F_{DF} \\ F_{DE} \\ F_{EF} \end{pmatrix} = \begin{pmatrix} 13.13 \\ 17.50 \\ 3.13 \\ 11.25 \\ -21.88 \\ -13.13 \\ -9.37 \\ -3.13 \\ 12.50 \\ -15.62 \\ 9.37 \end{pmatrix} \text{kN}$$

Positive (T)
Negative (C)

Problem 6-131

The spring has an unstretched length δ. Determine the angle θ for equilibrium if the uniform links each have a mass m_{link}.

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Given:

$m_{link} = 5\ \text{kg}$

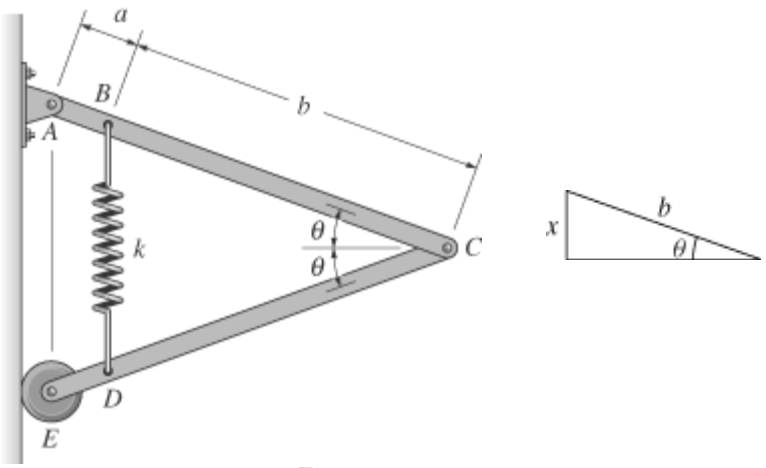
$\delta = 0.3\ \text{m}$

$k = 400\ \frac{\text{N}}{\text{m}}$

$a = 0.1\ \text{m}$

$b = 0.6\ \text{m}$

$g = 9.81\ \frac{\text{m}}{\text{s}^2}$



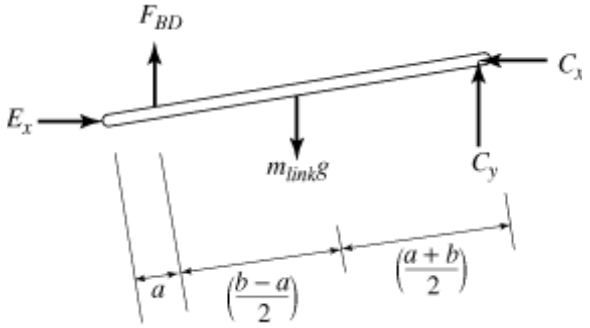
Solution:

Guesses

$\theta = 10\ \text{deg}$

$F_{BD} = 1\ \text{N}$

$E_x = 1\ \text{N}$



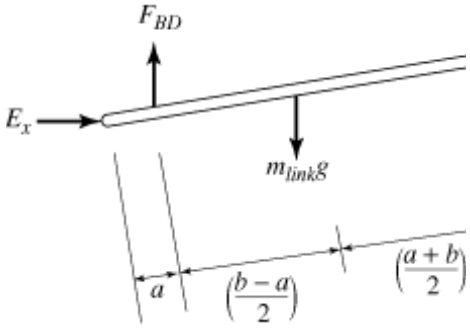
Given

$$m_{link}\, g\, \frac{a+b}{2}\cos(\theta) - F_{BD}\, b\cos(\theta) + E_x\, b\sin(\theta) = 0$$

$$-2\ m_{link}\, g\, \frac{a+b}{2}\cos(\theta) + E_x\ 2\ b\sin(\theta) = 0$$

$$F_{BD} = k(2\ b\sin(\theta) - \delta)$$

$$\begin{pmatrix} F_{BD} \\ E_x \\ \theta \end{pmatrix} = \text{Find}(F_{BD}, E_x, \theta) \qquad \theta = 21.7\ \text{deg}$$



Problem 6-132

The spring has an unstretched length δ. Determine the mass m_{link} of each uniform link if the angle for equilibrium is θ.

Given:

$\delta = 0.3\ \text{m}$

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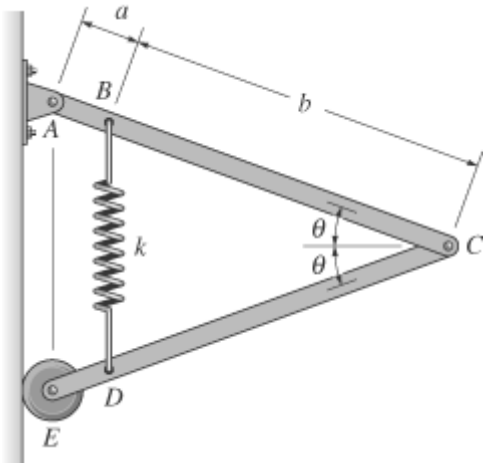
$\theta = 20 \text{ deg}$

$k = 400 \dfrac{\text{N}}{\text{m}}$

$a = 0.1 \text{ m}$

$b = 0.6 \text{ m}$

$g = 9.81 \dfrac{\text{m}}{\text{s}^2}$



Solution:

Guesses $E_y = 1 \text{ N}$ $m_{link} = 1 \text{ kg}$

$F_s = 1 \text{ N}$

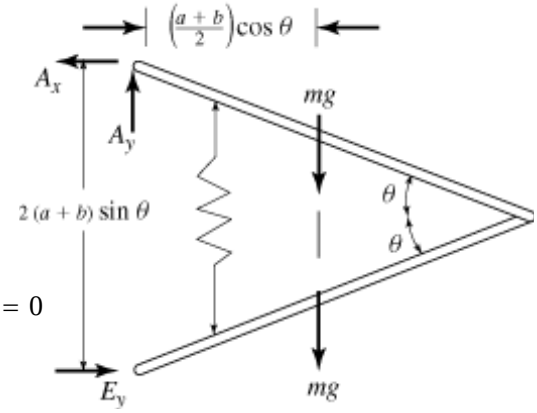
Given

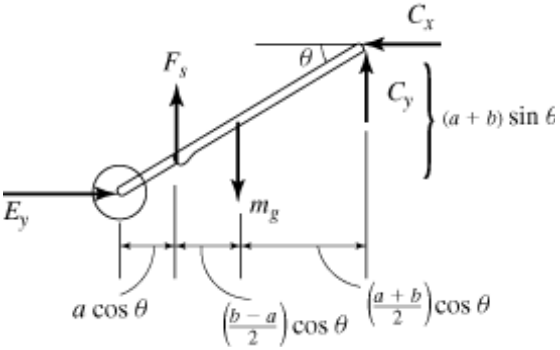
$F_s = (2\, b \sin(\theta) - \delta) k$

$m_{link}\, g \dfrac{a+b}{2} \cos(\theta) - F_s\, b \cos(\theta) + E_y\, b \sin(\theta) = 0$

$-2 m_{link}\, g \dfrac{a+b}{2} \cos(\theta) + E_y 2b \sin(\theta) = 0$

$\begin{pmatrix} m_{link} \\ F_s \\ E_y \end{pmatrix} = \text{Find}(m_{link}, F_s, E_y)$





$m_{link} = 3.859 \text{ kg}$

$\dfrac{y}{2(b)} = \sin(\theta)$

$y = 2\, b \sin(\theta)$

$F_s = (y - \delta)(k)$

$F_s = 44.17 \text{ N}$

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$\circlearrowleft + \Sigma M_A = 0;$ $\left[E_y(2)(a+b)\sin(\theta) - 2(M)(g)\right]\frac{a+b}{2}(\cos(\theta)) = 0$

$\left[E_y(2)\sin(\theta) - 2\,(m)\,\frac{g}{2}\right](\cos(\theta) = 0$

$\Rightarrow\ E_y(2\,\sin(\theta)) = m(g)(\cos(\theta))$

$E_y = \frac{m(g)(\cos(\theta))}{2\,\sin(\theta)}$

$\circlearrowleft + \Sigma M_C = 0;$ $\frac{m(g)(\cos(\theta))}{2\,\sin(\theta)}(a+b)\sin(\theta) + m(g)\left(\frac{a+b}{2}\right)\cos(\theta) - F_S(b\cos(\theta)) = 0$

$m = F_S\,\frac{b}{g\,(a+b)}$

$m = 3.859$ kg

Problem 6-133

Determine the horizontal and vertical components of force that the pins *A* and *B* exert on the two-member frame.

Given:

$w = 400\ \frac{\text{N}}{\text{m}}$

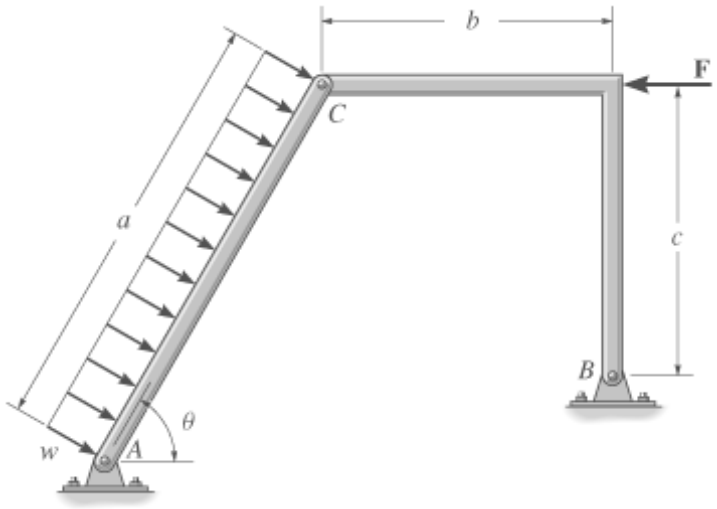
$a = 1.5$ m

$b = 1$ m

$c = 1$ m

$F = 0$ N

$\theta = 60$ deg



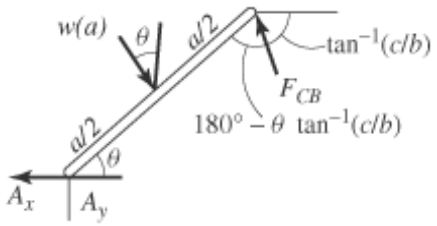
Solution:

Guesses

$A_x = 1$ N $\quad A_y = 1$ N $\quad B_x = 1$ N

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$B_y = 1$ N $\quad C_x = 1$ N $\quad C_y = 1$ N



Given

$$-w a \frac{a}{2} + C_x a \sin(\theta) - C_y a \cos(\theta) = 0$$

$$F c - C_x c - C_y b = 0 \qquad A_y - C_y - w a \cos(\theta) = 0$$

$$-A_x - C_x + w a \sin(\theta) = 0 \qquad C_x - B_x - F = 0$$

$$C_y + B_y = 0$$

$$\begin{pmatrix} A_x \\ A_y \\ B_x \\ B_y \\ C_x \\ C_y \end{pmatrix} = \text{Find}(A_x, A_y, B_x, B_y, C_x, C_y) \qquad \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 300.0 \\ 80.4 \end{pmatrix} \text{N} \qquad \begin{pmatrix} B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 220 \\ 220 \end{pmatrix} \text{N}$$

Problem 6-134

Determine the horizontal and vertical components of force that the pins *A* and *B* exert on the two-member frame.

Given:

$w = 400 \ \frac{\text{N}}{\text{m}}$

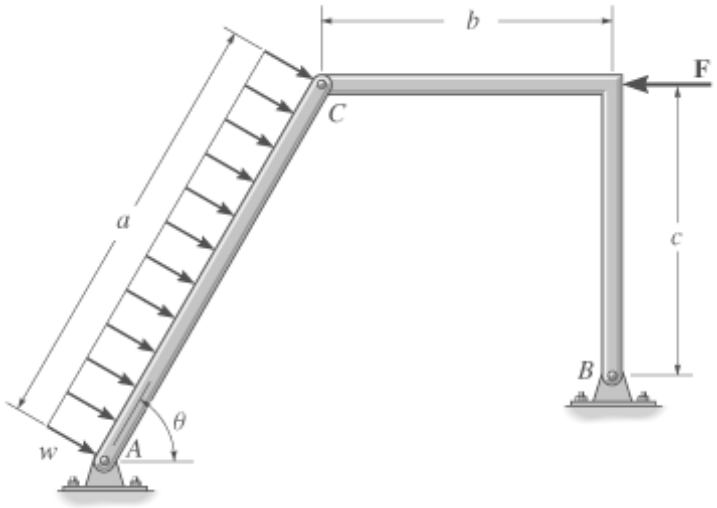
$a = 1.5$ m

$b = 1$ m

$c = 1$ m

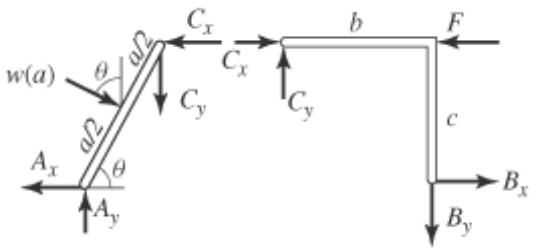
$F = 500$ N

$\theta = 60$ deg



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Solution:



Guesses

$A_x = 1\ \text{N}$ $A_y = 1\ \text{N}$ $B_x = 1\ \text{N}$

$B_y = 1\ \text{N}$ $C_x = 1\ \text{N}$ $C_y = 1\ \text{N}$

Given

$$-w a \frac{a}{2} + C_x a \sin(\theta) - C_y a \cos(\theta) = 0 \qquad A_y - C_y - w a \cos(\theta) = 0$$

$$F c - C_x c - C_y b = 0 \qquad C_x + B_x - F = 0$$

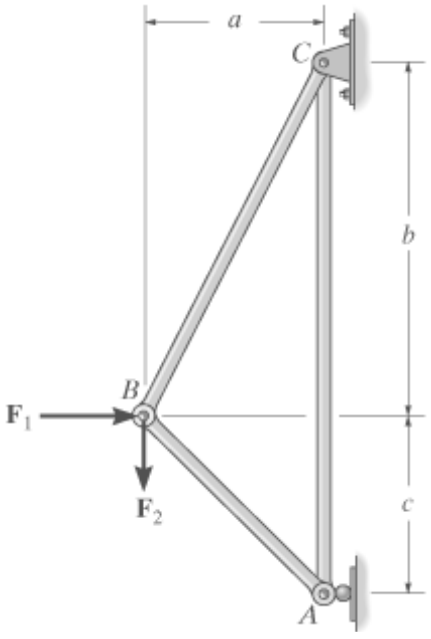
$$-A_x - C_x + w a \sin(\theta) = 0 \qquad C_y - B_y = 0$$

$$\begin{pmatrix} A_x \\ A_y \\ B_x \\ B_y \\ C_x \\ C_y \end{pmatrix} = \text{Find}(A_x, A_y, B_x, B_y, C_x, C_y)$$

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 117.0 \\ 397.4 \end{pmatrix} \text{N} \qquad \begin{pmatrix} B_x \\ B_y \end{pmatrix} = \begin{pmatrix} 97.4 \\ 97.4 \end{pmatrix} \text{N}$$

Problem 6-135

Determine the force in each member of the truss and indicate whether the members are in tension or compression.



Units Used:

kip = 1000 lb

Given:

$F_1 = 1000\ \text{lb}$ $b = 8\ \text{ft}$

$F_2 = 500\ \text{lb}$ $c = 4\ \text{ft}$

$a = 4\ \text{ft}$

Solution:

Joint B:

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Initial Guesses:

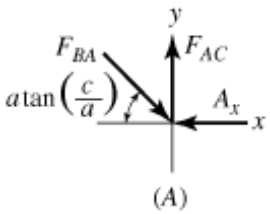
Given

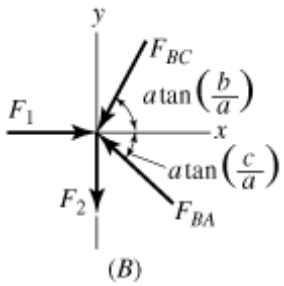
$F_{BC} = 100 \text{ lb} \quad F_{BA} = 150 \text{ lb}$

$\Sigma F_x = 0; \quad F_1 - F_{BC}\cos\left(\text{atan}\left(\frac{b}{a}\right)\right) - F_{BA}\cos\left(\text{atan}\left(\frac{c}{a}\right)\right) = 0$

$\Sigma F_y = 0; \quad -F_{BC}\sin\left(\text{atan}\left(\frac{b}{a}\right)\right) + F_{BA}\sin\left(\text{atan}\left(\frac{c}{a}\right)\right) - F_2 = 0$

$\begin{pmatrix} F_{BC} \\ F_{BA} \end{pmatrix} = \text{Find}\left(F_{BC}, F_{BA}\right)$





$F_{BC} = 373 \text{ lb (C)}$

$F_{BA} = 1178.51 \text{ lb}$

$F_{BA} = 1.179 \text{ kip (C)}$

Joint *A*:

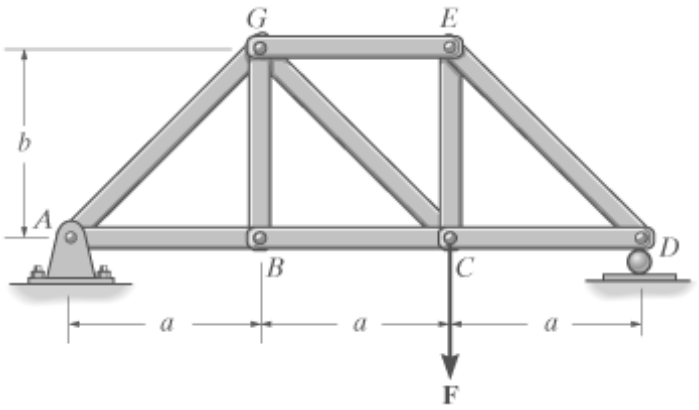
$\Sigma F_y = 0; \quad F_{AC} - \left(F_{BA}\right)\sin\left(\text{atan}\left(\frac{c}{a}\right)\right) = 0$

$$F_{AC} = F_{BA}\frac{c}{a\sqrt{\frac{a^2 + c^2}{a^2}}}$$

$F_{AC} = 833 \text{ lb (T)}$

Problem 6-136

Determine the force in each member of the truss and state if the members are in tension or compression.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$F = 1000 \text{ lb}$

$a = 10 \text{ ft}$

$b = 10 \text{ ft}$

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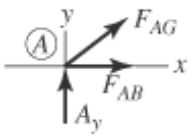
Solution: $\theta = \operatorname{atan}\left(\frac{b}{a}\right)$

Guesses

$F_{AB} = 1\ \text{lb}$ $F_{AG} = 1\ \text{lb}$ $F_{BC} = 1\ \text{lb}$ $F_{BG} = 1\ \text{lb}$

$F_{CD} = 1\ \text{lb}$ $F_{CE} = 1\ \text{lb}$ $F_{CG} = 1\ \text{lb}$ $F_{DE} = 1\ \text{lb}$

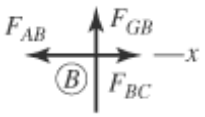
$F_{EG} = 1\ \text{lb}$



Given

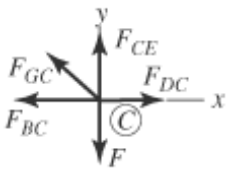
$F_{AB} + F_{AG}\cos(\theta) = 0$

$-F_{AB} + F_{BC} = 0$



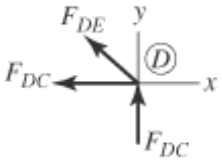
$F_{BG} = 0$

$F_{CD} - F_{BC} - F_{CG}\cos(\theta) = 0$



$F_{CE} - F + F_{CG}\sin(\theta) = 0$

$-F_{CD} - F_{DE}\cos(\theta) = 0$

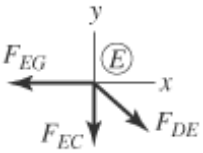


$F_{DE}\cos(\theta) - F_{EG} = 0$

$-F_{CE} - F_{DE}\sin(\theta) = 0$

$F_{EG} - F_{AG}\cos(\theta) + F_{CG}\cos(\theta) = 0$

$-F_{AG}\sin(\theta) - F_{BG} - F_{CG}\sin(\theta) = 0$



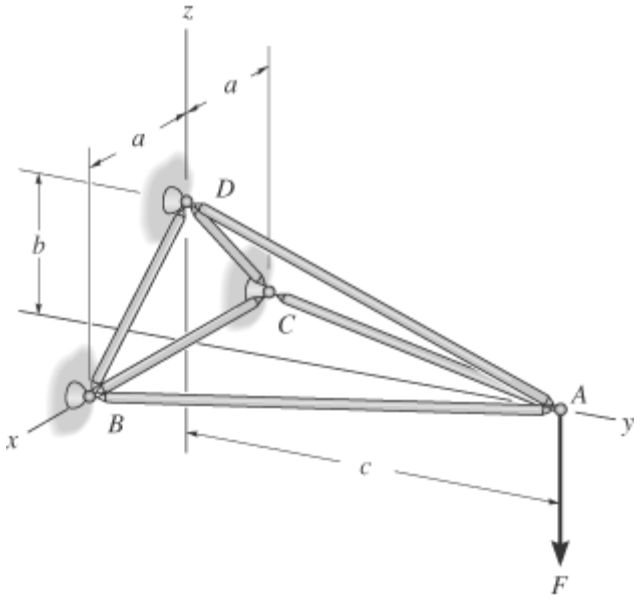
$$\begin{pmatrix} F_{AB} \\ F_{AG} \\ F_{BC} \\ F_{BG} \\ F_{CD} \\ F_{CE} \\ F_{CG} \\ F_{DE} \\ F_{EG} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AG}, F_{BC}, F_{BG}, F_{CD}, F_{CE}, F_{CG}, F_{DE}, F_{EG}\right)$$

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$$\begin{pmatrix} F_{AB} \\ F_{AG} \\ F_{BC} \\ F_{BG} \\ F_{CD} \\ F_{CE} \\ F_{CG} \\ F_{DE} \\ F_{EG} \end{pmatrix} = \begin{pmatrix} 333 \\ -471 \\ 333 \\ 0 \\ 667 \\ 667 \\ 471 \\ -943 \\ -667 \end{pmatrix} \text{lb} \qquad \text{Positive (T), Negative (C)}$$

Problem 6-137

Determine the force in members *AB*, *AD*, and *AC* of the space truss and state if the members are in tension or compression. The force **F** is vertical.



Units Used:

$\text{kip} = 10^3 \text{ lb}$

Given:

$F = 600 \text{ lb}$

$a = 1.5 \text{ ft}$

$b = 2 \text{ ft}$

$c = 8 \text{ ft}$

Solution:

$$\mathbf{AB} = \begin{pmatrix} a \\ -c \\ 0 \end{pmatrix} \qquad \mathbf{AC} = \begin{pmatrix} -a \\ -c \\ 0 \end{pmatrix}$$

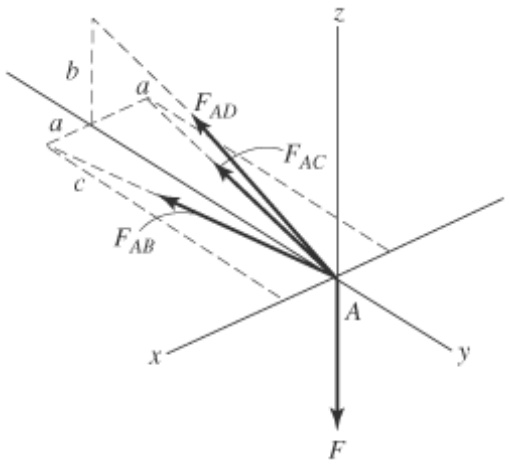
$$\mathbf{AD} = \begin{pmatrix} 0 \\ -c \\ b \end{pmatrix}$$

Guesses $F_{AB} = 1 \text{ lb}$ $F_{AC} = 1 \text{ lb}$ $F_{AD} = 1 \text{ lb}$

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Given $F_{AB}\frac{\mathbf{AB}}{|\mathbf{AB}|} + F_{AC}\frac{\mathbf{AC}}{|\mathbf{AC}|} + F_{AD}\frac{\mathbf{AD}}{|\mathbf{AD}|} + \begin{pmatrix} 0 \\ 0 \\ -F \end{pmatrix} = \mathbf{0}$

$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \text{Find}\left(F_{AB}, F_{AC}, F_{AD}\right)$$



$$\begin{pmatrix} F_{AB} \\ F_{AC} \\ F_{AD} \end{pmatrix} = \begin{pmatrix} -1.221 \\ -1.221 \\ 2.474 \end{pmatrix} \text{kip}$$

Positive (T)
Negative (C)

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